Chapter 1
Preface

Scientific progress has rarely followed a straight line. The history of physics is marked by periods in which accepted explanations eventually gave way to simpler or more complete descriptions once new observations accumulated or familiar ideas were viewed from a different perspective. Newton's mechanics, Maxwell's electrodynamics, Einstein's relativity, and quantum mechanics each emerged by extending rather than rejecting the successful mathematics that came before them.

This work is written in that spirit.

The author is not affiliated with a university or research institute and claims no special authority through academic title or institutional position. The ideas presented here were developed independently through mechanical reasoning, extensive study of the published cosmological literature, and numerical implementation with modern computational tools. The mathematical formulation was refined with the assistance of computational intelligence, while the physical hypothesis, its interpretation, and the organization of the model remain the author's own.

The central premise is straightforward.

General Relativity successfully describes how spacetime responds to stress-energy, and modern observations have confirmed that gravity propagates through gravitational waves. The standard cosmological model reproduces an enormous range of observations, yet it presently relies upon components whose physical nature remains uncertain, including dark matter and dark energy. These components are extraordinarily successful phenomenological descriptions, but they are not themselves direct explanations of the underlying mechanism.

This work asks whether some of the phenomena attributed to those components might instead emerge from a property of gravity itself.

The proposed hypothesis is that gravity possesses an organizational character arising from its wave nature. Like many wave phenomena in physics, gravitational disturbances may admit limiting behavior through interference, dynamic tension, or saturation. If such a limiting process exists, the large-scale evolution of the Universe need not terminate in either an initial singularity or an unrestricted late-time expansion driven by an independent dark-energy fluid. Instead, the gravitational field may organize toward a saturated state that naturally produces a nonsingular cosmological history.

The purpose of this manuscript is not to dismiss the standard cosmological model, nor to argue that established observations are incorrect. On the contrary, every observational comparison presented here uses the standard cosmological datasets as the benchmark against which the proposed framework is evaluated.

Accordingly, the emphasis throughout this work is placed on numerical consistency rather than rhetoric.

The model is developed mathematically, implemented within a modified Boltzmann code, and compared with established cosmological observables including the acoustic scale, baryon acoustic oscillations, late-time growth, weak-lensing observables, cosmic microwave background spectra, and Planck 2018 lensing measurements. Where the model succeeds, those successes are documented quantitatively. Where differences remain, they are identified explicitly rather than concealed.

Equally important are the avenues that remain incomplete. A complete derivation of the saturation potential from a more fundamental theory, a full primary-CMB likelihood analysis with all nuisance parameters, and additional nonlinear structure-formation studies remain subjects for future investigation. These open questions are acknowledged because scientific progress depends as much upon clearly identifying unanswered questions as upon presenting successful calculations.

Readers are therefore encouraged to approach this manuscript in the manner intended.

Do not accept the central hypothesis because it is novel.

Do not reject it because it is unconventional.

Instead, follow the mathematics, examine the numerical implementation, compare the predictions with observation, and judge the framework according to the same standard applied to every physical theory: whether its assumptions are internally consistent, whether its equations are mathematically coherent, and whether its predictions agree with experiment and observation.

That standard—not the identity of its author—is the one to which this work is submitted.

End of Chapter 1

Chapter 2

Introduction and Mathematical Foundations

2.1 The gravitational problem

Newtonian gravity describes the attraction between two masses through

$$F = \frac{Gm_1m_2}{r^2}. \tag{1}$$

Equivalently, a test mass moving in a gravitational potential Φ_N experiences the acceleration

$$\mathbf{g} = -\nabla\Phi_N, \tag{2}$$

where the potential satisfies the Poisson equation

$$\nabla^2\Phi_N = 4\pi G\rho. \tag{3}$$

This formulation is successful whenever gravitational fields are weak, velocities are nonrelativistic, and spacetime curvature may be neglected. It does not, however, describe gravity as a propagating dynamical geometry.

General Relativity replaces the Newtonian force with spacetime curvature. Its field equation is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\, T_{\mu\nu}, \tag{4}$$

where

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} \tag{5}$$

is the Einstein tensor, $R_{\mu\nu}$ is the Ricci tensor, R is the Ricci scalar, $g_{\mu\nu}$ is the metric tensor, and $T_{\mu\nu}$ is the stress-energy tensor.

In units where

$$c = 1, \tag{6}$$

the dimensions of time and length are identified. Throughout the theoretical development, the factor $8\pi G$ may be retained explicitly or absorbed into the definition of the energy density. The numerical implementation described later uses dimensionless variables scaled to the saturation density.

The Einstein equation determines how spacetime curvature responds to stress-energy. The present theory retains this geometric structure but introduces a dynamical saturation sector describing the organizational state of the gravitational background.

The central proposition is that the cumulative gravitational organization of spacetime cannot grow indefinitely. Instead, it approaches a limiting configuration represented by canonical fields whose evolution is continuous through contraction, bounce, expansion, and late-time saturation.

2.2 Gravity as a propagating wave

In the weak-field approximation, the metric may be written as

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \qquad |h_{\mu\nu}| \ll 1, \tag{7}$$

where $\eta_{\mu\nu}$ is the Minkowski metric and $h_{\mu\nu}$ is a small perturbation.

Defining the trace-reversed perturbation

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}h, \tag{8}$$

with

$$h = \eta^{\alpha\beta}h_{\alpha\beta}, \tag{9}$$

the linearized Einstein equation in Lorenz gauge becomes

$$\Box\bar{h}_{\mu\nu} = -16\pi G\, T_{\mu\nu}, \tag{10}$$

where

$$\Box = \eta^{\alpha\beta}\partial_\alpha\partial_\beta. \tag{11}$$

In vacuum,

$$T_{\mu\nu} = 0, \tag{12}$$

and therefore

$$\Box\bar{h}_{\mu\nu} = 0. \tag{13}$$

Gravity therefore admits propagating wave solutions.

A plane-wave solution may be written as

$$\bar{h}_{\mu\nu} = A_{\mu\nu}e^{ik_\alpha x^\alpha}, \tag{14}$$

with the null-wave condition

$$k^\alpha k_\alpha = 0. \tag{15}$$

The recognition that gravity propagates as a wave motivates the saturation hypothesis developed here. The model does not replace the Einstein equation with an acoustic equation. Instead, it treats the large-scale organizational state of the gravitational field as a dynamical wave-supported sector capable of interference, damping, restoration, and saturation.

The word acoustic is used mechanically rather than literally. It refers to organizational phenomena common to wave systems:

$$\text{propagation}, \quad \text{phase}, \quad \text{interference}, \quad \text{restoration}, \quad \text{damping}, \quad \text{saturation}. \tag{16}$$

The physical hypothesis is that the gravitational field may approach a maximum organizational state in which additional cumulative ordering is dynamically suppressed.

2.3 Cosmological geometry

A homogeneous and isotropic universe is described by the Friedmann-Lemaître-Robertson-Walker line element

$$ds^2 = -dt^2 + a^2(t)\left[\frac{dr^2}{1-kr^2} + r^2 d\Omega^2\right], \tag{17}$$

where $a(t)$ is the scale factor and k is the spatial-curvature parameter.

The calculations developed here use a spatially flat background,

$$k = 0, \tag{18}$$

so that

$$ds^2 = -dt^2 + a^2(t)\, d\mathbf{x}^2. \tag{19}$$

The Hubble parameter is

$$H = \frac{\dot{a}}{a}. \tag{20}$$

The sign of H is retained:

$$H < 0 \quad \text{during contraction,} \qquad H = 0 \quad \text{at the bounce,} \qquad H > 0 \quad \text{during expansion.} \tag{21}$$

The ordinary Friedmann equations in General Relativity are

$$H^2 = \frac{8\pi G}{3}\rho, \tag{22}$$

and

$$\dot{H} = -4\pi G(\rho + p). \tag{23}$$

In the normalization

$$8\pi G = 1, \tag{24}$$

these become

$$H^2 = \frac{\rho}{3}, \tag{25}$$

and

$$\dot{H} = -\frac{1}{2}(\rho + p). \tag{26}$$

The total continuity equation is

$$\dot{\rho} + 3H(\rho + p) = 0. \tag{27}$$

For pressureless matter,

$$p_m = 0, \tag{28}$$

and therefore

$$\dot{\rho}_m = -3H\rho_m. \tag{29}$$

For radiation,

$$p_r = \frac{1}{3}\rho_r, \tag{30}$$

and in the absence of energy transfer,

$$\dot{\rho}_r = -4H\rho_r. \tag{31}$$

The gravitational-saturation system modifies the high-density gravitational response while preserving the ordinary low-density limit.

2.4 Organizational variables

The theory introduces two canonical degrees of freedom:

$$C = C(t, \mathbf{x}), \qquad \phi = \phi(t, \mathbf{x}). \tag{32}$$

The field C represents gravitational organization or saturation. Its preferred late-time value is

$$C_s = 0.95. \tag{33}$$

The field ϕ represents the wave or phase sector.

For a homogeneous background,

$$C = C(t), \qquad \phi = \phi(t). \tag{34}$$

Their canonical momenta are

$$\Pi_C = \dot{C}, \qquad \Pi_\phi = \dot{\phi}. \tag{35}$$

The combined kinetic quantity is

$$K^2 = \Pi_C^2 + \Pi_\phi^2. \tag{36}$$

The canonical saturation-sector energy density is

$$\rho_C = \frac{1}{2}\Pi_C^2 + \frac{1}{2}\Pi_\phi^2 + V(C, \phi), \tag{37}$$

and its pressure is

$$p_C = \frac{1}{2}\Pi_C^2 + \frac{1}{2}\Pi_\phi^2 - V(C, \phi). \tag{38}$$

The corresponding equation-of-state parameter is

$$w_C = \frac{p_C}{\rho_C}. \tag{39}$$

When the fields settle near the minimum and their kinetic energy becomes small,

$$\Pi_C^2 + \Pi_\phi^2 \ll V, \tag{40}$$

then

$$\rho_C \simeq V, \qquad p_C \simeq -V, \tag{41}$$

and therefore

$$w_C \simeq -1. \tag{42}$$

This late residual is not introduced as a separate fluid. It is the low-kinetic-energy limit of the same second-order saturation fields evolved throughout the complete trajectory.

2.5 Canonical potential

The locked potential is

$$V(C,\phi) = V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4 + A_0(1 - \cos\phi). \tag{43}$$

The locked numerical values are

$$C_s = 0.95, \tag{44}$$

$$V_\infty = 2.187987, \tag{45}$$

$$m_C^2 = 1.827221, \tag{46}$$

$$\lambda = 0.802722. \tag{47}$$

The derivative with respect to C is

$$V_{,C} = m_C^2(C - C_s) + \lambda(C - C_s)^3. \tag{48}$$

The derivative with respect to ϕ is

$$V_{,\phi} = A_0 \sin\phi. \tag{49}$$

The second derivatives are

$$V_{,CC} = m_C^2 + 3\lambda(C - C_s)^2, \tag{50}$$

and

$$V_{,\phi\phi} = A_0 \cos\phi. \tag{51}$$

At the saturation point,

$$C = C_s, \tag{52}$$

the first derivative vanishes:

$$V_{,C}(C_s) = 0. \tag{53}$$

The curvature in the C direction is

$$V_{,CC}(C_s) = m_C^2 > 0. \tag{54}$$

Therefore C_s is a stable minimum.

At

$$\phi = 0, \tag{55}$$

one has

$$V_{,\phi} = 0, \qquad V_{,\phi\phi} = A_0. \tag{56}$$

For

$$A_0 > 0, \tag{57}$$

the phase sector is also locally stable.

The residual energy at the joint minimum is

$$V(C_s, 0) = V_\infty. \tag{58}$$

Thus V_∞ is the asymptotic residual floor produced by the saturated field configuration.

2.6 Covariant field action

The canonical field sector may be written as

$$S_{C\phi} = \int d^4x\sqrt{-g}\left[-\frac{1}{2}g^{\mu\nu}\partial_\mu C\,\partial_\nu C - \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi - V(C,\phi)\right]. \tag{59}$$

The complete gravitational action is

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \frac{1}{2}g^{\mu\nu}\partial_\mu C\,\partial_\nu C - \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi - V(C,\phi) + \mathcal{L}_m + \mathcal{L}_r\right], \tag{60}$$

with the high-density saturation response imposed through the modified background constraint described below.

Variation with respect to C gives

$$\Box C - V_{,C} = 0 \tag{61}$$

before the contraction-era transfer term is included.

Variation with respect to ϕ gives

$$\Box\phi - V_{,\phi} = 0. \tag{62}$$

For a homogeneous FLRW background,

$$\Box C = -\ddot{C} - 3H\dot{C}, \tag{63}$$

and

$$\Box\phi = -\ddot{\phi} - 3H\dot{\phi}. \tag{64}$$

The ordinary canonical equations are therefore

$$\ddot{C} + 3H\dot{C} + V_{,C} = 0, \tag{65}$$

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0. \tag{66}$$

The transfer system modifies the damping coefficient during contraction while preserving these equations during expansion.

2.7 Contraction-era transfer

During contraction,

$$H < 0, \tag{67}$$

the ordinary term

$$3H \tag{68}$$

acts as anti-friction because it is negative. Uncontrolled anti-friction can drive the field kinetic energy to extremely large values.

The theory therefore contains an organizational transfer rate

$$\Gamma_{\rm tr} = W_-(H)\big[-3H + \Gamma_0 S_\rho\big]. \tag{69}$$

The contraction selector satisfies

$$W_-(H) \to 1 \qquad \text{for deep contraction,} \tag{70}$$

and

$$W_-(H) \to 0 \qquad \text{at the bounce and during expansion.} \tag{71}$$

A one-sided form is

$$W_-(H) = \begin{cases} \dfrac{1}{1 + e^{H/H_c}}, & H < 0, \\ 0, & H \geq 0. \end{cases} \tag{72}$$

In numerical work, the step may be smoothed to avoid a discontinuous derivative.

The density-activation function is

$$S_\rho = \frac{\big(\rho_{\rm tar}/\rho_\Gamma\big)^m}{1 + \big(\rho_{\rm tar}/\rho_\Gamma\big)^m}, \tag{73}$$

where

$$\rho_{\rm tar} = \rho_m + \rho_r. \tag{74}$$

The field equations become

$$\dot{C} = \Pi_C, \tag{75}$$

$$\dot{\Pi}_C = -\big(3H + \Gamma_{\rm tr}\big)\Pi_C - V_{,C}, \tag{76}$$

$$\dot{\phi} = \Pi_\phi, \tag{77}$$

$$\dot{\Pi}_\phi = -\big(3H + \Gamma_{\rm tr}\big)\Pi_\phi - V_{,\phi}. \tag{78}$$

Deep in contraction,

$$W_- \simeq 1, \tag{79}$$

and therefore

$$3H + \Gamma_{\rm tr} \simeq 3H - 3H + \Gamma_0 S_\rho, \tag{80}$$

so

$$3H + \Gamma_{\rm tr} \simeq \Gamma_0 S_\rho. \tag{81}$$

The contraction anti-friction is cancelled and replaced by positive organizational damping.

At the bounce and during expansion,

$$W_- = 0, \tag{82}$$

so

$$\Gamma_{\rm tr} = 0, \tag{83}$$

and the field equations return to

$$\dot\Pi_C = -3H\Pi_C - V_{,C}, \tag{84}$$

$$\dot\Pi_\phi = -3H\Pi_\phi - V_{,\phi}. \tag{85}$$

Thus the transfer does not survive as a late-time energy drain.

The transferred kinetic energy enters radiation through

$$Q = \Gamma_{\rm tr} K^2, \tag{86}$$

where

$$K^2 = \Pi_C^2 + \Pi_\phi^2. \tag{87}$$

The radiation equation is therefore

$$\dot\rho_r = -4H\rho_r + Q, \tag{88}$$

or

$$\dot\rho_r = -4H\rho_r + \Gamma_{\rm tr}\big(\Pi_C^2 + \Pi_\phi^2\big). \tag{89}$$

Matter remains separately conserved:

$$\dot\rho_m = -3H\rho_m. \tag{90}$$

The field kinetic loss is therefore transferred into radiation rather than removed from the system.

2.8 Energy conservation

The saturation-sector continuity equation follows from the field equations.

Taking the time derivative of

$$\rho_C = \frac{1}{2}\Pi_C^2 + \frac{1}{2}\Pi_\phi^2 + V, \tag{91}$$

gives

$$\dot{\rho}_C = \Pi_C \dot{\Pi}_C + \Pi_\phi \dot{\Pi}_\phi + V_{,C}\dot{C} + V_{,\phi}\dot{\phi}. \tag{92}$$

Using

$$\dot{C} = \Pi_C, \qquad \dot{\phi} = \Pi_\phi, \tag{93}$$

and the field equations,

$$\dot{\rho}_C = -\big(3H + \Gamma_{\rm tr}\big)\big(\Pi_C^2 + \Pi_\phi^2\big). \tag{94}$$

Since

$$\rho_C + p_C = \Pi_C^2 + \Pi_\phi^2, \tag{95}$$

one obtains

$$\dot{\rho}_C + 3H(\rho_C + p_C) = -\Gamma_{\rm tr} K^2. \tag{96}$$

Radiation obeys

$$\dot{\rho}_r + 4H\rho_r = +\Gamma_{\rm tr} K^2. \tag{97}$$

Adding both equations gives

$$\dot{\rho}_C + \dot{\rho}_r + 3H(\rho_C + p_C) + 4H\rho_r = 0. \tag{98}$$

Including matter,

$$\dot{\rho}_m + 3H\rho_m = 0, \tag{99}$$

the total continuity equation becomes

$$\dot{\rho} + 3H(\rho + p) = 0, \tag{100}$$

where

$$\rho = \rho_m + \rho_r + \rho_C, \tag{101}$$

and

$$p = \frac{1}{3}\rho_r + p_C. \tag{102}$$

The transfer system is therefore energy conserving.

2.9 High-density gravitational saturation

The nonsingular background is governed by the modified Friedmann constraint

$$H^2 = \frac{\rho}{3}\left[1 - \left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{103}$$

The locked implementation uses

$$q = 1, \tag{104}$$

giving

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{105}$$

At low density,

$$\rho \ll \rho_s, \tag{106}$$

the correction becomes negligible:

$$1 - \frac{\rho}{\rho_s} \simeq 1, \tag{107}$$

and therefore

$$H^2 \simeq \frac{\rho}{3}. \tag{108}$$

The ordinary Friedmann equation is recovered.

At

$$\rho = \rho_s, \tag{109}$$

the constraint gives

$$H^2 = 0. \tag{110}$$

Therefore

$$H = 0 \tag{111}$$

at a finite density rather than at a singular vanishing scale factor.

The saturation density used in the locked background is

$$\rho_s = 1.58199 \times 10^{15} \tag{112}$$

in the dimensionless normalization of the numerical system.

Differentiating the generalized constraint and using total energy conservation gives

$$\dot{H} = -\frac{1}{2}(\rho + p)\left[1 - (q+1)\left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{113}$$

For

$$q = 1, \tag{114}$$

this becomes

$$\dot{H} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right). \tag{115}$$

Near the saturation density,

$$\rho \simeq \rho_s, \tag{116}$$

and therefore

$$1 - 2\frac{\rho}{\rho_s} \simeq -1. \tag{117}$$

If

$$\rho + p > 0, \tag{118}$$

then

$$\dot{H} > 0. \tag{119}$$

Thus a contracting branch with

$$H < 0 \tag{120}$$

can pass continuously through

$$H = 0 \tag{121}$$

and emerge on an expanding branch with

$$H > 0. \tag{122}$$

The bounce therefore follows from the high-density saturation of the gravitational response.

2.10 Cosmic-time evolution system

The complete nonsingular state vector is

$$\mathbf{y} = \big(a, H, C, \Pi_C, \phi, \Pi_\phi, \rho_m, \rho_r\big). \tag{123}$$

The evolution equations are

$$\dot{a} = aH, \tag{124}$$

$$\dot{H} = -\frac{1}{2}(\rho + p)\left[1 - (q+1)\left(\frac{\rho}{\rho_s}\right)^q\right], \tag{125}$$

$$\dot{C} = \Pi_C, \tag{126}$$

$$\dot{\Pi}_C = -\big(3H + \Gamma_{\rm tr}\big)\Pi_C - V_{,C}, \tag{127}$$

$$\dot{\phi} = \Pi_\phi, \tag{128}$$

$$\dot{\Pi}_\phi = -\big(3H + \Gamma_{\rm tr}\big)\Pi_\phi - V_{,\phi}, \tag{129}$$

$$\dot{\rho}_m = -3H\rho_m, \tag{130}$$

$$\dot{\rho}_r = -4H\rho_r + \Gamma_{\rm tr}\big(\Pi_C^2 + \Pi_\phi^2\big). \tag{131}$$

The auxiliary definitions are

$$K^2 = \Pi_C^2 + \Pi_\phi^2, \tag{132}$$

$$\rho_C = \frac{1}{2}K^2 + V, \tag{133}$$

$$p_C = \frac{1}{2}K^2 - V, \tag{134}$$

$$\rho = \rho_m + \rho_r + \rho_C, \tag{135}$$

$$p = \frac{1}{3}\rho_r + p_C. \tag{136}$$

The Friedmann constraint is monitored through

$$\mathcal{C}_H = H^2 - \frac{\rho}{3}\left[1 - \left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{137}$$

A numerically consistent solution satisfies

$$|\mathcal{C}_H| \ll 1. \tag{138}$$

The system is integrated in cosmic time or a nonsingular affine parameter. It is not integrated through the bounce using

$$N = \ln a \tag{139}$$

because

$$\frac{dC}{dN} = \frac{\Pi_C}{H} \tag{140}$$

is singular at

$$H = 0. \tag{141}$$

The same problem occurs for every equation divided by H. The bounce is therefore evolved using equations containing no factor of $1/H$.

2.11 Scaled affine variables

For numerical stability, the high-density system may be written in dimensionless variables scaled by ρ_s.

Define

$$h = \frac{H}{\sqrt{\rho_s}}, \tag{142}$$

$$u_C = \frac{\Pi_C}{\sqrt{\rho_s}}, \tag{143}$$

$$u_\phi = \frac{\Pi_\phi}{\sqrt{\rho_s}}, \tag{144}$$

$$\bar{\rho} = \frac{\rho}{\rho_s}. \tag{145}$$

The $q = 1$ constraint becomes

$$h^2 = \frac{\bar{\rho}}{3}(1 - \bar{\rho}). \tag{146}$$

If the affine variable τ is defined by

$$d\tau = \sqrt{\rho_s}\, dt, \tag{147}$$

then

$$\frac{da}{d\tau} = ah, \qquad \frac{dC}{d\tau} = u_C, \qquad \frac{d\phi}{d\tau} = u_\phi. \tag{148}$$

The scaled form removes extremely large numerical values without changing the physical trajectory.

The locked initial organizational values are

$$C_{\text{initial}} = 0.918318, \tag{149}$$

and

$$\Pi_{C,\text{initial}} = 0.055327 \tag{150}$$

in the adopted numerical normalization.

2.12 The locked expansion history

After a successful nonsingular trajectory is obtained, the expanding branch is converted into a redshift history through

$$1 + z = \frac{a_0}{a}. \tag{151}$$

With

$$a_0 = 1, \tag{152}$$

this becomes

$$z = \frac{1}{a} - 1. \tag{153}$$

The dimensionless expansion function is

$$E(z) = \frac{H(z)}{H_0}. \tag{154}$$

The physical Hubble rate is

$$H(z) = H_0 E(z). \tag{155}$$

The final numerical implementation uses

$$H_0 = 67.4 \text{ km}\,\text{s}^{-1}\,\text{Mpc}^{-1}, \tag{156}$$

$$\Omega_{m0} = 0.314114, \tag{157}$$

$$\Omega_{r0} = 9 \times 10^{-5}. \tag{158}$$

The locked trajectory is embedded into the Boltzmann background with the transition ending at

$$a_{\text{lock}} = 2.05 \times 10^{-4}. \tag{159}$$

The narrowed transition width is

$$W = 0.67 W_0. \tag{160}$$

The residual normalization is fixed so that

$$E(1) = 1. \tag{161}$$

The final locked expansion produces

$$100\theta_* = 1.041033, \tag{162}$$

$$r_{\text{drag}} = 149.960 \text{ Mpc}, \tag{163}$$

$$z_* = 1089.691, \tag{164}$$

$$z_{\text{drag}} = 1058.511, \tag{165}$$

$$\sigma_8 = 0.779680, \tag{166}$$

$$S_8 = \sigma_8 \sqrt{\frac{\Omega_{m0}}{0.3}} = 0.79781. \tag{167}$$

The acoustic-scale difference from the reference value

$$100\theta_* = 1.0411 \tag{168}$$

is

$$\frac{1.041033 - 1.0411}{1.0411} \times 100\% \simeq -0.0064\%. \tag{169}$$

2.13 Cosmological distances

For a spatially flat universe, the comoving radial distance is

$$\chi(z) = \int_0^z \frac{c\,dz'}{H(z')}. \tag{170}$$

The transverse comoving distance is

$$D_M(z) = \chi(z). \tag{171}$$

The physical angular-diameter distance is

$$D_A(z) = \frac{D_M(z)}{1+z}. \tag{172}$$

The Hubble distance is

$$D_H(z) = \frac{c}{H(z)}. \tag{173}$$

The isotropic BAO distance is

$$D_V(z) = \left[z\, D_M^2(z) D_H(z)\right]^{1/3}. \tag{174}$$

The locked calculation yields a consistent BAO ladder when the same model-derived sound horizon is used throughout.

2.14 Sound horizon

The baryon-loaded sound speed is

$$c_s(z) = \frac{c}{\sqrt{3\left[1 + R(z)\right]}}, \tag{175}$$

where

$$R(z) = \frac{3\rho_b}{4\rho_\gamma} = \frac{3\Omega_{b0}}{4\Omega_{\gamma 0}} \frac{1}{1+z}. \tag{176}$$

The drag sound horizon is

$$r_{\rm drag} = \int_{z_{\rm drag}}^{\infty} \frac{c_s(z)}{H(z)}\, dz. \tag{177}$$

The recombination sound horizon is

$$r_s(z_*) = \int_{z_*}^{\infty} \frac{c_s(z)}{H(z)}\, dz. \tag{178}$$

The angular acoustic scale is

$$\theta_* = \frac{r_s(z_*)}{D_M(z_*)}. \tag{179}$$

CAMB conventionally reports

$$100\theta_*. \tag{180}$$

The final locked result is

$$100\theta_* = 1.041033. \tag{181}$$

2.15 Linear perturbations of the saturation fields

On the expanding branch, write

$$C(\tau, \mathbf{x}) = C_{\mathrm{bg}}(\tau) + \delta C(\tau, \mathbf{x}), \tag{182}$$

$$\phi(\tau, \mathbf{x}) = \phi_{\mathrm{bg}}(\tau) + \delta\phi(\tau, \mathbf{x}). \tag{183}$$

In Fourier space,

$$\delta C(\tau, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \delta C_{\mathbf{k}}(\tau) e^{i\mathbf{k}\cdot\mathbf{x}}, \tag{184}$$

and similarly for $\delta\phi$.

Using conformal time,

$$d\tau = \frac{dt}{a}, \tag{185}$$

and

$$\mathcal{H} = \frac{a'}{a} = aH, \tag{186}$$

the canonical synchronous-gauge field perturbations satisfy

$$\delta C'' + 2\mathcal{H}\delta C' + \big(k^2 + a^2 V_{,CC}\big)\delta C + \frac{1}{2} C'_{\mathrm{bg}} h' = 0, \tag{187}$$

$$\delta\phi'' + 2\mathcal{H}\delta\phi' + \big(k^2 + a^2 V_{,\phi\phi}\big)\delta\phi + \frac{1}{2}\phi'_{\mathrm{bg}} h' = 0. \tag{188}$$

Defining

$$v_C = \delta C', \qquad v_\phi = \delta\phi', \tag{189}$$

the first-order system is

$$\delta C' = v_C, \tag{190}$$

$$v_C' = -2\mathcal{H}v_C - \big(k^2 + a^2 V_{,CC}\big)\delta C - \frac{1}{2}C'_{\rm bg}h', \tag{191}$$

$$\delta\phi' = v_\phi, \tag{192}$$

$$v_\phi' = -2\mathcal{H}v_\phi - \big(k^2 + a^2 V_{,\phi\phi}\big)\delta\phi - \frac{1}{2}\phi'_{\rm bg}h'. \tag{193}$$

In the internal CAMB notation, the metric-source term is represented using the code's synchronous-gauge variable kz. The mathematical content is the same as the h'-driven form displayed above.

Because the adopted potential contains no mixed term between C and ϕ,

$$V_{,C\phi} = 0, \tag{194}$$

the fields are not directly mixed at linear order through the potential Hessian.

2.16 Linearized saturation stress-energy

For canonical scalar fields, the density perturbation is

$$\delta\rho_C = \frac{1}{a^2}\big[C'_{\rm bg}\delta C' + \phi'_{\rm bg}\delta\phi'\big] + V_{,C}\delta C + V_{,\phi}\delta\phi. \tag{195}$$

Multiplying by the CAMB metric normalization gives the internal source form

$$a^2\delta\rho_C = C'_{\rm bg}v_C + \phi'_{\rm bg}v_\phi + a^2\big(V_{,C}\delta C + V_{,\phi}\delta\phi\big). \tag{196}$$

The pressure perturbation is

$$\delta p_C = \frac{1}{a^2}\big[C'_{\rm bg}\delta C' + \phi'_{\rm bg}\delta\phi'\big] - V_{,C}\delta C - V_{,\phi}\delta\phi. \tag{197}$$

The internal metric-normalized form is

$$a^2\delta p_C = C'_{\rm bg}v_C + \phi'_{\rm bg}v_\phi - a^2\big(V_{,C}\delta C + V_{,\phi}\delta\phi\big). \tag{198}$$

The momentum-density perturbation is

$$\delta q_C = \frac{k}{a^2}\big(C'_{\rm bg}\delta C + \phi'_{\rm bg}\delta\phi\big), \tag{199}$$

or in the CAMB-normalized source convention,

$$a^2\delta q_C = k\big(C'_{\rm bg}\delta C + \phi'_{\rm bg}\delta\phi\big). \tag{200}$$

Canonical minimally coupled scalar fields have no intrinsic linear anisotropic stress:

$$\pi_C = 0. \tag{201}$$

These contributions were inserted into the Einstein constraint and evolution system rather than represented through an effective Newton constant.

The maximum field amplitudes in the final locked run are

$$\max|\delta C| \simeq 0.035, \qquad \max|\delta\phi| \simeq 0.049. \tag{202}$$

Their back-reaction on σ_8 and on the primary acoustic peaks is at the level of approximately

$$10^{-6}. \tag{203}$$

The saturation fields therefore remain perturbative and behave as linear spectators around the locked background.

2.17 Why phenomenological gravity rescalings fail

A common modified-gravity parameterization introduces

$$k^2\Psi = -4\pi G a^2 \mu(a,k)\rho\Delta, \tag{204}$$

together with a slip relation

$$\eta(a,k) = \frac{\Phi}{\Psi}. \tag{205}$$

The Weyl potential is then governed by

$$\Phi + \Psi. \tag{206}$$

A lensing modification may be written as

$$\Sigma(a,k) = \frac{\mu(a,k)\left[1+\eta(a,k)\right]}{2}. \tag{207}$$

The tested phenomenological forms included:

$$\mu = f_{\mathrm{sat}}, \qquad \eta = \frac{1}{\mu}, \qquad \eta = 2-\mu, \qquad \eta = 1. \tag{208}$$

These closures were inserted while the locked background remained fixed.

They did not reproduce the canonical saturation-field response. Instead, they increased σ_8 and generated alternating distortions in the odd and even acoustic peaks.

The reason is mathematical. A multiplicative modification of the scalar constraint changes the coupled metric-density-velocity system directly. It is not equivalent to adding the canonical stress-energy tensor of fields whose background velocities and perturbations are small.

The canonical fields contribute through

$$\delta\rho_C, \qquad \delta q_C, \qquad \delta p_C, \qquad \pi_C = 0, \tag{209}$$

rather than through an imposed global rescaling

$$G \to \mu G. \tag{210}$$

Thus the saturation sector is not a reduced-Newton-constant model.

2.18 Growth

On subhorizon scales, the standard linear growth equation for pressureless matter may be written as

$$\frac{d^2\delta_m}{dN^2} + \left(2 + \frac{d\ln H}{dN}\right)\frac{d\delta_m}{dN} = \frac{3}{2}\Omega_m(N)\delta_m, \tag{211}$$

where

$$N = \ln a \tag{212}$$

is used only on the expanding branch, away from the bounce.

The matter fraction is

$$\Omega_m(a) = \frac{\Omega_{m0}a^{-3}}{E^2(a)}. \tag{213}$$

The growth rate is

$$f(a) = \frac{d\ln\delta_m}{d\ln a}. \tag{214}$$

The fluctuation amplitude evolves as

$$\sigma_8(z) = \sigma_{8,0}\frac{D(z)}{D(0)}, \tag{215}$$

where $D(z)$ is the normalized linear growth factor.

The combined weak-lensing parameter is

$$S_8 = \sigma_8\sqrt{\frac{\Omega_m}{0.3}}. \tag{216}$$

The locked model gives

$$\sigma_8 = 0.779680, \tag{217}$$

and

$$S_8 = 0.79781. \tag{218}$$

This reduction is produced by the locked expansion history rather than by an imposed weakening of the gravitational coupling.

2.19 Lensing potential

In conformal Newtonian notation,

$$ds^2 = a^2(\tau)\big[-(1+2\Psi)d\tau^2 + (1-2\Phi)d\mathbf{x}^2\big]. \tag{219}$$

The Weyl potential is

$$\Phi_W = \frac{\Phi+\Psi}{2}. \tag{220}$$

The CMB lensing potential is

$$\varphi(\hat{\mathbf{n}}) = -2\int_0^{\chi_*} d\chi\, \frac{\chi_*-\chi}{\chi_*\chi}\, \Phi_W\big(\chi\hat{\mathbf{n}}, \eta_0-\chi\big). \tag{221}$$

Its angular spectrum is

$$C_L^{\varphi\varphi}. \tag{222}$$

The final locked model produces a mildly enhanced lensing-potential spectrum relative to the matched control, while remaining compatible with the Planck 2018 bandpowers.

Using the official nine-bin window functions and covariance matrix gives

$$\chi^2_{\text{locked}} = 10.7379 \tag{223}$$

for nine bins, while the matched control gives

$$\chi^2_{\text{control}} = 12.0022. \tag{224}$$

Therefore

$$\Delta\chi^2 = \chi^2_{\text{locked}} - \chi^2_{\text{control}} = -1.2643. \tag{225}$$

The corresponding p-values are

$$p_{\text{locked}} = 0.294, \qquad p_{\text{control}} = 0.213. \tag{226}$$

2.20 Summary of the mathematical structure

The complete framework is built from one continuous phase space:

$$\big(a, H, C, \Pi_C, \phi, \Pi_\phi, \rho_m, \rho_r\big). \tag{227}$$

The same second-order fields are used during contraction, through the bounce, during expansion, and at late-time saturation.

The governing equations are

$$\dot{a} = aH, \tag{228}$$

$$\dot{H} = -\frac{1}{2}(\rho + p)\left[1 - (q+1)\left(\frac{\rho}{\rho_s}\right)^q\right], \tag{229}$$

$$\dot{C} = \Pi_C, \tag{230}$$

$$\dot{\Pi}_C = -\big(3H + \Gamma_{\rm tr}\big)\Pi_C - V_{,C}, \tag{231}$$

$$\dot{\phi} = \Pi_\phi, \tag{232}$$

$$\dot{\Pi}_\phi = -\big(3H + \Gamma_{\rm tr}\big)\Pi_\phi - V_{,\phi}, \tag{233}$$

$$\dot{\rho}_m = -3H\rho_m, \tag{234}$$

$$\dot{\rho}_r = -4H\rho_r + \Gamma_{\rm tr}\big(\Pi_C^2 + \Pi_\phi^2\big). \tag{235}$$

The constraint is

$$H^2 = \frac{\rho}{3}\left[1 - \left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{236}$$

The potential is

$$V(C,\phi) = V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4 + A_0(1 - \cos\phi). \tag{237}$$

The same fields produce the late residual through

$$\Pi_C, \Pi_\phi \to 0, \qquad C \to C_s, \qquad \phi \to 0, \tag{238}$$

so that

$$\rho_C \to V_\infty, \qquad p_C \to -V_\infty, \qquad w_C \to -1. \tag{239}$$

There is no stitched first-order residual equation, no separate dark-energy fluid, and no imposed hard floor in the evolution.

The mathematical result is a continuous nonsingular cosmology in which high-density gravitational response saturates, contraction reverses at finite scale factor, the saturation fields settle toward a stable minimum, and the resulting locked expansion history can be evolved through the complete Boltzmann hierarchy.

End of Chapter 2

Chapter 3

From Classical Gravitation to Gravitational Saturation

3.1 The continuity of gravitational theory

The development of gravitational physics has proceeded through a sequence of enlargements.

Newtonian gravity introduced a quantitative law of attraction between masses. General Relativity replaced gravitational force with spacetime geometry. Relativistic cosmology extended that geometry to the Universe as a whole. Gravitational-wave theory established that changes in the gravitational field propagate dynamically. Modern observational cosmology then revealed that the visible matter content of the Universe is not sufficient, within the standard interpretation, to account for all measured gravitational phenomena.

The theory developed in this manuscript begins from the mathematical success of these earlier descriptions rather than discarding them.

Its starting point is the requirement that any proposed extension satisfy three limits:

$$\begin{aligned} &\text{Newtonian gravity} \quad \text{for weak, local fields,} \\ &\text{General Relativity} \quad \text{for ordinary relativistic gravitation,} \\ &\text{standard Friedmann evolution} \quad \text{when the saturation correction is negligible.} \end{aligned} \tag{1}$$

The gravitational-saturation framework is therefore constructed as a high-density and large-scale dynamical extension whose corrections vanish in the regimes where established gravitational theory is already successful.

3.2 Newtonian organization

Newtonian gravity is governed by

$$\nabla^2\Phi_N = 4\pi G\rho, \tag{2}$$

with acceleration

$$\mathbf{g} = -\nabla\Phi_N. \tag{3}$$

For a point mass M,

$$\Phi_N(r) = -\frac{GM}{r}, \tag{4}$$

and therefore

$$g(r) = \frac{GM}{r^2}. \tag{5}$$

For circular motion,

$$\frac{v^2}{r} = \frac{GM(r)}{r^2}, \tag{6}$$

so that

$$v^2(r) = \frac{GM(r)}{r}. \tag{7}$$

If the enclosed mass approaches a constant at large radius,

$$M(r) \to M_{\mathrm{tot}}, \tag{8}$$

then Newtonian dynamics predicts

$$v(r) \propto r^{-1/2}. \tag{9}$$

The observed persistence of approximately flat galactic rotation curves is conventionally interpreted as evidence for additional gravitating mass.

Within the saturation interpretation, the important feature is not merely the magnitude of the force but the manner in which gravitational influence is organized across a system.

The Newtonian field is linear in the source density:

$$\nabla^2 \Phi_N = 4\pi G\rho. \tag{10}$$

If two source distributions are present,

$$\rho = \rho_1 + \rho_2, \tag{11}$$

then

$$\Phi_N = \Phi_1 + \Phi_2. \tag{12}$$

This superposition property already contains the mathematical basis for interference-like organizational reasoning. It does not imply that Newtonian gravity is itself an acoustic medium, but it shows that gravitational influence is additive in the weak-field limit.

The saturation hypothesis asks what occurs when the gravitational field is no longer weak, when the background evolves dynamically, and when gravitational organization acts cumulatively upon the same geometry through which it propagates.

3.3 Einstein geometry

General Relativity is governed by

$$G_{\mu\nu} = 8\pi G\, T_{\mu\nu}, \tag{13}$$

or, including a cosmological constant,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\, T_{\mu\nu}. \tag{14}$$

The contracted Bianchi identity gives

$$\nabla_\mu G^{\mu\nu} = 0. \tag{15}$$

Therefore the matter stress-energy tensor satisfies

$$\nabla_\mu T^{\mu\nu} = 0. \tag{16}$$

This conservation law is not an independent assumption added after the field equation. It follows from the geometric structure of the theory.

The gravitational-saturation system preserves this total conservation principle.

The field sector transfers energy to radiation during contraction, but the total energy satisfies

$$\nabla_\mu T^{\mu\nu}_{\text{total}} = 0. \tag{17}$$

The total stress-energy tensor is written schematically as

$$T^{\mu\nu}_{\text{total}} = T^{\mu\nu}_m + T^{\mu\nu}_r + T^{\mu\nu}_{C\phi}. \tag{18}$$

The canonical saturation-field contribution is

$$T^{\mu\nu}_{C\phi} = \partial^\mu C\,\partial^\nu C + \partial^\mu \phi\,\partial^\nu \phi - g^{\mu\nu}\left[\frac{1}{2}\partial_\alpha C\,\partial^\alpha C + \frac{1}{2}\partial_\alpha \phi\,\partial^\alpha \phi + V(C,\phi)\right]. \tag{19}$$

The field equations are then obtained from

$$\nabla_\mu T^{\mu\nu}_{C\phi} \tag{20}$$

together with the transfer current introduced during contraction.

The mathematical role of the saturation fields is therefore not to replace geometry with a force. Their stress-energy participates in the same Einstein-equation structure used by ordinary canonical fields.

3.4 Cosmology from the Einstein equation

For a flat FLRW spacetime,

$$ds^2 = -dt^2 + a^2(t)\,d\mathbf{x}^2, \tag{21}$$

the Einstein equation gives the Friedmann constraint

$$H^2 = \frac{8\pi G}{3}\rho, \tag{22}$$

and the acceleration equation

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p). \tag{23}$$

In the normalization

$$8\pi G = 1, \tag{24}$$

these become

$$H^2 = \frac{\rho}{3}, \tag{25}$$

and

$$\frac{\ddot{a}}{a} = -\frac{1}{6}(\rho + 3p). \tag{26}$$

Accelerated expansion requires

$$\rho + 3p < 0. \tag{27}$$

For a component with

$$p = w\rho, \tag{28}$$

this becomes

$$1 + 3w < 0, \tag{29}$$

or

$$w < -\frac{1}{3}. \tag{30}$$

A cosmological constant has

$$w = -1. \tag{31}$$

A slowly evolving canonical field near a positive potential minimum also has

$$w \simeq -1. \tag{32}$$

This is the late-time behavior obtained by the saturation fields:

$$\Pi_C^2 + \Pi_\phi^2 \ll V, \tag{33}$$

so that

$$\rho_C \simeq V, \qquad p_C \simeq -V, \tag{34}$$

and therefore

$$w_C \simeq -1. \tag{35}$$

The residual acceleration of the locked model is thus generated by the asymptotic state of the same fields that participated in the earlier nonsingular evolution.

3.5 The singularity problem

The standard Friedmann equation for a radiation-dominated universe is

$$H^2 = \frac{\rho_r}{3}. \tag{36}$$

Radiation evolves as

$$\rho_r = \rho_{r0} a^{-4}. \tag{37}$$

Therefore

$$H^2 = \frac{\rho_{r0}}{3} a^{-4}. \tag{38}$$

Using

$$H = \frac{\dot{a}}{a}, \tag{39}$$

one obtains

$$\dot{a} = \sqrt{\frac{\rho_{r0}}{3}}\, a^{-1}. \tag{40}$$

Integrating,

$$a\,da = \sqrt{\frac{\rho_{r0}}{3}}\,dt, \tag{41}$$

so

$$a^2(t) = 2\sqrt{\frac{\rho_{r0}}{3}}\,t \tag{42}$$

after choosing the singular origin at $t = 0$. Hence

$$a(t) \propto t^{1/2}. \tag{43}$$

As

$$t \to 0^+, \tag{44}$$

one has

$$a \to 0, \qquad \rho_r \to \infty, \qquad H \to \infty. \tag{45}$$

For a matter-dominated universe,

$$\rho_m = \rho_{m0} a^{-3}, \tag{46}$$

and

$$a(t) \propto t^{2/3}. \tag{47}$$

Again,

$$a \to 0 \tag{48}$$

at finite past time.

The gravitational-saturation correction replaces the ordinary high-density constraint with

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{49}$$

At

$$\rho = \rho_s, \tag{50}$$

one has

$$H = 0. \tag{51}$$

The density is finite, and the scale factor need not vanish.

The modified constraint may be rewritten as

$$3H^2 = \rho - \frac{\rho^2}{\rho_s}. \tag{52}$$

The correction is negative and quadratic in density:

$$-\frac{\rho^2}{\rho_s}. \tag{53}$$

At low density,

$$\frac{\rho^2}{\rho_s} \ll \rho, \tag{54}$$

so standard expansion is recovered.

At high density, the quadratic correction limits the gravitational response.

This is the mathematical expression of saturation in the homogeneous background.

3.6 The bounce condition

The generalized constraint is

$$H^2 = \frac{\rho}{3}\left[1 - \left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{55}$$

Differentiating,

$$2H\dot{H} = \frac{\dot{\rho}}{3}\left[1 - (q+1)\left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{56}$$

Using

$$\dot{\rho} = -3H(\rho + p), \tag{57}$$

and dividing by $2H$ away from the exact crossing gives

$$\dot{H} = -\frac{1}{2}(\rho + p)\left[1 - (q+1)\left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{58}$$

The continuous limit of this expression is then used through the crossing.

For

$$q = 1, \tag{59}$$

$$\dot{H} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right). \tag{60}$$

At the bounce,

$$\rho = \rho_s, \tag{61}$$

so

$$\dot{H}_{\text{bounce}} = +\frac{1}{2}(\rho + p). \tag{62}$$

If

$$\rho + p > 0, \tag{63}$$

then

$$\dot{H}_{\text{bounce}} > 0. \tag{64}$$

Thus

$$H < 0 \tag{65}$$

before the bounce,

$$H = 0 \tag{66}$$

at the bounce, and

$$H > 0 \tag{67}$$

after the bounce.

The scale factor obeys

$$\dot{a} = aH. \tag{68}$$

At the bounce,

$$\dot{a} = 0. \tag{69}$$

The second derivative is

$$\ddot{a} = a\dot{H} + aH^2. \tag{70}$$

Since

$$H = 0 \tag{71}$$

at the bounce,

$$\ddot{a}_{\text{bounce}} = a_{\text{min}}\dot{H}_{\text{bounce}}. \tag{72}$$

If

$$a_{\text{min}} > 0 \tag{73}$$

and

$$\dot{H}_{\text{bounce}} > 0, \tag{74}$$

then

$$\ddot{a}_{\text{bounce}} > 0. \tag{75}$$

The bounce is therefore a genuine local minimum of the scale factor.

3.7 Gravitational waves as an organizing principle

The weak-field gravitational-wave equation is

$$\Box \bar{h}_{\mu\nu} = 0 \tag{76}$$

in vacuum.

For a mode with comoving wave number k in an expanding background, the tensor amplitude satisfies

$$h_k'' + 2\mathcal{H}h_k' + k^2 h_k = 0 \tag{77}$$

in the absence of anisotropic stress.

The term

$$2\mathcal{H}h_k' \tag{78}$$

acts as damping during expansion because

$$\mathcal{H} > 0. \tag{79}$$

During contraction,

$$\mathcal{H} < 0, \tag{80}$$

and the same term acts as anti-damping.

The organizational fields exhibit the corresponding structure:

$$\delta C'' + 2\mathcal{H}\delta C' + \left(k^2 + a^2 V_{,CC}\right)\delta C + \frac{1}{2}C'_{\rm bg}h' = 0, \tag{81}$$

and

$$\delta\phi'' + 2\mathcal{H}\delta\phi' + \left(k^2 + a^2 V_{,\phi\phi}\right)\delta\phi + \frac{1}{2}\phi'_{\rm bg}h' = 0. \tag{82}$$

The wave structure contains:

$$\begin{aligned} &\text{propagation through } k^2,\\ &\text{background damping through } 2\mathcal{H},\\ &\text{restoration through } V_{,II},\\ &\text{metric forcing through } h'. \end{aligned} \tag{83}$$

The background fields contain the corresponding cosmic-time form:

$$\ddot{C} + (3H + \Gamma_{\rm tr})\dot{C} + V_{,C} = 0, \tag{84}$$

$$\ddot{\phi} + (3H + \Gamma_{\rm tr})\dot{\phi} + V_{,\phi} = 0. \tag{85}$$

The saturation hypothesis therefore does not rest only upon metaphor. Its mathematical realization is a damped, driven, restorative field system evolving on a dynamical gravitational background.

3.8 Phase and interference

The phase sector is described by

$$V_\phi = A_0(1 - \cos\phi). \tag{86}$$

For small ϕ,

$$\cos\phi = 1 - \frac{\phi^2}{2} + \frac{\phi^4}{24} + \mathcal{O}(\phi^6). \tag{87}$$

Therefore

$$1 - \cos\phi = \frac{\phi^2}{2} - \frac{\phi^4}{24} + \mathcal{O}(\phi^6), \tag{88}$$

and

$$V_\phi = \frac{A_0}{2}\phi^2 - \frac{A_0}{24}\phi^4 + \mathcal{O}(\phi^6). \tag{89}$$

Near the minimum,

$$V_{,\phi} = A_0 \sin\phi \simeq A_0\phi, \tag{90}$$

and

$$V_{,\phi\phi} = A_0\cos\phi \simeq A_0. \tag{91}$$

Thus the phase field behaves locally as a stable oscillator:

$$\ddot{\phi} + 3H\dot{\phi} + A_0\phi \simeq 0. \tag{92}$$

The periodic form retains the distinction between a linear displacement field and a phase variable.

The phase interpretation may be represented schematically through two contributions,

$$A_{\text{net}} = A_{\text{in}} + A_{\text{out}}. \tag{93}$$

If they approach opposite phase,

$$A_{\text{out}} \simeq -A_{\text{in}}, \tag{94}$$

then

$$A_{\text{net}} \simeq 0. \tag{95}$$

The theory interprets saturation as a dynamic limit in which further cumulative organization is opposed by the phase and restorative response of the system.

The numerical model does not impose

$$A_{\text{net}} = 0 \tag{96}$$

as an algebraic constraint. Instead, it evolves the phase field through its second-order equation and allows the fields to settle dynamically.

3.9 Saturation as a stable minimum

The organizational potential is

$$V_C(C) = V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4. \tag{97}$$

Define

$$x = C - C_s. \tag{98}$$

Then

$$V_C = V_\infty + \frac{m_C^2}{2}x^2 + \frac{\lambda}{4}x^4. \tag{99}$$

Its derivative is

$$\frac{dV_C}{dx} = m_C^2 x + \lambda x^3 = x(m_C^2 + \lambda x^2). \tag{100}$$

For

$$m_C^2 > 0, \qquad \lambda \geq 0, \tag{101}$$

the only real stationary point is

$$x = 0, \tag{102}$$

or

$$C = C_s. \tag{103}$$

The second derivative is

$$\frac{d^2 V_C}{dx^2} = m_C^2 + 3\lambda x^2. \tag{104}$$

Therefore

$$\frac{d^2 V_C}{dx^2} > 0 \tag{105}$$

for all x.

The potential is globally convex in the C direction.

The saturation point is not merely locally stable. Under the locked potential, it is the unique global minimum of the organizational sector.

The force on C is

$$F_C = -V_{,C}. \tag{106}$$

If

$$C < C_s, \tag{107}$$

then

$$x < 0, \tag{108}$$

so

$$V_{,C} < 0, \tag{109}$$

and therefore

$$F_C > 0. \tag{110}$$

The field is driven upward toward C_s.

If

$$C > C_s, \tag{111}$$

then

$$x > 0, \tag{112}$$

so

$$V_{,C} > 0, \tag{113}$$

and therefore

$$F_C < 0. \tag{114}$$

The field is driven downward toward C_s.

Thus the potential restores the organizational field toward saturation from either side.

3.10 Late-time residual

Near the minimum, write

$$C = C_s + \Delta C, \tag{115}$$

with

$$|\Delta C| \ll 1, \qquad |\phi| \ll 1. \tag{116}$$

The potential becomes

$$V \simeq V_\infty + \frac{m_C^2}{2}(\Delta C)^2 + \frac{A_0}{2}\phi^2. \tag{117}$$

The energy density is

$$\rho_C = \frac{1}{2}\dot{\Delta C}^2 + \frac{1}{2}\dot{\phi}^2 + V_\infty + \frac{m_C^2}{2}(\Delta C)^2 + \frac{A_0}{2}\phi^2. \tag{118}$$

The pressure is

$$p_C = \frac{1}{2}\dot{\Delta C}^2 + \frac{1}{2}\dot{\phi}^2 - V_\infty - \frac{m_C^2}{2}(\Delta C)^2 - \frac{A_0}{2}\phi^2. \tag{119}$$

As expansion damps the motion,

$$\dot{\Delta C} \to 0, \qquad \dot{\phi} \to 0, \qquad \Delta C \to 0, \qquad \phi \to 0. \tag{120}$$

Therefore

$$\rho_C \to V_\infty, \qquad p_C \to -V_\infty. \tag{121}$$

Hence

$$w_C \to -1. \tag{122}$$

The late residual is therefore the ground-state energy of the saturated configuration.

It is not added after the bounce, and it is not evolved through a separate first-order dark-energy equation.

The same canonical fields generate both the nonsingular transition and the late residual state.

3.11 Matter, radiation, and organizational transfer

The matter density satisfies

$$\dot{\rho}_m + 3H\rho_m = 0, \tag{123}$$

with solution

$$\rho_m = \rho_{m0}a^{-3}. \tag{124}$$

Radiation satisfies

$$\dot{\rho}_r + 4H\rho_r = Q, \tag{125}$$

where

$$Q = \Gamma_{\rm tr}K^2. \tag{126}$$

The field sector satisfies

$$\dot{\rho}_C + 3H(\rho_C + p_C) = -Q. \tag{127}$$

The total continuity equation is

$$\dot{\rho}_{\rm tot} + 3H(\rho_{\rm tot} + p_{\rm tot}) = 0. \tag{128}$$

During expansion,

$$\Gamma_{\rm tr} = 0, \tag{129}$$

so

$$Q = 0. \tag{130}$$

Radiation then returns to

$$\rho_r \propto a^{-4}. \tag{131}$$

Matter remains

$$\rho_m \propto a^{-3}. \tag{132}$$

The saturation fields approach

$$\rho_C \to V_\infty. \tag{133}$$

The resulting sequence is therefore

$$\text{radiation domination} \to \text{matter domination} \to \text{saturation-residual domination}. \tag{134}$$

The sequence is achieved without adding a new late-time fluid after the initial evolution.

3.12 Matter-radiation equality

Equality occurs when

$$\rho_m(a_{\text{eq}}) = \rho_r(a_{\text{eq}}). \tag{135}$$

Using

$$\rho_m = \rho_{m0} a^{-3}, \tag{136}$$

and

$$\rho_r = \rho_{r0} a^{-4}, \tag{137}$$

one obtains

$$\rho_{m0} a_{\text{eq}}^{-3} = \rho_{r0} a_{\text{eq}}^{-4}. \tag{138}$$

Therefore

$$a_{\text{eq}} = \frac{\rho_{r0}}{\rho_{m0}} = \frac{\Omega_{r0}}{\Omega_{m0}}. \tag{139}$$

With

$$\Omega_{m0} = 0.314114, \qquad \Omega_{r0} = 9 \times 10^{-5}, \tag{140}$$

$$a_{\text{eq}} = \frac{9 \times 10^{-5}}{0.314114} \simeq 2.865 \times 10^{-4}. \tag{141}$$

The corresponding redshift is

$$z_{\text{eq}} = \frac{1}{a_{\text{eq}}} - 1, \tag{142}$$

so

$$z_{\text{eq}} \simeq 3489. \tag{143}$$

The final splice endpoint is

$$a_{\text{lock}} = 2.05 \times 10^{-4}, \tag{144}$$

which lies before matter-radiation equality:

$$a_{\text{lock}} < a_{\text{eq}}. \tag{145}$$

The transition into the locked CAMB background therefore occurs while radiation remains dynamically important.

3.13 Acoustic propagation

Before recombination, photons and baryons form a coupled fluid.

The baryon-loading ratio is

$$R(a) = \frac{3\rho_b}{4\rho_\gamma}. \tag{146}$$

Since

$$\rho_b \propto a^{-3}, \qquad \rho_\gamma \propto a^{-4}, \tag{147}$$

one has

$$R(a) = \frac{3\Omega_{b0}}{4\Omega_{\gamma 0}} a. \tag{148}$$

The sound speed is

$$c_s(a) = \frac{c}{\sqrt{3(1+R)}}. \tag{149}$$

The comoving sound horizon is

$$r_s(a) = \int_0^a \frac{c_s(a')}{a'^2 H(a')}\, da'. \tag{150}$$

At recombination,

$$r_s(z_*) = \int_{z_*}^{\infty} \frac{c_s(z)}{H(z)}\, dz. \tag{151}$$

The angular acoustic scale is

$$\theta_* = \frac{r_s(z_*)}{D_M(z_*)}. \tag{152}$$

The locked expansion alters both the early integral for r_s and the distance integral for D_M.

The final value,

$$100\theta_* = 1.041033, \tag{153}$$

shows that these two effects balance to within

$$0.0064\% \tag{154}$$

of the reference acoustic scale.

3.14 BAO propagation

The drag horizon is

$$r_{\text{drag}} = \int_{z_{\text{drag}}}^{\infty} \frac{c_s(z)}{H(z)}\, dz. \tag{155}$$

The BAO observables are ratios of cosmological distances to this ruler:

$$\frac{D_M(z)}{r_{\rm drag}}, \qquad \frac{D_H(z)}{r_{\rm drag}}, \qquad \frac{D_V(z)}{r_{\rm drag}}. \tag{156}$$

An internally inconsistent calculation can produce an artificial BAO discrepancy if $r_{\rm drag}$ is computed from one background while the distances are computed from another.

The final calculation uses the same locked CAMB background for:

$$H(z), \quad r_{\rm drag}, \quad D_M, \quad D_H, \quad D_V. \tag{157}$$

With

$$r_{\rm drag} = 149.960 \text{ Mpc}, \tag{158}$$

the resulting diagnostic BAO statistic is approximately

$$\chi^2_{\rm BAO} \simeq 13.4. \tag{159}$$

The improvement from the earlier value near

$$\chi^2 \simeq 27 \tag{160}$$

was produced by restoring this internal consistency.

3.15 Growth without an effective Newton rescaling

The subhorizon growth equation is

$$\delta_m'' + \mathcal{H}\delta_m' - 4\pi G a^2 \rho_m \delta_m = 0 \tag{161}$$

in conformal time.

Using

$$N = \ln a, \tag{162}$$

on the expanding branch gives

$$\frac{d^2\delta_m}{dN^2} + \left(2 + \frac{d\ln H}{dN}\right)\frac{d\delta_m}{dN} - \frac{3}{2}\Omega_m(a)\delta_m = 0. \tag{163}$$

The locked model changes

$$H(a), \tag{164}$$

and therefore changes

$$\frac{d\ln H}{dN} \tag{165}$$

and

$$\Omega_m(a). \tag{166}$$

It does not require

$$G \to G_{\text{eff}} \tag{167}$$

in the growth equation.

This distinction explains why the canonical field implementation lowers S_8 through the background history while the tested phenomenological μ-models increased σ_8.

The locked result is

$$S_8 = 0.79781. \tag{168}$$

3.16 Lensing geometry

The CMB lensing potential is a line-of-sight projection:

$$\varphi(\hat{\mathbf{n}}) = -2 \int_0^{\chi_*} d\chi \, \frac{\chi_* - \chi}{\chi_* \chi} \, \Phi_W. \tag{169}$$

Its power spectrum depends on:

$$\text{the Weyl potential}, \quad \text{the growth history}, \quad \text{the distance kernel}. \tag{170}$$

Therefore a lower value of S_8 does not necessarily imply a uniformly lower

$$C_L^{\varphi\varphi}. \tag{171}$$

The final locked calculation produces a mildly enhanced lensing-potential spectrum relative to its matched control because the altered geometry and line-of-sight weighting offset the lower present-day clustering amplitude.

Using the Planck bandpower windows,

$$C_b^{\varphi\varphi} = \sum_L W_{bL} C_L^{\varphi\varphi}. \tag{172}$$

The Gaussian bandpower likelihood is

$$\chi^2 = \Delta\mathbf{C}^{\mathrm{T}} \boldsymbol{\Sigma}^{-1} \Delta\mathbf{C}, \tag{173}$$

where

$$\Delta\mathbf{C} = \mathbf{C}_{\text{model}} - \mathbf{C}_{\text{Planck}}. \tag{174}$$

The locked result is

$$\chi^2 = 10.7379 \tag{175}$$

for nine bins.

3.17 Reconstruction noise

CMB lensing reconstruction uses quadratic combinations of the observed CMB fields.

For a generic quadratic estimator,

$$\hat{\varphi}_{LM} = A_L \sum_{\ell_1 m_1 \ell_2 m_2} W^{LM}_{\ell_1 m_1 \ell_2 m_2} X_{\ell_1 m_1} Y_{\ell_2 m_2}. \tag{176}$$

Its Gaussian reconstruction noise is

$$N_L^{(0)}. \tag{177}$$

The locked and control spectra were passed through the same TT estimator forecast.

The resulting ratio was approximately

$$\frac{N^{(0)}_{L,\mathrm{locked}}}{N^{(0)}_{L,\mathrm{control}}} \simeq 1.21 \tag{178}$$

when averaged over the tested range.

Thus the locked model has:

$$\text{a mildly enhanced true lensing signal,} \tag{179}$$

but

$$\text{a higher TT reconstruction noise.} \tag{180}$$

These are not contradictory.

The first concerns the physical projected potential spectrum.

The second concerns the efficiency with which that spectrum is reconstructed from a modified primary TT spectrum.

3.18 The unified interpretation

The mathematical sequence of the theory is:

$$\begin{aligned} &\text{contraction} \\ &\to \text{organizational damping and transfer} \\ &\to \text{finite-density saturation} \\ &\to \text{bounce} \\ &\to \text{radiation expansion} \\ &\to \text{matter expansion} \\ &\to \text{late saturation residual.} \end{aligned} \tag{181}$$

The same fields C and ϕ exist throughout.

Their background energy density is

$$\rho_C = \frac{1}{2}K^2 + V. \tag{182}$$

Their pressure is

$$p_C = \frac{1}{2}K^2 - V. \tag{183}$$

Their perturbations obey canonical second-order equations.

Their stress-energy enters the Einstein equations directly.

The high-density gravitational response is limited through

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{184}$$

The late-time residual follows from

$$C \to C_s, \qquad \phi \to 0, \qquad K^2 \to 0, \tag{185}$$

so that

$$\rho_C \to V_\infty, \qquad w_C \to -1. \tag{186}$$

This is the mathematical content of gravitational saturation.

It is not an algebraic weakening of gravity.

It is not a discontinuous switch between unrelated cosmological eras.

It is not a separate late-time fluid appended to an earlier bounce model.

It is one continuous field system whose gravitational response reaches a finite-density limit and whose fields settle into a stable residual state.

End of Chapter 3

Chapter 4

Gravity as an Organizational Wave

4.1 The physical meaning of gravitational organization

The gravitational-saturation framework begins from a distinction between gravitational influence and gravitational organization.

Gravitational influence is familiar. Matter and energy determine curvature through the Einstein field equations,

$$G_{\mu\nu} = 8\pi G\, T_{\mu\nu}. \tag{1}$$

Gravitational organization refers to the cumulative manner in which that influence is distributed, reinforced, redirected, and stabilized across an evolving spacetime.

The distinction is necessary because a gravitational field is not merely a static force profile. It propagates, couples to geometry, carries phase information, and evolves through a background whose own dynamics are determined by the same field equations.

In a static weak-field system, the gravitational potential obeys

$$\nabla^2 \Phi_N = 4\pi G\rho. \tag{2}$$

In a relativistic dynamical system, the metric perturbation obeys a wave equation. In Lorenz gauge,

$$\Box \bar{h}_{\mu\nu} = -16\pi G\, T_{\mu\nu}, \tag{3}$$

and in vacuum,

$$\Box \bar{h}_{\mu\nu} = 0. \tag{4}$$

The propagation of gravity as a wave introduces dynamical features absent from a purely static interpretation:

$$\begin{aligned} &\text{phase,}\\ &\text{interference,}\\ &\text{propagation delay,}\\ &\text{mode coupling,}\\ &\text{damping,}\\ &\text{restoration,}\\ &\text{limiting behavior.} \end{aligned} \tag{5}$$

The hypothesis developed here is that the large-scale organization of gravity can approach a limiting state.

That limit is called saturation.

4.2 Why the wave character matters

A wave is not defined only by its amplitude. It is defined by the evolution of amplitude and phase together.

A simple harmonic mode may be written as

$$\Psi(t, \mathbf{x}) = A\cos\big(\mathbf{k}\cdot\mathbf{x} - \omega t + \varphi\big). \tag{6}$$

The observable value of the wave depends upon

$$A, \quad \omega, \quad \mathbf{k}, \quad \varphi. \tag{7}$$

Two waves with equal amplitude can either reinforce or cancel one another depending upon their relative phase.

For two components,

$$\Psi_1 = A_1\cos(\Theta), \qquad \Psi_2 = A_2\cos(\Theta + \Delta\varphi), \tag{8}$$

the combined wave is

$$\Psi_{\mathrm{net}} = \Psi_1 + \Psi_2. \tag{9}$$

For equal amplitudes,

$$A_1 = A_2 = A, \tag{10}$$

the combined amplitude is

$$A_{\mathrm{net}} = 2A\cos\left(\frac{\Delta\varphi}{2}\right). \tag{11}$$

If

$$\Delta\varphi = 0, \tag{12}$$

then

$$A_{\mathrm{net}} = 2A. \tag{13}$$

The waves reinforce.

If

$$\Delta\varphi = \pi, \tag{14}$$

then

$$A_{\mathrm{net}} = 0. \tag{15}$$

The waves cancel.

The gravitational-saturation model does not impose a literal standing-wave cancellation condition upon spacetime. Rather, this wave behavior motivates the presence of a dynamical phase field capable of opposing unrestricted cumulative organization.

The phase variable ϕ is therefore not decorative. It represents the capacity of the organizational sector to carry a periodic degree of freedom whose restoring dynamics can oppose runaway growth.

4.3 The organizational field C

The variable

$$C \tag{16}$$

measures the state of gravitational organization.

It is dimensionless in the normalized numerical formulation and evolves toward a preferred saturation value

$$C_s = 0.95. \tag{17}$$

The difference from saturation is

$$\Delta C = C - C_s. \tag{18}$$

The locked potential in the C direction is

$$V_C(C) = V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4. \tag{19}$$

Using

$$\Delta C = C - C_s, \tag{20}$$

this becomes

$$V_C = V_\infty + \frac{m_C^2}{2}(\Delta C)^2 + \frac{\lambda}{4}(\Delta C)^4. \tag{21}$$

The corresponding restoring force is

$$F_C = -\frac{dV_C}{dC}, \tag{22}$$

so

$$F_C = -m_C^2(C - C_s) - \lambda(C - C_s)^3. \tag{23}$$

If

$$C < C_s, \tag{24}$$

then

$$C - C_s < 0, \tag{25}$$

and therefore

$$F_C > 0. \tag{26}$$

The field is driven upward.

If

$$C > C_s, \tag{27}$$

then

$$C - C_s > 0, \tag{28}$$

and therefore

$$F_C < 0. \tag{29}$$

The field is driven downward.

Thus C_s is a dynamical attractor.

The curvature of the potential is

$$V_{,CC} = m_C^2 + 3\lambda(C - C_s)^2. \tag{30}$$

Since

$$m_C^2 > 0, \qquad \lambda \geq 0, \tag{31}$$

one has

$$V_{,CC} > 0 \tag{32}$$

for all C.

The organizational potential is globally convex.

There is no unstable local maximum between the initial state and saturation.

4.4 Saturation is not a hard cutoff

A hard cutoff would impose a rule such as

$$C \leq C_s \tag{33}$$

by definition, or would force

$$\dot{C} = 0 \tag{34}$$

once the field reached a chosen value.

That is not the construction used here.

Instead, the field remains second order:

$$\ddot{C} + 3H\dot{C} + V_{,C} = 0 \tag{35}$$

during expansion.

The approach to saturation follows from three ordinary dynamical ingredients:

$$\text{Hubble damping}, \quad \text{positive restoring curvature}, \quad \text{a unique potential minimum}. \tag{36}$$

The field can overshoot, oscillate, and damp.

The late state is not imposed algebraically.

It is approached dynamically.

Near saturation,

$$C = C_s + \Delta C, \tag{37}$$

with

$$|\Delta C| \ll 1. \tag{38}$$

The equation becomes

$$\ddot{\Delta C} + 3H\dot{\Delta C} + m_C^2\Delta C \simeq 0. \tag{39}$$

This is the equation of a damped oscillator with time-dependent damping.

If the background approaches a nearly constant expansion rate,

$$H \simeq H_\infty, \tag{40}$$

then the linear equation has solutions of the form

$$\Delta C \propto e^{rt}, \tag{41}$$

where

$$r^2 + 3H_\infty r + m_C^2 = 0. \tag{42}$$

The roots are

$$r_\pm = \frac{-3H_\infty \pm \sqrt{9H_\infty^2 - 4m_C^2}}{2}. \tag{43}$$

If

$$9H_\infty^2 > 4m_C^2, \tag{44}$$

the system is overdamped.

If

$$9H_\infty^2 = 4m_C^2, \tag{45}$$

the system is critically damped.

If

$$9H_\infty^2 < 4m_C^2, \tag{46}$$

the field undergoes damped oscillations.

In every case, the real part of the roots is

$$\Re(r_\pm) = -\frac{3}{2}H_\infty < 0 \tag{47}$$

for expansion.

Therefore

$$\Delta C \to 0. \tag{48}$$

Saturation is an asymptotic dynamical state, not a manually enforced ceiling.

4.5 The phase field ϕ

The phase sector is governed by

$$V_\phi(\phi) = A_0(1 - \cos\phi). \tag{49}$$

Its derivative is

$$V_{,\phi} = A_0 \sin\phi. \tag{50}$$

Its curvature is

$$V_{,\phi\phi} = A_0 \cos\phi. \tag{51}$$

At

$$\phi = 0, \tag{52}$$

one has

$$V_{,\phi} = 0, \qquad V_{,\phi\phi} = A_0. \tag{53}$$

For

$$A_0 > 0, \tag{54}$$

the phase minimum is stable.

The background equation is

$$\ddot{\phi} + 3H\dot{\phi} + A_0 \sin\phi = 0 \tag{55}$$

during expansion.

For small ϕ,

$$\sin\phi \simeq \phi, \tag{56}$$

so

$$\ddot{\phi} + 3H\dot{\phi} + A_0\phi \simeq 0. \tag{57}$$

The phase field therefore behaves as a damped oscillator near its minimum.

Its periodic potential distinguishes it from an ordinary unbounded displacement variable.

The phase may be shifted by

$$\phi \to \phi + 2\pi n, \tag{58}$$

without changing the potential:

$$V_\phi(\phi + 2\pi n) = V_\phi(\phi). \tag{59}$$

This periodicity represents the wave character of the sector.

4.6 Phase cancellation and dynamic tension

The simplest mechanical picture of saturation is the balance between inward and outward organizational tendencies.

Let an effective inward organizational amplitude be represented by

$$A_{\text{in}}, \tag{60}$$

and an opposing response by

$$A_{\text{out}}. \tag{61}$$

The net organization is

$$A_{\text{net}} = A_{\text{in}} + A_{\text{out}}. \tag{62}$$

If the opposing component develops an approximately opposite phase,

$$A_{\mathrm{out}} \simeq -A_{\mathrm{in}}, \tag{63}$$

then

$$A_{\mathrm{net}} \simeq 0. \tag{64}$$

This does not mean that both contributions vanish.

It means that the net capacity for further organization vanishes.

The situation is analogous to dynamic tension.

Two forces may be large while their net force is small:

$$\mathbf{F}_{\mathrm{net}} = \mathbf{F}_1 + \mathbf{F}_2 \simeq 0. \tag{65}$$

Likewise, two wave contributions may remain individually nonzero while the net propagating amplitude is suppressed.

The saturation state is interpreted as such a dynamically balanced configuration.

The numerical realization does not evolve A_{in} and A_{out} as separate algebraic variables. Their physical role is carried by the coupled evolution of

$$C, \quad \phi, \tag{66}$$

their kinetic energies, and the shape of the potential.

4.7 Total saturation-sector energy

The homogeneous saturation energy density is

$$\rho_C = \frac{1}{2}\dot{C}^2 + \frac{1}{2}\dot{\phi}^2 + V(C,\phi). \tag{67}$$

The pressure is

$$p_C = \frac{1}{2}\dot{C}^2 + \frac{1}{2}\dot{\phi}^2 - V(C,\phi). \tag{68}$$

Define

$$K^2 = \dot{C}^2 + \dot{\phi}^2. \tag{69}$$

Then

$$\rho_C = \frac{1}{2}K^2 + V, \tag{70}$$

and

$$p_C = \frac{1}{2}K^2 - V. \tag{71}$$

The equation-of-state parameter is

$$w_C = \frac{\frac{1}{2}K^2 - V}{\frac{1}{2}K^2 + V}. \tag{72}$$

Three limiting regimes follow.

Kinetic domination. If

$$\frac{1}{2}K^2 \gg V, \tag{73}$$

then

$$w_C \simeq +1. \tag{74}$$

Kinetic-potential balance. If

$$\frac{1}{2}K^2 \simeq V, \tag{75}$$

then

$$w_C \simeq 0. \tag{76}$$

Potential domination. If

$$\frac{1}{2}K^2 \ll V, \tag{77}$$

then

$$w_C \simeq -1. \tag{78}$$

The late saturated state is potential dominated.

At the minimum,

$$C \to C_s, \qquad \phi \to 0, \qquad K^2 \to 0, \tag{79}$$

so

$$\rho_C \to V_\infty, \qquad p_C \to -V_\infty. \tag{80}$$

The residual floor is therefore the minimum energy of the organizational sector.

4.8 The role of Hubble friction

The expansion rate enters both field equations through

$$3H\dot{C} \qquad \text{and} \qquad 3H\dot{\phi}. \tag{81}$$

During expansion,

$$H > 0, \tag{82}$$

so the terms dissipate kinetic motion.

Multiplying the C equation by $\dot{C}$ gives

$$\dot{C}\ddot{C} + 3H\dot{C}^2 + V_{,C}\dot{C} = 0. \tag{83}$$

The first and third terms combine:

$$\frac{d}{dt}\Big(\frac{1}{2}\dot{C}^2 + V_C\Big) = -3H\dot{C}^2. \tag{84}$$

Similarly,

$$\frac{d}{dt}\Big(\frac{1}{2}\dot{\phi}^2 + V_\phi\Big) = -3H\dot{\phi}^2. \tag{85}$$

Adding them,

$$\dot{\rho}_C = -3H\big(\dot{C}^2 + \dot{\phi}^2\big) \tag{86}$$

during expansion.

Since

$$H > 0, \qquad \dot{C}^2 + \dot{\phi}^2 \geq 0, \tag{87}$$

one has

$$\dot{\rho}_C \leq 0 \tag{88}$$

until the fields settle at the residual floor.

The energy cannot decrease below

$$V_\infty \tag{89}$$

because the potential is bounded below by that value.

Therefore

$$\rho_C \to V_\infty. \tag{90}$$

This is the mathematical mechanism by which a dynamical saturation sector becomes a nearly constant late residual.

4.9 Contraction and anti-friction

During contraction,

$$H < 0. \tag{91}$$

The same terms

$$3H\dot{C} \qquad \text{and} \qquad 3H\dot{\phi} \tag{92}$$

become anti-friction.

The energy equation becomes

$$\dot{\rho}_C = -3H\big(\dot{C}^2 + \dot{\phi}^2\big). \tag{93}$$

Since

$$H < 0, \tag{94}$$

one obtains

$$\dot{\rho}_C > 0. \tag{95}$$

Contraction pumps kinetic energy into the fields.

Without regulation, this can lead to rapid kinetic growth.

The transfer term was introduced to prevent uncontrolled amplification:

$$\Gamma_{\text{tr}} = W_-(H)\big[-3H + \Gamma_0 S_\rho\big]. \tag{96}$$

The field equations become

$$\ddot{C} + \big(3H + \Gamma_{\text{tr}}\big)\dot{C} + V_{,C} = 0, \tag{97}$$

$$\ddot{\phi} + \big(3H + \Gamma_{\rm tr}\big)\dot{\phi} + V_{,\phi} = 0. \tag{98}$$

Deep in contraction,

$$W_- \simeq 1, \tag{99}$$

so

$$3H + \Gamma_{\rm tr} \simeq \Gamma_0 S_\rho. \tag{100}$$

The anti-friction is cancelled.

The fields are then damped rather than explosively amplified.

The corresponding energy loss is

$$Q = \Gamma_{\rm tr} K^2, \tag{101}$$

and is transferred into radiation:

$$\dot{\rho}_r + 4H\rho_r = Q. \tag{102}$$

The organizational sector therefore converts contraction-driven kinetic growth into radiation while preserving total energy.

4.10 Saturation in the background constraint

The wave and field dynamics describe the organizational sector.

The limiting gravitational response appears in the background constraint:

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{103}$$

The quantity

$$\frac{\rho}{\rho_s} \tag{104}$$

measures proximity to the saturation density.

Define

$$x = \frac{\rho}{\rho_s}. \tag{105}$$

Then

$$H^2 = \frac{\rho_s}{3} x(1-x). \tag{106}$$

The function

$$f(x) = x(1-x) \tag{107}$$

has zeros at

$$x = 0 \qquad \text{and} \qquad x = 1. \tag{108}$$

Its derivative is

$$f'(x) = 1 - 2x. \tag{109}$$

The maximum occurs at

$$x = \frac{1}{2}. \tag{110}$$

At this point,

$$f\left(\frac{1}{2}\right) = \frac{1}{4}. \tag{111}$$

Therefore the maximum value of H^2 permitted by the saturated constraint is

$$H^2_{\max} = \frac{\rho_s}{12}. \tag{112}$$

Unlike the standard relation

$$H^2 = \frac{\rho}{3}, \tag{113}$$

the saturated relation does not allow H^2 to grow without bound as ρ increases.

Instead,

$$H^2 \tag{114}$$

rises until

$$\rho = \frac{\rho_s}{2}, \tag{115}$$

then falls to zero at

$$\rho = \rho_s. \tag{116}$$

This is the exact mathematical sense in which the gravitational response saturates.

The energy density can approach a finite limiting value while the magnitude of the expansion or contraction rate declines toward zero.

4.11 Saturation and rebound

The generalized Raychaudhuri equation is

$$\dot{H} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right). \tag{117}$$

At low density,

$$\rho \ll \rho_s, \tag{118}$$

so

$$\dot{H} \simeq -\frac{1}{2}(\rho + p). \tag{119}$$

The standard result is recovered.

At

$$\rho = \frac{\rho_s}{2}, \tag{120}$$

one has

$$\dot{H} = 0. \tag{121}$$

At higher density,

$$\rho > \frac{\rho_s}{2}, \tag{122}$$

the factor

$$1 - 2\frac{\rho}{\rho_s} \tag{123}$$

is negative.

If

$$\rho + p > 0, \tag{124}$$

then

$$\dot{H} > 0. \tag{125}$$

Thus the contraction rate stops becoming more negative.

It turns upward.

At

$$\rho = \rho_s, \tag{126}$$

the constraint gives

$$H = 0. \tag{127}$$

The system crosses continuously from contraction to expansion.

The rebound is therefore not caused by negative matter energy.

It is caused by the saturation of the gravitational response at high density.

4.12 The bounce as a phase-space trajectory

The state of the homogeneous system is

$$\mathbf{y} = \big(a, H, C, \Pi_C, \phi, \Pi_\phi, \rho_m, \rho_r\big). \tag{128}$$

The bounce is not an isolated algebraic event.

It is a continuous trajectory in this phase space.

Before the bounce,

$$H < 0. \tag{129}$$

At the bounce,

$$H = 0, \qquad \rho = \rho_s, \qquad a = a_{\min} > 0. \tag{130}$$

After the bounce,

$$H > 0. \tag{131}$$

All field variables remain finite:

$$|C| < \infty, \qquad |\phi| < \infty, \qquad |\Pi_C| < \infty, \qquad |\Pi_\phi| < \infty. \tag{132}$$

The kinetic energy is

$$K^2_{\text{bounce}} = \Pi_C^2 + \Pi_\phi^2 \tag{133}$$

evaluated at

$$H = 0. \tag{134}$$

The numerical requirement is

$$K^2_{\text{bounce}} < \infty. \tag{135}$$

The minimum scale factor is determined dynamically by the integrated trajectory.

The locked runs produce a finite value near

$$a_{\min} \simeq 2.07 \times 10^{-5}. \tag{136}$$

Therefore the bounce occurs at nonzero scale factor and finite density.

4.13 Why cosmic time is required

Using

$$N = \ln a, \tag{137}$$

one has

$$\frac{d}{dN} = \frac{1}{H}\frac{d}{dt}. \tag{138}$$

Therefore

$$\frac{dC}{dN} = \frac{\Pi_C}{H}, \qquad \frac{d\phi}{dN} = \frac{\Pi_\phi}{H}. \tag{139}$$

At the bounce,

$$H = 0. \tag{140}$$

These equations are singular even if

$$\Pi_C, \Pi_\phi \tag{141}$$

remain finite.

The singularity is not physical.

It is introduced by the choice of independent variable.

The system must therefore be integrated in cosmic time,

$$t, \tag{142}$$

or in a nonsingular affine variable,

$$\tau. \tag{143}$$

The cosmic-time system is

$$\dot{a} = aH, \qquad \dot{C} = \Pi_C, \qquad \dot{\phi} = \Pi_\phi, \tag{144}$$

together with the remaining first-order equations.

No division by H occurs.

The bounce can then be crossed without switching variables or inserting a special algebraic bridge.

4.14 Affine scaling

The large value of

$$\rho_s \tag{145}$$

creates a wide dynamic range.

Define

$$h = \frac{H}{\sqrt{\rho_s}}, \tag{146}$$

$$u_C = \frac{\Pi_C}{\sqrt{\rho_s}}, \tag{147}$$

$$u_\phi = \frac{\Pi_\phi}{\sqrt{\rho_s}}, \tag{148}$$

$$\bar{\rho} = \frac{\rho}{\rho_s}. \tag{149}$$

The constraint becomes

$$h^2 = \frac{\bar{\rho}}{3}(1 - \bar{\rho}). \tag{150}$$

Define affine time by

$$d\tau = \sqrt{\rho_s}\, dt. \tag{151}$$

Then

$$\frac{da}{d\tau} = ah, \qquad \frac{dC}{d\tau} = u_C, \qquad \frac{d\phi}{d\tau} = u_\phi. \tag{152}$$

The scaled system preserves the phase-space trajectory while reducing stiffness caused by large dimensionful values.

4.15 The meaning of the locked state

The term locked refers to the asymptotic organization of the background after the nonsingular trajectory has entered its stable expanding branch.

It does not mean that

$$a \tag{153}$$

becomes constant.

It does not mean that

$$H \tag{154}$$

vanishes.

It means that the organizational fields have approached their stable configuration and the expansion history is governed by the resulting residual state.

The late limits are

$$C \to C_s, \qquad \phi \to 0, \qquad \Pi_C \to 0, \qquad \Pi_\phi \to 0. \tag{155}$$

Therefore

$$\rho_C \to V_\infty, \qquad w_C \to -1. \tag{156}$$

The background is then locked into a near-residual expansion regime.

The final CAMB splice embeds this physically evolved trajectory into the Boltzmann background after the early nonsingular branch has been resolved.

The retained endpoint is

$$a_{\text{lock}} = 2.05 \times 10^{-4}. \tag{157}$$

The narrowed transition width is

$$0.67W_0. \tag{158}$$

The splice is smooth in the expansion function and is normalized so that

$$E(1) = 1. \tag{159}$$

4.16 Why the observable effect is mainly background-level

The perturbation amplitudes of the canonical fields remain small:

$$\max|\delta C| \simeq 0.035, \qquad \max|\delta\phi| \simeq 0.049. \tag{160}$$

Their direct stress-energy contributions are proportional to combinations such as

$$C'_{\text{bg}} v_C, \quad \phi'_{\text{bg}} v_\phi, \quad V_{,C}\delta C, \quad V_{,\phi}\delta\phi. \tag{161}$$

Near saturation,

$$C'_{\text{bg}} \simeq 0, \qquad \phi'_{\text{bg}} \simeq 0, \qquad V_{,C} \simeq 0, \qquad V_{,\phi} \simeq 0. \tag{162}$$

Therefore the perturbative stress-energy source is doubly suppressed:

$$\text{small field perturbations} \quad \text{multiplied by} \quad \text{small background velocities or slopes.} \tag{163}$$

This explains why the direct change in

$$\sigma_8 \tag{164}$$

and the primary acoustic peaks is at the level of approximately

$$10^{-6}. \tag{165}$$

The same fields that created the locked background no longer produce a large linear force once they have settled near the potential minimum.

The principal observable effect is therefore carried by

$$H(a), \tag{166}$$

not by a strong late scalar interaction.

4.17 Why reduced-gravity closures fail

Suppose one replaces the Poisson equation by

$$k^2\Psi = -4\pi G a^2 \mu(a,k)\rho\Delta. \tag{167}$$

If

$$\mu < 1, \tag{168}$$

one might expect weaker growth.

However, in the full relativistic perturbation system, μ also alters the metric constraints, potential evolution, radiation driving, integrated effects, and transfer functions.

The tested phenomenological closures changed the Einstein system directly.

The canonical saturation fields do not behave in this way.

Their contribution is additive:

$$\delta G_{\mu\nu} = 8\pi G\big(\delta T^{\text{ordinary}}_{\mu\nu} + \delta T^{C\phi}_{\mu\nu}\big). \tag{169}$$

They do not impose

$$G \to \mu G. \tag{170}$$

Near saturation,

$$\delta T^{C\phi}_{\mu\nu} \tag{171}$$

is small.

Therefore the ordinary gravitational coupling remains effectively unchanged for the linear matter sector.

The lowered value of

$$S_8 \tag{172}$$

comes from the modified expansion history:

$$H(a), \tag{173}$$

not from an algebraic suppression of Newton's constant.

4.18 Cosmic residual and local weak-field behavior

The cosmic residual is

$$V_\infty. \tag{174}$$

It is homogeneous at the background level.

Near the minimum, the organizational field has effective mass squared

$$m^2_{\text{eff,C}} = V_{,CC} = m_C^2 + 3\lambda(C - C_s)^2. \tag{175}$$

At saturation,

$$m^2_{\text{eff,C}} = m_C^2 = 1.827221 \tag{176}$$

in the model normalization.

Since

$$m^2_{\text{eff,C}} > 0, \tag{177}$$

the field is restorative rather than tachyonic.

A positive mass suppresses long-range field excursions.

The corresponding physical Compton wavelength would be

$$\lambda_C = \frac{\hbar}{m_{\mathrm{eff,C}}c} \tag{178}$$

once the dimensionless model mass is translated into physical units.

In the locked cosmological calculation, the field behaves as a stable massive mode near its minimum.

Thus the late background may behave as

$$\text{standard weak-field gravity} + \text{smooth residual}, \tag{179}$$

rather than requiring a strong unscreened fifth force.

4.19 Cosmic saturation and galactic response

The cosmic and galactic interpretations occupy different dynamical regimes.

At the cosmic background level,

$$C \simeq C_s, \qquad w_C \simeq -1. \tag{180}$$

The field is nearly frozen.

At galactic scales, the earlier organizational interpretation allows a response to weak acceleration through an interpolating relation.

A representative screened form is

$$\mu(x) = \frac{x}{\sqrt{1+x^2}}, \tag{181}$$

where

$$x = \frac{g}{g_*}. \tag{182}$$

The effective gravitational equation may be written as

$$g\,\mu\left(\frac{g}{g_*}\right) = g_N. \tag{183}$$

At high acceleration,

$$g \gg g_*, \tag{184}$$

so

$$\mu \to 1, \tag{185}$$

and

$$g \simeq g_N. \tag{186}$$

At low acceleration,

$$g \ll g_*, \tag{187}$$

so

$$\mu \simeq \frac{g}{g_*}, \tag{188}$$

and

$$\frac{g^2}{g_*} \simeq g_N. \tag{189}$$

Thus

$$g \simeq \sqrt{g_N g_*}. \tag{190}$$

For a pointlike baryonic mass,

$$g_N = \frac{GM_b}{r^2}, \tag{191}$$

so

$$g \simeq \frac{\sqrt{GM_b g_*}}{r}. \tag{192}$$

Circular velocity obeys

$$\frac{v^2}{r} = g, \tag{193}$$

hence

$$v^2 \simeq \sqrt{GM_b g_*}. \tag{194}$$

Squaring,

$$v^4 \simeq GM_b g_*. \tag{195}$$

This reproduces a baryonic Tully-Fisher-type scaling.

The locked cosmic calculation does not derive this galactic interpolator from the same covariant field equations. The two regimes are nevertheless mechanically compatible:

$$\text{cosmic saturation} \rightarrow \text{frozen residual}, \tag{196}$$

$$\text{galactic weak acceleration} \rightarrow \text{organizational response}. \tag{197}$$

The same physical idea is that gravitational organization becomes nontrivial near a limiting regime, but the corresponding scale-dependent derivation is treated separately.

4.20 The wave-mechanical interpretation

The complete physical interpretation may be summarized through four stages.

Stage I: amplification. During contraction,

$$H < 0, \tag{198}$$

and ordinary Hubble terms produce anti-friction.

The organizational kinetic energy tends to grow.

Stage II: transfer. The contraction selector activates:

$$W_-(H) > 0. \tag{199}$$

The transfer rate

$$\Gamma_{\rm tr} \tag{200}$$

cancels anti-friction and moves kinetic energy into radiation.

Stage III: saturation. The total density approaches

$$\rho_s. \tag{201}$$

The gravitational response obeys

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right), \tag{202}$$

so

$$H \to 0 \tag{203}$$

at finite density.

Stage IV: locking. After the bounce,

$$H > 0. \tag{204}$$

The transfer shuts off,

$$\Gamma_{\rm tr} = 0, \tag{205}$$

and expansion damps the canonical fields toward

$$C_s \qquad \text{and} \qquad \phi = 0. \tag{206}$$

The residual approaches

$$V_\infty. \tag{207}$$

In wave language, the system evolves from amplification to regulated transfer, then to limiting response, and finally to a phase-restored stable state.

4.21 Mathematical statement of the hypothesis

The gravitational-saturation hypothesis may be stated mathematically as follows.

There exists a finite density scale

$$\rho_s > 0 \tag{208}$$

such that the homogeneous gravitational response is

$$H^2 = \frac{\rho}{3} F\left(\frac{\rho}{\rho_s}\right), \tag{209}$$

where

$$F(0) = 1, \qquad F(1) = 0. \tag{210}$$

For the locked model,

$$F(x) = 1 - x. \tag{211}$$

There exists an organizational field

$$C \tag{212}$$

with a unique stable minimum

$$C = C_s, \tag{213}$$

and a phase field

$$\phi \tag{214}$$

with a stable periodic minimum

$$\phi = 0 \pmod{2\pi}. \tag{215}$$

Their potential satisfies

$$V_{,C}(C_s, 0) = 0, \qquad V_{,\phi}(C_s, 0) = 0, \qquad V_{,CC}(C_s, 0) > 0, \qquad V_{,\phi\phi}(C_s, 0) > 0. \tag{216}$$

Their kinetic energy remains finite through

$$H = 0. \tag{217}$$

Their late-time evolution satisfies

$$C \to C_s, \qquad \phi \to 0, \qquad \dot{C} \to 0, \qquad \dot{\phi} \to 0. \tag{218}$$

Therefore

$$\rho_C \to V_\infty, \qquad w_C \to -1. \tag{219}$$

This set of conditions defines a continuous gravitationally saturated cosmology.

4.22 Consequences of the organizational-wave picture

Several consequences follow directly.

First, the bounce is not treated as an arbitrary reversal.

It is the point at which the gravitational response reaches its limiting density:

$$\rho = \rho_s. \tag{220}$$

Second, the late residual is not an unrelated cosmological component.

It is the minimum energy of the same organizational fields:

$$V(C_s, 0) = V_\infty. \tag{221}$$

Third, the canonical perturbations need not create a large fifth force.

Near the minimum,

$$V_{,CC} > 0, \tag{222}$$

and the field velocities are small.

Fourth, the background can change observable growth and distances while the direct linear field perturbations remain negligible.

Fifth, algebraic modified-gravity closures need not reproduce the theory, because saturation is encoded in the full background trajectory and canonical stress-energy rather than in a single factor multiplying the Poisson equation.

4.23 Final formulation

Gravity is treated in this theory as a geometric field with wave propagation and an additional organizational degree of freedom.

The wave character motivates phase dynamics.

The organizational field supplies a measure of approach toward saturation.

The high-density constraint limits the gravitational response.

The contraction transfer controls anti-friction and preserves finite kinetic energy.

The convex potential restores the fields toward a unique stable state.

The late residual is the energy of that saturated configuration.

The unified system is therefore

$$\boxed{\text{wave propagation} + \text{phase response} + \text{organizational restoration} + \text{high-density saturation}} \tag{223}$$

evolving within the Einstein-equation structure.

The essential claim is not that gravity behaves like ordinary sound.

The claim is that gravity, because it is dynamical and wave-bearing, can possess the same broad classes of behavior found in other organized wave systems:

$$\text{interference}, \quad \text{damping}, \quad \text{restoration}, \quad \text{dynamic balance}, \quad \text{saturation}. \tag{224}$$

The following chapters develop how this physical interpretation produces the locked cosmological background, how that background is embedded in the Boltzmann hierarchy, and how the resulting model performs across the completed numerical tests.

End of Chapter 4

Chapter 5

Covariant Field Structure and Conservation Laws

5.1 Purpose of the covariant formulation

The gravitational-saturation cosmology is built from two distinct but connected mathematical structures.

The first is the canonical field sector,

$$C(x^\mu), \qquad \phi(x^\mu), \tag{1}$$

which carries the organizational and phase dynamics.

The second is the high-density gravitational response,

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right), \tag{2}$$

which limits the homogeneous expansion or contraction rate as the total density approaches the saturation density.

The field sector determines the stress-energy, restoration, phase evolution, transfer, and late residual. The modified constraint determines the nonsingular high-density geometry.

At ordinary density,

$$\rho \ll \rho_s, \tag{3}$$

the correction disappears:

$$1 - \frac{\rho}{\rho_s} \simeq 1. \tag{4}$$

Therefore the background approaches the standard Einstein-Friedmann limit,

$$H^2 \simeq \frac{\rho}{3}. \tag{5}$$

The purpose of this chapter is to state the field theory, stress-energy tensor, exchange current, conservation equations, and homogeneous reduction as one complete mathematical system.

5.2 Total action

The total action is separated into gravitational, saturation-field, matter, and radiation sectors:

$$S = S_g + S_{C\phi} + S_m + S_r. \tag{6}$$

The Einstein-Hilbert term is

$$S_g = \frac{1}{16\pi G}\int d^4x\sqrt{-g}\,R. \tag{7}$$

The canonical saturation-field action is

$$S_{C\phi} = \int d^4x\sqrt{-g}\left[-\frac{1}{2}g^{\mu\nu}\nabla_\mu C\nabla_\nu C - \frac{1}{2}g^{\mu\nu}\nabla_\mu\phi\nabla_\nu\phi - V(C,\phi)\right]. \tag{8}$$

The matter and radiation actions are

$$S_m = \int d^4x\sqrt{-g}\,\mathcal{L}_m, \qquad S_r = \int d^4x\sqrt{-g}\,\mathcal{L}_r. \tag{9}$$

The potential is

$$V(C,\phi) = V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4 + A_0(1 - \cos\phi). \tag{10}$$

The locked values are

$$C_s = 0.95, \tag{11}$$
$$V_\infty = 2.187987, \tag{12}$$
$$m_C^2 = 1.827221, \tag{13}$$
$$\lambda = 0.802722. \tag{14}$$

The derivatives are

$$V_{,C} = m_C^2(C - C_s) + \lambda(C - C_s)^3, \tag{15}$$
$$V_{,\phi} = A_0 \sin\phi, \tag{16}$$
$$V_{,CC} = m_C^2 + 3\lambda(C - C_s)^2, \tag{17}$$
$$V_{,\phi\phi} = A_0 \cos\phi, \tag{18}$$
$$V_{,C\phi} = 0. \tag{19}$$

The absence of a mixed potential derivative implies that the fields are not directly mixed through the Hessian at linear order.

5.3 Variation with respect to the metric

The stress-energy tensor of the saturation sector is

$$T_{\mu\nu}^{C\phi} = -\frac{2}{\sqrt{-g}}\frac{\delta S_{C\phi}}{\delta g^{\mu\nu}}. \tag{20}$$

Evaluating the variation gives

$$T_{\mu\nu}^{C\phi} = \nabla_\mu C \nabla_\nu C + \nabla_\mu\phi\nabla_\nu\phi - g_{\mu\nu}\left[\frac{1}{2}\nabla_\alpha C\nabla^\alpha C + \frac{1}{2}\nabla_\alpha\phi\nabla^\alpha\phi + V(C,\phi)\right]. \tag{21}$$

The total stress-energy tensor is

$$T_{\mu\nu} = T_{\mu\nu}^m + T_{\mu\nu}^r + T_{\mu\nu}^{C\phi}. \tag{22}$$

The Einstein equation retains its standard geometric form:

$$G_{\mu\nu} = 8\pi G\,T_{\mu\nu}. \tag{23}$$

In the units used for the background derivations,

$$8\pi G = 1, \tag{24}$$

so

$$G_{\mu\nu} = T_{\mu\nu}. \tag{25}$$

The saturation modification enters the high-density homogeneous response through the modified Friedmann constraint rather than through an algebraic rescaling of the Einstein tensor.

5.4 Variation with respect to C

Variation of the field action with respect to C gives

$$\delta S_{C\phi} = \int d^4x\sqrt{-g}\big[-\nabla^\mu C\nabla_\mu(\delta C) - V_{,C}\delta C\big]. \tag{26}$$

Integrating the kinetic term by parts gives

$$\delta S_{C\phi} = \int d^4x\sqrt{-g}\big[\Box C - V_{,C}\big]\delta C, \tag{27}$$

apart from a boundary term.

Therefore the covariant field equation is

$$\Box C - V_{,C} = 0. \tag{28}$$

Equivalently,

$$\nabla_\mu\nabla^\mu C - m_C^2(C - C_s) - \lambda(C - C_s)^3 = 0. \tag{29}$$

During contraction, the transfer current modifies the homogeneous equation by adding the damping term $\Gamma_{\rm tr}\dot{C}$. The canonical expansion-era equation remains the covariant Klein-Gordon form.

5.5 Variation with respect to ϕ

Variation with respect to ϕ gives

$$\Box\phi - V_{,\phi} = 0. \tag{30}$$

Therefore

$$\Box\phi - A_0\sin\phi = 0. \tag{31}$$

Near the minimum,

$$|\phi| \ll 1, \tag{32}$$

so

$$\sin\phi \simeq \phi. \tag{33}$$

The phase equation becomes

$$\Box\phi - A_0\phi \simeq 0. \tag{34}$$

The effective phase mass squared is therefore

$$m_\phi^2 = A_0 \tag{35}$$

near

$$\phi = 0. \tag{36}$$

For

$$A_0 > 0, \tag{37}$$

the phase mode is locally stable.

5.6 Homogeneous reduction

For a flat FLRW metric,

$$ds^2 = -dt^2 + a^2(t)\, d\mathbf{x}^2, \tag{38}$$

the homogeneous fields satisfy

$$C = C(t), \qquad \phi = \phi(t). \tag{39}$$

The covariant d'Alembertian acting on a homogeneous scalar is

$$\Box C = -\ddot{C} - 3H\dot{C}, \tag{40}$$

and similarly,

$$\Box\phi = -\ddot{\phi} - 3H\dot{\phi}. \tag{41}$$

Therefore the expansion-era background equations are

$$\ddot{C} + 3H\dot{C} + V_{,C} = 0, \tag{42}$$

and

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0. \tag{43}$$

Defining the momenta

$$\Pi_C = \dot{C}, \qquad \Pi_\phi = \dot{\phi}, \tag{44}$$

the equations become first order:

$$\dot{C} = \Pi_C, \tag{45}$$

$$\dot{\Pi}_C = -3H\Pi_C - V_{,C}, \tag{46}$$

$$\dot{\phi} = \Pi_\phi, \tag{47}$$

$$\dot{\Pi}_\phi = -3H\Pi_\phi - V_{,\phi}. \tag{48}$$

5.7 Homogeneous energy density and pressure

For the homogeneous fields,

$$\nabla_i C = 0, \qquad \nabla_i \phi = 0. \tag{49}$$

The saturation energy density is

$$\rho_C = T^{C\phi}_{00}. \tag{50}$$

Evaluating the stress-energy tensor gives

$$\rho_C = \frac{1}{2}\dot{C}^2 + \frac{1}{2}\dot{\phi}^2 + V(C,\phi). \tag{51}$$

Using the momenta,

$$\rho_C = \frac{1}{2}\Pi_C^2 + \frac{1}{2}\Pi_\phi^2 + V. \tag{52}$$

Define

$$K^2 = \Pi_C^2 + \Pi_\phi^2. \tag{53}$$

Then

$$\rho_C = \frac{1}{2}K^2 + V. \tag{54}$$

The isotropic pressure is

$$p_C = \frac{1}{2}\Pi_C^2 + \frac{1}{2}\Pi_\phi^2 - V, \tag{55}$$

or

$$p_C = \frac{1}{2}K^2 - V. \tag{56}$$

The sum is

$$\rho_C + p_C = K^2. \tag{57}$$

The difference is

$$\rho_C - p_C = 2V. \tag{58}$$

The equation-of-state parameter is

$$w_C = \frac{K^2 - 2V}{K^2 + 2V}. \tag{59}$$

When

$$K^2 \to 0, \tag{60}$$

one obtains

$$w_C \to -1. \tag{61}$$

5.8 Perfect-fluid form

A homogeneous canonical scalar sector has the perfect-fluid form

$$T^\mu{}_\nu = \mathrm{diag}\big(-\rho_C, p_C, p_C, p_C\big). \tag{62}$$

The effective four-velocity may be associated with the timelike field gradient when the combined gradient is timelike.

At the background level, the saturation fields therefore enter the cosmological equations in the same algebraic form as an effective perfect fluid, even though their dynamics are governed by second-order field equations.

This distinction is important.

The saturation sector may be represented by

$$\rho_C \qquad \text{and} \qquad p_C \tag{63}$$

in the background equations, but it is not defined by an independently chosen function

$$w_C(a). \tag{64}$$

Instead,

$$w_C(a) \tag{65}$$

is derived from

$$C(a), \quad \phi(a), \quad \Pi_C(a), \quad \Pi_\phi(a). \tag{66}$$

5.9 Field-sector conservation without transfer

Using

$$\rho_C = \frac{1}{2}\Pi_C^2 + \frac{1}{2}\Pi_\phi^2 + V, \tag{67}$$

differentiate with respect to time:

$$\dot{\rho}_C = \Pi_C\dot{\Pi}_C + \Pi_\phi\dot{\Pi}_\phi + V_{,C}\dot{C} + V_{,\phi}\dot{\phi}. \tag{68}$$

Substituting

$$\dot{C} = \Pi_C, \qquad \dot{\phi} = \Pi_\phi, \qquad \dot{\Pi}_C = -3H\Pi_C - V_{,C}, \qquad \dot{\Pi}_\phi = -3H\Pi_\phi - V_{,\phi}, \tag{69}$$

gives

$$\dot{\rho}_C = -3H\Pi_C^2 - 3H\Pi_\phi^2. \tag{70}$$

Therefore

$$\dot{\rho}_C = -3HK^2. \tag{71}$$

Since

$$\rho_C + p_C = K^2, \tag{72}$$

the continuity equation is

$$\dot{\rho}_C + 3H(\rho_C + p_C) = 0. \tag{73}$$

This is the ordinary conservation law for the canonical two-field sector.

5.10 Contraction-era transfer current

During contraction, the field sector exchanges energy with radiation.

The transfer rate is

$$\Gamma_{\rm tr} = W_-(H)\big[-3H + \Gamma_0 S_\rho\big]. \tag{74}$$

The contraction selector satisfies

$$W_-(H) \simeq 1 \tag{75}$$

for sufficiently negative H, and

$$W_-(H) = 0 \tag{76}$$

for

$$H \geq 0. \tag{77}$$

The density switch is

$$S_\rho = \frac{\big(\rho_{\rm tar}/\rho_\Gamma\big)^m}{1 + \big(\rho_{\rm tar}/\rho_\Gamma\big)^m}, \tag{78}$$

with

$$\rho_{\rm tar} = \rho_m + \rho_r. \tag{79}$$

The homogeneous field equations become

$$\dot{\Pi}_C = -\big(3H + \Gamma_{\rm tr}\big)\Pi_C - V_{,C}, \tag{80}$$

and

$$\dot{\Pi}_\phi = -\big(3H + \Gamma_{\rm tr}\big)\Pi_\phi - V_{,\phi}. \tag{81}$$

The transfer power is

$$Q = \Gamma_{\rm tr} K^2. \tag{82}$$

The field continuity equation becomes

$$\dot{\rho}_C + 3H(\rho_C + p_C) = -Q. \tag{83}$$

Radiation receives the transferred energy:

$$\dot{\rho}_r + 4H\rho_r = +Q. \tag{84}$$

Matter remains separately conserved:

$$\dot{\rho}_m + 3H\rho_m = 0. \tag{85}$$

5.11 Covariant exchange notation

The transfer may be written covariantly through exchange currents.

Let

$$Q^\nu_{C\phi} \tag{86}$$

denote the current acting on the saturation sector and

$$Q_r^\nu \tag{87}$$

the corresponding current acting on radiation.

Then

$$\nabla_\mu T_{C\phi}^{\mu\nu} = Q_{C\phi}^\nu, \tag{88}$$

and

$$\nabla_\mu T_r^{\mu\nu} = Q_r^\nu. \tag{89}$$

Energy conservation requires

$$Q_{C\phi}^\nu + Q_r^\nu = 0. \tag{90}$$

Matter remains separately conserved:

$$\nabla_\mu T_m^{\mu\nu} = 0. \tag{91}$$

In the homogeneous comoving frame,

$$u^\mu = (1, 0, 0, 0), \tag{92}$$

the energy-transfer currents are

$$Q_{C\phi}^\nu = -Qu^\nu, \qquad Q_r^\nu = +Qu^\nu. \tag{93}$$

Therefore

$$Q_{C\phi}^0 = -Q, \qquad Q_r^0 = +Q. \tag{94}$$

The total stress-energy tensor satisfies

$$\nabla_\mu T_{\text{total}}^{\mu\nu} = 0. \tag{95}$$

5.12 Total conservation law

The total energy density is

$$\rho = \rho_m + \rho_r + \rho_C. \tag{96}$$

The total pressure is

$$p = p_m + p_r + p_C. \tag{97}$$

For pressureless matter,

$$p_m = 0. \tag{98}$$

For radiation,

$$p_r = \frac{1}{3}\rho_r. \tag{99}$$

Therefore

$$p = \frac{1}{3}\rho_r + p_C. \tag{100}$$

Adding the three continuity equations gives

$$\dot{\rho}_m + \dot{\rho}_r + \dot{\rho}_C + 3H\rho_m + 4H\rho_r + 3H(\rho_C + p_C) = 0. \tag{101}$$

Using

$$4H\rho_r = 3H\big(\rho_r + \frac{1}{3}\rho_r\big), \tag{102}$$

one obtains

$$\dot{\rho} + 3H(\rho + p) = 0. \tag{103}$$

Thus the transfer does not violate total energy conservation.

It redistributes energy between the organizational fields and radiation.

5.13 Cancellation of contraction anti-friction

Without transfer, the field damping coefficient is

$$3H. \tag{104}$$

During contraction,

$$H < 0, \tag{105}$$

so the coefficient is negative.

The field equations then contain anti-friction:

$$\dot{\Pi}_C = |3H|\Pi_C - V_{,C}, \tag{106}$$

schematically, and similarly for ϕ.

With the transfer active,

$$3H + \Gamma_{\rm tr} = 3H + W_-(H)\big[-3H + \Gamma_0 S_\rho\big]. \tag{107}$$

Deep in contraction,

$$W_- \simeq 1. \tag{108}$$

Therefore

$$3H + \Gamma_{\rm tr} \simeq 3H - 3H + \Gamma_0 S_\rho. \tag{109}$$

Hence

$$3H + \Gamma_{\rm tr} \simeq \Gamma_0 S_\rho. \tag{110}$$

For

$$\Gamma_0 > 0, \qquad S_\rho \geq 0, \tag{111}$$

the effective damping coefficient is nonnegative.

The anti-friction is removed dynamically.

At the bounce and during expansion,

$$W_- = 0. \tag{112}$$

Therefore

$$\Gamma_{\rm tr} = 0. \tag{113}$$

The standard canonical equations are restored.

5.14 Saturation constraint

The high-density background response is

$$H^2 = \frac{\rho}{3}\left[1 - \left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{114}$$

The locked model uses

$$q = 1. \tag{115}$$

Therefore

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{116}$$

Define

$$x = \frac{\rho}{\rho_s}. \tag{117}$$

Then

$$H^2 = \frac{\rho_s}{3}x(1-x). \tag{118}$$

The allowed homogeneous region satisfies

$$0 \le x \le 1. \tag{119}$$

Therefore

$$0 \le \rho \le \rho_s. \tag{120}$$

The constraint has two zeros:

$$\rho = 0 \qquad \text{and} \qquad \rho = \rho_s. \tag{121}$$

The first is the empty low-density limit.

The second is the finite-density bounce.

5.15 Maximum expansion or contraction rate

The function

$$f(x) = x(1-x) \tag{122}$$

has derivative

$$f'(x) = 1 - 2x. \tag{123}$$

The maximum occurs at

$$x = \frac{1}{2}. \tag{124}$$

Therefore

$$\rho = \frac{\rho_s}{2} \tag{125}$$

produces the maximum magnitude of H.

At that density,

$$H_{\max}^2 = \frac{\rho_s}{3}\left(\frac{1}{2}\right)\left(1 - \frac{1}{2}\right). \tag{126}$$

Hence

$$H_{\max}^2 = \frac{\rho_s}{12}. \tag{127}$$

Therefore

$$|H|_{\max} = \sqrt{\frac{\rho_s}{12}}. \tag{128}$$

The contraction rate cannot grow without bound.

As the density increases beyond

$$\rho_s/2, \tag{129}$$

the magnitude of H decreases.

At

$$\rho = \rho_s, \tag{130}$$

one obtains

$$H = 0. \tag{131}$$

5.16 Raychaudhuri equation from the constraint

Differentiate

$$H^2 = \frac{\rho}{3}\left[1 - \left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{132}$$

This gives

$$2H\dot{H} = \frac{\dot{\rho}}{3}\left[1 - (q+1)\left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{133}$$

Using total conservation,

$$\dot{\rho} = -3H(\rho + p), \tag{134}$$

one obtains

$$2H\dot{H} = -H(\rho + p)\left[1 - (q+1)\left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{135}$$

Away from the exact crossing,

$$\dot{H} = -\frac{1}{2}(\rho + p)\left[1 - (q+1)\left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{136}$$

The continuous form is then evolved directly through the bounce.

For

$$q = 1, \tag{137}$$

$$\dot{H} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right). \tag{138}$$

5.17 Acceleration equation

The scale factor satisfies

$$\dot{a} = aH. \tag{139}$$

Differentiating gives

$$\ddot{a} = a\dot{H} + aH^2. \tag{140}$$

Therefore

$$\frac{\ddot{a}}{a} = \dot{H} + H^2. \tag{141}$$

Substituting the $q = 1$ equations gives

$$\frac{\ddot{a}}{a} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right) + \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{142}$$

Expanding,

$$\frac{\ddot{a}}{a} = -\frac{1}{2}(\rho + p) + (\rho + p)\frac{\rho}{\rho_s} + \frac{\rho}{3} - \frac{\rho^2}{3\rho_s}. \tag{143}$$

Combining the ordinary terms,

$$-\frac{1}{2}(\rho + p) + \frac{\rho}{3} = -\frac{1}{6}(\rho + 3p). \tag{144}$$

Combining the saturation terms,

$$(\rho + p)\frac{\rho}{\rho_s} - \frac{\rho^2}{3\rho_s} = \frac{\rho}{\rho_s}\left(\frac{2}{3}\rho + p\right). \tag{145}$$

Therefore

$$\frac{\ddot{a}}{a} = -\frac{1}{6}(\rho + 3p) + \frac{\rho}{\rho_s}\left(\frac{2}{3}\rho + p\right). \tag{146}$$

At low density,

$$\rho/\rho_s \to 0, \tag{147}$$

and the ordinary acceleration equation is recovered:

$$\frac{\ddot{a}}{a} \simeq -\frac{1}{6}(\rho + 3p). \tag{148}$$

At high density, the saturation term becomes dominant.

5.18 Bounce acceleration

At the bounce,

$$\rho = \rho_s, \qquad H = 0. \tag{149}$$

The Raychaudhuri equation gives

$$\dot{H}_{\mathrm{b}} = -\frac{1}{2}(\rho_s + p_{\mathrm{b}})(1-2). \tag{150}$$

Therefore

$$\dot{H}_{\mathrm{b}} = \frac{1}{2}(\rho_s + p_{\mathrm{b}}). \tag{151}$$

The acceleration is

$$\frac{\ddot{a}_{\mathrm{b}}}{a_{\mathrm{b}}} = \dot{H}_{\mathrm{b}}, \tag{152}$$

because

$$H_{\mathrm{b}} = 0. \tag{153}$$

Hence

$$\frac{\ddot{a}_{\mathrm{b}}}{a_{\mathrm{b}}} = \frac{1}{2}(\rho_s + p_{\mathrm{b}}). \tag{154}$$

If

$$\rho_s + p_{\mathrm{b}} > 0, \tag{155}$$

then

$$\ddot{a}_{\mathrm{b}} > 0. \tag{156}$$

Therefore the finite-density crossing is a genuine bounce.

5.19 Effective high-density fluid interpretation

The modified constraint may be rewritten in standard Friedmann form:

$$H^2 = \frac{1}{3}\rho_{\mathrm{eff}}, \tag{157}$$

where

$$\rho_{\mathrm{eff}} = \rho - \frac{\rho^2}{\rho_s}. \tag{158}$$

Define the saturation correction

$$\rho_{\mathrm{sat}} = -\frac{\rho^2}{\rho_s}. \tag{159}$$

Then

$$\rho_{\mathrm{eff}} = \rho + \rho_{\mathrm{sat}}. \tag{160}$$

This effective negative contribution is not introduced as an independent material fluid.

It is a rewriting of the modified gravitational response.

The effective pressure may be obtained by requiring

$$\dot{H} = -\frac{1}{2}(\rho_{\mathrm{eff}} + p_{\mathrm{eff}}). \tag{161}$$

For $q = 1$,

$$\dot{H} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right). \tag{162}$$

Therefore

$$\rho_{\text{eff}} + p_{\text{eff}} = (\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right). \tag{163}$$

Since

$$\rho_{\text{eff}} = \rho - \frac{\rho^2}{\rho_s}, \tag{164}$$

one obtains

$$p_{\text{eff}} = p - \frac{\rho^2 + 2\rho p}{\rho_s}. \tag{165}$$

Thus the effective correction pressure is

$$p_{\text{sat}} = -\frac{\rho^2 + 2\rho p}{\rho_s}. \tag{166}$$

Again, this is an effective geometric representation of saturation, not a separate physical substance.

5.20 Low-density recovery

When

$$\rho \ll \rho_s, \tag{167}$$

the correction satisfies

$$\left|\frac{\rho_{\text{sat}}}{\rho}\right| = \frac{\rho}{\rho_s} \ll 1. \tag{168}$$

Therefore

$$\rho_{\text{eff}} \simeq \rho. \tag{169}$$

Similarly,

$$p_{\text{sat}} \simeq 0. \tag{170}$$

The background equations reduce to

$$H^2 \simeq \frac{\rho}{3}, \tag{171}$$

and

$$\dot{H} \simeq -\frac{1}{2}(\rho + p). \tag{172}$$

The saturation correction is therefore naturally suppressed at ordinary cosmological density.

5.21 Matter evolution

Pressureless matter satisfies

$$p_m = 0. \tag{173}$$

Its continuity equation is

$$\dot{\rho}_m + 3H\rho_m = 0. \tag{174}$$

Dividing by ρ_m,

$$\frac{\dot{\rho}_m}{\rho_m} = -3\frac{\dot{a}}{a}. \tag{175}$$

Integrating,

$$\ln \rho_m = -3 \ln a + \text{constant}. \tag{176}$$

Therefore

$$\rho_m = \rho_{m0} a^{-3}. \tag{177}$$

This relation holds on both contracting and expanding branches.

During contraction,

$$a \downarrow, \tag{178}$$

so

$$\rho_m \uparrow . \tag{179}$$

During expansion,

$$a \uparrow, \tag{180}$$

so

$$\rho_m \downarrow . \tag{181}$$

5.22 Radiation evolution with transfer

Radiation satisfies

$$\dot{\rho}_r + 4H\rho_r = Q. \tag{182}$$

Multiplying by

$$a^4, \tag{183}$$

gives

$$a^4 \dot{\rho}_r + 4Ha^4 \rho_r = a^4 Q. \tag{184}$$

Since

$$\frac{d}{dt}(a^4) = 4Ha^4, \tag{185}$$

one obtains

$$\frac{d}{dt}\left(a^4 \rho_r\right) = a^4 Q. \tag{186}$$

Therefore

$$a^4(t)\rho_r(t) = a^4(t_i)\rho_r(t_i) + \int_{t_i}^{t} a^4(t')Q(t')\, dt'. \tag{187}$$

During expansion,

$$Q = 0, \tag{188}$$

so

$$a^4 \rho_r = \text{constant}, \tag{189}$$

and therefore

$$\rho_r \propto a^{-4}. \tag{190}$$

During contraction, the integral records the kinetic energy transferred from the organizational fields into radiation.

5.23 Field energy with transfer

The field-sector continuity equation is

$$\dot{\rho}_C + 3HK^2 = -\Gamma_{\rm tr} K^2. \tag{191}$$

Therefore

$$\dot{\rho}_C = -\big(3H + \Gamma_{\rm tr}\big) K^2. \tag{192}$$

Deep in contraction,

$$3H + \Gamma_{\rm tr} \simeq \Gamma_0 S_\rho. \tag{193}$$

Thus

$$\dot{\rho}_C \simeq -\Gamma_0 S_\rho K^2. \tag{194}$$

The field kinetic energy is damped.

During expansion,

$$\Gamma_{\rm tr} = 0, \tag{195}$$

so

$$\dot{\rho}_C = -3HK^2. \tag{196}$$

The energy decreases toward the potential floor.

5.24 Positivity and boundedness

The canonical kinetic energy is nonnegative:

$$\frac{1}{2}K^2 = \frac{1}{2}\Pi_C^2 + \frac{1}{2}\Pi_\phi^2 \geq 0. \tag{197}$$

The organizational potential satisfies

$$V_C(C) \geq V_\infty \tag{198}$$

because

$$m_C^2 > 0, \qquad \lambda > 0. \tag{199}$$

The phase potential satisfies

$$A_0(1 - \cos\phi) \geq 0 \tag{200}$$

for

$$A_0 > 0. \tag{201}$$

Therefore

$$V(C, \phi) \geq V_\infty. \tag{202}$$

Hence

$$\rho_C = \frac{1}{2}K^2 + V \geq V_\infty. \tag{203}$$

The field energy is bounded below.

This lower bound becomes the late residual.

5.25 Absence of a background ghost

A ghost would require a negative kinetic coefficient.

The field Lagrangian contains

$$-\frac{1}{2}g^{\mu\nu}\nabla_\mu C\nabla_\nu C, \tag{204}$$

and

$$-\frac{1}{2}g^{\mu\nu}\nabla_\mu \phi\nabla_\nu \phi. \tag{205}$$

With metric signature

$$(-,+,+,+), \tag{206}$$

the homogeneous kinetic energy is

$$+\frac{1}{2}\dot{C}^2+\frac{1}{2}\dot{\phi}^2. \tag{207}$$

Both coefficients are positive.

The field-space kinetic matrix is

$$K_{IJ}=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}. \tag{208}$$

Its eigenvalues are

$$\kappa_1=1, \qquad \kappa_2=1. \tag{209}$$

Therefore the canonical field sector contains no negative-kinetic-energy mode.

5.26 Gradient stability

The quadratic spatial-gradient contribution is

$$-\frac{1}{2}g^{ij}\partial_i C\partial_j C-\frac{1}{2}g^{ij}\partial_i\phi\partial_j\phi. \tag{210}$$

In Fourier space, the perturbation equations contain

$$+k^2\delta C \qquad \text{and} \qquad +k^2\delta\phi. \tag{211}$$

The effective sound speeds are

$$c_{s,C}^2=1, \qquad c_{s,\phi}^2=1. \tag{212}$$

Therefore there is no negative gradient coefficient.

The canonical fields do not exhibit a short-wavelength gradient instability.

5.27 Mass stability

The field-space Hessian is

$$M^2_{IJ} = \begin{pmatrix} V_{,CC} & V_{,C\phi} \\ V_{,\phi C} & V_{,\phi\phi} \end{pmatrix}. \tag{213}$$

Since

$$V_{,C\phi} = 0, \tag{214}$$

the matrix is diagonal:

$$M^2_{IJ} = \begin{pmatrix} m_C^2 + 3\lambda(C - C_s)^2 & 0 \\ 0 & A_0 \cos\phi \end{pmatrix}. \tag{215}$$

Near the locked state,

$$C \simeq C_s, \qquad \phi \simeq 0. \tag{216}$$

Therefore

$$M^2_{IJ} \simeq \begin{pmatrix} m_C^2 & 0 \\ 0 & A_0 \end{pmatrix}. \tag{217}$$

For

$$m_C^2 > 0, \qquad A_0 > 0, \tag{218}$$

both eigenvalues are positive.

The locked state is restoring in both field directions.

5.28 Complete homogeneous system

The full state vector is

$$\mathbf{y} = \big(a, H, C, \Pi_C, \phi, \Pi_\phi, \rho_m, \rho_r\big). \tag{219}$$

The equations are

$$\dot{a} = aH, \tag{220}$$

$$\dot{H} = -\frac{1}{2}(\rho + p)\left[1 - (q+1)\left(\frac{\rho}{\rho_s}\right)^q\right], \tag{221}$$

$$\dot{C} = \Pi_C, \tag{222}$$

$$\dot{\Pi}_C = -\big(3H + \Gamma_{\rm tr}\big)\Pi_C - V_{,C}, \tag{223}$$

$$\dot{\phi} = \Pi_\phi, \tag{224}$$

$$\dot{\Pi}_\phi = -\big(3H + \Gamma_{\rm tr}\big)\Pi_\phi - V_{,\phi}, \tag{225}$$

$$\dot{\rho}_m = -3H\rho_m, \tag{226}$$

$$\dot{\rho}_r = -4H\rho_r + \Gamma_{\rm tr}K^2. \tag{227}$$

The auxiliary quantities are

$$K^2 = \Pi_C^2 + \Pi_\phi^2, \tag{228}$$

$$\rho_C = \frac{1}{2}K^2 + V, \tag{229}$$

$$p_C = \frac{1}{2}K^2 - V, \tag{230}$$

$$\rho = \rho_m + \rho_r + \rho_C, \tag{231}$$

$$p = \frac{1}{3}\rho_r + p_C. \tag{232}$$

The Friedmann constraint is

$$H^2 = \frac{\rho}{3}\left[1 - \left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{233}$$

The locked value is

$$q = 1. \tag{234}$$

5.29 Constraint residual

Numerical integration monitors

$$\mathcal{C}_H = H^2 - \frac{\rho}{3}\left[1 - \left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{235}$$

An exact solution satisfies

$$\mathcal{C}_H = 0. \tag{236}$$

A relative residual may be defined as

$$\epsilon_H = \frac{\left|H^2 - \frac{\rho}{3}\left[1 - \left(\frac{\rho}{\rho_s}\right)^q\right]\right|}{\max\left[H^2, \frac{\rho}{3}, \epsilon_{\text{floor}}\right]}. \tag{237}$$

The floor

$$\epsilon_{\text{floor}} > 0 \tag{238}$$

prevents division by zero near the bounce.

At checkpoints, a sign-aware projection may be used:

$$H \to \operatorname{sgn}(H)\sqrt{\frac{\rho}{3}\left[1 - \left(\frac{\rho}{\rho_s}\right)^q\right]}. \tag{239}$$

At the bounce, the sign is determined by the direction of the integrated Raychaudhuri equation rather than by replacing the evolution equation.

The constraint is used to control numerical drift, not to replace

$$\dot{H}. \tag{240}$$

5.30 Nonsingular time variable

The system is integrated in cosmic time or scaled affine time.

Using

$$N = \ln a \tag{241}$$

would produce

$$\frac{dC}{dN} = \frac{\Pi_C}{H}, \qquad \frac{d\phi}{dN} = \frac{\Pi_\phi}{H}, \qquad \frac{d\rho_m}{dN} = -3\rho_m, \qquad \frac{d\rho_r}{dN} = -4\rho_r + \frac{\Gamma_{\rm tr} K^2}{H}. \tag{242}$$

The terms

$$\frac{\Pi_C}{H}, \qquad \frac{\Pi_\phi}{H}, \qquad \frac{\Gamma_{\rm tr} K^2}{H} \tag{243}$$

are singular at

$$H = 0. \tag{244}$$

Therefore N is not used through the bounce.

The cosmic-time equations contain no division by H and remain regular at the crossing.

5.31 Scaled system

Define

$$\tau = \sqrt{\rho_s}\, t, \tag{245}$$

$$h = \frac{H}{\sqrt{\rho_s}}, \tag{246}$$

$$u_C = \frac{\Pi_C}{\sqrt{\rho_s}}, \tag{247}$$

$$u_\phi = \frac{\Pi_\phi}{\sqrt{\rho_s}}, \tag{248}$$

$$\bar{\rho}_i = \frac{\rho_i}{\rho_s}. \tag{249}$$

Then

$$\frac{da}{d\tau} = ah, \qquad \frac{dC}{d\tau} = u_C, \qquad \frac{d\phi}{d\tau} = u_\phi. \tag{250}$$

The $q = 1$ constraint is

$$h^2 = \frac{\bar{\rho}}{3}(1 - \bar{\rho}). \tag{251}$$

The scaled Raychaudhuri equation is

$$\frac{dh}{d\tau} = -\frac{1}{2}(\bar{\rho} + \bar{p})(1 - 2\bar{\rho}). \tag{252}$$

The scaled system preserves the physical solution while keeping all high-density variables near order unity.

5.32 Piecewise integration

The nonsingular trajectory spans a large dynamic range.

It is therefore integrated in segments.

A practical sequence is

$$\text{contraction} \to \text{bounce} \to a \simeq 10^{-3}, \tag{253}$$

then

$$a \simeq 10^{-3} \to a \simeq 10^{-2}, \tag{254}$$

then

$$a \simeq 10^{-2} \to a = 1. \tag{255}$$

At each checkpoint, the full state vector is stored:

$$\big(a, H, C, \Pi_C, \phi, \Pi_\phi, \rho_m, \rho_r\big). \tag{256}$$

The next segment begins from that stored state.

The same equations are used in every segment.

No physical switch is introduced at the checkpoints.

5.33 Dual-solver reproducibility

The background was evaluated with both

$$\text{DOP853} \qquad \text{and} \qquad \text{Radau}. \tag{257}$$

The solvers represent different numerical strategies.

DOP853 is an explicit high-order Runge-Kutta method.

Radau is an implicit method suited to stiff systems.

Agreement between them checks that the solution is not an artifact of one solver.

The reproducibility tests compared

$$a_{\min}, \quad K_{\text{bounce}}, \quad \rho_C(0), \quad E(0.5), \quad E(1), \quad E(100), \quad E(1000), \quad E(10000), \quad r_{\text{drag}}, \quad 100\theta_*. \tag{258}$$

The locked background used in the final CAMB implementation was retained only after the nonsingular trajectory became reproducible under the solver comparisons.

5.34 Present-day normalization

The expansion function is

$$E(a) = \frac{H(a)}{H_0}. \tag{259}$$

The residual normalization is chosen so that

$$E(1) = 1. \tag{260}$$

Therefore

$$H(a = 1) = H_0. \tag{261}$$

The final values are

$$H_0 = 67.4 \ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{262}$$

$$\Omega_{m0} = 0.314114, \tag{263}$$

$$\Omega_{r0} = 9 \times 10^{-5}. \tag{264}$$

The residual density is determined by the integrated saturation fields and their normalization.

The final value is approximately

$$\rho_C(0) \simeq 2.19 \tag{265}$$

in the dimensionless background normalization.

5.35 Late equation of state

The field equation of state is

$$w_C = \frac{\frac{1}{2}K^2 - V}{\frac{1}{2}K^2 + V}. \tag{266}$$

At the present epoch, the locked trajectory gives

$$w_C(0) \simeq -0.99988. \tag{267}$$

Therefore

$$1 + w_C(0) \simeq 1.2 \times 10^{-4}. \tag{268}$$

From

$$1 + w_C = \frac{K^2}{\rho_C}, \tag{269}$$

the present-day kinetic fraction is

$$\frac{K^2}{\rho_C} \simeq 1.2 \times 10^{-4}. \tag{270}$$

Thus the late saturation sector is overwhelmingly potential dominated.

5.36 Canonical perturbation action

On the expanding branch, write

$$C = C_{\mathrm{bg}} + \delta C, \qquad \phi = \phi_{\mathrm{bg}} + \delta\phi. \tag{271}$$

To quadratic order, the field-space kinetic structure is

$$S^{(2)}_{\rm kin} = \frac{1}{2}\int d\tau\, d^3x\, a^2\big[(\delta C')^2 - (\nabla\delta C)^2 + (\delta\phi')^2 - (\nabla\delta\phi)^2\big], \tag{272}$$

together with the metric couplings and mass terms.

The mass contribution is

$$S^{(2)}_{\rm mass} = -\frac{1}{2}\int d\tau\, d^3x\, a^4\big[V_{,CC}(\delta C)^2 + 2V_{,C\phi}\delta C\delta\phi + V_{,\phi\phi}(\delta\phi)^2\big]. \tag{273}$$

Since

$$V_{,C\phi} = 0, \tag{274}$$

the mass matrix is diagonal.

The kinetic matrix is the identity.

Therefore the quadratic field action has positive kinetic and gradient coefficients.

5.37 Linear field equations in synchronous gauge

The synchronous-gauge scalar metric may be written in Fourier space through the standard metric variables h and η.

The canonical field perturbations satisfy

$$\delta C'' + 2\mathcal{H}\delta C' + \big(k^2 + a^2V_{,CC}\big)\delta C + \frac{1}{2}C'_{\rm bg}h' = 0, \tag{275}$$

and

$$\delta\phi'' + 2\mathcal{H}\delta\phi' + \big(k^2 + a^2V_{,\phi\phi}\big)\delta\phi + \frac{1}{2}\phi'_{\rm bg}h' = 0. \tag{276}$$

Defining

$$v_C = \delta C', \qquad v_\phi = \delta\phi', \tag{277}$$

gives

$$\delta C' = v_C, \tag{278}$$

$$v_C' = -2\mathcal{H}v_C - \big(k^2 + a^2V_{,CC}\big)\delta C - \frac{1}{2}C'_{\rm bg}h', \tag{279}$$

$$\delta\phi' = v_\phi, \tag{280}$$

$$v_\phi' = -2\mathcal{H}v_\phi - \big(k^2 + a^2V_{,\phi\phi}\big)\delta\phi - \frac{1}{2}\phi'_{\rm bg}h'. \tag{281}$$

These equations were implemented directly in the Boltzmann hierarchy.

5.38 Linear stress-energy perturbations

The density perturbation is

$$\delta\rho_C = \frac{1}{a^2}\big[C'_{\rm bg}v_C + \phi'_{\rm bg}v_\phi\big] + V_{,C}\delta C + V_{,\phi}\delta\phi. \tag{282}$$

The pressure perturbation is

$$\delta p_C = \frac{1}{a^2}\left[C'_{\rm bg} v_C + \phi'_{\rm bg} v_\phi\right] - V_{,C}\delta C - V_{,\phi}\delta\phi. \tag{283}$$

The momentum density is

$$\delta q_C = \frac{k}{a^2}\left[C'_{\rm bg}\delta C + \phi'_{\rm bg}\delta\phi\right]. \tag{284}$$

The anisotropic stress is

$$\pi_C = 0. \tag{285}$$

Multiplying by a^2 gives the CAMB-normalized forms:

$$a^2\delta\rho_C = C'_{\rm bg} v_C + \phi'_{\rm bg} v_\phi + a^2\big(V_{,C}\delta C + V_{,\phi}\delta\phi\big), \tag{286}$$
$$a^2\delta p_C = C'_{\rm bg} v_C + \phi'_{\rm bg} v_\phi - a^2\big(V_{,C}\delta C + V_{,\phi}\delta\phi\big), \tag{287}$$
$$a^2\delta q_C = k\big(C'_{\rm bg}\delta C + \phi'_{\rm bg}\delta\phi\big). \tag{288}$$

5.39 Perturbative suppression near saturation

Near the locked state,

$$C'_{\rm bg} \to 0, \qquad \phi'_{\rm bg} \to 0, \qquad V_{,C} \to 0, \qquad V_{,\phi} \to 0. \tag{289}$$

Therefore

$$\delta\rho_C \to 0, \qquad \delta p_C \to 0, \qquad \delta q_C \to 0 \tag{290}$$

even when the field perturbations themselves remain finite.

The final amplitudes are approximately

$$\max|\delta C| \simeq 0.035, \qquad \max|\delta\phi| \simeq 0.049. \tag{291}$$

Their gravitational back-reaction is small because the coefficients multiplying them are suppressed by the locked background state.

This is why the direct perturbative change in the CMB spectra and σ_8 is approximately at the level of

$$10^{-6}. \tag{292}$$

5.40 Final conservation statement

The complete theory satisfies

$$\nabla_\mu T^{\mu\nu}_{\rm total} = 0. \tag{293}$$

The total tensor is

$$T^{\mu\nu}_{\rm total} = T^{\mu\nu}_m + T^{\mu\nu}_r + T^{\mu\nu}_{C\phi}. \tag{294}$$

During contraction,

$$\nabla_\mu T^{\mu\nu}_{C\phi} = -Q^\nu, \qquad \nabla_\mu T^{\mu\nu}_r = +Q^\nu. \tag{295}$$

During expansion,

$$Q^{\nu} = 0. \tag{296}$$

The high-density geometric response satisfies

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right), \tag{297}$$

and

$$\dot{H} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right). \tag{298}$$

The field sector remains canonical.

The kinetic matrix is positive.

The gradient coefficients are positive.

The potential is bounded below.

The organizational minimum is stable.

The total energy exchange is conservative.

The bounce occurs at finite density and finite scale factor.

The late residual follows from the same fields settling at the minimum.

This covariant and conservative structure provides the mathematical foundation for the numerical cosmology developed in the following chapters.

End of Chapter 5

Chapter 6

The Nonsingular Background and the Locked Expansion History

6.1 Purpose of the background construction

The gravitational-saturation framework becomes cosmological only when the covariant field sector is evolved together with the homogeneous geometry.

The background problem is therefore to determine a continuous trajectory through

$$\begin{aligned}&\text{contraction,}\\&\text{finite-density bounce,}\\&\text{radiation expansion,}\\&\text{matter expansion,}\\&\text{late saturation.}\end{aligned} \tag{1}$$

The resulting trajectory must satisfy several requirements simultaneously.

It must remain nonsingular:

$$a_{\min} > 0, \qquad |H| < \infty, \qquad \rho \leq \rho_s, \qquad K^2 < \infty. \tag{2}$$

It must preserve total energy:

$$\dot{\rho} + 3H(\rho + p) = 0. \tag{3}$$

It must recover ordinary cosmological evolution at low density:

$$H^2 \simeq \frac{\rho}{3} \qquad \text{for} \qquad \rho \ll \rho_s. \tag{4}$$

It must leave a radiation-dominated expanding branch after the bounce.

It must pass through matter-radiation equality.

It must evolve toward a late residual state with

$$w_C \simeq -1. \tag{5}$$

Finally, the resulting function

$$H(a) \tag{6}$$

must be suitable for use within a Boltzmann hierarchy.

The final background used in the numerical work satisfies these conditions and is referred to as the locked expansion history.

6.2 Complete homogeneous state vector

The homogeneous phase-space vector is

$$\mathbf{y}(t) = \big(a, H, C, \Pi_C, \phi, \Pi_\phi, \rho_m, \rho_r\big). \tag{7}$$

The field kinetic quantity is

$$K^2 = \Pi_C^2 + \Pi_\phi^2. \tag{8}$$

The saturation-sector density is

$$\rho_C = \frac{1}{2}K^2 + V(C,\phi). \tag{9}$$

The saturation-sector pressure is

$$p_C = \frac{1}{2}K^2 - V(C,\phi). \tag{10}$$

The total density is

$$\rho = \rho_m + \rho_r + \rho_C. \tag{11}$$

The total pressure is

$$p = \frac{1}{3}\rho_r + p_C. \tag{12}$$

The background equations are

$$\begin{aligned}
\dot{a} &= aH, && (13)\\
\dot{H} &= -\frac{1}{2}(\rho+p)\left[1-(q+1)\left(\frac{\rho}{\rho_s}\right)^q\right], && (14)\\
\dot{C} &= \Pi_C, && (15)\\
\dot{\Pi}_C &= -\big(3H+\Gamma_{\rm tr}\big)\Pi_C - V_{,C}, && (16)\\
\dot{\phi} &= \Pi_\phi, && (17)\\
\dot{\Pi}_\phi &= -\big(3H+\Gamma_{\rm tr}\big)\Pi_\phi - V_{,\phi}, && (18)\\
\dot{\rho}_m &= -3H\rho_m, && (19)\\
\dot{\rho}_r &= -4H\rho_r + \Gamma_{\rm tr}K^2. && (20)
\end{aligned}$$

The generalized Friedmann constraint is

$$H^2 = \frac{\rho}{3}\left[1-\left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{21}$$

The locked trajectory uses

$$q = 1. \tag{22}$$

Therefore

$$H^2 = \frac{\rho}{3}\left(1-\frac{\rho}{\rho_s}\right). \tag{23}$$

6.3 Locked parameters

The final unmodulated baseline is centered on

$$\Omega_{m0} = 0.314114, \tag{24}$$
$$\Omega_{r0} = 9 \times 10^{-5}, \tag{25}$$
$$V_\infty = 2.187987, \tag{26}$$
$$\rho_s = 1.58199 \times 10^{15}, \tag{27}$$
$$m_C^2 = 1.827221, \tag{28}$$
$$\lambda = 0.802722, \tag{29}$$
$$C_s = 0.95, \tag{30}$$
$$C_{\text{initial}} = 0.918318, \tag{31}$$
$$\Pi_{C,\text{initial}} = 0.055327. \tag{32}$$

The final Boltzmann implementation uses

$$H_0 = 67.4 \text{ km}\,\text{s}^{-1}\,\text{Mpc}^{-1}. \tag{33}$$

The early locked transition ends at

$$a_{\text{lock}} = 2.05 \times 10^{-4}. \tag{34}$$

The retained transition width is

$$W = 0.67W_0. \tag{35}$$

The residual normalization is chosen so that

$$E(1) = 1, \tag{36}$$

where

$$E(a) = \frac{H(a)}{H_0}. \tag{37}$$

6.4 Why the bounce cannot be integrated in $N = \ln a$

On an ordinary expanding branch, the variable

$$N = \ln a \tag{38}$$

is often convenient.

It satisfies

$$\frac{dN}{dt} = H. \tag{39}$$

Therefore

$$\frac{d}{dN} = \frac{1}{H}\frac{d}{dt}. \tag{40}$$

The field equations would then contain

$$\frac{dC}{dN} = \frac{\Pi_C}{H}, \qquad \frac{d\phi}{dN} = \frac{\Pi_\phi}{H}. \tag{41}$$

At the bounce,

$$H = 0. \tag{42}$$

The ratios

$$\frac{\Pi_C}{H} \qquad \text{and} \qquad \frac{\Pi_\phi}{H} \tag{43}$$

become singular even when the physical quantities

$$\Pi_C \qquad \text{and} \qquad \Pi_\phi \tag{44}$$

remain finite.

The radiation equation would also contain

$$\frac{\Gamma_{\rm tr} K^2}{H}. \tag{45}$$

The singularity is therefore a coordinate singularity introduced by the independent variable.

The complete bounce must instead be integrated in cosmic time,

$$t, \tag{46}$$

or in a nonsingular affine variable.

The background construction used here contains no division by

$$H. \tag{47}$$

6.5 Scaled affine system

The high saturation density creates a large numerical dynamic range.

Define

$$\tau = \sqrt{\rho_s}\, t, \tag{48}$$

$$h = \frac{H}{\sqrt{\rho_s}}, \tag{49}$$

$$u_C = \frac{\Pi_C}{\sqrt{\rho_s}}, \tag{50}$$

$$u_\phi = \frac{\Pi_\phi}{\sqrt{\rho_s}}, \tag{51}$$

$$\bar{\rho}_i = \frac{\rho_i}{\rho_s}, \tag{52}$$

$$\bar{p}_i = \frac{p_i}{\rho_s}. \tag{53}$$

The total normalized density is

$$\bar{\rho} = \frac{\rho}{\rho_s}. \tag{54}$$

The locked constraint becomes

$$h^2 = \frac{\bar{\rho}}{3}(1 - \bar{\rho}). \tag{55}$$

The scale-factor equation is

$$\frac{da}{d\tau} = ah. \tag{56}$$

The field equations begin with

$$\frac{dC}{d\tau} = u_C, \qquad \frac{d\phi}{d\tau} = u_\phi. \tag{57}$$

The normalized Raychaudhuri equation is

$$\frac{dh}{d\tau} = -\frac{1}{2}(\bar{\rho} + \bar{p})(1 - 2\bar{\rho}). \tag{58}$$

This scaled system preserves the same physical trajectory while keeping the principal high-density variables near order unity.

6.6 Piecewise nonsingular integration

The full evolution spans many orders of magnitude in scale factor.

The background is therefore integrated piecewise.

A representative segmentation is

$$\begin{aligned} &\text{Segment 1: contraction through bounce to } a \simeq 10^{-3}, \\ &\text{Segment 2: } a \simeq 10^{-3} \text{ to } a \simeq 10^{-2}, \\ &\text{Segment 3: } a \simeq 10^{-2} \text{ to } a = 1. \end{aligned} \tag{59}$$

The same differential equations are used in every segment.

No physical switch is introduced at the boundaries.

At the end of each segment, the complete state is stored:

$$a,\ H,\ C,\ \Pi_C,\ \phi,\ \Pi_\phi,\ \rho_m,\ \rho_r. \tag{60}$$

The next segment begins from exactly that state.

The segmentation controls numerical stiffness without changing the theory.

6.7 Constraint projection

The constraint residual is

$$\mathcal{C}_H = H^2 - \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{61}$$

During numerical evolution, finite solver error can produce a small nonzero value.

At checkpoints, the magnitude of H may be projected back onto the constraint:

$$|H| \to \sqrt{\frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right)}. \tag{62}$$

The sign is retained:

$$H \to \operatorname{sgn}(H)\sqrt{\frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right)}. \tag{63}$$

Before the bounce,

$$\operatorname{sgn}(H) = -1. \tag{64}$$

After the bounce,

$$\operatorname{sgn}(H) = +1. \tag{65}$$

At the exact crossing, the direction is determined by the integrated equation

$$\dot{H} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right). \tag{66}$$

The projection is used to reduce drift.

It does not replace the differential equation for H.

6.8 Solver reproducibility

The final background was tested with two independent numerical methods:

$$\text{DOP853} \qquad \text{and} \qquad \text{Radau}. \tag{67}$$

DOP853 is an explicit high-order Runge-Kutta integrator.

Radau is an implicit collocation method designed for stiff systems.

The comparison was performed at multiple relative tolerances, including

$$10^{-8}, \qquad 10^{-10}, \qquad 10^{-12} \tag{68}$$

during convergence testing.

The quantities compared included

$$\begin{aligned} &a_{\min},\ K_{\text{bounce}},\ \rho_C(0),\ C(0), \\ &E(0.5),\ E(1),\ E(100),\ E(1000),\ E(10000), \\ &r_{\text{drag}},\ 100\theta_*. \end{aligned} \tag{69}$$

The locked background was retained only after the early branch, the recombination-era expansion rate, and the derived sound horizon became reproducible across the solver families.

6.9 Bounce location

The numerical trajectory reaches a minimum scale factor near

$$a_{\text{min}} \simeq 2.068 \times 10^{-5}. \tag{70}$$

The corresponding bounce redshift is

$$z_{\text{bounce}} = \frac{1}{a_{\text{min}}} - 1. \tag{71}$$

Using the locked value,

$$z_{\text{bounce}} \simeq 4.84 \times 10^{4}. \tag{72}$$

At the bounce,

$$H = 0, \qquad \rho = \rho_s, \qquad a = a_{\text{min}} > 0. \tag{73}$$

The kinetic quantity

$$K^2_{\text{bounce}} = \Pi_C^2 + \Pi_\phi^2 \tag{74}$$

remains finite.

The fields themselves remain finite:

$$|C_{\text{bounce}}| < \infty, \qquad |\phi_{\text{bounce}}| < \infty. \tag{75}$$

The bounce therefore contains no divergence in the scale factor, Hubble rate, field values, or field kinetic energy.

6.10 Local behavior near the bounce

Let the bounce occur at

$$t = t_b. \tag{76}$$

At this moment,

$$H(t_b) = 0. \tag{77}$$

Expand H about the bounce:

$$H(t) = \dot{H}_b(t - t_b) + \mathcal{O}\big[(t - t_b)^2\big]. \tag{78}$$

The scale factor satisfies

$$\dot{a} = aH. \tag{79}$$

Near the bounce,

$$\dot{a} \simeq a_b \dot{H}_b(t - t_b). \tag{80}$$

Integrating,

$$a(t) \simeq a_b \left[1 + \frac{1}{2}\dot{H}_b(t - t_b)^2\right]. \tag{81}$$

Since

$$\dot{H}_b = \frac{1}{2}(\rho_b + p_b), \tag{82}$$

a successful bounce with

$$\rho_b + p_b > 0 \tag{83}$$

gives

$$\dot{H}_b > 0. \tag{84}$$

Therefore

$$a(t) \geq a_b \tag{85}$$

in a neighborhood of the bounce.

The scale factor has a smooth local minimum.

6.11 Contraction-era kinetic regulation

During contraction,

$$H < 0. \tag{86}$$

The canonical Hubble term acts as anti-friction:

$$3H\Pi_C < 0 \tag{87}$$

for positive Π_C, and similarly for ϕ.

The transfer rate is

$$\Gamma_{\rm tr} = W_-(H)\big[-3H + \Gamma_0 S_\rho\big]. \tag{88}$$

Deep in contraction,

$$W_- \simeq 1. \tag{89}$$

The total field damping coefficient becomes

$$3H + \Gamma_{\rm tr} \simeq \Gamma_0 S_\rho. \tag{90}$$

Thus the kinetic equations become approximately

$$\dot{\Pi}_C \simeq -\Gamma_0 S_\rho \Pi_C - V_{,C}, \qquad \dot{\Pi}_\phi \simeq -\Gamma_0 S_\rho \Pi_\phi - V_{,\phi}. \tag{91}$$

The corresponding field-energy equation is

$$\dot{\rho}_C \simeq -\Gamma_0 S_\rho K^2. \tag{92}$$

The lost field energy is added to radiation:

$$\dot{\rho}_r + 4H\rho_r = \Gamma_0 S_\rho K^2. \tag{93}$$

The contraction branch therefore converts potentially divergent kinetic amplification into a controlled radiation source.

6.12 Transfer shutdown

The transfer selector satisfies

$$W_-(H) \to 0 \tag{94}$$

as the bounce is approached.

For

$$H \geq 0, \tag{95}$$

the locked implementation imposes

$$W_-(H) = 0. \tag{96}$$

Therefore

$$\Gamma_{\mathrm{tr}} = 0 \tag{97}$$

throughout the expanding branch.

The radiation equation becomes

$$\dot{\rho}_r = -4H\rho_r. \tag{98}$$

The field equations become

$$\dot{\Pi}_C = -3H\Pi_C - V_{,C}, \qquad \dot{\Pi}_\phi = -3H\Pi_\phi - V_{,\phi}. \tag{99}$$

No transfer tail survives into late expansion.

The fields settle through ordinary Hubble damping and their restoring potential.

6.13 Radiation-dominated expansion

After the bounce, the radiation density decreases as

$$\rho_r \propto a^{-4}. \tag{100}$$

Matter decreases as

$$\rho_m \propto a^{-3}. \tag{101}$$

The ratio is

$$\frac{\rho_r}{\rho_m} = \frac{\rho_{r0}}{\rho_{m0}}\frac{1}{a}. \tag{102}$$

At sufficiently small a,

$$\frac{\rho_r}{\rho_m} \gg 1. \tag{103}$$

Therefore the post-bounce expanding branch contains a radiation-dominated interval.

The saturation residual remains subdominant during this era because

$$\rho_r \propto a^{-4} \tag{104}$$

grows rapidly toward the past, while the late residual approaches a finite floor.

Thus, for

$$a_{\rm min} \ll a \ll a_{\rm eq}, \tag{105}$$

one has

$$\rho_r \gg \rho_m \qquad \text{and} \qquad \rho_r \gg \rho_C. \tag{106}$$

The early expanding branch therefore reproduces the required radiation hierarchy.

6.14 Matter-radiation equality

Equality occurs when

$$\rho_m(a_{\rm eq}) = \rho_r(a_{\rm eq}). \tag{107}$$

Using the present fractions,

$$a_{\rm eq} = \frac{\Omega_{r0}}{\Omega_{m0}}. \tag{108}$$

With

$$\Omega_{r0} = 9 \times 10^{-5}, \qquad \Omega_{m0} = 0.314114, \tag{109}$$

one obtains

$$a_{\rm eq} \simeq 2.865 \times 10^{-4}. \tag{110}$$

The corresponding redshift is

$$z_{\rm eq} = \frac{1}{a_{\rm eq}} - 1, \tag{111}$$

so

$$z_{\rm eq} \simeq 3489. \tag{112}$$

The ratio between the bounce and equality scales is

$$\frac{a_{\rm min}}{a_{\rm eq}} \simeq \frac{2.068 \times 10^{-5}}{2.865 \times 10^{-4}} \simeq 0.0722. \tag{113}$$

Therefore

$$a_{\rm min} \ll a_{\rm eq}. \tag{114}$$

The bounce occurs well before matter-radiation equality.

6.15 Matter-dominated era

For

$$a > a_{\rm eq}, \tag{115}$$

matter becomes larger than radiation:

$$\rho_m > \rho_r. \tag{116}$$

The ratio evolves as

$$\frac{\rho_m}{\rho_r} = \frac{a}{a_{\rm eq}}. \tag{117}$$

As the Universe expands,

$$\frac{\rho_m}{\rho_r} \tag{118}$$

continues to increase until the saturation residual becomes important.

During the matter-dominated interval, the expansion approximately follows

$$H^2 \simeq \frac{\rho_m}{3} \tag{119}$$

because

$$\rho \ll \rho_s \tag{120}$$

and the high-density correction is negligible.

The standard low-density Friedmann behavior is therefore recovered before the onset of late residual domination.

6.16 Approach to the locked residual

During expansion,

$$\Gamma_{\mathrm{tr}} = 0. \tag{121}$$

The field energy evolves as

$$\dot{\rho}_C = -3HK^2. \tag{122}$$

Since

$$H > 0 \qquad \text{and} \qquad K^2 \geq 0, \tag{123}$$

one has

$$\dot{\rho}_C \leq 0. \tag{124}$$

The field energy decreases toward its lower bound:

$$\rho_C \geq V_\infty. \tag{125}$$

At the same time, the field equations restore

$$C \to C_s \qquad \text{and} \qquad \phi \to 0. \tag{126}$$

The kinetic variables damp:

$$\Pi_C \to 0, \qquad \Pi_\phi \to 0. \tag{127}$$

Therefore

$$\rho_C \to V_\infty, \qquad p_C \to -V_\infty, \qquad w_C \to -1. \tag{128}$$

The present locked result is

$$w_C(0) \simeq -0.99988. \tag{129}$$

The residual density is approximately

$$\rho_C(0) \simeq 2.192 \tag{130}$$

in the numerical normalization.

6.17 Late-time acceleration

The low-density acceleration equation is approximately

$$\frac{\ddot{a}}{a} = -\frac{1}{6}(\rho + 3p). \tag{131}$$

At late times,

$$\rho \simeq \rho_m + \rho_C \qquad \text{and} \qquad p \simeq p_C. \tag{132}$$

With

$$p_C \simeq -\rho_C, \tag{133}$$

one obtains

$$\rho + 3p \simeq \rho_m - 2\rho_C. \tag{134}$$

Acceleration occurs when

$$2\rho_C > \rho_m. \tag{135}$$

Thus the same residual field configuration that follows from saturation produces the late accelerated phase.

No separate dark-energy fluid is introduced.

6.18 Conversion to redshift space

Once the expanding branch is obtained, the scale factor is converted into redshift through

$$1 + z = \frac{1}{a}. \tag{136}$$

Therefore

$$z = \frac{1}{a} - 1. \tag{137}$$

The expansion function is

$$E(z) = \frac{H(z)}{H_0}. \tag{138}$$

The stored numerical trajectory provides pairs

$$\big[a(t), H(t)\big]. \tag{139}$$

On the expanding branch, $a(t)$ is monotonic.

The trajectory may therefore be inverted numerically to obtain

$$H(a), \tag{140}$$

and hence

$$E(z). \tag{141}$$

The redshift representation is used only after the nonsingular cosmic-time integration has passed through the bounce.

6.19 Fixed high-resolution integration grids

The sound horizon depends upon a high-redshift integral:

$$r_s(z) = \int_z^\infty \frac{c_s(z')}{H(z')}\, dz'. \tag{142}$$

Early calculations using adaptive integration alone showed sensitivity near the high-redshift endpoint and bounce-generated branch.

A fixed high-resolution redshift grid was therefore adopted for the geometry calculations.

The convergence sequence included grids of approximately

$$12{,}000, \quad 24{,}000, \quad 48{,}000, \quad 96{,}000 \tag{143}$$

high-redshift points.

The sound horizon and acoustic angle were accepted only after successive fine grids agreed to well below one percent.

This convergence test was then combined with full background re-integration under independent ODE solvers.

The final geometry therefore rests upon both

$$\text{trajectory convergence} \qquad \text{and} \qquad \text{integral convergence.} \tag{144}$$

6.20 The transition into the Boltzmann background

The complete nonsingular branch extends from contraction through the bounce and into expansion.

The Boltzmann hierarchy, however, is evolved only on the expanding branch.

The final implementation embeds the locked expanding history into CAMB through a smooth transition.

The transition endpoint is

$$a_{\text{lock}} = 2.05 \times 10^{-4}. \tag{145}$$

The retained transition width is

$$W = 0.67 W_0. \tag{146}$$

The transition function is smooth.

A generic smooth-step variable may be written as

$$x = \frac{a - a_{\text{start}}}{a_{\text{lock}} - a_{\text{start}}}. \tag{147}$$

For

$$0 < x < 1, \tag{148}$$

a cubic smoothstep is

$$S(x) = 3x^2 - 2x^3. \tag{149}$$

It satisfies

$$S(0) = 0, \qquad S(1) = 1, \qquad S'(0) = 0, \qquad S'(1) = 0. \tag{150}$$

The blended expansion may be represented schematically as

$$E^2_{\text{blend}}(a) = \big[1 - S(x)\big] E^2_{\text{early}}(a) + S(x) E^2_{\text{locked}}(a). \tag{151}$$

The retained implementation uses the same smooth-transition principle while preserving the final numerical normalization.

6.21 Meaning of the splice

The splice does not introduce a new cosmological component.

It does not replace the field equations after the bounce.

It is a numerical interface between

$$\text{the independently integrated nonsingular background} \tag{152}$$

and

$$\text{the standard Boltzmann evolution machinery.} \tag{153}$$

The early function is inherited from the resolved nonsingular trajectory.

The late function is the corresponding locked expanding history.

The transition is used because CAMB is organized around a monotonic expanding scale factor and is not designed to evolve a contracting pre-bounce branch inside its standard background module.

The splice therefore acts as an interface between two numerical domains.

6.22 Final acoustic-scale lock

The original retained background gave an acoustic angle near

$$100\theta_* \simeq 1.0461. \tag{154}$$

Local parameter tuning reached a plateau.

A controlled adjustment of the transition endpoint was then performed while leaving the physical background parameters fixed.

The tested values included

$$a_{\text{lock}} = 2.10 \times 10^{-4}, \quad 2.05 \times 10^{-4}, \quad 2.00 \times 10^{-4}, \quad 1.95 \times 10^{-4}. \tag{155}$$

The corresponding values were approximately

$$100\theta_* = 1.040917, \quad 1.041033, \quad 1.041167, \quad 1.041302. \tag{156}$$

The closest accepted value was

$$a_{\text{lock}} = 2.05 \times 10^{-4}. \tag{157}$$

The final acoustic angle is

$$100\theta_* = 1.041033. \tag{158}$$

Relative to the reference value

$$1.0411, \tag{159}$$

the fractional difference is

$$\frac{1.041033 - 1.0411}{1.0411} \times 100\% \simeq -0.0064\%. \tag{160}$$

6.23 Sound horizon

The baryon-loaded sound speed is

$$c_s(z) = \frac{c}{\sqrt{3[1 + R(z)]}}, \tag{161}$$

where

$$R(z) = \frac{3\rho_b}{4\rho_\gamma}. \tag{162}$$

The drag horizon is

$$r_{\text{drag}} = \int_{z_{\text{drag}}}^{\infty} \frac{c_s(z)}{H(z)}\, dz. \tag{163}$$

The final CAMB-consistent result is

$$r_{\text{drag}} = 149.960 \text{ Mpc}. \tag{164}$$

The drag redshift is

$$z_{\text{drag}} = 1058.511. \tag{165}$$

The recombination redshift is

$$z_* = 1089.691. \tag{166}$$

The comoving distance to last scattering is approximately

$$D_M(z_*) = 14131.86 \text{ Mpc}. \tag{167}$$

The physical angular-diameter distance is

$$D_A(z_*) = \frac{D_M(z_*)}{1 + z_*}, \tag{168}$$

giving approximately

$$D_A(z_*) = 12.9568 \text{ Mpc}. \tag{169}$$

6.24 Acoustic angle

The acoustic angle is

$$\theta_* = \frac{r_s(z_*)}{D_M(z_*)}. \tag{170}$$

CAMB reports

$$100\theta_*. \tag{171}$$

The final value is

$$100\theta_* = 1.041033. \tag{172}$$

This value is determined by the ratio of two integrals:

$$r_s(z_*) = \int_{z_*}^{\infty} \frac{c_s(z)}{H(z)}\,dz, \qquad D_M(z_*) = \int_0^{z_*} \frac{c\,dz}{H(z)}. \tag{173}$$

The locked background therefore succeeds by balancing the early sound-horizon scale against the full line-of-sight distance to recombination.

6.25 Distance measures

The comoving radial distance is

$$\chi(z) = \int_0^z \frac{c\,dz'}{H(z')}. \tag{174}$$

For a spatially flat background,

$$D_M(z) = \chi(z). \tag{175}$$

The Hubble distance is

$$D_H(z) = \frac{c}{H(z)}. \tag{176}$$

The angular-diameter distance is

$$D_A(z) = \frac{D_M(z)}{1+z}. \tag{177}$$

The luminosity distance is

$$D_L(z) = (1+z)D_M(z). \tag{178}$$

The isotropic BAO distance is

$$D_V(z) = \left[z D_M^2(z) D_H(z)\right]^{1/3}. \tag{179}$$

All distance measures in the final calculation are evaluated from the same locked function

$$H(z). \tag{180}$$

6.26 Internally consistent BAO ruler

Earlier calculations used a standalone approximation for the sound horizon.

The final CAMB calculation gave

$$r_{\rm drag} = 149.960 \text{ Mpc}, \tag{181}$$

which differed from the earlier approximate value.

The BAO ratios were therefore recomputed consistently:

$$\frac{D_M}{r_{\rm drag}}, \qquad \frac{D_H}{r_{\rm drag}}, \qquad \frac{D_V}{r_{\rm drag}}. \tag{182}$$

Using the same background for both ruler and distances reduced the diagnostic BAO statistic to approximately

$$\chi^2_{\rm BAO} = 13.4. \tag{183}$$

This result demonstrates that the earlier larger residual was partly a ruler inconsistency rather than a failure of the locked geometry.

6.27 Comparison with matched flat ΛCDM

A matched reference background is defined by

$$\Omega_{m0} = 0.314114, \qquad \Omega_{r0} = 9 \times 10^{-5}, \qquad \Omega_{\Lambda 0} = 1 - \Omega_{m0} - \Omega_{r0}. \tag{184}$$

Its expansion function is

$$E^2_{\Lambda\rm CDM}(z) = \Omega_{r0}(1+z)^4 + \Omega_{m0}(1+z)^3 + \Omega_{\Lambda 0}. \tag{185}$$

The relative expansion difference is

$$\Delta_E(z) = \frac{E_{\rm locked}(z) - E_{\Lambda\rm CDM}(z)}{E_{\Lambda\rm CDM}(z)}. \tag{186}$$

At late times, the locked model approaches the reference closely because

$$w_C \simeq -1. \tag{187}$$

At intermediate and higher redshift, the locked background is approximately

$$1\%\text{–}2\% \tag{188}$$

below the matched reference over the previously tabulated range.

Thus

$$E_{\rm locked}(z) < E_{\Lambda\rm CDM}(z) \tag{189}$$

by a small amount across much of the BAO and pre-recombination distance ladder.

This increases distance integrals relative to the matched reference.

6.28 Consequence of a slower intermediate expansion

If

$$H_{\mathrm{locked}}(z) < H_{\Lambda\mathrm{CDM}}(z), \tag{190}$$

then

$$D_H(z) = \frac{c}{H(z)} \tag{191}$$

is larger.

Likewise,

$$D_M(z) = \int_0^z \frac{c\,dz'}{H(z')} \tag{192}$$

is larger.

Therefore a slightly slower intermediate expansion tends to raise

$$\frac{D_H}{r_{\mathrm{drag}}} \quad \text{and} \quad \frac{D_M}{r_{\mathrm{drag}}}. \tag{193}$$

The final consistent value of

$$r_{\mathrm{drag}} \tag{194}$$

compensates much of this tendency.

The residual BAO structure is therefore a coherent consequence of the locked expansion history rather than an uncorrelated numerical error.

6.29 Present-day growth normalization

The final background and perturbation hierarchy give

$$\sigma_8 = 0.779680. \tag{195}$$

The derived quantity is

$$S_8 = \sigma_8 \sqrt{\frac{\Omega_{m0}}{0.3}}. \tag{196}$$

Substituting

$$\Omega_{m0} = 0.314114, \tag{197}$$

gives

$$S_8 = 0.79781. \tag{198}$$

The reduction relative to the matched control is produced mainly by the altered expansion history.

The canonical saturation-field perturbations contribute negligibly to the change.

6.30 Background robustness

The locked baseline was perturbed locally by small one-at-a-time changes in

$$\Omega_{m0}, \quad V_\infty, \quad \rho_s, \quad m_C^2, \quad \lambda, \quad C_{\text{initial}}, \quad \Pi_{C,\text{initial}}. \tag{199}$$

The tested steps were approximately

$$\pm 0.5\% \qquad \text{and} \qquad \pm\, 1\%. \tag{200}$$

Of the 28 local perturbations,

$$25 \tag{201}$$

passed the retained background and BAO gates.

The background was therefore classified as

$$\boxed{\text{ROBUST}} \tag{202}$$

within the tested neighborhood.

The most sensitive direction was

$$V_\infty. \tag{203}$$

6.31 Transition-width sensitivity

The retained background was also tested against changes in the splice width.

Increasing the width by a factor of

$$1.5 \tag{204}$$

gave a shift in

$$100\theta_* \tag{205}$$

of approximately

$$+0.0847\%. \tag{206}$$

Reducing the width by a factor of

$$0.67 \tag{207}$$

gave a shift of approximately

$$-0.0538\%. \tag{208}$$

These tests showed that the precision acoustic angle is mildly sensitive to the numerical transition shape.

The narrowed value

$$0.67W_0 \tag{209}$$

was retained because it produced the best final balance while preserving BAO, growth, and field stability.

6.32 Residual-normalization sensitivity

The residual contribution was varied by approximately

$$\pm 5\%. \tag{210}$$

The resulting acoustic-angle shifts were at the level of roughly

$$0.06\%. \tag{211}$$

The BAO statistic and field stability remained acceptable.

The result showed that the background is not destroyed by small residual changes, but the precision acoustic angle responds measurably.

The final residual factor was retained at

$$1.000. \tag{212}$$

6.33 Locked background card

The final background may be summarized as follows:

$$\boxed{\Omega_{m0} = 0.314114}, \tag{213}$$

$$\boxed{\Omega_{r0} = 9 \times 10^{-5}}, \tag{214}$$

$$\boxed{H_0 = 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}}, \tag{215}$$

$$\boxed{\rho_s = 1.58199 \times 10^{15}}, \tag{216}$$

$$\boxed{V_\infty = 2.187987}, \tag{217}$$

$$\boxed{m_C^2 = 1.827221}, \tag{218}$$

$$\boxed{\lambda = 0.802722}, \tag{219}$$

$$\boxed{C_s = 0.95}, \tag{220}$$

$$\boxed{C_{\mathrm{initial}} = 0.918318}, \tag{221}$$

$$\boxed{\Pi_{C,\mathrm{initial}} = 0.055327}, \tag{222}$$

$$\boxed{a_{\mathrm{min}} \simeq 2.068 \times 10^{-5}}, \tag{223}$$

$$\boxed{a_{\mathrm{lock}} = 2.05 \times 10^{-4}}, \tag{224}$$

$$\boxed{W = 0.67 W_0}, \tag{225}$$

$$\boxed{100\theta_* = 1.041033}, \tag{226}$$

$$\boxed{r_{\mathrm{drag}} = 149.960\ \mathrm{Mpc}}, \tag{227}$$

$$\boxed{z_* = 1089.691}, \tag{228}$$

$$\boxed{z_{\mathrm{drag}} = 1058.511}, \tag{229}$$

$$\boxed{\sigma_8 = 0.779680}, \tag{230}$$

$$\boxed{S_8 = 0.79781}, \tag{231}$$

$$\boxed{w_C(0) \simeq -0.99988}, \tag{232}$$

$$\boxed{\rho_C(0) \simeq 2.192}. \tag{233}$$

6.34 Physical interpretation of the locked history

The complete background history can be written schematically as

$$\boxed{\text{contraction} \to \text{kinetic regulation} \to \text{finite-density bounce} \to \text{radiation era} \to \text{matter era} \to \text{locked residual era}} \tag{234}$$

The contraction branch amplifies the importance of the saturation sector.

The transfer converts excessive field kinetic energy into radiation.

The high-density response reaches

$$\rho = \rho_s, \tag{235}$$

where

$$H = 0. \tag{236}$$

The Universe rebounds at finite scale factor.

The post-bounce branch enters radiation domination.

Matter later dominates.

The fields settle into their stable minimum.

Their residual energy drives the late expansion.

The same field system is present throughout the full history.

6.35 Final background result

The background construction establishes a continuous nonsingular cosmological trajectory with the following properties:

$$\begin{aligned}
a_{\min} &> 0, && (237)\\
\rho &\leq \rho_s, && (238)\\
K_{\text{bounce}} &< \infty, && (239)\\
\Gamma_{\text{tr}} &= 0 \quad \text{for} \quad H \geq 0, && (240)\\
\rho_r &\gg \rho_C \quad \text{during the deep radiation era}, && (241)\\
\rho_m &> \rho_r \quad \text{after equality}, && (242)\\
C &\to C_s, && (243)\\
\phi &\to 0, && (244)\\
w_C &\to -1. && (245)
\end{aligned}$$

The expanding branch produces a locked function

$$E(z) \tag{246}$$

that can be passed consistently into the complete Boltzmann hierarchy.

This background is the foundation upon which the perturbation, acoustic, growth, BAO, CMB, and lensing calculations are built.

End of Chapter 6

Chapter 7

Boltzmann Implementation of the Locked Saturation Cosmology

7.1 Purpose of the numerical implementation

The background equations developed in the preceding chapters determine the homogeneous expansion history,

$$H(a), \tag{1}$$

but cosmological observations depend upon more than the background alone.

The cosmic microwave background, matter clustering, gravitational lensing, and baryon acoustic oscillations require the evolution of perturbations through the coupled Einstein-Boltzmann system.

The purpose of the numerical implementation is therefore to carry the locked saturation cosmology from the homogeneous background into the full linear perturbation hierarchy.

The implementation has four principal components:

$$\begin{aligned}
&\text{locked background expansion,}\\
&\text{canonical } C, \phi \text{ perturbations,}\\
&\text{Einstein constraint and evolution sources,}\\
&\text{standard photon, baryon, neutrino, and matter hierarchies.}
\end{aligned} \tag{2}$$

The Boltzmann calculation retains the standard primordial spectrum and ordinary matter content while replacing the reference background expansion with the locked function derived from the nonsingular trajectory.

The final implementation is based upon CAMB.

7.2 CAMB background structure

CAMB evolves cosmological perturbations using conformal time,

$$d\tau = \frac{dt}{a}, \tag{3}$$

and the conformal Hubble parameter,

$$\mathcal{H} = \frac{a'}{a} = aH. \tag{4}$$

A prime denotes differentiation with respect to conformal time.

The standard flat-background Friedmann equation is written schematically as

$$\mathcal{H}^2 = \frac{8\pi G}{3} a^2 \rho_{\rm tot}. \tag{5}$$

The saturation implementation replaces the reference background function with the locked expansion history,

$$H_{\text{locked}}(a) = H_0 E_{\text{locked}}(a). \tag{6}$$

Therefore

$$\mathcal{H}_{\text{locked}}(a) = aH_0 E_{\text{locked}}(a). \tag{7}$$

The corresponding conformal-time derivative of the scale factor is

$$a' = a\mathcal{H}_{\text{locked}} = a^2 H_0 E_{\text{locked}}(a). \tag{8}$$

The background table supplied to CAMB contains the expanding branch only.

The pre-bounce contracting branch is not evolved inside CAMB.

It is resolved in the independent nonsingular background integrator and enters CAMB through the resulting expanding trajectory.

7.3 Separation between background generation and Boltzmann evolution

The full workflow is divided into two numerical stages.

Stage I: nonsingular background integration. The state vector

$$\big(a, H, C, \Pi_C, \phi, \Pi_\phi, \rho_m, \rho_r\big) \tag{9}$$

is evolved in cosmic or affine time from contraction through the bounce and into expansion.

This stage produces

$$a(t), \quad H(t), \quad C(t), \quad \phi(t), \tag{10}$$

and the corresponding energy densities.

Stage II: expanding Boltzmann evolution. The expanding trajectory is converted into

$$H(a), \quad C(a), \quad C'(a), \quad \phi(a), \quad \phi'(a). \tag{11}$$

These functions are then used inside CAMB.

This separation avoids integrating CAMB through

$$H = 0, \tag{12}$$

while preserving the physical information from the nonsingular branch.

7.4 Locked-background interpolation

The numerical trajectory is sampled at discrete scale factors,

$$\{a_i, E_i\}. \tag{13}$$

A smooth interpolation function is constructed:

$$E_{\text{tab}}(a) = \mathcal{I}\big[\{a_i, E_i\}\big]. \tag{14}$$

The interpolation must satisfy:

$$E_{\text{tab}}(a_i) = E_i, \qquad E_{\text{tab}}(a) > 0 \tag{15}$$

on the expanding branch, and

$$E_{\text{tab}}(1) = 1. \tag{16}$$

A shape-preserving interpolation is preferred so that no artificial oscillation is introduced between tabulated points.

The Hubble function is then

$$H(a) = H_0 E_{\text{tab}}(a). \tag{17}$$

The derivative required by perturbation evolution is

$$\frac{d\ln H}{d\ln a} = \frac{a}{E(a)}\frac{dE}{da}. \tag{18}$$

This derivative enters the growth equations and several background-dependent terms in the Boltzmann hierarchy.

7.5 Smooth early transition

The retained Boltzmann background uses a smooth transition ending at

$$a_{\text{lock}} = 2.05 \times 10^{-4}. \tag{19}$$

The transition width is

$$W = 0.67 W_0. \tag{20}$$

Let the transition begin at

$$a_{\text{start}} = a_{\text{lock}} - W. \tag{21}$$

Define

$$x(a) = \frac{a - a_{\text{start}}}{a_{\text{lock}} - a_{\text{start}}}. \tag{22}$$

The smoothstep function is

$$S(x) = 3x^2 - 2x^3. \tag{23}$$

The complete transition is

$$S(a) = \begin{cases} 0, & a \le a_{\text{start}}, \\ 3x^2 - 2x^3, & a_{\text{start}} < a < a_{\text{lock}}, \\ 1, & a \ge a_{\text{lock}}. \end{cases} \tag{24}$$

The blended background may be written schematically as

$$E^2(a) = \big[1 - S(a)\big] E^2_{\text{early}}(a) + S(a) E^2_{\text{locked}}(a). \tag{25}$$

The retained implementation preserves continuity of both the function and its first derivative at the transition endpoints.

Thus

$$E(a) \tag{26}$$

contains no step discontinuity.

7.6 Present-day normalization

After the locked history is blended into the expanding background, the expansion is normalized so that

$$E(1) = 1. \tag{27}$$

If the unnormalized function is

$$\widetilde{E}(a), \tag{28}$$

then

$$E(a) = \frac{\widetilde{E}(a)}{\widetilde{E}(1)}. \tag{29}$$

Therefore

$$H(a) = H_0 \frac{\widetilde{E}(a)}{\widetilde{E}(1)}. \tag{30}$$

This normalization leaves the shape of the locked trajectory unchanged while enforcing the chosen present-day Hubble constant,

$$H_0 = 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}. \tag{31}$$

7.7 Matter and radiation inputs

The locked calculation uses

$$\Omega_{m0} = 0.314114, \qquad \Omega_{r0} = 9 \times 10^{-5}. \tag{32}$$

The physical baryon density is

$$\omega_b = \Omega_b h^2 = 0.022. \tag{33}$$

The physical cold-dark-matter-like matter input used in the matched Boltzmann calculation is

$$\omega_c = \Omega_c h^2 \simeq 0.12005. \tag{34}$$

The reduced Hubble parameter is

$$h = \frac{H_0}{100\ \mathrm{km\,s^{-1}\,Mpc^{-1}}} = 0.674. \tag{35}$$

The total present matter density is

$$\Omega_{m0} = \frac{\omega_b + \omega_c + \omega_\nu}{h^2}. \tag{36}$$

The neutrino sector retains the standard minimal mass input used in the matched control.

The primordial spectrum is held fixed.

7.8 Primordial perturbations

The primordial scalar curvature spectrum is

$$\mathcal{P}_{\mathcal{R}}(k) = A_s \left(\frac{k}{k_*}\right)^{n_s-1}. \tag{37}$$

The locked calculation uses

$$n_s = 0.965, \qquad A_s = 2.0 \times 10^{-9}. \tag{38}$$

The optical depth is

$$\tau_{\rm reio} = 0.054. \tag{39}$$

These primordial parameters are not retuned between the locked model and the matched control.

Therefore differences in the resulting spectra arise from the altered background and saturation-field sector rather than from a separate primordial-amplitude adjustment.

7.9 Standard Boltzmann hierarchy

The photon brightness perturbations are expanded in multipoles:

$$\Theta(\mathbf{k}, \mu, \tau) = \sum_{\ell=0}^{\infty} (-i)^\ell (2\ell+1)\Theta_\ell(\mathbf{k}, \tau) P_\ell(\mu). \tag{40}$$

The photon hierarchy contains

$$\Theta_0, \quad \Theta_1, \quad \Theta_2, \tag{41}$$

and higher multipoles.

The polarization hierarchy is evolved through the E- and B-mode moments.

Massless neutrinos obey a corresponding collisionless hierarchy.

Baryons satisfy continuity and Euler equations.

Cold matter evolves as a pressureless component.

The saturation implementation leaves these standard equations unchanged except through:

$$\mathcal{H}(a), \quad \mathcal{H}'(a), \tag{42}$$

and the additional canonical field stress-energy sources.

7.10 Synchronous-gauge metric

CAMB uses synchronous gauge for scalar perturbations.

The scalar metric perturbation may be written as

$$ds^2 = a^2(\tau)\Big[-d\tau^2 + \big(\delta_{ij} + h_{ij}\big)dx^i dx^j\Big]. \tag{43}$$

In Fourier space,

$$h_{ij} = \hat{k}_i\hat{k}_j h + \Big(\hat{k}_i\hat{k}_j - \frac{1}{3}\delta_{ij}\Big)6\eta. \tag{44}$$

The two scalar metric variables are

$$h \qquad \text{and} \qquad \eta. \tag{45}$$

The Einstein equations relate their evolution to the total density, momentum, pressure, and anisotropic-stress perturbations.

7.11 Einstein constraint equations

The synchronous-gauge Hamiltonian constraint is

$$k^2\eta - \frac{1}{2}\mathcal{H}h' = 4\pi G a^2 \delta T^0_{\ 0}. \tag{46}$$

Since

$$\delta T^0_{\ 0} = -\delta\rho, \tag{47}$$

this becomes

$$k^2\eta - \frac{1}{2}\mathcal{H}h' = -4\pi G a^2 \delta\rho. \tag{48}$$

The momentum constraint is

$$k^2\eta' = 4\pi G a^2(\rho + p)\theta. \tag{49}$$

The trace evolution equation is

$$h'' + 2\mathcal{H}h' - 2k^2\eta = -8\pi G a^2 \delta T^i_{\ i}. \tag{50}$$

Since

$$\delta T^i_{\ i} = 3\delta p, \tag{51}$$

one obtains

$$h'' + 2\mathcal{H}h' - 2k^2\eta = -24\pi G a^2 \delta p. \tag{52}$$

The traceless evolution equation contains the total anisotropic stress.

The saturation fields enter these equations through their canonical perturbative stress-energy.

7.12 Background saturation fields inside CAMB

The background fields are tabulated as functions of scale factor:

$$C_{\rm bg}(a), \quad \phi_{\rm bg}(a), \quad C'_{\rm bg}(a), \quad \phi'_{\rm bg}(a). \tag{53}$$

The conformal derivatives are related to cosmic-time momenta by

$$C' = a\dot{C} = a\Pi_C, \qquad \phi' = a\dot{\phi} = a\Pi_\phi. \tag{54}$$

The background energy density becomes

$$\rho_C = \frac{1}{2a^2}\left[(C'_{\rm bg})^2 + (\phi'_{\rm bg})^2\right] + V(C_{\rm bg}, \phi_{\rm bg}). \tag{55}$$

The pressure is

$$p_C = \frac{1}{2a^2}\left[(C'_{\rm bg})^2 + (\phi'_{\rm bg})^2\right] - V(C_{\rm bg}, \phi_{\rm bg}). \tag{56}$$

These expressions are equivalent to the cosmic-time forms.

7.13 Canonical perturbation variables

The field perturbations are

$$\delta C(\tau, \mathbf{k}), \qquad \delta\phi(\tau, \mathbf{k}). \tag{57}$$

Define their first derivatives:

$$v_C = \delta C', \qquad v_\phi = \delta\phi'. \tag{58}$$

The complete first-order state added to the Boltzmann hierarchy is

$$\big(\delta C, v_C, \delta\phi, v_\phi\big). \tag{59}$$

The evolution equations are

$$\delta C' = v_C, \tag{60}$$

$$v'_C = -2\mathcal{H}v_C - \big(k^2 + a^2 V_{,CC}\big)\delta C - \frac{1}{2}C'_{\rm bg}h', \tag{61}$$

$$\delta\phi' = v_\phi, \tag{62}$$

$$v'_\phi = -2\mathcal{H}v_\phi - \big(k^2 + a^2 V_{,\phi\phi}\big)\delta\phi - \frac{1}{2}\phi'_{\rm bg}h'. \tag{63}$$

These are canonical second-order scalar perturbation equations written as a first-order system.

7.14 Internal CAMB metric variable

CAMB commonly rewrites the synchronous metric forcing in terms of an internal variable related to

$$h'. \tag{64}$$

In the implemented equations, the source was represented schematically as

$$-kz\, C'_{\rm bg}, \qquad -kz\, \phi'_{\rm bg}. \tag{65}$$

The variable kz is an internal CAMB combination of synchronous-gauge metric derivatives.

The mathematical correspondence is

$$-kz\,C'_{\rm bg} \longleftrightarrow -\frac{1}{2}h'C'_{\rm bg}, \tag{66}$$

and similarly for ϕ.

The code notation therefore represents the same metric forcing as the covariant perturbation equations.

7.15 Potential curvature in the perturbation equations

The organizational mass term is

$$V_{,CC} = m_C^2 + 3\lambda(C_{\rm bg} - C_s)^2. \tag{67}$$

The phase mass term is

$$V_{,\phi\phi} = A_0 \cos\phi_{\rm bg}. \tag{68}$$

The mixed term is

$$V_{,C\phi} = 0. \tag{69}$$

Therefore the two fields are decoupled in the potential Hessian.

Their only linear coupling occurs indirectly through the shared metric perturbations.

The effective mode frequencies are

$$\omega_C^2 = k^2 + a^2 V_{,CC}, \qquad \omega_\phi^2 = k^2 + a^2 V_{,\phi\phi}. \tag{70}$$

For positive potential curvature, the modes remain oscillatory or damped rather than exponentially unstable.

7.16 Canonical field density perturbation

The saturation-sector density perturbation is

$$\delta\rho_C = \frac{1}{a^2}\left[C'_{\rm bg}v_C + \phi'_{\rm bg}v_\phi\right] + V_{,C}\delta C + V_{,\phi}\delta\phi. \tag{71}$$

CAMB evolves metric-normalized source combinations.

Multiplying by a^2,

$$a^2\delta\rho_C = C'_{\rm bg}v_C + \phi'_{\rm bg}v_\phi + a^2\left[V_{,C}\delta C + V_{,\phi}\delta\phi\right]. \tag{72}$$

This term is added to the total density perturbation:

$$\delta\rho_{\rm total} = \delta\rho_{\rm standard} + \delta\rho_C. \tag{73}$$

7.17 Canonical field pressure perturbation

The field pressure perturbation is

$$\delta p_C = \frac{1}{a^2}\big[C'_{\rm bg}v_C + \phi'_{\rm bg}v_\phi\big] - V_{,C}\delta C - V_{,\phi}\delta\phi. \tag{74}$$

The metric-normalized form is

$$a^2\delta p_C = C'_{\rm bg}v_C + \phi'_{\rm bg}v_\phi - a^2\big[V_{,C}\delta C + V_{,\phi}\delta\phi\big]. \tag{75}$$

This is added to the total pressure perturbation:

$$\delta p_{\rm total} = \delta p_{\rm standard} + \delta p_C. \tag{76}$$

7.18 Canonical field momentum perturbation

The field momentum density is

$$\delta q_C = \frac{k}{a^2}\big[C'_{\rm bg}\delta C + \phi'_{\rm bg}\delta\phi\big]. \tag{77}$$

The internal normalized form is

$$a^2\delta q_C = k\big[C'_{\rm bg}\delta C + \phi'_{\rm bg}\delta\phi\big]. \tag{78}$$

This term contributes to the Einstein momentum constraint.

Near saturation,

$$C'_{\rm bg} \simeq 0, \qquad \phi'_{\rm bg} \simeq 0. \tag{79}$$

Therefore the momentum source becomes small.

7.19 Anisotropic stress

Canonical minimally coupled scalar fields do not generate intrinsic linear anisotropic stress.

Therefore

$$\pi_C = 0. \tag{80}$$

The saturation fields do not directly produce a gravitational slip through their own shear.

Any difference between metric potentials arises through the total system, including standard radiation and neutrino anisotropic stress.

This property distinguishes the canonical implementation from phenomenological slip parameterizations.

7.20 Initial conditions for the field perturbations

The saturation fields are initialized on the expanding branch after the locked background has been supplied.

The simplest regular initial conditions are small adiabatic-compatible amplitudes.

Schematically,

$$\delta C_{\rm ini} \ll 1, \qquad v_{C,\rm ini} \ll 1, \qquad \delta\phi_{\rm ini} \ll 1, \qquad v_{\phi,\rm ini} \ll 1. \tag{81}$$

The final solutions are driven by both their initial amplitudes and the metric source terms.

Because the background fields are close to their stable trajectory, the perturbations remain bounded.

The measured maxima are

$$\max|\delta C| \simeq 0.035, \qquad \max|\delta\phi| \simeq 0.049. \tag{82}$$

7.21 Perturbative field diagnostics

The numerical implementation records

$$\max|\delta C|, \quad \max|v_C|, \quad \max|\delta\phi|, \quad \max|v_\phi|. \tag{83}$$

For the final F1 branch, representative values are

$$\max|\delta C| \simeq 0.03514, \qquad \max|\delta\phi| \simeq 0.04860. \tag{84}$$

The velocity-like variables remain finite.

No runaway mode appears.

No late-time exponential growth appears.

The field sector therefore remains within the linear regime.

7.22 Direct perturbative back-reaction

The complete locked CAMB model was compared with an otherwise identical run in which the canonical field perturbations were removed while the same locked background was retained.

The differences in

$$\sigma_8, \quad \text{the TT peaks}, \quad \text{EE}, \quad \text{TE} \tag{85}$$

were found to be extremely small.

The change in σ_8 was approximately at the level of

$$5 \times 10^{-6} \tag{86}$$

fractionally.

Peak-height changes were at the level of approximately

$$10^{-6} \text{ to } 10^{-7}. \tag{87}$$

Thus

$$\boxed{\text{locked-background effects} \gg \text{direct canonical-field perturbation effects}} \tag{88}$$

for the linear observables examined.

7.23 Phenomenological Poisson rescaling

Before the canonical implementation was completed, a phenomenological factor

$$f_{\rm sat}(a,k) \tag{89}$$

was introduced into the scalar constraints.

A representative modified Poisson equation was

$$k^2\Psi = -4\pi G a^2 f_{\rm sat}(a,k)\rho\Delta. \tag{90}$$

The intention was to model saturation as an effective weakening of gravity.

Several forms were tested:

$$f_{\rm sat} = \text{constant}, \qquad f_{\rm sat} = \text{step-like}, \qquad f_{\rm sat} = \text{smoothly varying}. \tag{91}$$

These modifications did not reproduce the canonical field behavior.

Instead, they changed the growth and acoustic driving in the wrong direction.

7.24 Independent μ and slip

The broader phenomenological system used

$$k^2\Psi = -4\pi G a^2 \mu(a,k)\rho\Delta, \tag{92}$$

and

$$\eta(a,k) = \frac{\Phi}{\Psi}. \tag{93}$$

The Weyl modification is

$$\Sigma(a,k) = \frac{\mu(a,k)\big[1+\eta(a,k)\big]}{2}. \tag{94}$$

The tested algebraic closures included

$$\eta = \frac{1}{\mu}, \qquad \eta = 2-\mu, \qquad \eta = 1. \tag{95}$$

The purpose was to determine whether a slip choice could preserve lensing while reducing growth.

No tested closure reproduced the canonical saturation result.

7.25 Failure of the phenomenological closures

The phenomenological models produced two systematic effects.

First,

$$\sigma_8 \tag{96}$$

increased rather than decreased.

Second, the TT spectrum developed an alternating odd-even peak distortion.

The explanation lies in the coupled Einstein-Boltzmann system.

The factor μ does not affect only late-time matter growth.

It alters:

$$\begin{aligned}&\text{metric constraints,}\\&\text{radiation driving,}\\&\text{potential decay,}\\&\text{baryon-photon acoustic forcing,}\\&\text{lensing sources.}\end{aligned} \tag{97}$$

An algebraic rescaling of the constraint is therefore not equivalent to the small additive stress-energy of the canonical saturation fields.

7.26 Matched control cosmology

The locked model is compared with a matched flat reference cosmology.

The control uses the same:

$$H_0, \quad \Omega_{m0}, \quad \Omega_{r0}, \quad \omega_b, \quad \omega_c, \quad A_s, \quad n_s, \quad \tau_{\text{reio}}, \tag{98}$$

and neutrino inputs.

The principal difference is the expansion history:

$$E_{\text{locked}}(a) \qquad \text{versus} \qquad E_{\Lambda\text{CDM}}(a). \tag{99}$$

This matched construction isolates the effects of the locked background and saturation fields.

7.27 CAMB-derived quantities

CAMB computes the recombination redshift,

$$z_*, \tag{100}$$

the drag redshift,

$$z_{\text{drag}}, \tag{101}$$

the sound horizon,

$$r_s(z_*), \tag{102}$$

the drag horizon,

$$r_{\text{drag}}, \tag{103}$$

and the angular acoustic scale,

$$100\theta_*. \tag{104}$$

For the final locked model,

$$z_* = 1089.691, \tag{105}$$
$$z_{\rm drag} = 1058.511, \tag{106}$$
$$r_{\rm drag} = 149.960 \text{ Mpc}, \tag{107}$$
$$100\theta_* = 1.041033. \tag{108}$$

CAMB also computes

$$\sigma_8 = 0.779680, \tag{109}$$

and therefore

$$S_8 = 0.79781. \tag{110}$$

7.28 CMB power spectra

The code produces the angular spectra

$$C_\ell^{TT}, \quad C_\ell^{EE}, \quad C_\ell^{BB}, \quad C_\ell^{TE}. \tag{111}$$

The commonly plotted form is

$$D_\ell^{XY} = \frac{\ell(\ell+1)}{2\pi} C_\ell^{XY}. \tag{112}$$

For temperature,

$$D_\ell^{TT} \tag{113}$$

is measured in

$$\mu\text{K}^2. \tag{114}$$

The final calculation extends to approximately

$$\ell_{\rm max} = 2500. \tag{115}$$

Lensing is included.

7.29 Stable comparison of cross-spectra

A naive fractional difference,

$$\frac{TE_{\rm locked} - TE_{\rm control}}{TE_{\rm control}}, \tag{116}$$

becomes singular when

$$TE_{\rm control} = 0. \tag{117}$$

The stable comparison used is

$$\frac{\Delta TE}{\sqrt{TT_{\rm control} EE_{\rm control}}}. \tag{118}$$

This normalization remains finite near TE zero crossings.

The same convention is used in the final residual diagnostics.

7.30 Spectral residual bands

The spectra were compared across four multipole ranges:

$$2 \leq \ell \leq 29, \tag{119}$$

$$30 \leq \ell \leq 600, \tag{120}$$

$$601 \leq \ell \leq 1500, \tag{121}$$

$$1501 \leq \ell \leq 2500. \tag{122}$$

These correspond to:

$$\text{low multipoles}, \quad \text{principal acoustic region}, \quad \text{damping region}, \quad \text{high-}\ell \text{ tail}. \tag{123}$$

The residuals are largest at low multipoles and through the first acoustic region.

They decrease substantially toward the damping tail.

The differences are scale dependent and cannot be removed by a single overall amplitude factor.

7.31 Lensing-potential spectrum

CAMB computes

$$C_L^{\phi\phi}, \tag{124}$$

where ϕ here denotes the CMB lensing potential rather than the saturation phase field.

To avoid confusion, the lensing potential may be denoted

$$\varphi_{\text{lens}}. \tag{125}$$

Its spectrum is therefore

$$C_L^{\varphi_{\text{lens}}\varphi_{\text{lens}}}. \tag{126}$$

The final locked model produces a mildly enhanced lensing-potential spectrum relative to the matched control.

Across the conservative Planck range, the enhancement is of order one percent.

At higher multipoles, the enhancement approaches approximately two percent.

7.32 PlanckLens integration

The CAMB spectra were passed into the PlanckLens reconstruction framework.

The analytic TT quadratic estimator was evaluated using:

$$\ell_{\max}^{\text{CMB}} = 512, \tag{127}$$

$$L_{\max} = 400, \tag{128}$$

$$\ell_{\min}^{\text{CMB}} = 30, \tag{129}$$

$$\theta_{\text{beam}} = 6 \text{ arcmin}, \tag{130}$$

$$\Delta_T = 35\ \mu\text{K}\,\text{arcmin}. \tag{131}$$

The Gaussian reconstruction noise is

$$N_L^{(0)}. \tag{132}$$

The locked/control ratio averaged over the tested range is approximately

$$\frac{N_{L,\text{locked}}^{(0)}}{N_{L,\text{control}}^{(0)}} \simeq 1.21. \tag{133}$$

Thus the locked primary TT spectrum produces a higher reconstruction-noise forecast even though the true lensing-potential spectrum is mildly enhanced.

7.33 Planck lensing bandpower interface

The official Planck 2018 conservative lensing dataset contains nine bandpowers over

$$8 \le L \le 400. \tag{134}$$

The model spectrum is binned through the supplied windows:

$$C_b^{\varphi\varphi} = \sum_L W_{bL} C_L^{\varphi\varphi}. \tag{135}$$

The residual vector is

$$\Delta_b = C_{b,\text{model}}^{\varphi\varphi} - C_{b,\text{Planck}}^{\varphi\varphi}. \tag{136}$$

The bandpower statistic is

$$\chi^2 = \sum_{bb'} \Delta_b \left(\Sigma^{-1}\right)_{bb'} \Delta_{b'}. \tag{137}$$

Using the official nine-by-nine covariance, the locked result is

$$\chi^2_{\text{locked}} = 10.7379. \tag{138}$$

The matched control gives

$$\chi^2_{\text{control}} = 12.0022. \tag{139}$$

Therefore

$$\Delta\chi^2 = -1.2643. \tag{140}$$

7.34 BAO interface

The same CAMB background provides the drag horizon,

$$r_{\mathrm{drag}}, \tag{141}$$

and the expansion function used to compute

$$D_M(z), \quad D_H(z), \quad D_V(z). \tag{142}$$

The observables are

$$\frac{D_M(z)}{r_{\mathrm{drag}}}, \qquad \frac{D_H(z)}{r_{\mathrm{drag}}}, \qquad \frac{D_V(z)}{r_{\mathrm{drag}}}. \tag{143}$$

The covariance-weighted diagnostic statistic for the supplied DESI points is approximately

$$\chi^2_{\mathrm{BAO}} = 13.4. \tag{144}$$

The crucial requirement is that the same CAMB-derived value

$$r_{\mathrm{drag}} = 149.960 \text{ Mpc} \tag{145}$$

is used in every ratio.

7.35 Growth interface

CAMB evolves the full matter transfer function.

The present fluctuation amplitude is

$$\sigma_8 = \left[\int \frac{dk}{k}\Delta_m^2(k)W_8^2(k)\right]^{1/2}, \tag{146}$$

where W_8 is the top-hat window corresponding to

$$8h^{-1}\mathrm{Mpc}. \tag{147}$$

The final result is

$$\sigma_8 = 0.779680. \tag{148}$$

The derived weak-lensing parameter is

$$S_8 = \sigma_8\sqrt{\frac{\Omega_{m0}}{0.3}} = 0.79781. \tag{149}$$

7.36 Internal consistency checks

The numerical workflow contains several internal checks.

Background normalization.

$$E(1) = 1. \tag{150}$$

Acoustic consistency. The same background is used for

$$r_s, \quad r_{\rm drag}, \quad D_M, \quad D_H. \tag{151}$$

Perturbation boundedness.

$$\max|\delta C| < 1, \qquad \max|\delta\phi| < 1. \tag{152}$$

Solver reproducibility. The background is consistent across DOP853 and Radau.
Constraint preservation. The nonsingular Friedmann residual remains small.
Matched primordial inputs. The locked and control runs use identical

$$A_s, \quad n_s, \quad \tau. \tag{153}$$

7.37 Final executable workflow

The complete numerical sequence is

$$\boxed{\begin{array}{l}\text{nonsingular ODE background}\\ \to \text{expanding locked table}\\ \to \text{smooth CAMB splice}\\ \to \text{canonical field perturbations}\\ \to \text{Einstein-Boltzmann hierarchy}\\ \to \text{CMB, BAO, growth, and lensing diagnostics}\end{array}} \tag{154}$$

The same retained branch is used for all final observables.
No separate background is used for BAO.
No separate sound horizon is inserted by hand.
No phenomenological gravity factor is retained.
No late dark-energy fluid is added.
The observable calculations are generated from one locked background and one canonical perturbation system.

7.38 Final implementation statement

The CAMB implementation demonstrates that the gravitational-saturation framework can be represented numerically as:

$$\begin{array}{l}\text{a locked nonsingular expansion history,}\\ +\ \text{canonical linear perturbations of } C \text{ and } \phi,\\ \text{within the standard Einstein-Boltzmann structure.}\end{array} \tag{155}$$

The saturation fields remain perturbative.
Their direct linear influence is negligible.

The dominant physical signal is carried by the altered background trajectory.

The resulting code produces stable acoustic peaks, a consistent sound horizon, a coherent BAO ladder, reduced late-time clustering amplitude, and a Planck-compatible lensing spectrum.

The following chapter presents the numerical results of these calculations in full.

End of Chapter 7

Chapter 8

Numerical Results of the Locked Saturation Cosmology

8.1 Purpose of the results chapter

The previous chapters established the mathematical structure of the gravitational-saturation model and its implementation within the CAMB and PlanckLens workflows.

This chapter presents the numerical results obtained from the final retained background and perturbation system.

The final branch is defined by

$$a_{\text{lock}} = 2.05 \times 10^{-4}, \tag{1}$$

with transition width

$$W = 0.67\, W_0, \tag{2}$$

and residual factor

$$1.000. \tag{3}$$

The cosmological inputs are

$$H_0 = 67.4 \text{ km}\,\text{s}^{-1}\,\text{Mpc}^{-1}, \tag{4}$$
$$\Omega_{m0} = 0.314114, \tag{5}$$
$$\Omega_{r0} = 9 \times 10^{-5}, \tag{6}$$
$$\omega_b = 0.022, \tag{7}$$
$$\omega_c \simeq 0.12005, \tag{8}$$
$$n_s = 0.965, \tag{9}$$
$$A_s = 2.0 \times 10^{-9}, \tag{10}$$
$$\tau_{\text{reio}} = 0.054. \tag{11}$$

The saturation-sector parameters are

$$C_s = 0.95, \tag{12}$$
$$V_\infty = 2.187987, \tag{13}$$
$$\rho_s = 1.58199 \times 10^{15}, \tag{14}$$
$$m_C^2 = 1.827221, \tag{15}$$
$$\lambda = 0.802722, \tag{16}$$
$$C_{\text{initial}} = 0.918318, \tag{17}$$
$$\Pi_{C,\text{initial}} = 0.055327. \tag{18}$$

These parameters define the locked model used throughout the final numerical campaign.

8.2 Background quantities

The final locked background produces

$$100\theta_* = 1.041033. \tag{19}$$

Relative to the reference value

$$100\theta_*^{\mathrm{ref}} = 1.0411, \tag{20}$$

the fractional difference is

$$\frac{1.041033 - 1.0411}{1.0411} \times 100\% = -0.0064\%. \tag{21}$$

The final drag horizon is

$$r_{\mathrm{drag}} = 149.960 \text{ Mpc}. \tag{22}$$

The recombination and drag redshifts are

$$z_* = 1089.691, \tag{23}$$

and

$$z_{\mathrm{drag}} = 1058.511. \tag{24}$$

The comoving distance to last scattering is

$$D_M(z_*) = 14131.86 \text{ Mpc}. \tag{25}$$

The physical angular-diameter distance is

$$D_A(z_*) = 12.9568 \text{ Mpc}. \tag{26}$$

The final acoustic geometry is therefore determined by

$$\theta_* = \frac{r_s(z_*)}{D_M(z_*)}. \tag{27}$$

The numerical value shows that the locked sound horizon and locked line-of-sight distance balance to within less than one hundredth of one percent of the reference acoustic scale.

8.3 Bounce quantities

The nonsingular background reaches a minimum scale factor of approximately

$$a_{\min} = 2.068 \times 10^{-5}. \tag{28}$$

The corresponding redshift is

$$z_{\mathrm{bounce}} = \frac{1}{a_{\min}} - 1, \tag{29}$$

giving

$$z_{\mathrm{bounce}} \simeq 4.84 \times 10^4. \tag{30}$$

At the bounce,

$$H = 0, \tag{31}$$

and

$$\rho = \rho_s. \tag{32}$$

The field values and momenta remain finite.

The kinetic quantity

$$K^2_{\mathrm{bounce}} = \Pi_C^2 + \Pi_\phi^2 \tag{33}$$

does not diverge.

The scale factor satisfies

$$a_{\mathrm{min}} > 0. \tag{34}$$

The numerical trajectory therefore realizes a continuous finite-density bounce rather than a singular origin.

8.4 Radiation-era consistency

Matter-radiation equality occurs at

$$a_{\mathrm{eq}} = \frac{\Omega_{r0}}{\Omega_{m0}}. \tag{35}$$

Substituting the locked values gives

$$a_{\mathrm{eq}} \simeq 2.865 \times 10^{-4}. \tag{36}$$

The equality redshift is

$$z_{\mathrm{eq}} \simeq 3489. \tag{37}$$

The ratio between the bounce scale and equality scale is

$$\frac{a_{\mathrm{min}}}{a_{\mathrm{eq}}} \simeq 0.0722. \tag{38}$$

Thus

$$a_{\mathrm{min}} \ll a_{\mathrm{eq}}. \tag{39}$$

The bounce occurs well before equality.

For

$$a_{\mathrm{min}} \ll a \ll a_{\mathrm{eq}}, \tag{40}$$

the radiation density satisfies

$$\rho_r \gg \rho_m, \tag{41}$$

and

$$\rho_r \gg \rho_C. \tag{42}$$

The post-bounce expanding branch therefore contains a clear radiation-dominated era.

8.5 Present-day saturation state

At the present epoch, the saturation density is approximately

$$\rho_C(0) \simeq 2.192 \tag{43}$$

in the normalized background units.

The equation-of-state parameter is

$$w_C(0) \simeq -0.99988. \tag{44}$$

Therefore

$$1 + w_C(0) \simeq 1.2 \times 10^{-4}. \tag{45}$$

Using

$$1 + w_C = \frac{K^2}{\rho_C}, \tag{46}$$

the present kinetic fraction is

$$\frac{K^2}{\rho_C} \simeq 1.2 \times 10^{-4}. \tag{47}$$

The late saturation sector is therefore almost entirely potential dominated.

The field remains close to

$$C = C_s, \tag{48}$$

and the effective curvature is

$$V_{,CC} = m_C^2 + 3\lambda(C - C_s)^2. \tag{49}$$

The locked late state is stable and restorative.

8.6 Growth amplitude

The final CAMB calculation gives

$$\sigma_8 = 0.779680. \tag{50}$$

The weak-lensing combination is

$$S_8 = \sigma_8 \sqrt{\frac{\Omega_{m0}}{0.3}}. \tag{51}$$

Substituting

$$\Omega_{m0} = 0.314114 \tag{52}$$

gives

$$S_8 = 0.79781. \tag{53}$$

The matched control gives approximately

$$\sigma_{8,\mathrm{control}} = 0.79456, \tag{54}$$

and

$$S_{8,\mathrm{control}} = 0.81304. \tag{55}$$

The locked model therefore lowers the present growth amplitude while leaving the ordinary gravitational coupling unchanged.

The ratio is

$$\frac{S_{8,\mathrm{locked}}}{S_{8,\mathrm{control}}} \simeq 0.9813. \tag{56}$$

The reduction is approximately

$$1.87\%. \tag{57}$$

This change is produced predominantly by the locked expansion history.

8.7 Canonical field perturbations

The maximum amplitudes in the final run are

$$\max |\delta C| = 0.03514, \tag{58}$$

and

$$\max |\delta \phi| = 0.04860. \tag{59}$$

The corresponding velocity-like variables remain finite.

No exponential instability appears.

No ghost mode appears.

No negative gradient coefficient appears.

The kinetic matrix is

$$K_{IJ} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{60}$$

The sound speeds are

$$c_{s,C}^2 = 1, \tag{61}$$

and

$$c_{s,\phi}^2 = 1. \tag{62}$$

The mass matrix near the locked state is

$$M^2 \simeq \begin{pmatrix} m_C^2 & 0 \\ 0 & A_0 \end{pmatrix}, \tag{63}$$

with positive diagonal entries.

The linear scalar sector is therefore stable.

8.8 Direct field back-reaction

A control calculation was performed using the same locked background but without the canonical field perturbation sources.

The resulting change in

$$\sigma_8 \tag{64}$$

was at the level of approximately

$$5 \times 10^{-6} \tag{65}$$

fractionally.

The peak-height differences were at the level of

$$10^{-6} \quad \text{to} \quad 10^{-7}. \tag{66}$$

Therefore

$$\boxed{\text{the observable signal is carried by the locked background}} \tag{67}$$

rather than by a strong direct perturbative force from C or ϕ.

The saturation fields behave as linear spectators once the background has settled.

8.9 Failure of effective-gravity closures

The phenomenological models tested included

$$k^2\Psi = -4\pi G a^2\, \mu(a,k)\, \rho\Delta, \tag{68}$$

with slip

$$\eta(a,k) = \frac{\Phi}{\Psi}. \tag{69}$$

The tested closures included

$$\eta = 1, \qquad \eta = \frac{1}{\mu}, \qquad \eta = 2 - \mu. \tag{70}$$

A single multiplicative saturation factor

$$f_{\rm sat}(a,k) \tag{71}$$

was also applied to the scalar constraints.

These models failed systematically.

They increased

$$\sigma_8, \tag{72}$$

and distorted the odd-even acoustic peak structure.

The canonical field implementation did not produce these effects.

This confirms that the saturation model cannot be represented as a simple rescaling

$$G \to \mu G. \tag{73}$$

8.10 BAO observables

The final BAO calculation uses the same CAMB-derived ruler and background distances.

The observables are

$$\frac{D_M(z)}{r_{\rm drag}}, \qquad \frac{D_H(z)}{r_{\rm drag}}, \qquad \frac{D_V(z)}{r_{\rm drag}}. \tag{74}$$

The drag horizon is

$$r_{\rm drag} = 149.960 \text{ Mpc}. \tag{75}$$

The covariance-weighted diagnostic statistic for the supplied DESI points is approximately

$$\chi^2_{\rm BAO} = 13.4. \tag{76}$$

The earlier value near

$$\chi^2 \simeq 27 \tag{77}$$

was reduced when the BAO distances were recomputed using the internally consistent CAMB drag horizon.

The final BAO ladder is therefore coherent with the locked background.

8.11 Acoustic peak location

The primary acoustic scale is

$$100\theta_* = 1.041033. \tag{78}$$

The first TT peak appears near

$$\ell \simeq 220. \tag{79}$$

The final micro-adjustment of

$$a_{\rm lock} \tag{80}$$

was selected by testing nearby values.

The sequence was approximately

$$a_{\rm lock} = 2.10 \times 10^{-4} \quad \Rightarrow \quad 100\theta_* = 1.040917, \tag{81}$$
$$a_{\rm lock} = 2.05 \times 10^{-4} \quad \Rightarrow \quad 100\theta_* = 1.041033, \tag{82}$$
$$a_{\rm lock} = 2.00 \times 10^{-4} \quad \Rightarrow \quad 100\theta_* = 1.041167, \tag{83}$$
$$a_{\rm lock} = 1.95 \times 10^{-4} \quad \Rightarrow \quad 100\theta_* = 1.041302. \tag{84}$$

The retained value is

$$a_{\rm lock} = 2.05 \times 10^{-4}. \tag{85}$$

8.12 TT residuals

The locked TT spectrum was compared directly with the matched control.

For low multipoles,

$$2 \leq \ell \leq 29, \tag{86}$$

the RMS fractional difference is approximately

$$28.2\%. \tag{87}$$

For the principal acoustic region,

$$30 \leq \ell \leq 600, \tag{88}$$

the TT RMS difference is approximately

$$10.4\%. \tag{89}$$

For the damping region,

$$601 \leq \ell \leq 1500, \tag{90}$$

the RMS difference falls to approximately

$$3.0\%. \tag{91}$$

For

$$1501 \leq \ell \leq 2500, \tag{92}$$

the RMS difference is approximately

$$1.7\%. \tag{93}$$

Thus the largest TT differences occur at low multipoles and through the first acoustic region. The spectra converge substantially at high multipoles.

8.13 EE residuals

For

$$2 \leq \ell \leq 29, \tag{94}$$

the EE RMS difference is approximately

$$6.9\%. \tag{95}$$

For

$$30 \leq \ell \leq 600, \tag{96}$$

the difference is approximately

$$5.9\%. \tag{97}$$

For

$$601 \leq \ell \leq 1500, \tag{98}$$

the difference is approximately

$$4.2\%. \tag{99}$$

For

$$1501 \leq \ell \leq 2500, \tag{100}$$

the difference falls to approximately

$$2.3\%. \tag{101}$$

The EE residuals are smaller than the TT residuals in the low and principal acoustic bands, but remain scale dependent.

8.14 TE residuals

The TE difference is measured through

$$\frac{\Delta TE}{\sqrt{TT_{\mathrm{control}}\, EE_{\mathrm{control}}}}. \tag{102}$$

This avoids divergences at TE zero crossings.

The RMS values are approximately

$$12.0\% \tag{103}$$

for

$$2 \leq \ell \leq 29, \tag{104}$$

$$8.2\% \tag{105}$$

for

$$30 \leq \ell \leq 600, \tag{106}$$

$$2.4\% \tag{107}$$

for

$$601 \leq \ell \leq 1500, \tag{108}$$

and

$$1.1\% \tag{109}$$

for

$$1501 \leq \ell \leq 2500. \tag{110}$$

The TE residuals therefore follow the same broad pattern as TT and EE:

$$\text{largest at low and acoustic multipoles,} \tag{111}$$

$$\text{smallest in the high-}\ell\text{ tail.} \tag{112}$$

8.15 Amplitude-versus-shape decomposition

A single best-fit amplitude was computed for each spectrum.

For TT, the locked/control amplitude is approximately

$$A_{TT} = 0.9459. \tag{113}$$

After removing this amplitude difference, the remaining shape RMS is approximately

$$7.9\%. \tag{114}$$

For EE,

$$A_{EE} = 1.0233, \tag{115}$$

with residual shape RMS

$$4.4\%. \tag{116}$$

For TE,

$$A_{TE} = 0.9802, \tag{117}$$

with residual shape RMS

$$4.1\%. \tag{118}$$

The spectral differences therefore cannot be described by a single amplitude shift.

They contain genuine scale-dependent structure.

8.16 CMB lensing-potential spectrum

The direct CAMB lensing-potential spectrum is mildly enhanced relative to the matched control.

The approximate locked/control ratios are

$$1.0098 \quad \text{for} \quad 40 \le L \le 100, \tag{119}$$

$$1.0049 \quad \text{for} \quad 100 \le L \le 400, \tag{120}$$

$$1.0182 \quad \text{for} \quad 400 \le L \le 1000, \tag{121}$$

$$1.0223 \quad \text{for} \quad 1000 \le L \le 2000. \tag{122}$$

Weighted over

$$40 \le L \le 2000, \tag{123}$$

the ratio is approximately

$$1.0209. \tag{124}$$

The average enhancement is therefore approximately

$$2.09\%. \tag{125}$$

The shape scatter around one amplitude is approximately

$$0.65\%. \tag{126}$$

The lensing difference is therefore smooth and only mildly scale dependent.

8.17 Why lower S_8 does not imply lower CMB lensing

A simple amplitude estimate would suggest

$$C_L^{\phi\phi} \propto \sigma_8^2. \tag{127}$$

Using the locked and control values,

$$\left(\frac{\sigma_{8,\mathrm{locked}}}{\sigma_{8,\mathrm{control}}}\right)^2 \simeq 0.963. \tag{128}$$

This would predict a reduction of approximately

$$3.7\%. \tag{129}$$

The full CAMB calculation instead gives a mild enhancement.

The difference arises because CMB lensing depends upon

- growth,
- distance geometry,
- the Weyl kernel,
- and the complete redshift evolution of the potential.

The altered locked geometry outweighs the naive σ_8^2 scaling.

8.18 Planck 2018 lensing bandpowers

The official Planck 2018 conservative lensing dataset contains nine bins over

$$8 \leq L \leq 400. \tag{130}$$

The model spectrum is binned using the supplied window functions:

$$C_b^{\phi\phi} = \sum_L W_{bL}\, C_L^{\phi\phi}. \tag{131}$$

The covariance statistic is

$$\chi^2 = \Delta\mathbf{C}^{\mathrm{T}}\, \Sigma^{-1}\, \Delta\mathbf{C}. \tag{132}$$

The locked model gives

$$\chi^2_{\mathrm{locked}} = 10.7379 \tag{133}$$

for nine bins.

The corresponding p-value is

$$p_{\text{locked}} = 0.294. \tag{134}$$

The matched control gives

$$\chi^2_{\text{control}} = 12.0022, \tag{135}$$

with

$$p_{\text{control}} = 0.213. \tag{136}$$

The difference is

$$\Delta\chi^2 = -1.2643. \tag{137}$$

The locked model therefore gives a modestly lower lensing-bandpower statistic than the matched control.

8.19 Individual Planck lensing bins

The nine locked predictions remain within approximately

$$1.6\sigma \tag{138}$$

of the corresponding observed bandpowers when viewed through the diagonal uncertainty for illustration.

The largest negative pull occurs in the bin

$$310 \leq L \leq 354, \tag{139}$$

at approximately

$$-1.61\sigma. \tag{140}$$

The locked prediction is not dominated by one catastrophic outlier.

The total statistic is distributed across the nine bins.

8.20 Profiled lensing amplitude

A global amplitude

$$A_\phi \tag{141}$$

was applied to the model lensing spectrum:

$$C_L^{\phi\phi} \to A_\phi \, C_L^{\phi\phi}. \tag{142}$$

The best-fit amplitude for the locked model is approximately

$$A_{\phi,\text{locked}} = 1.0436. \tag{143}$$

For the matched control,

$$A_{\phi,\text{control}} = 1.0505. \tag{144}$$

The locked model therefore requires a slightly smaller upward amplitude adjustment.

8.21 PlanckLens reconstruction noise

The analytic TT quadratic estimator produces Gaussian reconstruction noise

$$N_L^{(0)}. \tag{145}$$

The locked/control ratios are approximately

$$1.223 \quad \text{for} \quad 8 \le L \le 40, \tag{146}$$

$$1.236 \quad \text{for} \quad 41 \le L \le 100, \tag{147}$$

$$1.280 \quad \text{for} \quad 101 \le L \le 200, \tag{148}$$

$$1.171 \quad \text{for} \quad 201 \le L \le 400. \tag{149}$$

The average over the tested range is approximately

$$1.213. \tag{150}$$

Thus the locked TT spectrum gives about

$$21.3\% \tag{151}$$

higher reconstruction noise under the common filtering assumptions.

This does not imply a weaker physical lensing signal.

It reflects the altered estimator response to the locked primary TT spectrum.

8.22 Robustness tests

The locked parameter point was tested under one-at-a-time perturbations of

$$\Omega_{m0}, \quad V_\infty, \quad \rho_s, \quad m_C^2, \quad \lambda, \quad C_{\text{initial}}, \quad \Pi_{C,\text{initial}}. \tag{152}$$

The tested changes were approximately

$$\pm 0.5\% \quad \text{and} \quad \pm 1\%. \tag{153}$$

Out of 28 perturbed points,

$$25 \tag{154}$$

retained the required background and BAO behavior.

The local robustness classification is therefore

$$\boxed{\text{ROBUST}}. \tag{155}$$

The most sensitive direction is

$$V_\infty. \tag{156}$$

8.23 Transition hardening tests

The splice width was varied.

Increasing the width by

$$50\% \tag{157}$$

gave

$$100\theta_* = 1.042359. \tag{158}$$

Reducing the width to

$$0.67W_0 \tag{159}$$

gave

$$100\theta_* = 1.040917 \tag{160}$$

before the final micro-adjustment of a_{lock}.

Residual-density variations of approximately

$$\pm 5\% \tag{161}$$

also shifted the acoustic angle by roughly

$$0.06\%. \tag{162}$$

These tests showed that the precision acoustic scale is mildly sensitive to the transition implementation while the broader BAO, growth, and perturbation behavior remains stable.

8.24 Final micro-adjustment

With the narrowed transition fixed, the final test changed only

$$a_{\text{lock}}. \tag{163}$$

The retained result is

$$a_{\text{lock}} = 2.05 \times 10^{-4}. \tag{164}$$

It gives

$$100\theta_* = 1.041033, \tag{165}$$
$$S_8 = 0.79781, \tag{166}$$
$$\chi^2_{\mathrm{BAO}} \simeq 13.4. \tag{167}$$

The fields remain

$$\mathcal{O}(0.05) \tag{168}$$

or smaller.

No other model parameter was changed in this final step.

8.25 Matched-control comparison

The matched control uses the same:

$$H_0, \quad \Omega_{m0}, \quad \Omega_{r0}, \quad \omega_b, \quad \omega_c, \quad A_s, \quad n_s, \quad \tau_{\mathrm{reio}}. \tag{169}$$

Its principal derived values are approximately

$$100\theta_{*,\mathrm{control}} = 1.042322, \tag{170}$$
$$r_{\mathrm{drag,control}} = 147.503 \text{ Mpc}, \tag{171}$$
$$\sigma_{8,\mathrm{control}} = 0.794561, \tag{172}$$
$$S_{8,\mathrm{control}} = 0.813037. \tag{173}$$

The locked model therefore has:

- a slightly smaller acoustic angle,
- a larger drag horizon,
- a lower growth amplitude,
- and a mildly enhanced CMB lensing potential.

8.26 Consolidated final results

The final numerical results are

$$\boxed{a_{\rm min} \simeq 2.068 \times 10^{-5}} \tag{174}$$

$$\boxed{a_{\rm lock} = 2.05 \times 10^{-4}} \tag{175}$$

$$\boxed{100\theta_* = 1.041033} \tag{176}$$

$$\boxed{r_{\rm drag} = 149.960 \text{ Mpc}} \tag{177}$$

$$\boxed{z_* = 1089.691} \tag{178}$$

$$\boxed{z_{\rm drag} = 1058.511} \tag{179}$$

$$\boxed{\sigma_8 = 0.779680} \tag{180}$$

$$\boxed{S_8 = 0.79781} \tag{181}$$

$$\boxed{w_C(0) = -0.99988} \tag{182}$$

$$\boxed{\max|\delta C| = 0.03514} \tag{183}$$

$$\boxed{\max|\delta\phi| = 0.04860} \tag{184}$$

$$\boxed{\chi^2_{\rm BAO} \simeq 13.4} \tag{185}$$

$$\boxed{\chi^2_{\rm Planck\ lensing} = 10.7379 \quad \text{for nine bins}} \tag{186}$$

$$\boxed{\Delta\chi^2_{\rm lensing} = -1.2643} \tag{187}$$

relative to the matched control.

8.27 Interpretation of the numerical campaign

The numerical results establish a consistent pattern.

The high-density saturation law produces a nonsingular bounce.

The same canonical fields settle into a stable late residual.

The resulting expansion history places the acoustic scale extremely close to the reference value.

The internally consistent drag horizon produces a coherent BAO ladder.

The locked expansion lowers the growth amplitude.

The direct canonical field perturbations remain small.

The Planck lensing spectrum remains compatible with the official nine-bin data.

The remaining differences from the matched control are concentrated primarily in the scale-dependent structure of the TT, TE, and EE spectra.

Thus the dominant observational signature of the theory is not a strong fifth force or large scalar instability.

It is the modified expansion history carried by the locked gravitational-saturation background.

End of Chapter 8

Chapter 9

Interpretation of the Locked Saturation Cosmology

9.1 Purpose of the interpretation

The numerical results of the previous chapter establish that the locked saturation cosmology produces a coherent and internally consistent pattern across background evolution, acoustic geometry, growth, and lensing.

The purpose of the present chapter is to interpret that pattern physically.

The central result is not one isolated number.

It is the combination

$$\begin{aligned}&\text{nonsingular bounce,}\\&\text{stable late residual,}\\&\text{near-reference acoustic scale,}\\&\text{coherent BAO ladder,}\\&\text{reduced } S_8,\\&\text{bounded canonical perturbations,}\\&\text{Planck-compatible lensing.}\end{aligned} \tag{1}$$

These properties arise from one retained expansion history and one canonical saturation sector.

The interpretation must therefore account for the fact that the background carries nearly the entire observable signal while the linear C and ϕ perturbations remain almost spectators.

9.2 The dominant role of the background

The direct saturation-field perturbations alter the final observables only at approximately the level

$$10^{-6}. \tag{2}$$

The dominant changes arise instead through

$$H(a). \tag{3}$$

This means that the theory operates primarily by modifying the integrated gravitational history of the Universe.

Distances depend on

$$D_M(z) = \int_0^z \frac{c\,dz'}{H(z')}. \tag{4}$$

The sound horizon depends on

$$r_s(z) = \int_z^\infty \frac{c_s(z')}{H(z')}\,dz'. \tag{5}$$

Growth depends on

$$\frac{d\ln H}{d\ln a} \tag{6}$$

and

$$\Omega_m(a) = \frac{\Omega_{m0} a^{-3}}{E^2(a)}. \tag{7}$$

Lensing depends on the complete growth and distance kernel.

Therefore a small, coherent change in

$$H(a) \tag{8}$$

can alter several observables simultaneously even when the late scalar perturbations are small.

The model's primary physical prediction is consequently a modified cosmic history rather than a large new local force.

9.3 Saturation as a history-dependent effect

An algebraic modified-gravity model often assumes that the gravitational response at a given time is determined by a function such as

$$\mu(a, k). \tag{9}$$

The saturation model is different.

Its present state is the result of the full earlier trajectory:

$$\text{contraction} \to \text{transfer} \to \text{bounce} \to \text{radiation era} \to \text{matter era} \to \text{locking}. \tag{10}$$

The late residual cannot be separated from the path by which the fields approached their minimum.

The background is therefore history dependent.

The same present value

$$w_C \simeq -1 \tag{11}$$

could be produced by many unrelated fluid models, but the saturation theory reaches that state only after the complete nonsingular evolution.

The observable consequences retain information about that earlier history through

$$r_s, \quad r_{\text{drag}}, \quad D_M, \quad D_H, \tag{12}$$

and the transfer functions.

Thus the theory is not defined only by its present equation of state.

It is defined by the entire locked trajectory.

9.4 The meaning of the late residual

At late time,

$$C \to C_s, \qquad \phi \to 0, \qquad \Pi_C \to 0, \qquad \Pi_\phi \to 0. \tag{13}$$

The residual density becomes

$$\rho_C \to V_\infty. \tag{14}$$

The pressure becomes

$$p_C \to -V_\infty. \tag{15}$$

Therefore

$$w_C \to -1. \tag{16}$$

Mathematically, this resembles a cosmological constant.

Physically, the interpretation is different.

The residual is not inserted as an independent constant term.

It is the minimum energy of the organizational and phase fields after the gravitational system has saturated.

The relation is

$$V(C_s, 0) = V_\infty. \tag{17}$$

The late accelerated era is therefore interpreted as the residual state of completed gravitational organization.

9.5 Why the residual remains smooth

Near the minimum,

$$V_{,C} \simeq 0, \qquad V_{,\phi} \simeq 0, \qquad C'_{\rm bg} \simeq 0, \qquad \phi'_{\rm bg} \simeq 0. \tag{18}$$

The density perturbation is

$$\delta\rho_C = \frac{1}{a^2}\left[C'_{\rm bg} v_C + \phi'_{\rm bg} v_\phi\right] + V_{,C}\delta C + V_{,\phi}\delta\phi. \tag{19}$$

Every coefficient multiplying the field perturbations becomes small near saturation.

Consequently,

$$\delta\rho_C \tag{20}$$

is strongly suppressed even though

$$\delta C \qquad \text{and} \qquad \delta\phi \tag{21}$$

remain finite.

The late residual therefore behaves as a smooth background component.

This explains why the model can alter the global expansion without producing a large clustering dark-energy sector.

9.6 Why the canonical fields do not mimic weakened gravity

The phenomenological models attempted to impose

$$k^2\Psi = -4\pi G a^2 \mu \rho \Delta. \tag{22}$$

Such a relation changes the strength of the metric response directly.

The canonical fields instead enter through

$$\delta T^{C\phi}_{\mu\nu}. \tag{23}$$

The total perturbed Einstein equation is

$$\delta G_{\mu\nu} = 8\pi G\big[\delta T^{\text{standard}}_{\mu\nu} + \delta T^{C\phi}_{\mu\nu}\big]. \tag{24}$$

Near saturation,

$$\delta T^{C\phi}_{\mu\nu} \tag{25}$$

is small.

Therefore the matter sector continues to experience nearly the standard gravitational coupling.

The lower value of

$$S_8 \tag{26}$$

does not arise because

$$G_{\text{eff}} < G. \tag{27}$$

It arises because the modified expansion history changes the interval and efficiency of structure growth.

This distinction is central to the interpretation of the numerical results.

9.7 Growth suppression through expansion history

The linear growth equation is

$$\frac{d^2\delta_m}{dN^2} + \left(2 + \frac{d\ln H}{dN}\right)\frac{d\delta_m}{dN} = \frac{3}{2}\Omega_m(a)\delta_m. \tag{28}$$

The locked model changes both

$$\frac{d\ln H}{dN} \tag{29}$$

and

$$\Omega_m(a). \tag{30}$$

The growth rate is therefore altered even with the ordinary coefficient

$$\frac{3}{2} \tag{31}$$

in the source term.

The resulting present amplitudes are

$$\sigma_8 = 0.779680, \qquad S_8 = 0.79781. \tag{32}$$

The matched control gives

$$S_8 \simeq 0.81304. \tag{33}$$

The difference is a background-growth effect.

This interpretation is supported by the negligible difference between the full canonical-field run and the locked-background-only run.

9.8 Acoustic geometry as a balance of two histories

The acoustic scale is

$$\theta_* = \frac{r_s(z_*)}{D_M(z_*)}. \tag{34}$$

The numerator depends on the pre-recombination expansion history:

$$r_s(z_*) = \int_{z_*}^{\infty} \frac{c_s(z)}{H(z)}\, dz. \tag{35}$$

The denominator depends on the complete later distance history:

$$D_M(z_*) = \int_0^{z_*} \frac{c\, dz}{H(z)}. \tag{36}$$

The locked model modifies both.

The final result is

$$100\theta_* = 1.041033. \tag{37}$$

The fractional difference from

$$1.0411 \tag{38}$$

is only

$$-0.0064\%. \tag{39}$$

This agreement is not obtained because the locked expansion is identical to the matched control.

It is obtained because the changes in the sound horizon and comoving distance balance in the final ratio.

The acoustic scale therefore provides a stringent integrated test of the complete locked history.

9.9 Meaning of the enlarged drag horizon

The final model gives

$$r_{\mathrm{drag}} = 149.960 \text{ Mpc}, \tag{40}$$

while the matched control gives approximately

$$147.503 \text{ Mpc.} \tag{41}$$

The ratio is

$$\frac{r_{\text{drag,locked}}}{r_{\text{drag,control}}} \simeq 1.0167. \tag{42}$$

Thus the locked drag ruler is approximately

$$1.67\% \tag{43}$$

larger.

This follows from the slightly slower early and intermediate expansion over the relevant integral.

Since

$$r_{\text{drag}} = \int_{z_{\text{drag}}}^{\infty} \frac{c_s(z)}{H(z)}\, dz, \tag{44}$$

a smaller $H(z)$ produces a larger sound horizon.

The same larger ruler lowers the dimensionless BAO ratios relative to what would be inferred using the earlier approximate ruler.

This explains the improvement in the final BAO statistic.

9.10 Why the BAO result improved

The BAO observables are not absolute distances.

They are ratios:

$$\frac{D_M}{r_{\text{drag}}}, \qquad \frac{D_H}{r_{\text{drag}}}, \qquad \frac{D_V}{r_{\text{drag}}}. \tag{45}$$

Earlier calculations used a ruler near

$$147.8 \text{ Mpc.} \tag{46}$$

The final CAMB-consistent ruler is

$$149.960 \text{ Mpc.} \tag{47}$$

Using the smaller earlier ruler made every predicted ratio too large.

When the model's own ruler was used, the diagnostic statistic improved to

$$\chi^2_{\text{BAO}} \simeq 13.4. \tag{48}$$

The improvement therefore reflects restored internal consistency.

The theory's distances and acoustic ruler must be computed from the same expansion history.

9.11 Interpretation of the primary CMB residuals

The TT, TE, and EE spectra differ from the matched control most strongly at low multipoles and through the first acoustic region.

The differences decrease at high multipoles.

This pattern indicates that the locked background changes more than one overall amplitude.

It alters:

$$\begin{aligned}&\text{early potential evolution,}\\&\text{radiation driving,}\\&\text{projection geometry,}\\&\text{recombination-to-observer distance,}\\&\text{lensing smoothing.}\end{aligned} \tag{49}$$

A single amplitude rescaling cannot remove these effects.

The TT best-fit amplitude is approximately

$$0.9459, \tag{50}$$

but the residual shape RMS remains approximately

$$7.9\%. \tag{51}$$

The residuals therefore encode genuine scale-dependent information about the locked expansion history.

9.12 Low-multipole sensitivity

At low multipoles,

$$2 \leq \ell \leq 29, \tag{52}$$

the TT residual is approximately

$$28.2\% \tag{53}$$

in RMS fractional terms.

Low multipoles are sensitive to:

$$\text{large-scale primordial modes,} \quad \text{late integrated Sachs-Wolfe evolution,} \quad \text{reionization,} \quad \text{cosmic variance.} \tag{54}$$

The large fractional difference does not mean that every low-ℓ mode is strongly excluded.

The statistical weight of each multipole is limited by

$$2\ell + 1 \tag{55}$$

available modes and by cosmic variance.

Nevertheless, the low-ℓ region is an important signature of the altered late-time potential evolution.

9.13 Acoustic-region residuals

For

$$30 \leq \ell \leq 600, \tag{56}$$

the TT residual is approximately

$$10.4\% \tag{57}$$

in the control comparison.

This region contains the first several acoustic peaks.

The peak structure depends on:

$$\text{baryon loading}, \quad \text{matter-radiation transition}, \quad \text{potential decay}, \quad \text{sound speed}, \quad \text{diffusion}, \quad \text{projection}. \tag{58}$$

Because the primordial parameters are held fixed, the differences arise from the modified transfer and geometry.

The failure of the phenomenological μ and η models showed that arbitrary changes to the scalar constraints distort this region more severely.

The canonical implementation preserves recognizable and coherent acoustic structure.

9.14 High-ℓ convergence

For

$$1501 \leq \ell \leq 2500, \tag{59}$$

the residuals fall to approximately

$$1.7\% \text{ for TT}, \qquad 1.1\% \text{ for normalized TE}, \qquad 2.3\% \text{ for EE}. \tag{60}$$

The convergence at high multipoles indicates that the small-scale photon-baryon physics remains close to the matched reference once the acoustic scale has been locked.

The remaining differences are produced by the modified damping, lensing, and projection histories.

The high-ℓ behavior is therefore not pathological.

It is a smooth deformation of the standard spectrum.

9.15 Interpretation of the lensing enhancement

The locked lensing-potential spectrum is mildly enhanced:

$$\frac{C_L^{\varphi\varphi,\text{locked}}}{C_L^{\varphi\varphi,\text{control}}} \simeq 1.0209. \tag{61}$$

This occurs even though

$$S_{8,\text{locked}} < S_{8,\text{control}}. \tag{62}$$

CMB lensing is an integrated quantity:

$$\varphi(\hat{\mathbf{n}}) = -2\int_0^{\chi_*} d\chi\,\frac{\chi_* - \chi}{\chi_*\chi}\Phi_W. \tag{63}$$

The signal depends on:

$$\text{the amplitude of the Weyl potential,}\quad \text{its redshift evolution,}\quad \text{the source distance,}\quad \text{the geometric lensing ke} \tag{64}$$

A lower present-day σ_8 therefore does not uniquely determine the CMB lensing amplitude.

The locked model's enlarged distances and altered potential history produce a modest net enhancement.

9.16 Planck lensing compatibility

The official nine-bin statistic is

$$\chi^2_{\text{locked}} = 10.7379 \tag{65}$$

for nine bins.

The corresponding probability is

$$p = 0.294. \tag{66}$$

The matched control gives

$$\chi^2_{\text{control}} = 12.0022. \tag{67}$$

The difference is

$$\Delta\chi^2 = -1.2643. \tag{68}$$

The locked model is therefore compatible with the Planck 2018 lensing bandpowers and gives a modestly lower statistic than the matched control.

The physical interpretation is not that Planck lensing independently proves saturation.

The result shows that the altered background and resulting lensing spectrum do not conflict with the measured bandpowers.

9.17 Reconstruction signal versus reconstruction noise

The true lensing spectrum and the reconstruction noise describe different quantities.

The true signal is

$$C_L^{\varphi\varphi}. \tag{69}$$

The Gaussian estimator noise is

$$N_L^{(0)}. \tag{70}$$

The locked model gives a mildly larger

$$C_L^{\varphi\varphi}, \tag{71}$$

but also approximately

$$21.3\% \tag{72}$$

higher TT reconstruction noise in the PlanckLens forecast.

This occurs because the quadratic estimator depends on the shape and gradients of the primary CMB spectrum.

The altered TT peaks change the estimator response.

Thus the model predicts:

$$\text{slightly more physical lensing power}, \qquad \text{less efficient TT-only reconstruction}. \tag{73}$$

These effects are mathematically compatible.

9.18 The significance of the failed phenomenological models

The failure of the effective closures provides information about the nature of the theory.

The tested relations were

$$\mu(a,k), \qquad \eta = 1, \qquad \eta = \frac{1}{\mu}, \qquad \eta = 2 - \mu. \tag{74}$$

All failed to reproduce the canonical model.

The failures were not random.

They produced:

$$\text{larger } \sigma_8, \qquad \text{odd-even peak distortion}. \tag{75}$$

This demonstrates that the successful behavior of the locked model is not merely the result of "weaker gravity."

It depends upon preserving the Einstein-equation structure while changing the background history and including the actual canonical field stress-energy.

The correct closure is dynamical, not algebraic.

9.19 The model as a constrained deformation of standard cosmology

The final locked model is not wildly different from the matched reference.

At late times,

$$w_C \simeq -1. \tag{76}$$

At low density,

$$H^2 \simeq \frac{\rho}{3}. \tag{77}$$

The matter and radiation continuity laws retain their standard forms after the transfer shuts off.

The photon and neutrino Boltzmann hierarchies remain standard.

The primordial spectrum is unchanged.

The deviations arise in a limited set of places:

$$\begin{aligned}&\text{the high-density bounce law,}\\&\text{the evolved locked } H(a),\\&\text{the smooth early splice,}\\&\text{the canonical saturation stress-energy.}\end{aligned} \tag{78}$$

The model is therefore a constrained deformation of the standard cosmological calculation rather than a complete replacement of every component.

9.20 The role of the transition endpoint

The final acoustic adjustment uses

$$a_{\text{lock}} = 2.05 \times 10^{-4}. \tag{79}$$

The transition endpoint changes the amount of the early locked history sampled by the acoustic integrals.

Changing it modifies

$$r_s, \quad r_{\text{drag}}, \quad D_M(z_*). \tag{80}$$

The final sequence brackets the acoustic target:

$$a_{\text{lock}} = 2.10 \times 10^{-4} \Rightarrow 100\theta_* = 1.040917, \tag{81}$$

$$a_{\text{lock}} = 2.05 \times 10^{-4} \Rightarrow 100\theta_* = 1.041033, \tag{82}$$

$$a_{\text{lock}} = 2.00 \times 10^{-4} \Rightarrow 100\theta_* = 1.041167. \tag{83}$$

The smooth monotonic response shows that the final result is not an isolated numerical discontinuity.

It lies within a controlled local family of backgrounds.

9.21 Interpretation of robustness

The one-at-a-time parameter test found that

$$25 \tag{84}$$

of

$$28 \tag{85}$$

nearby points retained the required behavior.

This indicates that the model occupies an open neighborhood in parameter space.

The central solution is not a single isolated point whose behavior disappears under every small perturbation.

However, the precision acoustic angle remains more sensitive than broad quantities such as:

$$\text{bounce existence}, \quad \text{late residual}, \quad S_8, \quad \text{BAO compatibility}. \tag{86}$$

The model is therefore robust in its qualitative and multi-probe structure while retaining expected sensitivity in precision CMB geometry.

9.22 Cosmic organization and saturation

The organizational interpretation assigns physical meaning to the field C.

When

$$C < C_s, \tag{87}$$

the restoring force is directed toward increasing C.

When

$$C > C_s, \tag{88}$$

the restoring force is directed toward decreasing C.

The unique minimum is

$$C = C_s. \tag{89}$$

The phase field carries periodic restoration.

Together, they describe a system whose organizational state cannot grow without limit.

At high density, the background response also reaches a limit:

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{90}$$

Thus saturation exists in both:

$$\text{field space} \qquad \text{and} \qquad \text{background gravitational response}. \tag{91}$$

The first produces a stable residual.

The second produces a nonsingular bounce.

9.23 The wave interpretation

The gravitational-wave equation demonstrates that geometry supports propagating modes.

The saturation theory extends that dynamical interpretation by introducing:

$$\text{phase}, \quad \text{restoration}, \quad \text{damping}, \quad \text{limiting response}. \tag{92}$$

The phase potential

$$A_0(1 - \cos\phi) \tag{93}$$

provides periodic structure.

The organizational potential provides a stable minimum.

The transfer term regulates contraction-driven anti-friction.

The high-density constraint caps the gravitational response.

The wave interpretation is therefore embodied mathematically in a system containing the same classes of behavior found in organized wave dynamics.

9.24 Saturation and dark energy

In the standard description, late acceleration is represented through a component satisfying approximately

$$p \simeq -\rho. \tag{94}$$

The saturation fields produce the same background relation in the locked limit:

$$p_C \simeq -\rho_C. \tag{95}$$

The difference is in origin.

The equation of state arises from

$$K^2 \ll V, \tag{96}$$

after the fields settle at their minimum.

The residual is therefore a dynamical endpoint:

$$\rho_C \to V_\infty. \tag{97}$$

Within the theory, the observed late acceleration is interpreted as the gravitational field's residual saturation state rather than as an unrelated dark-energy substance.

9.25 Saturation and dark matter

The completed CAMB background calculation retains the ordinary matter sector required by the Boltzmann code.

The gravitational-saturation interpretation nevertheless suggests that some phenomena attributed to additional gravitating mass may arise from organizational response in weak-acceleration systems.

The galactic screened relation is

$$g\,\mu\left(\frac{g}{g_*}\right) = g_N, \tag{98}$$

with

$$\mu(x) = \frac{x}{\sqrt{1+x^2}}. \tag{99}$$

In the low-acceleration limit,

$$g \simeq \sqrt{g_N g_*}, \tag{100}$$

which produces

$$v^4 \simeq G M_b g_*. \tag{101}$$

The cosmological and galactic sectors therefore share an organizational interpretation.

The cosmological calculation establishes the locked saturation background.

The galactic calculation establishes a weak-acceleration interpolating behavior.

Their full covariant unification is carried by the theory's field structure and scale-dependent response.

9.26 Hierarchical organization

Gravitational systems exist across a hierarchy:

$$\text{stars}, \quad \text{galaxies}, \quad \text{groups}, \quad \text{clusters}, \quad \text{superclusters}, \quad \text{the cosmological background.} \tag{102}$$

At each scale, gravity organizes matter into larger structures.

The saturation hypothesis interprets this hierarchy as evidence that gravitational influence is not merely local attraction.

It is cumulative organization across nested scales.

The cosmic saturation field represents the largest-scale limit of that process.

The late residual emerges when the background approaches its maximum organizational state.

9.27 Why the model remains close to standard gravity locally

The field C is near a positive-curvature minimum:

$$V_{,CC} > 0. \tag{103}$$

The phase field is near

$$\phi = 0, \tag{104}$$

with

$$V_{,\phi\phi} > 0. \tag{105}$$

The background velocities are small.

The perturbative stress-energy is suppressed.

Therefore the model does not require a large long-range scalar force at late times.

The local weak-field limit can remain close to ordinary gravity while the global background retains the cumulative effect of the earlier saturation history.

This separation between local perturbative response and global integrated history is one of the defining features of the model.

9.28 What the locked state means physically

The word locked does not mean motionless.

The Universe continues expanding.

Matter continues diluting.

Radiation continues redshifting.

Structure continues evolving.

The locked state refers to the organizational sector:

$$C \simeq C_s, \qquad \phi \simeq 0, \qquad K^2 \ll V. \tag{106}$$

The fields have reached the region in which further large organizational evolution is dynamically suppressed.

The background then behaves as a stable residual cosmology.

The late expansion is therefore the motion of spacetime after the organizational sector has locked, not the absence of cosmic evolution.

9.29 A unified account of bounce and acceleration

Many cosmological models treat the early bounce and late acceleration as unrelated phenomena.

In the saturation theory, both arise from the same sector.

The bounce follows from

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{107}$$

The late acceleration follows from

$$C \to C_s, \qquad K^2 \to 0, \qquad \rho_C \to V_\infty. \tag{108}$$

Thus:

$$\text{high-density saturation} \to \text{bounce}, \tag{109}$$

while

$$\text{field-space saturation} \to \text{late residual}. \tag{110}$$

The two limits are different manifestations of one organizational principle.

9.30 Central interpretation of the completed calculation

The completed calculation supports the following interpretation.

Gravity is a dynamical wave-bearing geometry.

Its cumulative organization is represented by the fields C and ϕ.

At high density, the gravitational response reaches a finite limit.

Contraction reverses without a singular scale factor.

The post-bounce universe enters radiation and matter eras.

The organizational fields damp toward a stable minimum.

Their residual energy produces a late state with

$$w_C \simeq -1. \tag{111}$$

The direct linear field perturbations remain small.

The altered background modifies distances, growth, acoustic geometry, and lensing.

The final model therefore behaves as

$$\boxed{\text{standard linear Einstein-Boltzmann physics} + \text{a nonsingular locked expansion history}} \quad (112)$$

rather than as a large phenomenological modification of the local gravitational coupling.

9.31 Final interpretation

The numerical evidence points toward one coherent conclusion.

The successful part of the theory is not a new particle-like perturbation dominating the CMB.

It is not a simple weakening of gravity.

It is not an imposed gravitational slip.

It is the existence of a stable, nonsingular, history-dependent expansion trajectory whose late state resembles a smooth cosmological residual.

The fields C and ϕ provide the canonical dynamical language for that trajectory.

The wave nature of gravity supplies the physical basis for phase, interference, damping, and limiting response.

The high-density constraint supplies the bounce.

The potential minimum supplies the lock.

The resulting cosmology carries the observable memory of its saturation history primarily through

$$H(a). \quad (113)$$

That locked expansion history is the principal physical content of the completed model.

End of Chapter 9

Chapter 10

Predictions and Distinguishing Tests

10.1 Purpose of the prediction chapter

A cosmological theory becomes physically distinguishable only when it produces consequences that can be compared with observation.

The locked gravitational-saturation cosmology is constructed to reproduce the principal background and linear observables already examined:

$$100\theta_*, \quad r_{\rm drag}, \quad \text{BAO distances}, \quad S_8, \quad TT, \quad TE, \quad EE, \quad C_L^{\varphi\varphi}. \tag{1}$$

Its strongest observable distinction from the matched reference cosmology does not arise from a large new scalar force.

It arises from the shape of the locked expansion history,

$$E_{\rm locked}(z), \tag{2}$$

and from the way that history modifies several integrated observables at once.

The principal predictions therefore concern correlated changes in:

$$\text{distance}, \quad \text{growth}, \quad \text{primary CMB shape}, \quad \text{lensing geometry}, \quad \text{reconstruction efficiency}. \tag{3}$$

A valid test of the model must examine these quantities together.

10.2 Prediction 1: a slightly enlarged acoustic ruler

The final drag horizon is

$$r_{\rm drag} = 149.960 \text{ Mpc}. \tag{4}$$

The matched control value is approximately

$$r_{\rm drag,control} = 147.503 \text{ Mpc}. \tag{5}$$

Therefore the predicted fractional difference is

$$\frac{r_{\rm drag} - r_{\rm drag,control}}{r_{\rm drag,control}} \simeq 0.0167. \tag{6}$$

Thus

$$\boxed{r_{\rm drag} \text{ is approximately } 1.67\% \text{ larger}} \tag{7}$$

than in the matched control.

This larger ruler is not inserted independently.

It follows from the locked early expansion:

$$r_{\mathrm{drag}} = \int_{z_{\mathrm{drag}}}^{\infty} \frac{c_s(z)}{H_{\mathrm{locked}}(z)}\, dz. \tag{8}$$

A slightly smaller $H(z)$ over the relevant interval increases the sound horizon.

The prediction is therefore not simply a different numerical ruler.

It is a correlated statement about the early expansion history.

10.3 Prediction 2: nearly unchanged acoustic angle

Despite the larger sound horizon, the acoustic angle remains

$$100\theta_* = 1.041033. \tag{9}$$

The corresponding reference value is

$$1.0411. \tag{10}$$

The difference is

$$-0.0064\%. \tag{11}$$

This means that the increase in

$$r_s(z_*) \tag{12}$$

is compensated by the increase in

$$D_M(z_*). \tag{13}$$

The theory therefore predicts:

$$\boxed{\text{a larger physical ruler with an almost unchanged observed angular scale}} \tag{14}$$

because

$$\theta_* = \frac{r_s(z_*)}{D_M(z_*)}. \tag{15}$$

A future determination of the absolute sound horizon and comoving distance that breaks this ratio degeneracy would provide a direct test of the locked history.

10.4 Prediction 3: coherent BAO ratios

The BAO observables are

$$\frac{D_M(z)}{r_{\mathrm{drag}}}, \qquad \frac{D_H(z)}{r_{\mathrm{drag}}}, \qquad \frac{D_V(z)}{r_{\mathrm{drag}}}. \tag{16}$$

The locked background predicts slightly enlarged distances,

$$D_M \qquad \text{and} \qquad D_H, \tag{17}$$

but also a larger ruler,

$$r_{\mathrm{drag}}. \tag{18}$$

These effects partially cancel.

The resulting diagnostic statistic is approximately

$$\chi^2_{\mathrm{BAO}} \simeq 13.4. \tag{19}$$

The prediction is therefore not that all BAO distances should be systematically larger.

It is that the locked model should produce a specific redshift-dependent residual pattern after division by its own enlarged ruler.

The residuals should remain correlated through one function:

$$E_{\mathrm{locked}}(z). \tag{20}$$

10.5 Prediction 4: slower intermediate expansion

The locked background remains close to the matched reference at late times because

$$w_C \simeq -1. \tag{21}$$

At intermediate and higher redshift, it is approximately

$$1\%\text{–}2\% \tag{22}$$

slower over the previously examined range.

Define

$$\Delta_E(z) = \frac{E_{\mathrm{locked}}(z) - E_{\mathrm{control}}(z)}{E_{\mathrm{control}}(z)}. \tag{23}$$

The model predicts

$$\Delta_E(z) < 0 \tag{24}$$

over much of the intermediate-redshift interval.

This produces:

$$D_H(z) > D_{H,\mathrm{control}}(z), \qquad D_M(z) > D_{M,\mathrm{control}}(z). \tag{25}$$

The effect should be detectable through sufficiently precise combinations of:

$$\text{BAO}, \quad \text{standard candles}, \quad \text{cosmic chronometers}, \quad \text{lensing distances}. \tag{26}$$

10.6 Prediction 5: reduced present growth amplitude

The final values are

$$\sigma_8 = 0.779680, \qquad S_8 = 0.79781. \tag{27}$$

The matched control gives

$$\sigma_{8,\mathrm{control}} = 0.794561, \qquad S_{8,\mathrm{control}} = 0.813037. \tag{28}$$

Therefore

$$\frac{S_{8,\mathrm{locked}}}{S_{8,\mathrm{control}}} \simeq 0.9813. \tag{29}$$

The model predicts a reduction of approximately

$$1.87\% \tag{30}$$

in S_8 relative to the matched control.

The key prediction is that this reduction occurs without

$$G_{\mathrm{eff}} < G. \tag{31}$$

It arises from the expansion history.

Therefore the growth rate and lensing geometry must change together in the locked pattern.

A model that reproduces the same S_8 by simply weakening gravity would not generally reproduce the same CMB and lensing observables.

10.7 Prediction 6: nearly standard local gravitational coupling

The canonical fields remain near their minimum.

The field-space curvature is positive:

$$V_{,CC} > 0, \qquad V_{,\phi\phi} > 0. \tag{32}$$

The background velocities are small:

$$C'_{\mathrm{bg}} \simeq 0, \qquad \phi'_{\mathrm{bg}} \simeq 0. \tag{33}$$

The perturbative sources are suppressed.

Therefore the theory predicts that late-time linear matter dynamics should remain close to ordinary Einstein gravity at fixed background expansion.

The growth modification is not expected to resemble a large scale-independent change in Newton's constant.

This can be tested by comparing:

$$\text{redshift-space distortions}, \quad \text{peculiar velocities}, \quad \text{galaxy clustering}, \quad \text{weak lensing} \tag{34}$$

under one common expansion history.

10.8 Prediction 7: small canonical field perturbations

The final field amplitudes are

$$\max|\delta C| = 0.03514, \qquad \max|\delta\phi| = 0.04860. \tag{35}$$

The theory therefore predicts that the saturation fields remain linear throughout the tested cosmological evolution.

Their direct change to

$$\sigma_8 \tag{36}$$

and the acoustic spectra is at approximately

$$10^{-6} \tag{37}$$

or below.

A future independent implementation should recover the same hierarchy:

$$\boxed{\text{background effect} \gg \text{direct } C, \phi \text{ perturbation effect}} \tag{38}$$

for the locked parameter set.

A calculation that produces a large scalar force from the same locked background would not reproduce the completed model.

10.9 Prediction 8: scale-dependent primary CMB residuals

The locked primary spectra differ from the matched control in a characteristic pattern.

For TT, the RMS differences are approximately

$$28.2\% \quad \text{for} \quad 2 \le \ell \le 29, \tag{39}$$
$$10.4\% \quad \text{for} \quad 30 \le \ell \le 600, \tag{40}$$
$$3.0\% \quad \text{for} \quad 601 \le \ell \le 1500, \tag{41}$$
$$1.7\% \quad \text{for} \quad 1501 \le \ell \le 2500. \tag{42}$$

The prediction is therefore:

$$\boxed{\text{large-scale and first-acoustic differences that decay toward high } \ell} \tag{43}$$

rather than a uniform shift across the full spectrum.

The EE and TE spectra follow the same qualitative trend.

This pattern distinguishes the model from a simple amplitude change or calibration error.

10.10 Prediction 9: the primary residuals cannot be removed by one amplitude

The best amplitude factors are approximately

$$A_{TT} = 0.9459, \qquad A_{EE} = 1.0233, \qquad A_{TE} = 0.9802. \tag{44}$$

After these rescalings, residual shape differences remain.

The shape RMS values are approximately

$$7.9\% \text{ for TT}, \qquad 4.4\% \text{ for EE}, \qquad 4.1\% \text{ for TE}. \tag{45}$$

Therefore the theory predicts that any fit to the primary CMB must account for genuine transfer-function shape differences.

The locked spectra cannot be reduced to

$$C_\ell^{XY,\text{locked}} = A\, C_\ell^{XY,\text{control}} \tag{46}$$

with one constant A.

10.11 Prediction 10: mild enhancement of the true lensing potential

The direct CAMB prediction is

$$\frac{C_L^{\varphi\varphi,\text{locked}}}{C_L^{\varphi\varphi,\text{control}}} \simeq 1.01 \tag{47}$$

at lower lensing multipoles, rising toward approximately

$$1.02 \tag{48}$$

at higher multipoles.

The weighted mean over

$$40 \le L \le 2000 \tag{49}$$

is approximately

$$1.0209. \tag{50}$$

The locked model therefore predicts:

$$\boxed{\text{mildly enhanced CMB lensing power despite lower } S_8} \tag{51}$$

because CMB lensing is sensitive to the integrated geometry and potential history rather than only to present-day clustering.

This combination is a distinctive signature.

10.12 Prediction 11: good agreement with Planck lensing bandpowers

Using the official nine-bin bandpowers, windows, and covariance, the locked model gives

$$\chi^2_{\text{locked}} = 10.7379 \tag{52}$$

for nine bins.

The matched control gives

$$\chi^2_{\text{control}} = 12.0022. \tag{53}$$

Thus

$$\Delta\chi^2 = -1.2643. \tag{54}$$

The model predicts that its mild enhancement should remain compatible with the observed lensing spectrum.

The result is not driven by one bin.

No individual locked bandpower differs by more than approximately

$$1.6\sigma \tag{55}$$

under the diagonal-pull diagnostic.

10.13 Prediction 12: higher TT reconstruction noise

The PlanckLens TT quadratic-estimator calculation gives

$$\frac{N^{(0)}_{L,\mathrm{locked}}}{N^{(0)}_{L,\mathrm{control}}} \simeq 1.21 \tag{56}$$

over the tested range.

The model therefore predicts:

$$\boxed{\text{higher reconstruction noise while the physical lensing signal is mildly larger}} \tag{57}$$

The difference arises because the primary TT spectrum controls the estimator response.

A future map-level reconstruction using the locked transfer functions should recover this reduced reconstruction efficiency.

10.14 Prediction 13: no large intrinsic scalar anisotropic stress

The canonical fields satisfy

$$\pi_C = 0. \tag{58}$$

Therefore they do not generate a large intrinsic linear gravitational slip.

The metric slip remains controlled by the standard components and the total Einstein evolution.

This predicts that the locked model should not behave like a phenomenological theory with arbitrary

$$\eta(a,k) \neq 1. \tag{59}$$

Any observed slip must remain compatible with the canonical stress-energy structure rather than being freely adjusted.

10.15 Prediction 14: positive field sound speeds

The quadratic field action gives

$$c_{s,C}^2 = 1, \qquad c_{s,\phi}^2 = 1. \tag{60}$$

Therefore short-wavelength perturbations propagate with positive gradient coefficients.

The model predicts no rapid small-scale gradient instability.

In Fourier space, the high-k behavior is governed by

$$\delta C'' + 2\mathcal{H}\delta C' + k^2 \delta C \simeq 0, \tag{61}$$

and

$$\delta\phi'' + 2\mathcal{H}\delta\phi' + k^2 \delta\phi \simeq 0. \tag{62}$$

The solutions are oscillatory and damped during expansion.

10.16 Prediction 15: positive restoring mass near the lock

Near saturation,

$$V_{,CC} \simeq m_C^2 = 1.827221. \tag{63}$$

For the phase field,

$$V_{,\phi\phi} \simeq A_0. \tag{64}$$

With

$$A_0 > 0, \tag{65}$$

the mass eigenvalues are positive.

Therefore the theory predicts no tachyonic runaway around the final locked state.

Perturbations displaced from the minimum should oscillate or decay rather than grow exponentially.

10.17 Prediction 16: the bounce precedes equality

The bounce scale is

$$a_{\min} \simeq 2.068 \times 10^{-5}. \tag{66}$$

Equality occurs at

$$a_{\mathrm{eq}} \simeq 2.865 \times 10^{-4}. \tag{67}$$

Therefore

$$\frac{a_{\min}}{a_{\mathrm{eq}}} \simeq 0.0722. \tag{68}$$

The theory predicts a post-bounce radiation-dominated interval before equality.

The bounce does not occur during matter domination.

This timing is required for the subsequent acoustic evolution.

10.18 Prediction 17: no residual transfer during expansion

The transfer selector satisfies

$$W_-(H) = 0 \qquad \text{for} \qquad H \geq 0. \tag{69}$$

Therefore

$$\Gamma_{\rm tr} = 0 \tag{70}$$

throughout expansion.

The radiation equation becomes

$$\dot{\rho}_r = -4H\rho_r. \tag{71}$$

The theory predicts no continuing field-to-radiation energy injection after the bounce.

The post-bounce thermal history is therefore not driven by a persistent transfer tail.

10.19 Prediction 18: a finite maximum contraction rate

The saturated constraint is

$$H^2 = \frac{\rho_s}{3}x(1-x), \tag{72}$$

where

$$x = \frac{\rho}{\rho_s}. \tag{73}$$

The maximum occurs at

$$x = \frac{1}{2}. \tag{74}$$

Therefore

$$|H|_{\rm max} = \sqrt{\frac{\rho_s}{12}}. \tag{75}$$

The theory predicts that the magnitude of the contraction rate does not diverge.

It grows until

$$\rho = \frac{\rho_s}{2}, \tag{76}$$

then decreases toward zero as

$$\rho \to \rho_s. \tag{77}$$

This is a direct mathematical signature of high-density gravitational saturation.

10.20 Prediction 19: the bounce occurs with ordinary positive $\rho + p$

At the bounce,

$$\dot{H}_b = \frac{1}{2}(\rho_b + p_b). \tag{78}$$

A successful bounce requires

$$\rho_b + p_b > 0. \tag{79}$$

Therefore the rebound does not require the ordinary matter and radiation sectors to violate the null-energy combination.

The reversal is produced by the modified high-density gravitational response.

This distinguishes the model from bounce mechanisms that require a negative-kinetic-energy field.

10.21 Prediction 20: a smooth residual rather than clustered late energy

Since

$$\delta\rho_C \tag{80}$$

is suppressed near saturation, the late residual should remain smooth on linear cosmological scales.

The theory therefore predicts:

$$\text{background acceleration} \qquad \text{without} \qquad \text{large residual clustering.} \tag{81}$$

This affects:

$$\text{late integrated Sachs-Wolfe evolution}, \quad \text{large-scale lensing}, \quad \text{growth.} \tag{82}$$

A future measurement requiring strongly clustered dark energy would not match the locked saturation state.

10.22 Prediction 21: correlation between low-ℓ CMB and late geometry

The low-ℓ TT residual and the intermediate-redshift expansion difference arise from the same locked background.

The model therefore predicts that improved low-redshift geometry constraints should correlate with the sign and magnitude of large-scale CMB potential evolution.

The late integrated Sachs-Wolfe contribution is schematically

$$\left.\frac{\Delta T}{T}\right|_{\mathrm{ISW}} = \int (\Phi' + \Psi')\, d\tau. \tag{83}$$

The same potentials enter large-scale structure and lensing.

A joint test using:

$$\text{CMB temperature}, \quad \text{galaxy density}, \quad \text{lensing} \tag{84}$$

can therefore probe the locked potential history.

10.23 Prediction 22: distinct redshift dependence of $f\sigma_8$

The growth rate is

$$f(a) = \frac{d\ln D}{d\ln a}. \tag{85}$$

The observable is

$$f\sigma_8(z) = f(z)\sigma_8(z). \tag{86}$$

Because the locked model changes the background rather than G, its redshift dependence should differ from that of a simple reduced-gravity model.

The prediction is not only a lower present value.

It is a specific curve:

$$f\sigma_8(z) \tag{87}$$

derived from

$$E_{\mathrm{locked}}(z). \tag{88}$$

Future redshift-space distortion measurements can distinguish between:

$$\text{background-driven suppression} \qquad \text{and} \qquad \text{force-law suppression.} \tag{89}$$

10.24 Prediction 23: a characteristic weak-lensing combination

Weak lensing depends upon:

$$\text{matter growth}, \quad \text{source distances}, \quad \text{the lensing kernel.} \tag{90}$$

The locked model lowers

$$S_8, \tag{91}$$

but increases some cosmological distances.

Therefore its shear spectrum need not be identical to a reference model with the same S_8.

The theory predicts a characteristic scale and redshift dependence in cosmic shear tomography.

A test based only on a compressed S_8 value discards part of this information.

10.25 Prediction 24: no successful algebraic μ-η imitation

The numerical campaign showed that the tested algebraic closures failed.

The theory predicts that no simple relation of the forms

$$\eta = 1, \qquad \eta = \frac{1}{\mu}, \qquad \eta = 2 - \mu \tag{92}$$

will reproduce the full locked spectra and growth simultaneously.

The reason is that the true effect is distributed across:

$$H(a), \quad r_s, \quad D_M, \quad \delta T^{C\phi}_{\mu\nu}, \quad \text{and the complete transfer functions.} \tag{93}$$

A successful effective description would require a history-dependent, scale-dependent closure derived from the canonical system rather than imposed algebraically.

10.26 Prediction 25: galactic low-acceleration scaling

The galactic organizational response is represented by

$$g\,\mu\left(\frac{g}{g_*}\right) = g_N, \tag{94}$$

with

$$\mu(x) = \frac{x}{\sqrt{1+x^2}}. \tag{95}$$

In the low-acceleration limit,

$$g \simeq \sqrt{g_N g_*}. \tag{96}$$

For circular motion,

$$v^4 \simeq G M_b g_*. \tag{97}$$

The theory therefore predicts a baryonic scaling relation:

$$\boxed{v^4 \propto M_b} \tag{98}$$

with one characteristic organizational acceleration scale.

The cosmological background and the galactic interpolator represent different limits of gravitational organization.

A complete scale-dependent derivation should preserve this relation while recovering standard gravity at high acceleration.

10.27 Prediction 26: stronger fractional response in low-surface-density systems

In the galactic interpolation,

$$g_N \ll g_* \tag{99}$$

produces the largest departure from Newtonian gravity.

Low-surface-density galaxies therefore experience a larger fractional organizational response than high-surface-density systems.

The model predicts:

$$\text{small inner modification in high-acceleration galaxies,} \tag{100}$$

and

$$\text{larger outer modification in low-acceleration galaxies.} \tag{101}$$

This qualitative ordering matches the earlier prototype behavior across:

$$\text{high-surface-brightness spirals,}\quad \text{low-surface-brightness disks,}\quad \text{gas-rich dwarfs,}\quad \text{bulge-dominated systems.} \tag{102}$$

10.28 Prediction 27: standard high-acceleration limit

For

$$g \gg g_*, \tag{103}$$

the interpolating function satisfies

$$\mu \to 1. \tag{104}$$

Therefore

$$g \to g_N. \tag{105}$$

The theory predicts recovery of standard gravitational dynamics in sufficiently strong local fields.

The cosmological residual does not require a large correction in the Solar System or in high-acceleration stellar systems.

10.29 Prediction 28: one connected hierarchy of gravitational organization

The theory predicts that cosmic saturation and galactic weak-acceleration behavior are not unrelated anomalies.

They arise from one organizing principle expressed in different regimes.

At cosmological saturation,

$$C \to C_s, \qquad \rho_C \to V_\infty. \tag{106}$$

At galactic weak acceleration,

$$g \ll g_*, \tag{107}$$

and the organizational response modifies the relation between baryonic and effective acceleration.

A complete formulation should therefore connect:

$$\rho_s, \quad C_s, \quad V_\infty, \quad g_*. \tag{108}$$

The present theory predicts that these scales are not fundamentally independent.

10.30 Falsification conditions

The gravitational-saturation model would fail its completed form if any of the following occurred.

Background failure. If no nonsingular trajectory exists with

$$a_{\min} > 0, \qquad K_{\text{bounce}} < \infty, \qquad E(1) = 1, \tag{109}$$

the background construction fails.

Acoustic failure. If the model cannot maintain

$$100\theta_* \tag{110}$$

near the observed value while preserving its own

$$r_{\text{drag}}, \tag{111}$$

the locked geometry fails.

BAO failure. If future BAO data reject the correlated locked ratios

$$\frac{D_M}{r_{\text{drag}}} \quad \text{and} \quad \frac{D_H}{r_{\text{drag}}}, \tag{112}$$

the background is excluded.

Growth failure. If precise growth data require a significantly larger

$$S_8 \tag{113}$$

or a force-law modification incompatible with the locked history, the model fails.

Lensing failure. If the observed lensing-potential spectrum excludes the predicted mild enhancement, the model fails.

Stability failure. If an independent implementation finds a ghost, negative gradient coefficient, or runaway canonical mode, the field theory fails.

Local-limit failure. If the theory cannot recover standard high-acceleration gravity, its organizational interpretation fails.

10.31 Independent numerical replication

The most direct test is replication of the full pipeline.

An independent calculation should use:

$$\Omega_{m0} = 0.314114, \tag{114}$$

$$\Omega_{r0} = 9 \times 10^{-5}, \tag{115}$$

$$V_\infty = 2.187987, \tag{116}$$

$$\rho_s = 1.58199 \times 10^{15}, \tag{117}$$

$$m_C^2 = 1.827221, \tag{118}$$

$$\lambda = 0.802722, \tag{119}$$

$$C_s = 0.95, \tag{120}$$

$$C_{\text{initial}} = 0.918318, \tag{121}$$

$$\Pi_{C,\text{initial}} = 0.055327, \tag{122}$$

$$a_{\text{lock}} = 2.05 \times 10^{-4}, \tag{123}$$

$$W = 0.67 W_0. \tag{124}$$

The replicated calculation should recover approximately

$$
\begin{aligned}
100\theta_* &= 1.041033, && (125)\\
r_{\mathrm{drag}} &= 149.960 \text{ Mpc}, && (126)\\
S_8 &= 0.79781, && (127)\\
\max|\delta C| &\simeq 0.035, && (128)\\
\max|\delta\phi| &\simeq 0.049, && (129)
\end{aligned}
$$

and the reported lensing bandpower statistic.

Replication would test both the mathematics and the numerical implementation.

10.32 Near-term observational tests

The most direct near-term tests are:

$$
\begin{aligned}
&\text{high-precision BAO across multiple redshifts,}\\
&\text{redshift-space distortion measurements,}\\
&\text{tomographic weak lensing,}\\
&\text{CMB lensing reconstruction,}\\
&\text{improved large-scale polarization,}\\
&\text{joint CMB-large-scale-structure analyses.}
\end{aligned}
\tag{130}
$$

The model should be tested through its correlated predictions rather than one observable at a time.

10.33 Long-term tests

Longer-term tests include:

$$
\begin{aligned}
&\text{nonlinear structure formation,}\\
&\text{cluster abundance,}\\
&\text{void statistics,}\\
&\text{strong and weak galactic lensing,}\\
&\text{galaxy rotation-curve universality,}\\
&\text{high-redshift standard rulers,}\\
&\text{primordial gravitational-wave propagation.}
\end{aligned}
\tag{131}
$$

The model predicts that the background saturation history will leave correlated signatures across these domains.

10.34 Central predictive pattern

The theory's primary predictions may be summarized as

$$\boxed{\begin{aligned}&\text{larger } r_{\text{drag}},\\&\text{nearly unchanged } \theta_*,\\&\text{slightly slower intermediate } E(z),\\&\text{lower } S_8,\\&\text{small canonical field perturbations},\\&\text{scale-dependent primary CMB residuals},\\&\text{mildly enhanced true CMB lensing},\\&\text{higher TT reconstruction noise},\\&\text{standard local linear gravitational coupling}.\end{aligned}} \tag{132}$$

This combination is more restrictive than any one number.

A competing model may reproduce one or two of these features.

The locked saturation cosmology predicts the entire correlated pattern from one expansion history.

10.35 Final prediction statement

The gravitational-saturation theory predicts that the Universe carries the observable memory of a finite-density bounce and a later organizational lock primarily through its expansion history.

The canonical fields themselves remain stable and weakly perturbative.

The resulting cosmic history produces:

$$\begin{aligned}&\text{a larger acoustic ruler},\\&\text{an accurately placed acoustic angle},\\&\text{a coherent BAO ladder},\\&\text{reduced late growth},\\&\text{distinct primary CMB shape residuals},\\&\text{Planck-compatible lensing}.\end{aligned} \tag{133}$$

These predictions define the observational identity of the model.

They also define the conditions under which it can be independently confirmed or rejected.

End of Chapter 10

Chapter 11

Comparative Cosmology and the Structure of the Saturation Model

11.1 Purpose of the comparison

The locked gravitational-saturation cosmology is not constructed in isolation from established cosmology.

Its background, perturbations, acoustic observables, growth history, and lensing predictions are all evaluated through the same mathematical framework used to study standard relativistic cosmology.

The purpose of this chapter is to compare the saturation model directly with the corresponding structures of a matched flat reference cosmology.

The comparison is not limited to one quantity such as

$$H_0, \qquad S_8, \qquad \text{or} \qquad 100\theta_*. \tag{1}$$

It concerns the complete relationship between

$$\begin{aligned} &\text{background evolution,}\\ &\text{field dynamics,}\\ &\text{acoustic geometry,}\\ &\text{matter growth,}\\ &\text{primary CMB spectra,}\\ &\text{lensing.} \end{aligned} \tag{2}$$

The central result is that the locked model remains close to the matched reference at late times while preserving a distinct earlier expansion history.

11.2 Matched reference cosmology

The matched flat reference uses

$$\Omega_{m0} = 0.314114, \tag{3}$$
$$\Omega_{r0} = 9 \times 10^{-5}, \tag{4}$$
$$\Omega_{\Lambda 0} = 1 - \Omega_{m0} - \Omega_{r0}. \tag{5}$$

Its expansion function is

$$E^2_{\Lambda\text{CDM}}(z) = \Omega_{r0}(1+z)^4 + \Omega_{m0}(1+z)^3 + \Omega_{\Lambda 0}. \tag{6}$$

The reference uses the same

$$H_0, \quad \omega_b, \quad \omega_c, \quad A_s, \quad n_s, \quad \tau_{\text{reio}}, \tag{7}$$

and neutrino content as the locked model.

Therefore the comparison isolates the effect of the locked expansion history and saturation sector.

The reference is not independently reoptimized against the locked model.

It is matched in the parameters required to reveal the consequences of replacing the standard late residual history with the evolved saturation background.

11.3 Background expansion comparison

Define the relative expansion difference

$$\Delta_E(z) = \frac{E_{\text{locked}}(z) - E_{\Lambda\text{CDM}}(z)}{E_{\Lambda\text{CDM}}(z)}. \tag{8}$$

At

$$z = 0, \tag{9}$$

both models satisfy

$$E(0) = 1. \tag{10}$$

At late time, the saturation field obeys

$$w_C \simeq -1, \tag{11}$$

so the locked model approaches the reference closely.

At intermediate and higher redshift, the locked model expands more slowly by approximately

$$1\%\text{–}2\% \tag{12}$$

over the examined range.

Therefore

$$\Delta_E(z) < 0 \tag{13}$$

through much of the interval that contributes to BAO distances, growth, and CMB projection.

This small offset is sufficient to modify integrated observables while leaving the broad expansion history recognizable.

11.4 Comparison of the residual sectors

In the matched reference cosmology, the late residual is represented by

$$\rho_\Lambda = \text{constant}, \tag{14}$$

with

$$p_\Lambda = -\rho_\Lambda. \tag{15}$$

The equation-of-state parameter is exactly

$$w_\Lambda = -1. \tag{16}$$

In the saturation model,

$$\rho_C = \frac{1}{2}K^2 + V(C,\phi), \qquad p_C = \frac{1}{2}K^2 - V(C,\phi). \tag{17}$$

Therefore

$$w_C = \frac{\frac{1}{2}K^2 - V}{\frac{1}{2}K^2 + V}. \tag{18}$$

At late time,

$$K^2 \to 0, \qquad V \to V_\infty. \tag{19}$$

Thus

$$w_C \to -1. \tag{20}$$

The two backgrounds become nearly indistinguishable in the asymptotic limit. Their difference lies in origin:

$$\text{reference residual} = \text{constant component}, \tag{21}$$

whereas

$$\text{saturation residual} = \text{minimum energy of evolved fields}. \tag{22}$$

11.5 Comparison of early-time behavior

The standard reference extrapolates backward according to

$$H^2 = \frac{\rho}{3}. \tag{23}$$

For radiation,

$$\rho_r \propto a^{-4}, \tag{24}$$

so

$$H^2 \propto a^{-4}. \tag{25}$$

As

$$a \to 0, \tag{26}$$

one obtains

$$H \to \infty. \tag{27}$$

The locked model instead uses

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{28}$$

At

$$\rho = \rho_s, \tag{29}$$

one obtains

$$H = 0. \tag{30}$$

Thus the central early-time distinction is

$$\boxed{\text{singular extrapolation} \quad \text{versus} \quad \text{finite-density rebound}} \tag{31}$$

while the low-density radiation era remains standard in form after the bounce.

11.6 Comparison of the acoustic ruler

The matched control gives approximately

$$r_{\text{drag,control}} = 147.503 \text{ Mpc}. \tag{32}$$

The locked model gives

$$r_{\text{drag,locked}} = 149.960 \text{ Mpc}. \tag{33}$$

The difference is

$$\Delta r_{\text{drag}} = 2.457 \text{ Mpc}. \tag{34}$$

The fractional increase is

$$\frac{2.457}{147.503} \simeq 1.67\%. \tag{35}$$

The larger locked ruler follows from

$$H_{\text{locked}}(z) < H_{\text{control}}(z) \tag{36}$$

over part of the pre-drag integration.

This difference propagates into every BAO ratio.

11.7 Comparison of the acoustic angle

The matched control gives approximately

$$100\theta_{*,\text{control}} = 1.042322. \tag{37}$$

The locked model gives

$$100\theta_{*,\text{locked}} = 1.041033. \tag{38}$$

The difference between the two calculations is

$$\Delta(100\theta_*) = -0.001289. \tag{39}$$

Relative to the control,

$$\frac{1.041033 - 1.042322}{1.042322} \times 100\% \simeq -0.124\%. \tag{40}$$

The locked result nevertheless lies extremely close to the reference target

$$1.0411. \tag{41}$$

The comparison shows that the matched control and the locked model reach similar angular acoustic scales through different combinations of sound horizon and distance.

11.8 Comparison of last-scattering distances

The locked comoving distance is

$$D_{M,\text{locked}}(z_*) = 14131.86 \text{ Mpc}. \tag{42}$$

The matched control gives approximately

$$D_{M,\text{control}}(z_*) = 13883.67 \text{ Mpc}. \tag{43}$$

The difference is

$$\Delta D_M = 248.19 \text{ Mpc}. \tag{44}$$

The fractional increase is

$$\frac{248.19}{13883.67} \simeq 1.79\%. \tag{45}$$

The locked sound horizon is also larger.

The acoustic angle remains close because both numerator and denominator increase together.

11.9 Comparison of matter growth

The locked model gives

$$\sigma_{8,\text{locked}} = 0.779680. \tag{46}$$

The matched control gives

$$\sigma_{8,\text{control}} = 0.794561. \tag{47}$$

The difference is

$$\Delta\sigma_8 = -0.014881. \tag{48}$$

The fractional change is

$$\frac{-0.014881}{0.794561} \times 100\% \simeq -1.87\%. \tag{49}$$

Since both models have the same

$$\Omega_{m0}, \tag{50}$$

the same fractional difference appears in

$$S_8. \tag{51}$$

The locked result is

$$S_{8,\text{locked}} = 0.79781, \tag{52}$$

while the control gives

$$S_{8,\mathrm{control}} = 0.81304. \tag{53}$$

The comparison demonstrates that a percent-level change in the expansion history can produce a nearly two-percent change in the final growth amplitude.

11.10 Comparison of growth mechanisms

In the matched reference, the growth equation is

$$\frac{d^2\delta_m}{dN^2} + \left(2 + \frac{d\ln H_{\Lambda\mathrm{CDM}}}{dN}\right)\frac{d\delta_m}{dN} = \frac{3}{2}\Omega_m^{\Lambda\mathrm{CDM}}(a)\delta_m. \tag{54}$$

In the locked model,

$$\frac{d^2\delta_m}{dN^2} + \left(2 + \frac{d\ln H_{\mathrm{locked}}}{dN}\right)\frac{d\delta_m}{dN} = \frac{3}{2}\Omega_m^{\mathrm{locked}}(a)\delta_m. \tag{55}$$

The gravitational source coefficient remains

$$\frac{3}{2}. \tag{56}$$

The difference is carried by

$$H(a) \qquad \text{and} \qquad \Omega_m(a). \tag{57}$$

Thus the locked model alters growth without introducing

$$\mu(a, k) \tag{58}$$

into the source term.

11.11 Comparison with phenomenological modified gravity

A phenomenological modified-gravity model often writes

$$k^2\Psi = -4\pi G a^2 \mu(a, k)\rho\Delta. \tag{59}$$

The gravitational slip is

$$\eta(a, k) = \frac{\Phi}{\Psi}. \tag{60}$$

The lensing combination is

$$\Sigma(a, k) = \frac{\mu(1 + \eta)}{2}. \tag{61}$$

The saturation model does not use these as fundamental functions.

Its metric sources are obtained from

$$\delta T^{C\phi}_{\mu\nu} \tag{62}$$

and the standard Einstein equations.

Near the lock,

$$\delta T_{\mu\nu}^{C\phi} \tag{63}$$

is small.

The numerical failures of the tested μ-η closures confirm that these algebraic parameters do not reproduce the canonical model.

The comparison is therefore:

$$\boxed{\text{algebraic force modification} \neq \text{history-dependent saturation background}} \tag{64}$$

11.12 Comparison of field content

The matched reference contains standard matter, radiation, neutrinos, and a constant residual component.

The locked model contains standard matter and radiation together with canonical fields:

$$C \qquad \text{and} \qquad \phi. \tag{65}$$

Their energy density is

$$\rho_C = \frac{1}{2}\Pi_C^2 + \frac{1}{2}\Pi_\phi^2 + V(C,\phi). \tag{66}$$

Their pressure is

$$p_C = \frac{1}{2}\Pi_C^2 + \frac{1}{2}\Pi_\phi^2 - V(C,\phi). \tag{67}$$

The phase-space description is therefore larger than that of a constant residual.

However, once the fields settle,

$$\Pi_C, \Pi_\phi \to 0, \qquad C, \phi \to C_s, 0, \tag{68}$$

their background behavior approaches the constant-residual limit.

11.13 Comparison of perturbative behavior

A cosmological constant has no propagating perturbation.

The saturation sector contains propagating perturbations:

$$\delta C \qquad \text{and} \qquad \delta\phi. \tag{69}$$

Their equations are

$$\delta C'' + 2\mathcal{H}\delta C' + \big(k^2 + a^2 V_{,CC}\big)\delta C + \frac{1}{2}C'_{\rm bg}h' = 0, \tag{70}$$

and

$$\delta\phi'' + 2\mathcal{H}\delta\phi' + \big(k^2 + a^2 V_{,\phi\phi}\big)\delta\phi + \frac{1}{2}\phi'_{\rm bg}h' = 0. \tag{71}$$

The important numerical result is that these additional modes remain small:

$$\max|\delta C| = 0.03514, \qquad \max|\delta\phi| = 0.04860. \tag{72}$$

Their direct contribution is negligible.

Thus the saturation model contains more dynamical structure than the reference but approaches it closely in the late perturbative regime.

11.14 Comparison of stability

The reference cosmology has no instability associated with a dynamical residual field because its residual is constant.

The saturation model must demonstrate:

$$\text{positive kinetic coefficients}, \quad \text{positive gradient coefficients}, \quad \text{positive mass curvature near the lock.} \tag{73}$$

The kinetic matrix is

$$K_{IJ} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{74}$$

The sound speeds are

$$c_{s,C}^2 = 1, \qquad c_{s,\phi}^2 = 1. \tag{75}$$

The late mass matrix is approximately

$$M^2 = \begin{pmatrix} m_C^2 & 0 \\ 0 & A_0 \end{pmatrix}. \tag{76}$$

For positive A_0, both eigenvalues are positive.

The numerical modes remain bounded.

The dynamical residual therefore reproduces the stability expected of the reference late state.

11.15 Comparison of BAO behavior

The matched and locked models differ in both

$$D_M \qquad \text{and} \qquad r_{\text{drag}}. \tag{77}$$

A comparison using the same external ruler for both would be inconsistent.

Each model must use its own

$$r_{\text{drag}}. \tag{78}$$

The locked model gives a diagnostic BAO statistic near

$$13.4. \tag{79}$$

The value is produced by the combined effect of:

$$\text{larger distances,} \qquad \text{a larger ruler.} \tag{80}$$

The BAO test therefore constrains the complete early-to-late background history rather than one isolated distance.

11.16 Comparison of primary TT spectra

The locked TT spectrum differs from the matched control most strongly at low and acoustic multipoles.

The bandwise RMS differences are:

$$28.2\% \quad \text{for} \quad 2 \leq \ell \leq 29, \tag{81}$$
$$10.4\% \quad \text{for} \quad 30 \leq \ell \leq 600, \tag{82}$$
$$3.0\% \quad \text{for} \quad 601 \leq \ell \leq 1500, \tag{83}$$
$$1.7\% \quad \text{for} \quad 1501 \leq \ell \leq 2500. \tag{84}$$

The comparison shows that the locked cosmology is not simply a rescaled reference spectrum. Its early potential evolution and projection geometry produce a distinctive scale dependence.

11.17 Comparison of polarization spectra

The EE differences are smaller than TT at low and intermediate multipoles but remain nonzero.

The normalized TE comparison also reveals a scale-dependent pattern.

The high-ℓ convergence shows that the small-scale acoustic and damping physics remain close.

The larger low-ℓ and first-peak differences show where the locked expansion history leaves its strongest primary-CMB imprint.

11.18 Comparison of lensing signals

The locked model has lower

$$S_8, \tag{85}$$

but a mildly higher

$$C_L^{\varphi\varphi}. \tag{86}$$

The matched control has higher present-day clustering amplitude but slightly lower projected CMB lensing over the tested range.

This demonstrates that

$$S_8 \qquad \text{and} \qquad C_L^{\varphi\varphi} \tag{87}$$

are not interchangeable observables.

The locked geometry modifies the full line-of-sight kernel:

$$\frac{\chi_* - \chi}{\chi_* \chi}. \tag{88}$$

The altered potential history and geometry overcome the lower present clustering amplitude.

11.19 Comparison with Planck lensing data

The direct bandpower statistics are

$$\chi^2_{\text{locked}} = 10.7379, \qquad \chi^2_{\text{control}} = 12.0022. \tag{89}$$

The difference is

$$\Delta\chi^2 = -1.2643. \tag{90}$$

The locked model gives a modestly lower statistic.

Both are compatible with the nine measured bandpowers.

The result demonstrates that the locked model's altered growth and geometry do not create a lensing conflict.

11.20 Comparison of reconstruction efficiency

The locked/control PlanckLens noise ratio is approximately

$$1.21. \tag{91}$$

The matched control therefore permits a more efficient TT-only reconstruction under the common forecast assumptions.

The locked model has a modified primary spectrum, which changes the response weights used by the quadratic estimator.

The comparison distinguishes

$$\text{physical lensing power} \qquad \text{from} \qquad \text{reconstruction noise.} \tag{92}$$

The locked model predicts more of the first and more of the second.

11.21 Comparison of singular and nonsingular phase space

In the standard backward extrapolation,

$$a \to 0, \qquad \rho \to \infty, \qquad |H| \to \infty. \tag{93}$$

The phase-space trajectory terminates at a singular boundary.

In the saturation model,

$$\rho \to \rho_s, \qquad H \to 0, \qquad a \to a_{\min} > 0. \tag{94}$$

The trajectory continues from

$$H < 0 \qquad \text{to} \qquad H > 0. \tag{95}$$

Thus the locked theory replaces the singular phase-space endpoint with a regular crossing.

11.22 Comparison with a simple bounce-only model

A bounce-only model could modify the early constraint but leave the late universe unchanged.

The saturation theory contains more structure.

Its high-density response gives the bounce:

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{96}$$

Its field minimum gives the late residual:

$$V(C_s, 0) = V_\infty. \tag{97}$$

The same sector therefore connects:

$$\text{early nonsingularity} \qquad \text{and} \qquad \text{late acceleration}. \tag{98}$$

This unification distinguishes it from models in which the bounce and dark-energy eras arise from unrelated mechanisms.

11.23 Comparison with a scalar dark-energy model

A conventional scalar dark-energy field may use

$$\rho_\varphi = \frac{1}{2}\dot{\varphi}^2 + V(\varphi), \qquad p_\varphi = \frac{1}{2}\dot{\varphi}^2 - V(\varphi). \tag{99}$$

The saturation model shares this canonical late-time structure.

Its distinction is that the fields also participate in:

$$\text{contraction regulation}, \quad \text{radiation transfer}, \quad \text{high-density saturation}, \quad \text{the nonsingular background}. \tag{100}$$

The late scalar-like residual is one limit of a broader history rather than the entire theory.

11.24 Comparison with a cosmological constant

A cosmological constant is exactly constant:

$$\dot{\rho}_\Lambda = 0. \tag{101}$$

The saturation residual approaches constancy dynamically:

$$\dot{\rho}_C = -3HK^2. \tag{102}$$

As

$$K^2 \to 0, \tag{103}$$

one obtains

$$\dot{\rho}_C \to 0. \tag{104}$$

The present equation of state is

$$w_C = -0.99988. \tag{105}$$

The saturation background is therefore extremely close to a cosmological constant today, while retaining a different earlier origin and evolution.

11.25 Comparison with a reduced-gravity explanation of low S_8

A reduced-gravity model may lower growth through

$$G_{\text{eff}} < G. \tag{106}$$

The saturation model lowers growth through

$$H_{\text{locked}}(a) \neq H_{\text{control}}(a). \tag{107}$$

This distinction affects:

$$f\sigma_8, \quad C_L^{\varphi\varphi}, \quad \text{potential evolution}, \quad \text{the primary CMB}. \tag{108}$$

The failed phenomenological closures demonstrate that the reduced-gravity route does not reproduce the full saturation pattern.

11.26 Comparison with an added early-energy component

An added early component would contribute an independent density

$$\rho_X(a) \tag{109}$$

to the Friedmann equation.

The locked model instead changes the complete expansion trajectory produced by the saturation system.

Its early acoustic effects are not represented by simply adding

$$\Omega_X(a) \tag{110}$$

to an otherwise standard background.

The transition carries information from the nonsingular trajectory and is tied to the same field sector that produces the late residual.

11.27 Comparison of parameter sensitivity

The model is robust under most small one-at-a-time parameter perturbations.

The local test gave

$$25/28 \tag{111}$$

passing points.

The most sensitive physical direction is

$$V_\infty. \tag{112}$$

The most sensitive implementation quantities are

$$a_{\text{lock}} \qquad \text{and} \qquad \text{the transition width.} \tag{113}$$

This division is physically meaningful.

The late residual amplitude controls the low-redshift geometry.

The transition controls the amount of early locked history entering the acoustic integrals.

11.28 The role of the transition as a physical matching scale

The retained value is

$$a_{\text{lock}} = 2.05 \times 10^{-4}. \tag{114}$$

Equality occurs at

$$a_{\text{eq}} \simeq 2.865 \times 10^{-4}. \tag{115}$$

Therefore the lock endpoint occurs before equality.

The transition is positioned in a regime where radiation remains important.

Its effect is therefore strongest on:

$$r_s, \qquad r_{\text{drag}}, \qquad \text{and the early transfer functions.} \tag{116}$$

The smooth matching scale is part of the model's numerical realization of the post-bounce branch.

11.29 Internal hierarchy of the theory

The saturation model contains a natural hierarchy.

At the deepest level is the physical premise:

$$\text{gravity is wave-bearing and organizational.} \tag{117}$$

The field representation is:

$$C, \phi. \tag{118}$$

The high-density response is:

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{119}$$

The background consequence is:

$$\text{bounce and lock.} \tag{120}$$

The perturbative consequence is:

$$\text{small canonical field sources.} \tag{121}$$

The observational consequence is:

$$\text{a modified } H(a) \text{ across several integrated probes.} \tag{122}$$

This hierarchy explains why the strongest observable effects appear in the background rather than in a large direct fifth force.

11.30 The model's minimal distinct content

Relative to the matched reference, the essential new content can be summarized as:

$$\boxed{\begin{aligned}
&H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right),\\
&V(C,\phi) = V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4 + A_0(1 - \cos\phi),\\
&\Gamma_{\rm tr} = W_-(H)\big[-3H + \Gamma_0 S_\rho\big],\\
&\text{canonical } C, \phi \text{ perturbations},\\
&\text{locked expanding } H(a).
\end{aligned}} \tag{123}$$

Everything else follows through the ordinary Einstein-Boltzmann machinery.

11.31 Why the theory remains empirically constrained

The model is not free to alter one observable independently.

Changing

$$H(a) \tag{124}$$

changes:

$$r_s, \quad r_{\rm drag}, \quad D_M, \quad D_H, \quad \theta_*, \quad f\sigma_8, \quad C_\ell^{TT}, \quad C_\ell^{EE}, \quad C_\ell^{TE}, \quad C_L^{\varphi\varphi}. \tag{125}$$

The same locked history must satisfy them all.

This interdependence is what makes the numerical agreement meaningful.

The acoustic angle cannot be corrected without affecting BAO.

Growth cannot be changed without affecting lensing.

Lensing cannot be changed without altering the primary spectra and reconstruction response.

The model is therefore tightly constrained by its own internal consistency.

11.32 Comparative summary

The matched reference and locked model share:

$$\begin{aligned}&\text{flat geometry,}\\&\text{the same present } H_0,\\&\text{the same matter and radiation fractions,}\\&\text{the same primordial spectrum,}\\&\text{the same ordinary Boltzmann hierarchy,}\\&\text{the same Einstein-equation perturbative structure.}\end{aligned} \tag{126}$$

They differ in:

$$\begin{aligned}&\text{early singular versus nonsingular history,}\\&\text{constant versus dynamically reached residual,}\\&\text{standard versus locked } H(a),\\&\text{absence versus presence of canonical saturation fields,}\\&\text{the resulting integrated observables.}\end{aligned} \tag{127}$$

The locked model is therefore close enough to preserve successful cosmological structure while distinct enough to generate measurable departures.

11.33 Final comparative statement

The gravitational-saturation cosmology is best understood neither as a radically unrelated universe nor as a trivial reparameterization of the matched reference.

It is a specific dynamical extension.

At low density and late time, it approaches the familiar background.

At high density, it replaces singular growth with saturation.

Across cosmic history, it evolves canonical organizational fields toward a stable residual.

Its primary observational identity is carried by a slightly slower intermediate expansion, a larger acoustic ruler, a lower growth amplitude, distinct primary-spectrum shapes, and compatible lensing.

The comparison with the matched reference therefore reveals the central character of the model:

$$\boxed{\begin{aligned}&\text{near-standard late cosmology}\\&+\ \text{nonsingular saturation history}\\&+\ \text{correlated multi-probe deviations}\end{aligned}} \tag{128}$$

This combination defines the theory's relationship to established cosmology and provides the basis for its broader physical implications.

End of Chapter 11

Chapter 12

Broader Physical Implications of Gravitational Saturation

12.1 Purpose of the chapter

The preceding chapters developed the gravitational-saturation framework from its mathematical foundations through its numerical implementation and cosmological tests.

The resulting model contains three linked elements:

$$\text{a finite-density gravitational limit,} \tag{1}$$

$$\text{canonical organizational and phase fields,} \tag{2}$$

and

$$\text{a locked post-bounce expansion history.} \tag{3}$$

Together, these elements produce:

$$\text{a nonsingular rebound,} \tag{4}$$

$$\text{a radiation- and matter-dominated expanding branch,} \tag{5}$$

$$\text{a stable late residual,} \tag{6}$$

and

$$\text{a coherent set of cosmological observables.} \tag{7}$$

The purpose of this chapter is to examine the broader physical meaning of that structure.

The discussion concerns not only the expansion of the Universe, but the interpretation of gravitation itself.

The theory proposes that gravity is not exhausted by the static statement that mass curves spacetime.

Gravity also propagates.

It carries phase.

It organizes matter hierarchically.

It exchanges energy through dynamical geometry.

It can therefore approach a limiting state.

The mathematical form of that limit is gravitational saturation.

12.2 Gravitation as geometry and process

The Einstein equation,

$$G_{\mu\nu} = 8\pi G\, T_{\mu\nu}, \tag{8}$$

is often summarized by saying that matter tells spacetime how to curve and spacetime tells matter how to move.

That statement is correct, but the gravitational field is also a dynamical process.

Changes in the source do not rearrange geometry instantaneously.

They propagate through gravitational waves.

In the weak-field limit,

$$\Box \bar{h}_{\mu\nu} = -16\pi G\, T_{\mu\nu}. \tag{9}$$

In vacuum,

$$\Box \bar{h}_{\mu\nu} = 0. \tag{10}$$

The field therefore possesses independent propagating degrees of freedom.

The saturation hypothesis adds that a propagating geometric field can have a cumulative organizational state.

This state is represented by

$$C, \tag{11}$$

while its periodic or phase response is represented by

$$\phi. \tag{12}$$

The theory therefore treats gravity as both:

$$\text{geometry,} \tag{13}$$

and

$$\text{evolving organization.} \tag{14}$$

The geometric structure remains Einsteinian.

The organizational state determines how the full cosmic history approaches its limiting regimes.

12.3 Why organization is a gravitational concept

Gravity differs from the other familiar interactions in one fundamental respect.

It universally attracts ordinary positive-energy matter and therefore builds structure across an enormous hierarchy.

Starting from small density variations, gravity produces:

$$\text{stars,} \tag{15}$$

$$\text{stellar systems,} \tag{16}$$

$$\text{galaxies,} \tag{17}$$

$$\text{groups,} \tag{18}$$

$$\text{clusters,} \tag{19}$$

and

$$\text{the cosmic web.} \tag{20}$$

This process is not merely motion toward local potential minima.

It is cumulative organization.

Each gravitationally bound structure becomes part of a larger gravitational environment.

The field generated by matter becomes part of the geometry through which later gravitational effects propagate.

The saturation theory interprets this cumulative character as a genuine physical degree of freedom.

The field C does not count individual structures.

It represents the collective state of gravitational organization in the cosmological background.

12.4 Saturation as the limit of organization

The organizational potential is

$$V_C = V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4. \tag{21}$$

The restoring force is

$$F_C = -m_C^2(C - C_s) - \lambda(C - C_s)^3. \tag{22}$$

The preferred state is

$$C = C_s. \tag{23}$$

The theory interprets this point as the saturation state.

Before saturation,

$$C < C_s, \tag{24}$$

and the restoring dynamics move the field toward greater organization.

After an overshoot,

$$C > C_s, \tag{25}$$

the restoring force reverses.

Thus the field does not grow without limit.

The quartic term strengthens the restoration away from the minimum:

$$|F_C| \sim \lambda |C - C_s|^3 \tag{26}$$

for sufficiently large displacement.

The theory therefore describes saturation as a dynamically maintained balance rather than an externally imposed maximum.

12.5 The phase field and gravitational interference

The phase potential is

$$V_\phi = A_0(1 - \cos\phi). \tag{27}$$

It is periodic:

$$V_\phi(\phi + 2\pi n) = V_\phi(\phi). \tag{28}$$

The force is

$$F_\phi = -A_0 \sin\phi. \tag{29}$$

Near the minimum,

$$F_\phi \simeq -A_0\phi. \tag{30}$$

The phase field therefore supplies a restoring response analogous to a pendulum or nonlinear wave phase.

The physical interpretation is that cumulative gravitational organization cannot be described by amplitude alone.

A wave-bearing gravitational system also carries relative phase.

Reinforcing configurations increase the net organizational response.

Opposing configurations reduce it.

The phase field provides the theory with a mathematical channel for cancellation, reversal, and dynamic balance.

12.6 Phase cancellation without disappearance of energy

Suppose two wave contributions have amplitudes

$$A_{\text{in}} \tag{31}$$

and

$$A_{\text{out}}. \tag{32}$$

Their net amplitude is

$$A_{\text{net}} = A_{\text{in}} + A_{\text{out}}. \tag{33}$$

If

$$A_{\text{out}} \simeq -A_{\text{in}}, \tag{34}$$

then

$$A_{\text{net}} \simeq 0. \tag{35}$$

This does not imply that the underlying energy is absent.

In an ordinary wave system, destructive interference can suppress displacement at a point while energy remains stored in motion, stress, or neighboring modes.

The saturation theory applies the same broad principle to gravitational organization.

At saturation, the capacity for additional net organization becomes small even though the field retains a nonzero residual energy:

$$V(C_s, 0) = V_\infty. \tag{36}$$

The residual is therefore not the absence of gravity.

It is the energy of a balanced gravitational state.

12.7 Dynamic tension

A static balance can be represented by

$$\mathbf{F}_1 + \mathbf{F}_2 = 0. \tag{37}$$

Each force may be nonzero while the net acceleration vanishes.

A dynamically balanced wave system behaves similarly.

Its internal stresses and motions may remain finite while the net macroscopic response becomes stable.

The saturation state is interpreted as gravitational dynamic tension.

The field is not inactive.

Its competing organizational and phase tendencies have reached a stable relationship.

This interpretation is consistent with the late limits

$$C \to C_s, \qquad \phi \to 0, \qquad \Pi_C \to 0, \qquad \Pi_\phi \to 0, \tag{38}$$

and

$$\rho_C \to V_\infty. \tag{39}$$

The kinetic evolution diminishes, but the residual field energy remains.

12.8 The bounce as saturation rather than repulsion

In many nonsingular models, a bounce is described as the result of a repulsive component.

The saturation model gives a different interpretation.

The modified constraint is

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{40}$$

As density rises from zero,

$$H^2 \tag{41}$$

first increases.

It reaches its maximum at

$$\rho = \frac{\rho_s}{2}. \tag{42}$$

It then decreases.

At

$$\rho = \rho_s, \tag{43}$$

one has

$$H = 0. \tag{44}$$

The gravitational response has reached its limit.

The contraction does not reverse because a new force suddenly pushes everything outward.

It reverses because the relation between density and gravitational rate ceases to increase and then turns over.

The rebound is therefore a consequence of saturation in the response law.

12.9 The meaning of finite density at the bounce

At the bounce,

$$\rho = \rho_s < \infty. \tag{45}$$

The scale factor is

$$a = a_{\min} > 0. \tag{46}$$

The kinetic energy is finite.

The mathematical trajectory remains inside the phase space.

This changes the interpretation of the beginning of the expanding branch.

The expanding Universe does not emerge from a point of infinite density.

It emerges from a finite limiting state reached by an earlier contracting phase.

The bounce is therefore not an initial boundary of the equations.

It is an interior event in the continuous trajectory.

12.10 Time through the bounce

Because the trajectory is continuous in cosmic time,

$$t, \tag{47}$$

the bounce does not require time to begin or reverse.

The scale factor passes through a minimum:

$$\dot{a} = 0, \tag{48}$$

while

$$\ddot{a} > 0. \tag{49}$$

The Hubble parameter crosses

$$H = 0. \tag{50}$$

The time coordinate remains regular.

This is one reason the system cannot be integrated through the bounce using

$$N = \ln a. \tag{51}$$

At the minimum, a is not monotonic over the complete trajectory.

Cosmic time preserves the continuity of the physical evolution.

12.11 The arrow of the expanding branch

During contraction, the transfer term converts field kinetic energy into radiation:

$$Q = \Gamma_{\mathrm{tr}} K^2. \tag{52}$$

The field sector loses energy:

$$\dot{\rho}_C + 3H(\rho_C + p_C) = -Q. \tag{53}$$

Radiation gains the same energy:

$$\dot{\rho}_r + 4H\rho_r = Q. \tag{54}$$

After the bounce,

$$Q = 0. \tag{55}$$

The resulting radiation-dominated branch contains the thermal content inherited from the regulated contraction.

This transfer supplies a natural directional distinction between the contracting and expanding branches.

The contraction organizes and transfers.

The expansion dilutes and damps.

The resulting asymmetry contributes to the physical arrow of the post-bounce cosmology.

12.12 Radiation as released organizational energy

The transfer equation gives

$$\frac{d}{dt}(a^4\rho_r) = a^4 Q. \tag{56}$$

During contraction,

$$Q > 0 \tag{57}$$

when the transfer is active.

Radiation therefore records the energy removed from the growing organizational kinetic sector.

Within the mechanical interpretation, radiation is partly the released energy of gravitational reorganization near the bounce.

The theory does not require this interpretation to alter the ordinary post-bounce radiation law.

Once

$$H \geq 0, \tag{58}$$

the transfer shuts off and

$$\rho_r \propto a^{-4}. \tag{59}$$

Thus the later radiation era follows the standard expansion law.

12.13 Matter as the target of gravitational organization

The transfer activation uses

$$\rho_{\rm tar} = \rho_m + \rho_r. \tag{60}$$

The density switch is

$$S_\rho = \frac{(\rho_{\rm tar}/\rho_\Gamma)^m}{1 + (\rho_{\rm tar}/\rho_\Gamma)^m}. \tag{61}$$

The use of target density reflects the physical idea that gravitational organization is meaningful only in relation to the material and radiative content being organized.

At low density,

$$S_\rho \to 0. \tag{62}$$

At high density,

$$S_\rho \to 1. \tag{63}$$

The transfer therefore becomes important only when the contraction has generated a sufficiently dense target environment.

12.14 Saturation and the cosmological constant form

The late residual satisfies

$$p_C \simeq -\rho_C. \tag{64}$$

This is the same algebraic relation produced by a cosmological constant.

The saturation theory explains the relation through the canonical field limit:

$$\rho_C = \frac{1}{2}K^2 + V, \qquad p_C = \frac{1}{2}K^2 - V. \tag{65}$$

When

$$K^2 \to 0, \tag{66}$$

one obtains

$$p_C \to -V, \qquad \rho_C \to V. \tag{67}$$

Therefore

$$p_C \to -\rho_C. \tag{68}$$

The negative pressure is not a separate assumption.

It is the mechanical consequence of stored potential energy dominating over motion.

12.15 Why the residual does not dilute like matter or radiation

Matter has

$$w_m = 0, \tag{69}$$

so

$$\rho_m \propto a^{-3}. \tag{70}$$

Radiation has

$$w_r = \frac{1}{3}, \tag{71}$$

so

$$\rho_r \propto a^{-4}. \tag{72}$$

The locked field has

$$w_C \simeq -1. \tag{73}$$

Its continuity equation is

$$\dot{\rho}_C + 3H(1 + w_C)\rho_C = 0. \tag{74}$$

Since

$$1 + w_C \simeq 0, \tag{75}$$

one obtains

$$\dot{\rho}_C \simeq 0. \tag{76}$$

The residual therefore remains nearly constant while matter and radiation dilute.

It eventually dominates the late expansion.

12.16 The residual as a boundary condition of organization

The value

$$V_\infty \tag{77}$$

is the floor of the potential.

It may be interpreted as the minimum energy required to maintain the saturated organizational state.

The Universe cannot relax below this value without leaving the field minimum.

Thus

$$V_\infty \tag{78}$$

acts as the energetic boundary of complete organization.

It is not zero because the saturated state is not an empty state.

It contains the stored energy of the balanced gravitational configuration.

12.17 Why the fields become spectators

The fields build the background history, but their linear perturbations become weak after locking.

This follows from the source structure.

The density perturbation contains

$$C'_{\rm bg} v_C, \qquad \phi'_{\rm bg} v_\phi, \qquad V_{,C}\delta C, \qquad V_{,\phi}\delta\phi. \tag{79}$$

Near the minimum,

$$C'_{\rm bg} \to 0, \qquad \phi'_{\rm bg} \to 0, \qquad V_{,C} \to 0, \qquad V_{,\phi} \to 0. \tag{80}$$

The perturbations are therefore present, but they have little stress-energy with which to alter the metric.

The model consequently remembers the fields through the background they created rather than through a large continuing perturbative interaction.

12.18 Memory in the expansion history

The present background contains information about the earlier trajectory.

The sound horizon remembers the expansion before recombination:

$$r_s = \int \frac{c_s}{H}\, dz. \tag{81}$$

The drag horizon remembers the expansion before baryon release:

$$r_{\rm drag} = \int_{z_{\rm drag}}^{\infty} \frac{c_s}{H}\, dz. \tag{82}$$

The distance to last scattering remembers the entire later expansion:

$$D_M(z_*) = \int_0^{z_*} \frac{c}{H}\, dz. \tag{83}$$

Growth remembers the friction and matter fraction at every epoch:

$$D'' + \Big(2 + \frac{d\ln H}{dN}\Big) D' = \frac{3}{2}\Omega_m D. \tag{84}$$

The locked observables are therefore records of the saturation history.

12.19 Cosmological observables as historical integrals

Many of the strongest cosmological measurements are not instantaneous.

They are integrals.

The luminosity distance is an integral of

$$1/H. \tag{85}$$

The BAO transverse distance is an integral of

$$1/H. \tag{86}$$

The sound horizon is an integral of

$$c_s/H. \tag{87}$$

CMB lensing is an integral of the Weyl potential weighted by geometry.

The saturation model alters these observables coherently because it changes the historical path of H.

A small change repeated across a long interval can produce a measurable integrated effect.

This explains how percent-level expansion differences create visible changes in distances, growth, and spectra.

12.20 Why the theory is not defined by one tension

The model produces a lower value of

$$S_8. \tag{88}$$

It also produces an accurately placed

$$\theta_*. \tag{89}$$

It gives a coherent BAO ruler and Planck-compatible lensing.

These results are connected.

The theory was not constructed merely to lower one parameter.

Its defining claim is the existence of a consistent gravitational history.

A model that lowers S_8 while disrupting the sound horizon would not reproduce the theory.

A model that matches θ_* while failing the bounce would not reproduce it.

A model that fits BAO by inserting an unrelated ruler would not reproduce it.

The complete correlated pattern is the theory.

12.21 Dark matter as an interpretation of organization

At galactic scales, observed accelerations often exceed those predicted from visible baryons under a simple Newtonian mass distribution.

The conventional interpretation introduces additional gravitating matter.

The saturation framework permits another interpretation.

The gravitational field may respond collectively in the weak-acceleration regime.

A representative relation is

$$g\,\mu\!\left(\frac{g}{g_*}\right) = g_N, \tag{90}$$

with

$$\mu(x) = \frac{x}{\sqrt{1+x^2}}. \tag{91}$$

At low acceleration,

$$g \simeq \sqrt{g_N g_*}. \tag{92}$$

This response can produce an apparent excess gravitational acceleration without requiring that the effect be interpreted only as unseen local mass.

The organizational field offers a physical reason why the weak-field response could depend upon the state of the larger gravitational system.

12.22 Dark energy as an interpretation of residual saturation

At cosmic scales, accelerated expansion is conventionally represented by a negative-pressure component.

The saturation model gives the same background behavior from the locked field minimum.

Thus phenomena conventionally divided into

$$\text{dark-matter-like organization} \tag{93}$$

and

$$\text{dark-energy-like residual expansion} \tag{94}$$

may represent different limits of one gravitational principle.

At galactic scales, the relevant limit is weak acceleration.

At cosmic scales, the relevant limit is completed organization.

The underlying concept is gravitational saturation.

12.23 Scale dependence of organization

A universal organizational theory must allow different responses at different scales.

The cosmic background is nearly homogeneous and locked:

$$C \simeq C_s. \tag{95}$$

A galaxy is locally inhomogeneous and contains strong gradients in density and acceleration.

The same field can therefore occupy different regimes depending on:

$$\text{scale}, \quad \text{density}, \quad \text{acceleration}, \quad \text{environment}, \quad \text{distance from saturation}. \tag{96}$$

The cosmological field may be nearly frozen while local organizational responses remain active around structured systems.

This is analogous to a material whose bulk state is stable while supporting local stresses and excitations.

12.24 The distinction between local saturation and universal saturation

No local structure contains all gravitational targets.

A galaxy remains embedded in a group, a cluster, a filament, and the cosmological background.

Its organization is therefore incomplete.

The Universe as a whole is the only system with no external gravitational environment.

This gives the cosmic saturation state a special status.

Local systems can approach partial organization.

The complete background can approach global saturation.

This distinction supports the interpretation that the residual floor is a cosmological property while galactic systems retain active organizational response.

12.25 External mass and incomplete cohesion

A mechanical measure of incomplete local organization may be written as

$$M_{\text{ext}} = M_{\text{total}}(1 - C). \tag{97}$$

If

$$C = 1, \tag{98}$$

then

$$M_{\text{ext}} = 0. \tag{99}$$

A completely saturated system has no remaining external target within the chosen hierarchy.

For a local structure,

$$0 < C_{\text{local}} < 1, \tag{100}$$

so

$$M_{\text{ext}} > 0. \tag{101}$$

The local gravitational environment remains dynamically relevant.

This provides a simple organizational interpretation for why local systems need not share the fully locked cosmic behavior.

12.26 External coupling

A schematic external coupling is

$$F_{\text{ext}} = \eta_C \frac{G M_{\text{coh}} M_{\text{total}}}{R^2}(1 - C). \tag{102}$$

As

$$C \to 1, \tag{103}$$

the external coupling decreases:

$$F_{\text{ext}} \to 0. \tag{104}$$

The expression represents the decreasing capacity of a system to gain further organization as it approaches complete cohesion.

At the universal level, the absence of an external target leaves the residual saturation state.

At local levels, incomplete cohesion permits continuing organizational dynamics.

12.27 Hierarchical gravitational scaling

The same broad gravitational principles appear across scales.

For a bound structure,

$$v^2 \sim \frac{GM}{R}. \tag{105}$$

The characteristic density is

$$\rho \sim \frac{M}{R^3}. \tag{106}$$

The characteristic dynamical time is

$$t_{\text{dyn}} \sim \frac{1}{\sqrt{G\rho}}. \tag{107}$$

Larger structures generally have lower average density and longer dynamical time.

This hierarchy means that gravitational organization is never completed simultaneously at every scale.

Small structures can be locally virialized while larger structures continue evolving.

The cosmic saturation field represents the limiting background state of this hierarchy, not the instantaneous completion of every local process.

12.28 Virialization and saturation

A virialized system satisfies approximately

$$2K + U = 0. \tag{108}$$

Its kinetic and potential energies remain nonzero, but their average relation is stable.

This provides another mechanical analogy for saturation.

A saturated gravitational state need not be static or empty.

It can contain continuing internal motion while its macroscopic organization remains bounded.

The late cosmic residual similarly represents a stable energy relationship rather than the disappearance of dynamics.

12.29 Gravitational ringdown as an analogy

When compact objects merge, the final object undergoes ringdown.

Its gravitational perturbations decay through damped modes:

$$h(t) \sim e^{-t/\tau} \cos(\omega t + \varphi). \tag{109}$$

The system approaches a stable geometry by radiating excess dynamical structure.

The saturation fields behave similarly near their minimum:

$$\Delta C \sim e^{-3Ht/2} \cos(\omega_C t + \varphi_C), \tag{110}$$

and

$$\phi \sim e^{-3Ht/2} \cos(\omega_\phi t + \varphi_\phi) \tag{111}$$

in the underdamped regime.

The cosmological lock can therefore be interpreted as a large-scale gravitational ringdown toward an organizational ground state.

12.30 The Universe as a self-organizing gravitational system

The Universe differs from an isolated laboratory wave system because the background geometry is itself dynamical.

The waves propagate through the geometry they help determine.

The fields influence

$$H, \tag{112}$$

while H controls their damping.

The coupled structure is

$$\text{fields} \to \rho, p \to H \to \text{field damping} \to \text{fields}. \tag{113}$$

This feedback loop allows self-organization.

During contraction, the feedback amplifies kinetic motion until the transfer regulates it.

Near the bounce, the high-density response limits H.

During expansion, positive Hubble friction damps the fields.

The system naturally approaches the locked state.

12.31 Feedback structure

The coupled equations can be summarized as

$$\dot{C} = \Pi_C, \tag{114}$$

$$\dot{\Pi}_C = -(3H + \Gamma_{\rm tr})\Pi_C - V_{,C}, \tag{115}$$

$$\dot{\phi} = \Pi_\phi, \tag{116}$$

$$\dot{\Pi}_\phi = -(3H + \Gamma_{\rm tr})\Pi_\phi - V_{,\phi}, \tag{117}$$

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{118}$$

The field motion contributes to ρ.

The total ρ determines H.

The sign and magnitude of H determine whether the fields are amplified or damped.

The transfer modifies the kinetic energy during contraction.

This closed feedback structure is the mathematical basis of gravitational self-organization.

12.32 The two saturation limits

The theory contains two related saturation limits.

High-density saturation

$$\rho \to \rho_s. \tag{119}$$

Then

$$H \to 0. \tag{120}$$

This produces the bounce.

Field-space saturation

$$C \to C_s, \qquad \phi \to 0, \qquad K^2 \to 0. \tag{121}$$

Then

$$w_C \to -1. \tag{122}$$

This produces the late residual.

The early and late limits are not identical.

One limits the gravitational rate at high density.

The other minimizes the organizational field energy during expansion.

Together they form one continuous cosmological mechanism.

12.33 A universe without unrestricted gravitational growth

In the ordinary Friedmann relation,

$$H^2 = \frac{\rho}{3}, \tag{123}$$

the gravitational rate grows without bound as density grows without bound.

In the saturation relation,

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right), \tag{124}$$

the rate is bounded:

$$|H| \le \sqrt{\frac{\rho_s}{12}}. \tag{125}$$

This boundedness is the core mathematical departure.

It replaces unrestricted gravitational growth with a finite response curve.

The theory therefore proposes that gravity contains its own limiting mechanism.

12.34 The meaning of cosmic acceleration after locking

After the fields lock,

$$\rho_C \simeq V_\infty. \tag{126}$$

Matter continues diluting:

$$\rho_m \propto a^{-3}. \tag{127}$$

Eventually,

$$2\rho_C > \rho_m. \tag{128}$$

The acceleration equation gives

$$\ddot{a} > 0. \tag{129}$$

The acceleration is therefore not interpreted as gravity becoming repulsive in the ordinary local sense.

Matter still attracts matter.

The background accelerates because the nearly constant residual becomes dominant in the cosmic energy balance.

The distinction between local attraction and global accelerated expansion is retained.

12.35 No runaway residual

The potential is bounded below:

$$V \geq V_\infty. \tag{130}$$

The field energy decreases during expansion:

$$\dot{\rho}_C = -3HK^2 \leq 0. \tag{131}$$

It cannot run below the floor.

It also does not grow without bound because the fields are damped toward the minimum.

The late residual is therefore stable:

$$\rho_C \to V_\infty. \tag{132}$$

The model predicts no phantom runaway in which

$$\rho_C \tag{133}$$

increases as the Universe expands.

12.36 No ghost-driven bounce

The kinetic matrix is positive:

$$K_{IJ} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{134}$$

The bounce therefore does not arise from a field with negative kinetic energy.

The reversal is carried by the high-density gravitational response.

This is physically significant because negative-kinetic modes would destabilize the vacuum.

The canonical saturation fields remain positive-energy degrees of freedom.

12.37 No gradient catastrophe

The field perturbations have

$$c_s^2 = 1. \tag{135}$$

Their short-wavelength equations contain

$$+k^2\delta C, \qquad +k^2\delta\phi. \tag{136}$$

Therefore high-k modes oscillate rather than grow exponentially.

The theory's large-scale departures do not require a pathological small-scale propagation law.

12.38 The weak-field late Universe

At late time, the fields are near the minimum.

Their effective mass is positive.

Their source terms are small.

The ordinary weak-field metric can therefore remain close to

$$\nabla^2\Phi = 4\pi G\rho \tag{137}$$

for local high-acceleration systems.

The cosmic residual affects the background but does not automatically produce a large local fifth force.

This allows the theory to combine:

$$\text{nonstandard cosmic history} \tag{138}$$

with

$$\text{nearly standard local linear gravity.} \tag{139}$$

12.39 Gravitational saturation and equivalence

The canonical fields couple through the metric and their stress-energy.

Matter continues following geodesics of the same spacetime metric.

The theory therefore preserves the universal geometric role of gravity in the implemented cosmological sector.

The saturation effect is not introduced by assigning different gravitational charges to different matter species.

Its principal action is through the common background geometry.

This preserves the equivalence of freely falling matter within the model's cosmological implementation.

12.40 Relation between wave propagation and cosmological background

Gravitational waves are perturbations of geometry.

The saturation background is a homogeneous gravitational state.

The two are connected by the fact that both are governed by the same dynamical spacetime.

The wave interpretation does not require the homogeneous background itself to be a single sinusoidal wave.

Rather, it asserts that the gravitational system possesses wave-compatible properties:

$$\text{phase}, \quad \text{propagation}, \quad \text{damping}, \quad \text{restoration}, \quad \text{saturation.} \tag{140}$$

The fields C and ϕ encode these properties at the background and perturbative levels.

12.41 The cosmological lock as gravitational ground state

At the minimum,

$$V(C_s, 0) = V_\infty. \tag{141}$$

The fields have no remaining restoring force:

$$V_{,C} = 0, \qquad V_{,\phi} = 0. \tag{142}$$

Their velocities approach zero.

This state may be regarded as the gravitational organizational ground state of the background.

It is not a zero-energy vacuum.

It is the lowest-energy state available to the saturated field configuration.

The residual cosmic acceleration is the observable consequence of living in that ground state.

12.42 Why the lock is not the end of structure formation

The cosmic background can be locked while matter perturbations continue growing.

The field condition

$$C \simeq C_s \tag{143}$$

describes the homogeneous organizational sector.

Matter still obeys its perturbation equations.

Galaxies, clusters, and local systems remain dynamically active.

The lock therefore limits the evolution of the cosmic organizational background, not every local gravitational process.

This distinction is necessary for a Universe that accelerates globally while continuing to form and evolve local structures.

12.43 Global saturation and local nonequilibrium

A thermodynamic system can be near global equilibrium while supporting local nonequilibrium processes.

Likewise, the cosmological background can be near organizational saturation while galaxies and clusters retain:

$$\text{density gradients}, \quad \text{velocity flows}, \quad \text{mergers}, \quad \text{accretion}, \quad \text{wave emission}. \tag{144}$$

The local dynamics occur as excitations within the globally locked state.

This provides a consistent physical picture of late cosmology.

12.44 The role of environment

If gravitational organization is hierarchical, local behavior may depend upon environment.

A galaxy in a dense cluster is embedded in a different external gravitational field than an isolated field galaxy.

The organizational state may therefore depend upon:

$$\text{local acceleration}, \quad \text{external field}, \quad \text{mass distribution}, \quad \text{hierarchical position}. \tag{145}$$

The saturation interpretation consequently allows environmental dependence without abandoning universal field equations.

The environment changes the solution, not the underlying law.

12.45 Bulk flows and large-scale organization

Large-scale coherent velocities arise from the gravitational potential generated by broad matter distributions.

Within the saturation picture, bulk flows are part of ongoing organization below the universal saturation scale.

They represent matter responding to the gravitational hierarchy of filaments, superclusters, and surrounding density fields.

The theory therefore expects that local and regional systems remain dynamically coupled to larger structures even when the homogeneous background is locked.

12.46 Gravitational waves as carriers of reorganization

When a system changes its mass quadrupole, gravitational waves carry information and energy away.

A merger reorganizes the gravitational field.

The emitted waveform contains:

$$\text{inspiral}, \quad \text{merger}, \quad \text{ringdown}. \tag{146}$$

This sequence mirrors the larger saturation logic:

$$\text{growth}, \quad \text{nonlinear transition}, \quad \text{relaxation}. \tag{147}$$

The observation of gravitational ringdown demonstrates that gravitating systems naturally approach stable states through damped wave emission.

The cosmological saturation model applies this organizational principle to the background history.

12.47 Phase coherence and loss of net response

A collection of incoherent waves can produce a smaller organized macroscopic amplitude than a coherent set of waves with the same total energy.

The distinction is between energy content and coherent response.

The saturation theory proposes that the late residual represents stored gravitational energy after the net capacity for additional coherent organization has diminished.

This is why the residual can remain nonzero even as the fields settle:

$$K^2 \to 0, \tag{148}$$

while

$$V \to V_\infty. \tag{149}$$

The energy remains.

The active organizational motion vanishes.

12.48 Saturation and information

A gravitational field encodes information about its sources and their history.

The cosmic expansion history records the integrated state of the Universe.

The locked function

$$H(a) \tag{150}$$

therefore acts as a compressed record of the earlier gravitational trajectory.

The values of

$$r_s, \quad r_{\rm drag}, \quad D_M, \quad S_8, \quad C_L^{\varphi\varphi} \tag{151}$$

are different projections of this stored history.

The model's observational coherence arises because all of these quantities are derived from the same record.

12.49 The absence of independent late-time tuning

The late residual, growth, lensing, and acoustic distances are not independently selected.

They follow from the locked parameter set.

Changing

$$V_\infty \tag{152}$$

changes late geometry.

Changing

$$\rho_s \tag{153}$$

changes the high-density response and early trajectory.

Changing

$$a_{\rm lock} \tag{154}$$

changes the acoustic integration.

Changing the field potential changes the approach to saturation.

The theory's successful numerical balance therefore reflects the coupled structure of the model rather than arbitrary independent fitting of each observable.

12.50 Gravitational saturation as a conservation mechanism

The ordinary continuity equation remains

$$\dot{\rho} + 3H(\rho + p) = 0. \tag{155}$$

The transfer satisfies

$$Q_{C\phi} + Q_r = 0. \tag{156}$$

Energy is redistributed, not destroyed.

The high-density saturation law changes how density produces expansion or contraction.

It does not remove energy from the system.

Thus saturation acts as a limit on gravitational response rather than a violation of conservation.

This distinction is essential.

12.51 Why density can be finite while the response vanishes

At the bounce,

$$\rho = \rho_s, \tag{157}$$

but

$$H = 0. \tag{158}$$

In the standard Friedmann equation, nonzero density implies nonzero H^2.

The saturation factor changes this:

$$H^2 = \frac{\rho}{3} - \frac{\rho^2}{3\rho_s}. \tag{159}$$

At

$$\rho = \rho_s, \tag{160}$$

the two terms cancel:

$$\frac{\rho_s}{3} - \frac{\rho_s^2}{3\rho_s} = 0. \tag{161}$$

The vanishing rate therefore results from cancellation between the ordinary response and the saturation correction.

The density itself remains finite and positive.

12.52 Mechanical interpretation of the quadratic correction

The correction is

$$-\frac{\rho^2}{3\rho_s}. \tag{162}$$

Its magnitude grows faster than the ordinary term

$$\frac{\rho}{3}. \tag{163}$$

At low density, it is negligible.

At high density, it opposes further increase in H^2.

This behavior resembles a nonlinear restoring effect.

The greater the density becomes, the stronger the limiting response.

The correction therefore acts like dynamic tension increasing with compression.

12.53 Saturation versus antigravity

The negative quadratic correction should not be interpreted simply as a new repulsive material component.

It is part of the gravitational response law.

The effective-fluid rewriting,

$$\rho_{\mathrm{sat}} = -\frac{\rho^2}{\rho_s}, \tag{164}$$

is mathematically useful, but the theory does not assign this term an independent particle content.

The physical statement is:

$$\text{gravity responds differently near its saturation density.} \tag{165}$$

This is not the same as introducing negative mass.

12.54 The significance of $q = 1$

The generalized constraint is

$$H^2 = \frac{\rho}{3}\left[1 - \left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{166}$$

The locked model uses

$$q = 1. \tag{167}$$

This gives the simplest nonlinear correction:

$$H^2 = \frac{\rho}{3} - \frac{\rho^2}{3\rho_s}. \tag{168}$$

The response is polynomial, continuous, and differentiable.

The bounce occurs at one finite density.

The derivative gives the simple Raychaudhuri factor

$$1 - 2\frac{\rho}{\rho_s}. \tag{169}$$

The $q = 1$ form therefore provides the minimal saturation law consistent with the required rebound behavior.

12.55 Why the potential has quadratic and quartic terms

The quadratic term,

$$\frac{m_C^2}{2}(C - C_s)^2, \tag{170}$$

provides linear restoration near the minimum.

The quartic term,

$$\frac{\lambda}{4}(C - C_s)^4, \tag{171}$$

provides stronger restoration farther away.

The combination ensures:

$$V_{,C}(C_s) = 0, \qquad V_{,CC}(C_s) = m_C^2 > 0, \tag{172}$$

and

$$V \to +\infty \tag{173}$$

for sufficiently large $|C - C_s|$ when

$$\lambda > 0. \tag{174}$$

The potential is therefore stable both locally and globally.

12.56 Why the phase potential is periodic

The phase field represents a variable whose physically equivalent states repeat after

$$2\pi. \tag{175}$$

The simplest smooth periodic potential with a stable minimum is

$$A_0(1 - \cos\phi). \tag{176}$$

It has minima at

$$\phi = 2\pi n. \tag{177}$$

It has maxima at

$$\phi = (2n + 1)\pi. \tag{178}$$

Near a minimum, it becomes quadratic.

Farther away, it retains nonlinear phase behavior.

This makes it appropriate for representing a wave-like organizational phase.

12.57 The meaning of the saturation mass

The curvature

$$m_C^2 \tag{179}$$

sets the restoring strength of the organizational field near the lock.

The linearized equation is

$$\delta C'' + 2\mathcal{H}\delta C' + \left(k^2 + a^2 m_C^2\right)\delta C \simeq -\frac{1}{2}C'_{\mathrm{bg}}h'. \tag{180}$$

A larger m_C^2 produces faster restoration and shorter characteristic response length.

A smaller value allows longer-lived deviations from saturation.

The locked value

$$m_C^2 = 1.827221 \tag{181}$$

places the field in a stable restorative regime.

12.58 The significance of $C_s = 0.95$

The saturation point is

$$C_s = 0.95. \tag{182}$$

It lies near, but not at, the normalized value unity.

This allows the field to represent a near-complete organizational state without imposing an exact hard boundary.

The field remains dynamical around the minimum.

The value also permits a residual separation between total mathematical completion and the physical equilibrium reached by the locked trajectory.

The saturation point is therefore a stable preferred state rather than an absolute algebraic maximum.

12.59 The late Universe as near-saturation rather than exact stasis

The present value

$$w_C = -0.99988 \tag{183}$$

is close to, but not exactly,

$$-1. \tag{184}$$

This indicates a very small remaining kinetic fraction.

The fields are nearly locked but not mathematically frozen at every finite time.

The exact limit is approached asymptotically:

$$t \to \infty. \tag{185}$$

The late Universe is therefore a near-saturated dynamical system rather than an exactly static field configuration.

12.60 The physical unity of the model

The theory unifies several cosmological behaviors through one organizational principle.

The high-density limit gives:

$$\text{bounded } H, \qquad \text{a bounce.} \tag{186}$$

The contraction transfer gives:

$$\text{finite kinetic energy}, \qquad \text{radiation production.} \tag{187}$$

The field potential gives:

$$\text{stability}, \qquad \text{late residual energy.} \tag{188}$$

The locked background gives:

$$\text{the acoustic ruler}, \qquad \text{the BAO distances}, \qquad \text{growth suppression}, \qquad \text{lensing geometry.} \tag{189}$$

The canonical perturbations preserve:

$$\text{linear stability}, \qquad \text{the Einstein-equation structure.} \tag{190}$$

The theory is therefore not a collection of independent fixes.

Its separate results follow from one coupled system.

12.61 Philosophical economy

A physical theory gains strength when multiple phenomena follow from one mechanism.

The saturation model uses:

$$C, \qquad \phi, \qquad V(C,\phi), \qquad \rho_s, \tag{191}$$

and the contraction transfer law.

From these, it produces both the early and late cosmological limits.

The same expansion history then determines the observable results.

This economy is one of the principal motivations for the theory.

The model does not require a separate bounce field, a separate late dark-energy fluid, and a separate phenomenological weakening of gravity.

It uses one organizational sector.

12.62 The role of computation in the theory

The coupled equations cannot be understood fully through isolated analytic limits.

The bounce, radiation transfer, field damping, acoustic integrals, growth, and lensing must be evolved together.

The numerical calculation therefore forms part of the physical argument.

It tests whether the conceptual mechanism survives the full set of coupled consequences.

The final retained model does so across the completed calculations.

The computational implementation is not a replacement for the theory.

It is the means by which the theory's internal coherence is exposed.

12.63 Observable reality of an integrated history

No observer directly sees the field C as a local laboratory meter reading in the completed cosmological calculation.

Its effects appear through the history it produces.

The sound horizon, growth factor, and lensing potential are observable remnants of that history.

This is common in cosmology.

Inflationary parameters are inferred from primordial spectra.

Matter density is inferred from gravitational effects.

The expansion history is inferred from distances and redshifts.

The saturation field is likewise inferred through the correlated observables generated by its trajectory.

12.64 Why independent replication matters physically

A history-dependent model can be sensitive to numerical implementation.

The use of:

$$\text{cosmic-time integration,} \quad \text{piecewise checkpoints,} \quad \text{constraint monitoring,} \quad \text{DOP853,} \quad \text{Radau} \tag{192}$$

was therefore essential.

Agreement between distinct solvers shows that the locked trajectory is not created by one numerical algorithm.

The physical interpretation rests upon a reproducible mathematical path through phase space.

12.65 The status of the locked model

The completed model establishes a concrete gravitational-saturation cosmology with:

$$a_{\min} > 0, \tag{193}$$

$$K_{\text{bounce}} < \infty, \tag{194}$$

$$w_C \to -1, \tag{195}$$

$$100\theta_* = 1.041033, \tag{196}$$

$$r_{\text{drag}} = 149.960 \text{ Mpc}, \tag{197}$$

$$S_8 = 0.79781, \tag{198}$$

and Planck-compatible lensing.

The numerical values are not separate claims.

They are the measurable expression of the same locked background.

The theory's physical significance lies in that unity.

12.66 Final physical interpretation

The gravitational-saturation framework describes the Universe as a self-organizing wave-bearing gravitational system.

During contraction, organizational motion grows and is regulated through transfer.

At finite density, the gravitational response reaches its maximum and declines to zero.

The Universe rebounds.

During expansion, radiation and matter evolve through their ordinary eras.

The canonical fields damp toward a stable minimum.

Their remaining potential energy becomes the late residual.

The background locks.

Local structures remain active within this globally saturated state.

The observable Universe therefore carries two complementary signatures of gravity:

$$\text{local organization into structure,} \tag{199}$$

and

$$\text{global organization into saturation.} \tag{200}$$

The first builds the cosmic hierarchy.

The second governs the full cosmological history.

That is the broader physical meaning of gravitational saturation.

End of Chapter 12

Chapter 13

Synthesis of the Theory and the Completed Cosmological Calculation

13.1 Purpose of the synthesis

The preceding chapters developed gravitational saturation from its physical premise through its mathematical construction, numerical implementation, observational tests, and broader implications.

The completed framework contains one continuous chain:

$$\text{wave-bearing gravity} \rightarrow \text{organizational dynamics} \rightarrow \text{high-density saturation} \rightarrow \text{nonsingular bounce} \rightarrow \text{locked e} \tag{1}$$

The purpose of this chapter is to assemble that chain into one unified statement of the theory.

The model is not defined by one equation, one numerical fit, or one observational result.

It is defined by the consistency of all of them together.

13.2 Central physical proposition

The central proposition is that gravitation possesses an organizational degree of freedom arising from its dynamical and wave-bearing character.

Gravity is described geometrically by the Einstein equation,

$$G_{\mu\nu} = 8\pi G\, T_{\mu\nu}. \tag{2}$$

In the weak-field limit, metric disturbances propagate as waves:

$$\Box \bar{h}_{\mu\nu} = -16\pi G\, T_{\mu\nu}. \tag{3}$$

The existence of propagating gravitational degrees of freedom permits:

$$\text{phase}, \quad \text{interference}, \quad \text{damping}, \quad \text{restoration}, \quad \text{saturation}. \tag{4}$$

The theory represents this organizational behavior through the fields

$$C(x^\mu) \qquad \text{and} \qquad \phi(x^\mu). \tag{5}$$

The field C describes the degree of gravitational organization.

The field ϕ describes the periodic phase sector.

Together, they evolve toward a stable limiting configuration.

13.3 Core potential

The canonical field potential is

$$V(C,\phi) = V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4 + A_0(1 - \cos\phi). \tag{6}$$

Its derivatives are

$$V_{,C} = m_C^2(C - C_s) + \lambda(C - C_s)^3, \tag{7}$$

and

$$V_{,\phi} = A_0 \sin\phi. \tag{8}$$

The curvatures are

$$V_{,CC} = m_C^2 + 3\lambda(C - C_s)^2, \tag{9}$$

and

$$V_{,\phi\phi} = A_0 \cos\phi. \tag{10}$$

At the locked minimum,

$$C = C_s, \qquad \phi = 0, \tag{11}$$

so

$$V_{,C} = 0, \qquad V_{,\phi} = 0, \qquad V_{,CC} = m_C^2 > 0, \qquad V_{,\phi\phi} = A_0 > 0. \tag{12}$$

The joint minimum is therefore stable.

Its residual energy is

$$V(C_s, 0) = V_\infty. \tag{13}$$

13.4 Canonical energy and pressure

The homogeneous field energy is

$$\rho_C = \frac{1}{2}\Pi_C^2 + \frac{1}{2}\Pi_\phi^2 + V(C, \phi). \tag{14}$$

The pressure is

$$p_C = \frac{1}{2}\Pi_C^2 + \frac{1}{2}\Pi_\phi^2 - V(C, \phi). \tag{15}$$

Defining

$$K^2 = \Pi_C^2 + \Pi_\phi^2, \tag{16}$$

one obtains

$$\rho_C = \frac{1}{2}K^2 + V, \qquad p_C = \frac{1}{2}K^2 - V. \tag{17}$$

The equation of state is

$$w_C = \frac{\frac{1}{2}K^2 - V}{\frac{1}{2}K^2 + V}. \tag{18}$$

As the fields lock,

$$K^2 \to 0, \qquad V \to V_\infty. \tag{19}$$

Therefore

$$w_C \to -1. \tag{20}$$

The late residual is thus the potential-dominated limit of the same fields that participated in the earlier cosmological evolution.

13.5 High-density saturation law

The homogeneous gravitational response is

$$H^2 = \frac{\rho}{3}\left[1 - \left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{21}$$

The locked model uses

$$q = 1. \tag{22}$$

Therefore

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{23}$$

At low density,

$$\rho \ll \rho_s, \tag{24}$$

the standard Friedmann relation is recovered:

$$H^2 \simeq \frac{\rho}{3}. \tag{25}$$

At

$$\rho = \rho_s, \tag{26}$$

the response vanishes:

$$H = 0. \tag{27}$$

The Universe reaches a finite-density limiting state.

The Raychaudhuri equation is

$$\dot{H} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right). \tag{28}$$

At the bounce,

$$\rho = \rho_s, \tag{29}$$

so

$$\dot{H}_{\mathrm{b}} = \frac{1}{2}(\rho_{\mathrm{b}} + p_{\mathrm{b}}). \tag{30}$$

For

$$\rho_{\mathrm{b}} + p_{\mathrm{b}} > 0, \tag{31}$$

one obtains

$$\dot{H}_{\mathrm{b}} > 0. \tag{32}$$

The trajectory passes continuously from contraction to expansion.

13.6 Finite maximum gravitational rate

Writing

$$x = \frac{\rho}{\rho_s}, \tag{33}$$

the constraint becomes

$$H^2 = \frac{\rho_s}{3}\, x(1-x). \tag{34}$$

The function has a maximum at

$$x = \frac{1}{2}. \tag{35}$$

Therefore

$$|H|_{\max} = \sqrt{\frac{\rho_s}{12}}. \tag{36}$$

The contraction rate is bounded.

It does not diverge as density increases.

Beyond

$$\rho = \frac{\rho_s}{2}, \tag{37}$$

the magnitude of H decreases until the bounce is reached.

This is the direct mathematical expression of gravitational saturation.

13.7 Contraction-era regulation

During contraction,

$$H < 0, \tag{38}$$

and the ordinary field damping term becomes anti-friction.

The transfer rate is

$$\Gamma_{\rm tr} = W_-(H)\big[-3H + \Gamma_0 S_\rho\big]. \tag{39}$$

The field equations are

$$\dot{\Pi}_C = -(3H + \Gamma_{\rm tr})\Pi_C - V_{,C}, \tag{40}$$

and

$$\dot{\Pi}_\phi = -(3H + \Gamma_{\rm tr})\Pi_\phi - V_{,\phi}. \tag{41}$$

Deep in contraction,

$$W_- \simeq 1, \tag{42}$$

and therefore

$$3H + \Gamma_{\rm tr} \simeq \Gamma_0 S_\rho. \tag{43}$$

The anti-friction is cancelled.

The kinetic transfer is

$$Q = \Gamma_{\rm tr} K^2. \tag{44}$$

The field sector loses energy:

$$\dot{\rho}_C + 3H(\rho_C + p_C) = -Q. \tag{45}$$

Radiation gains the same energy:

$$\dot{\rho}_r + 4H\rho_r = Q. \tag{46}$$

Total energy is conserved.

13.8 Shutdown after the bounce

For

$$H \geq 0, \tag{47}$$

the transfer selector satisfies

$$W_{-}(H) = 0. \tag{48}$$

Therefore

$$\Gamma_{\text{tr}} = 0. \tag{49}$$

The field equations return to their canonical expansion form:

$$\ddot{C} + 3H\dot{C} + V_{,C} = 0, \tag{50}$$

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0. \tag{51}$$

Radiation obeys

$$\dot{\rho}_r = -4H\rho_r. \tag{52}$$

No continuing transfer tail alters the later expansion.

The post-bounce radiation and matter eras are therefore governed by their ordinary dilution laws.

13.9 Continuous nonsingular trajectory

The state vector is

$$\mathbf{y} = \big(a, H, C, \Pi_C, \phi, \Pi_\phi, \rho_m, \rho_r\big). \tag{53}$$

The final trajectory satisfies

$$a_{\text{min}} > 0, \qquad |H| < \infty, \qquad K^2 < \infty, \qquad |C| < \infty, \qquad |\phi| < \infty, \qquad \rho \leq \rho_s. \tag{54}$$

The minimum scale factor is approximately

$$a_{\text{min}} = 2.068 \times 10^{-5}. \tag{55}$$

The corresponding redshift is

$$z_{\text{bounce}} \simeq 4.84 \times 10^4. \tag{56}$$

The bounce is an interior point of the trajectory rather than a boundary where the equations fail.

13.10 Radiation and matter eras

Matter-radiation equality occurs at

$$a_{\text{eq}} = \frac{\Omega_{r0}}{\Omega_{m0}}. \tag{57}$$

For

$$\Omega_{r0} = 9 \times 10^{-5}, \qquad \Omega_{m0} = 0.314114, \tag{58}$$

one obtains

$$a_{\rm eq} \simeq 2.865 \times 10^{-4}. \tag{59}$$

Since

$$a_{\rm min} < a_{\rm eq}, \tag{60}$$

the expanding branch contains a radiation-dominated interval.

After equality,

$$\rho_m > \rho_r, \tag{61}$$

and matter dominates until the saturation residual becomes dynamically important.

The full sequence is therefore

$$\text{bounce} \to \text{radiation domination} \to \text{matter domination} \to \text{residual domination.} \tag{62}$$

13.11 Locked late state

Expansion damps the field kinetic energy:

$$\dot{\rho}_C = -3HK^2. \tag{63}$$

Since

$$H > 0, \tag{64}$$

one has

$$\dot{\rho}_C \leq 0. \tag{65}$$

The energy approaches its lower bound:

$$\rho_C \to V_\infty. \tag{66}$$

The fields approach

$$C \to C_s, \qquad \phi \to 0. \tag{67}$$

The present equation of state is

$$w_C(0) \simeq -0.99988. \tag{68}$$

The present residual density is approximately

$$\rho_C(0) \simeq 2.192 \tag{69}$$

in the adopted normalization.

The cosmological background is therefore almost completely locked at the present epoch.

13.12 Locked numerical parameters

The retained background parameters are

$$\Omega_{m0} = 0.314114, \tag{70}$$
$$\Omega_{r0} = 9 \times 10^{-5}, \tag{71}$$
$$H_0 = 67.4 \text{ km}\,\text{s}^{-1}\,\text{Mpc}^{-1}, \tag{72}$$
$$V_\infty = 2.187987, \tag{73}$$
$$\rho_s = 1.58199 \times 10^{15}, \tag{74}$$
$$m_C^2 = 1.827221, \tag{75}$$
$$\lambda = 0.802722, \tag{76}$$
$$C_s = 0.95, \tag{77}$$
$$C_{\text{initial}} = 0.918318, \tag{78}$$
$$\Pi_{C,\text{initial}} = 0.055327. \tag{79}$$

The CAMB transition is fixed at

$$a_{\text{lock}} = 2.05 \times 10^{-4}, \tag{80}$$

with width

$$W = 0.67 W_0. \tag{81}$$

13.13 Nonsingular numerical integration

The bounce was integrated in cosmic time or scaled affine time.

The system was not integrated through the bounce using

$$N = \ln a, \tag{82}$$

because

$$\frac{d}{dN} = \frac{1}{H}\frac{d}{dt} \tag{83}$$

is singular at

$$H = 0. \tag{84}$$

The nonsingular integration was performed piecewise and verified using both

$$\text{DOP853} \qquad \text{and} \qquad \text{Radau}. \tag{85}$$

The resulting background was retained only after agreement across solver families and integration tolerances.

This dual-solver reproducibility establishes that the bounce and locked branch are not artifacts of one numerical method.

13.14 Embedding in CAMB

The expanding branch was translated into

$$E(a) = \frac{H(a)}{H_0}. \tag{86}$$

A smooth early transition connected the resolved nonsingular trajectory to the Boltzmann background.

The function was normalized so that

$$E(1) = 1. \tag{87}$$

The canonical field backgrounds

$$C_{\rm bg}(a), \qquad \phi_{\rm bg}(a), \tag{88}$$

and their conformal derivatives were passed into the perturbation system.

The standard matter, radiation, photon, neutrino, and recombination equations were retained.

The principal modification was therefore:

$$\boxed{\text{locked } H(a) + \text{canonical saturation stress-energy}} \tag{89}$$

inside the standard Einstein-Boltzmann hierarchy.

13.15 Linear field equations

The field perturbations satisfy

$$\delta C'' + 2\mathcal{H}\delta C' + \left(k^2 + a^2 V_{,CC}\right)\delta C + \frac{1}{2}C'_{\rm bg}h' = 0, \tag{90}$$

and

$$\delta\phi'' + 2\mathcal{H}\delta\phi' + \left(k^2 + a^2 V_{,\phi\phi}\right)\delta\phi + \frac{1}{2}\phi'_{\rm bg}h' = 0. \tag{91}$$

Their density perturbation is

$$\delta\rho_C = \frac{1}{a^2}\left[C'_{\rm bg}\delta C' + \phi'_{\rm bg}\delta\phi'\right] + V_{,C}\delta C + V_{,\phi}\delta\phi. \tag{92}$$

Their pressure perturbation is

$$\delta p_C = \frac{1}{a^2}\left[C'_{\rm bg}\delta C' + \phi'_{\rm bg}\delta\phi'\right] - V_{,C}\delta C - V_{,\phi}\delta\phi. \tag{93}$$

The momentum perturbation is

$$\delta q_C = \frac{k}{a^2}\left[C'_{\rm bg}\delta C + \phi'_{\rm bg}\delta\phi\right]. \tag{94}$$

Their intrinsic anisotropic stress is

$$\pi_C = 0. \tag{95}$$

13.16 Stability summary

The kinetic matrix is

$$K_{IJ} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{96}$$

Its eigenvalues are positive.

The field sound speeds are

$$c_{s,C}^2 = 1, \qquad c_{s,\phi}^2 = 1. \tag{97}$$

The mass matrix near the lock is

$$M^2 \simeq \begin{pmatrix} m_C^2 & 0 \\ 0 & A_0 \end{pmatrix}. \tag{98}$$

For positive A_0, both eigenvalues are positive.

The measured field amplitudes are

$$\max |\delta C| = 0.03514, \qquad \max |\delta\phi| = 0.04860. \tag{99}$$

No ghost, gradient instability, or runaway linear mode appears.

13.17 Spectator behavior

The canonical fields are responsible for the background trajectory, yet their direct perturbative back-reaction is extremely small after locking.

Near the field minimum,

$$C'_{\rm bg} \simeq 0, \qquad \phi'_{\rm bg} \simeq 0, \qquad V_{,C} \simeq 0, \qquad V_{,\phi} \simeq 0. \tag{100}$$

Their stress-energy perturbations are therefore suppressed.

The direct effect on

$$\sigma_8 \tag{101}$$

and the primary CMB peaks is at approximately the level of

$$10^{-6}. \tag{102}$$

The dominant observable effect is carried by the background function

$$H(a). \tag{103}$$

This is one of the defining numerical results of the completed model.

13.18 Acoustic geometry

The final acoustic quantities are

$$100\theta_* = 1.041033, \tag{104}$$
$$r_{\text{drag}} = 149.960 \text{ Mpc}, \tag{105}$$
$$z_* = 1089.691, \tag{106}$$
$$z_{\text{drag}} = 1058.511. \tag{107}$$

The difference from the reference acoustic value

$$1.0411 \tag{108}$$

is

$$-0.0064\%. \tag{109}$$

The comoving distance to last scattering is

$$D_M(z_*) = 14131.86 \text{ Mpc}. \tag{110}$$

The locked background therefore balances a larger sound horizon with a larger comoving distance.

13.19 BAO synthesis

The BAO observables are

$$\frac{D_M}{r_{\text{drag}}}, \qquad \frac{D_H}{r_{\text{drag}}}, \qquad \frac{D_V}{r_{\text{drag}}}. \tag{111}$$

All are computed from the same locked background and the same CAMB-derived ruler.

The diagnostic DESI statistic is approximately

$$\chi^2_{\text{BAO}} = 13.4. \tag{112}$$

The result demonstrates that the distance ladder is coherent when the sound horizon and distances are evaluated from one expansion history.

The earlier larger discrepancy was removed by eliminating the inconsistent external ruler.

13.20 Growth synthesis

The final growth amplitude is

$$\sigma_8 = 0.779680. \tag{113}$$

The derived weak-lensing parameter is

$$S_8 = 0.79781. \tag{114}$$

The matched control gives

$$S_{8,\mathrm{control}} = 0.81304. \tag{115}$$

The locked model lowers S_8 by approximately

$$1.87\%. \tag{116}$$

This reduction occurs without replacing

$$G \tag{117}$$

by a smaller effective gravitational constant.

It follows from the altered expansion history and matter fraction over time.

13.21 Primary CMB synthesis

The final TT, TE, and EE spectra retain recognizable acoustic structure.

Their differences from the matched control are scale dependent.

For TT, the RMS residuals are approximately

$$28.2\% \quad \text{for} \quad 2 \le \ell \le 29, \tag{118}$$

$$10.4\% \quad \text{for} \quad 30 \le \ell \le 600, \tag{119}$$

$$3.0\% \quad \text{for} \quad 601 \le \ell \le 1500, \tag{120}$$

$$1.7\% \quad \text{for} \quad 1501 \le \ell \le 2500. \tag{121}$$

The residuals decline toward high multipoles.

They cannot be removed by one amplitude factor.

The primary spectra therefore preserve a measurable imprint of the locked history.

13.22 Primary likelihood result

The locked TT, TE, and EE spectra were evaluated within the completed primary-CMB workflow.

The resulting likelihood treatment used the locked CAMB spectra and their corresponding covariance and data products.

The primary CMB calculation therefore extends beyond a visual comparison of peak locations.

It evaluates the model through the full numerical spectrum and likelihood structure used in the completed pipeline.

The result confirms that the acoustic, polarization, and damping-tail behavior remain compatible with the locked model's retained parameter set.

13.23 Lensing synthesis

The true CMB lensing potential is mildly enhanced relative to the matched control.

The weighted ratio over

$$40 \le L \le 2000 \tag{122}$$

is approximately

$$1.0209. \tag{123}$$

The enhancement is about

$$2.09\%. \tag{124}$$

Despite this enhancement, the present growth amplitude is lower.

The result demonstrates that CMB lensing cannot be inferred from S_8 alone.

The geometry and integrated potential history are essential.

13.24 Official Planck lensing comparison

Using the official Planck 2018 nine-bin bandpowers, window functions, and covariance, the locked model gives

$$\chi^2_{\text{locked}} = 10.7379 \tag{125}$$

for nine bins.

The matched control gives

$$\chi^2_{\text{control}} = 12.0022. \tag{126}$$

Therefore

$$\Delta\chi^2 = -1.2643. \tag{127}$$

The corresponding locked p-value is

$$p_{\text{locked}} = 0.294. \tag{128}$$

The model is compatible with the measured lensing bandpowers and gives a modestly lower statistic than the matched control.

13.25 Reconstruction synthesis

The PlanckLens TT quadratic-estimator forecast gives

$$\frac{N^{(0)}_{L,\text{locked}}}{N^{(0)}_{L,\text{control}}} \simeq 1.21. \tag{129}$$

The locked model therefore has higher TT reconstruction noise.

This does not contradict the enhanced physical lensing spectrum.

The two quantities describe different properties:

$$C_L^{\varphi\varphi} = \text{physical projected lensing power}, \tag{130}$$

while

$$N_L^{(0)} = \text{Gaussian estimator noise}. \tag{131}$$

The altered TT spectrum reduces reconstruction efficiency while the integrated geometry slightly enhances the actual signal.

13.26 Failure of phenomenological substitutes

The tested phenomenological modifications included:

$$f_{\text{sat}}(a,k), \qquad \mu(a,k), \tag{132}$$

and algebraic slip relations

$$\eta = 1, \qquad \eta = \frac{1}{\mu}, \qquad \eta = 2 - \mu. \tag{133}$$

These models increased

$$\sigma_8 \tag{134}$$

and distorted the odd-even acoustic peak pattern.

None reproduced the canonical saturation-field result.

The failed closures establish that the successful model is not equivalent to a simple rescaling of Newton's constant or an imposed gravitational slip.

The canonical field stress-energy and locked background must be retained together.

13.27 Robustness synthesis

The locked parameter point was tested through one-at-a-time changes of approximately

$$\pm 0.5\% \qquad \text{and} \qquad \pm 1\%. \tag{135}$$

The varied quantities were

$$\Omega_{m0}, \quad V_\infty, \quad \rho_s, \quad m_C^2, \quad \lambda, \quad C_{\text{initial}}, \quad \Pi_{C,\text{initial}}. \tag{136}$$

Of 28 nearby points,

$$25 \tag{137}$$

retained the required behavior.

The local model is therefore classified as

$$\boxed{\text{ROBUST}}. \tag{138}$$

The most sensitive direction is

$$V_\infty. \tag{139}$$

13.28 Internal consistency of the full theory

The theory satisfies several independent consistency conditions.

- Conservation: $\nabla_\mu T^{\mu\nu}_{\text{total}} = 0$.

- Nonsingularity: $a_{\min} > 0$.
- Finite field energy: $K^2_{\text{bounce}} < \infty$.
- Stable potential: $V_{,CC} > 0$, $V_{,\phi\phi} > 0$ near the lock.
- Positive kinetic matrix: $K_{IJ} > 0$.
- Positive gradient terms: $c_s^2 = 1$.
- Low-density recovery: $H^2 \simeq \rho/3$.
- Late residual: $w_C \to -1$.
- Acoustic coherence: $100\theta_* = 1.041033$.
- BAO coherence: $\chi^2_{\text{BAO}} \simeq 13.4$.
- Lensing compatibility: $\chi^2_{\text{lensing}} = 10.7379$.

These conditions are all satisfied by one locked parameter set.

13.29 Unification of early and late cosmology

The early and late limits follow from one saturation principle.

At high density,

$$\rho \to \rho_s, \qquad H \to 0. \tag{140}$$

This produces the bounce.

At late time,

$$C \to C_s, \qquad \phi \to 0, \qquad K^2 \to 0. \tag{141}$$

This produces the residual.

Thus

$$\boxed{\text{high-density saturation} \to \text{nonsingular origin of expansion}} \tag{142}$$

and

$$\boxed{\text{field-space saturation} \to \text{late accelerated expansion}} \tag{143}$$

are two limits of the same theory.

13.30 Relation to dark energy

The locked residual produces

$$p_C \simeq -\rho_C. \tag{144}$$

The background therefore resembles a cosmological constant.

Its physical interpretation is different.

The residual is not introduced independently.

It is the minimum energy

$$V_\infty \tag{145}$$

of the evolved organizational field system.

The theory therefore explains the cosmological-constant-like behavior as the energetic consequence of saturation.

13.31 Relation to dark matter-like behavior

The cosmological implementation retains the matter content required by the completed Boltzmann calculation.

At galactic scales, the organizational interpretation introduces the weak-acceleration response

$$g\mu\left(\frac{g}{g_*}\right) = g_N. \tag{146}$$

For

$$\mu(x) = \frac{x}{\sqrt{1+x^2}}, \tag{147}$$

the low-acceleration limit gives

$$g \simeq \sqrt{g_N g_*}. \tag{148}$$

For circular motion,

$$v^4 \simeq GM_b g_*. \tag{149}$$

This provides a gravitational-organizational interpretation of baryonic scaling relations.

The cosmic residual and galactic response represent different regimes of the same underlying principle.

13.32 Local and global gravity

The late cosmological field is near its minimum.

Its gradients and perturbative sources are small.

Therefore local high-acceleration gravity remains close to its ordinary form.

The cosmic background can differ from the matched reference without requiring a large local fifth force.

This creates a separation between:

$$\text{global integrated history} \tag{150}$$

and

$$\text{local weak-field response.} \tag{151}$$

The completed CAMB calculation confirms this separation at linear cosmological order.

13.33 Meaning of observational viability

The locked model was tested as one connected numerical system.

It produces:

$$\begin{aligned}&\text{a stable bounce,}\\&\text{a correct radiation-to-matter sequence,}\\&\text{a locked late residual,}\\&\text{a near-reference acoustic angle,}\\&\text{a coherent BAO ruler,}\\&\text{a lower growth amplitude,}\\&\text{bounded canonical perturbations,}\\&\text{primary CMB spectra,}\\&\text{Planck-compatible lensing.}\end{aligned} \tag{152}$$

Its observational viability therefore arises from agreement across several independent probes rather than from a single fitted parameter.

13.34 Scientific consequences

If gravitational saturation describes the physical Universe, several consequences follow.

1. The cosmological singularity is replaced by a finite-density bounce.
2. Late acceleration is the residual state of gravitational organization.
3. Reduced structure growth need not imply a weakened Newtonian coupling.
4. Cosmic lensing can increase even when present-day S_8 decreases.
5. Dark-energy-like and dark-matter-like phenomena may be different scale limits of gravitational organization.
6. The observable Universe retains memory of the earlier saturation trajectory primarily through its expansion history.

These consequences define the physical reach of the theory.

13.35 The theory's decisive feature

The decisive feature is not simply that the model can produce a bounce.

Many equations can be written that produce a bounce.

It is not simply that the model can reproduce

$$w \simeq -1. \tag{153}$$

Many scalar potentials can do so.

It is not simply that the model can lower

$$S_8. \tag{154}$$

Many modified histories can alter growth.

The decisive feature is that all three emerge from one organizational framework while preserving stable perturbations and coherent multi-probe observables.

The theory therefore possesses explanatory unity.

13.36 Final synthesized model

The complete model may be summarized by the following system:

$$\boxed{H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right)} \tag{155}$$

$$\boxed{V(C,\phi) = V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4 + A_0(1 - \cos\phi)} \tag{156}$$

$$\boxed{\dot{\Pi}_C = -(3H + \Gamma_{\rm tr})\Pi_C - V_{,C}} \tag{157}$$

$$\boxed{\dot{\Pi}_\phi = -(3H + \Gamma_{\rm tr})\Pi_\phi - V_{,\phi}} \tag{158}$$

$$\boxed{\dot{\rho}_r = -4H\rho_r + \Gamma_{\rm tr}K^2} \tag{159}$$

$$\boxed{\Gamma_{\rm tr} = 0 \quad \text{for} \quad H \geq 0} \tag{160}$$

together with canonical linear perturbations and the standard Einstein-Boltzmann hierarchy.

This system produces the locked cosmology examined throughout the manuscript.

13.37 Final chapter statement

The completed gravitational-saturation theory describes one continuous Universe.

It contracts.

Its gravitational response reaches a finite limit.

Its organizational kinetic energy is regulated and transferred.

It rebounds.

It enters radiation and matter domination.

Its canonical fields settle.

Their minimum energy remains as the late residual.

The expansion locks.

The resulting background produces the observed acoustic ruler, BAO geometry, reduced growth amplitude, stable scalar perturbations, primary CMB structure, and compatible Planck lensing.

The central conclusion is therefore:

$$\boxed{\text{gravity can organize, saturate, rebound, and lock}} \tag{161}$$

while retaining the mathematical structure required for a stable relativistic cosmology.

The theory's observable content is carried principally by the locked expansion history, while its physical unity is carried by the organizational wave dynamics of C and ϕ.

End of Chapter 13

Chapter 14

Conclusions

14.1 Purpose of the conclusion

The gravitational-saturation theory developed in this work begins from one physical proposition:

$$\boxed{\text{gravity is a dynamical, wave-bearing, self-organizing field whose response can reach a finite limit}} \tag{1}$$

That proposition was translated into a continuous cosmological system containing:

$$\begin{aligned}
&\text{a high-density saturation law,}\\
&\text{an organizational field } C,\\
&\text{a phase field } \phi,\\
&\text{a stable saturation potential,}\\
&\text{a contraction-era transfer mechanism,}\\
&\text{a finite-density bounce,}\\
&\text{a locked late-time residual.}
\end{aligned} \tag{2}$$

The resulting background was then carried into the Einstein-Boltzmann hierarchy, where its consequences were evaluated through the completed CAMB, BAO, growth, primary-CMB, PlanckLens, and Planck-lensing calculations.

The purpose of this chapter is to state the final conclusions of that work as one unified result.

14.2 The completed mathematical model

The background gravitational response is

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{3}$$

At low density,

$$\rho \ll \rho_s, \tag{4}$$

this reduces to

$$H^2 \simeq \frac{\rho}{3}. \tag{5}$$

At the saturation density,

$$\rho = \rho_s, \tag{6}$$

the expansion or contraction rate vanishes:

$$H = 0. \tag{7}$$

The corresponding Raychaudhuri equation is

$$\dot{H} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right). \tag{8}$$

At the bounce,

$$\dot{H}_{\rm b} = \frac{1}{2}(\rho_{\rm b} + p_{\rm b}). \tag{9}$$

For

$$\rho_{\rm b} + p_{\rm b} > 0, \tag{10}$$

the Universe crosses continuously from

$$H < 0 \tag{11}$$

to

$$H > 0. \tag{12}$$

The bounce therefore occurs without requiring a negative-kinetic-energy field.

14.3 The organizational fields

The saturation sector is described by

$$C(x^\mu) \qquad \text{and} \qquad \phi(x^\mu). \tag{13}$$

Their potential is

$$V(C,\phi) = V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4 + A_0(1 - \cos\phi). \tag{14}$$

The locked values are

$$C_s = 0.95, \tag{15}$$
$$V_\infty = 2.187987, \tag{16}$$
$$m_C^2 = 1.827221, \tag{17}$$
$$\lambda = 0.802722. \tag{18}$$

The derivatives are

$$V_{,C} = m_C^2(C - C_s) + \lambda(C - C_s)^3, \tag{19}$$

and

$$V_{,\phi} = A_0 \sin\phi. \tag{20}$$

At the locked minimum,

$$C = C_s, \qquad \phi = 0. \tag{21}$$

The curvatures satisfy

$$V_{,CC}(C_s) = m_C^2 > 0, \qquad V_{,\phi\phi}(0) = A_0 > 0. \tag{22}$$

The field configuration is therefore restoring and stable.

14.4 Energy and pressure

The saturation-sector density is

$$\rho_C = \frac{1}{2}\Pi_C^2 + \frac{1}{2}\Pi_\phi^2 + V(C,\phi). \tag{23}$$

The pressure is

$$p_C = \frac{1}{2}\Pi_C^2 + \frac{1}{2}\Pi_\phi^2 - V(C,\phi). \tag{24}$$

Defining

$$K^2 = \Pi_C^2 + \Pi_\phi^2, \tag{25}$$

one obtains

$$w_C = \frac{\frac{1}{2}K^2 - V}{\frac{1}{2}K^2 + V}. \tag{26}$$

During late expansion,

$$K^2 \to 0, \qquad V \to V_\infty. \tag{27}$$

Therefore

$$w_C \to -1. \tag{28}$$

The late accelerated background is thus produced by the minimum energy of the same fields that participated in the nonsingular history.

No separate late-time dark-energy fluid is added.

14.5 Contraction-era regulation

During contraction,

$$H < 0, \tag{29}$$

and the ordinary Hubble term acts as anti-friction.

The transfer rate is

$$\Gamma_{\rm tr} = W_-(H)\big[-3H + \Gamma_0 S_\rho\big]. \tag{30}$$

The field equations are

$$\dot{\Pi}_C = -(3H + \Gamma_{\rm tr})\Pi_C - V_{,C}, \tag{31}$$

and

$$\dot{\Pi}_\phi = -(3H + \Gamma_{\rm tr})\Pi_\phi - V_{,\phi}. \tag{32}$$

Deep in contraction,

$$3H + \Gamma_{\rm tr} \simeq \Gamma_0 S_\rho. \tag{33}$$

The anti-friction is cancelled.

The transferred power is

$$Q = \Gamma_{\rm tr} K^2. \tag{34}$$

The field sector loses this energy:

$$\dot{\rho}_C + 3H(\rho_C + p_C) = -Q. \tag{35}$$

Radiation receives it:

$$\dot{\rho}_r + 4H\rho_r = Q. \tag{36}$$

The total continuity equation remains

$$\dot{\rho} + 3H(\rho + p) = 0. \tag{37}$$

The transfer therefore regulates the contraction without violating energy conservation.

14.6 Transfer shutdown and ordinary expansion

For

$$H \geq 0, \tag{38}$$

the transfer vanishes:

$$\Gamma_{\rm tr} = 0. \tag{39}$$

Radiation returns to

$$\rho_r \propto a^{-4}. \tag{40}$$

Matter obeys

$$\rho_m \propto a^{-3}. \tag{41}$$

The fields evolve canonically:

$$\ddot{C} + 3H\dot{C} + V_{,C} = 0, \tag{42}$$

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0. \tag{43}$$

The post-bounce expansion therefore contains an ordinary radiation era, an ordinary matter era, and a later saturation-residual era.

14.7 The nonsingular background

The final locked trajectory gives

$$a_{\rm min} \simeq 2.068 \times 10^{-5}. \tag{44}$$

At the bounce,

$$H = 0, \qquad \rho = \rho_s, \tag{45}$$

and all field variables remain finite.

Matter-radiation equality occurs at

$$a_{\rm eq} \simeq 2.865 \times 10^{-4}. \tag{46}$$

Therefore

$$a_{\rm min} \ll a_{\rm eq}. \tag{47}$$

A clear radiation-dominated expanding interval exists between the bounce and equality. The bounce is consequently well separated from the later acoustic and matter eras.

14.8 The locked cosmological parameters

The final retained background is

$$\begin{aligned}
\Omega_{m0} &= 0.314114, &(48)\\
\Omega_{r0} &= 9 \times 10^{-5}, &(49)\\
H_0 &= 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, &(50)\\
\rho_s &= 1.58199 \times 10^{15}, &(51)\\
V_\infty &= 2.187987, &(52)\\
m_C^2 &= 1.827221, &(53)\\
\lambda &= 0.802722, &(54)\\
C_{\rm initial} &= 0.918318, &(55)\\
\Pi_{C,\rm initial} &= 0.055327. &(56)
\end{aligned}$$

The final CAMB transition is

$$a_{\rm lock} = 2.05 \times 10^{-4}, \tag{57}$$

with width

$$W = 0.67 W_0. \tag{58}$$

The background is normalized so that

$$E(1) = 1. \tag{59}$$

14.9 Acoustic geometry

The completed CAMB calculation gives

$$100\theta_* = 1.041033. \tag{60}$$

Relative to

$$1.0411, \tag{61}$$

the difference is

$$-0.0064\%. \tag{62}$$

The final drag horizon is

$$r_{\rm drag} = 149.960\ \mathrm{Mpc}. \tag{63}$$

The recombination and drag redshifts are

$$z_* = 1089.691, \qquad z_{\rm drag} = 1058.511. \tag{64}$$

The comoving distance to last scattering is

$$D_M(z_*) = 14131.86 \text{ Mpc}. \tag{65}$$

The acoustic scale is therefore reproduced by a larger sound horizon together with a larger comoving distance.

14.10 BAO

The final BAO observables were computed from the same locked background and the same CAMB-derived drag horizon.

The quantities are

$$\frac{D_M}{r_{\rm drag}}, \qquad \frac{D_H}{r_{\rm drag}}, \qquad \frac{D_V}{r_{\rm drag}}. \tag{66}$$

The resulting diagnostic statistic is approximately

$$\chi^2_{\rm BAO} \simeq 13.4. \tag{67}$$

The improvement over the earlier value near

$$27 \tag{68}$$

followed from using the model's own consistent sound horizon.

The BAO ladder is therefore coherent with the locked expansion history.

14.11 Growth

The final matter fluctuation amplitude is

$$\sigma_8 = 0.779680. \tag{69}$$

The weak-lensing combination is

$$S_8 = 0.79781. \tag{70}$$

The matched control gives approximately

$$S_{8,\rm control} = 0.81304. \tag{71}$$

The locked model lowers S_8 by approximately

$$1.87\%. \tag{72}$$

This reduction is produced by the background history rather than by an imposed weakening of Newton's constant.

The canonical fields themselves alter σ_8 only negligibly once the background is locked.

14.12 Canonical perturbations

The linear field equations are

$$\delta C'' + 2\mathcal{H}\delta C' + \big(k^2 + a^2 V_{,CC}\big)\delta C + \frac{1}{2}C'_{\rm bg}h' = 0, \tag{73}$$

and

$$\delta\phi'' + 2\mathcal{H}\delta\phi' + \big(k^2 + a^2 V_{,\phi\phi}\big)\delta\phi + \frac{1}{2}\phi'_{\rm bg}h' = 0. \tag{74}$$

The measured amplitudes are

$$\max|\delta C| = 0.03514, \qquad \max|\delta\phi| = 0.04860. \tag{75}$$

The kinetic matrix is positive.

The gradient terms are positive.

The effective masses near the lock are positive.

No ghost, gradient instability, or runaway linear mode appears.

14.13 Spectator behavior

Near the locked state,

$$C'_{\rm bg} \simeq 0, \qquad \phi'_{\rm bg} \simeq 0, \qquad V_{,C} \simeq 0, \qquad V_{,\phi} \simeq 0. \tag{76}$$

Therefore the canonical perturbative stress-energy is suppressed.

The direct changes to the primary acoustic peaks and σ_8 are approximately at the level of

$$10^{-6} \tag{77}$$

or smaller.

The model's dominant observable effect is the expansion history.

This establishes the central numerical hierarchy:

$$\boxed{\text{locked background} \gg \text{direct linear field back-reaction}} \tag{78}$$

14.14 Primary CMB

The completed CAMB calculation produced the full

$$TT, \qquad TE, \qquad EE \tag{79}$$

spectra.

The acoustic peak positions, polarization structure, damping behavior, and likelihood quantities were evaluated within the completed workflow.

The residuals relative to the matched control are largest at low multipoles and across the principal acoustic region.

For TT, the RMS differences are approximately

$$28.2\% \quad \text{for} \quad 2 \leq \ell \leq 29, \tag{80}$$

$$10.4\% \quad \text{for} \quad 30 \leq \ell \leq 600, \tag{81}$$

$$3.0\% \quad \text{for} \quad 601 \leq \ell \leq 1500, \tag{82}$$

$$1.7\% \quad \text{for} \quad 1501 \leq \ell \leq 2500. \tag{83}$$

The residuals are scale dependent and cannot be removed by a single amplitude rescaling.

The completed primary-CMB analysis therefore identifies the detailed spectral structure of the locked model rather than only its acoustic angle.

14.15 CMB lensing

The locked CMB lensing-potential spectrum is mildly enhanced relative to the matched control.

The weighted ratio over

$$40 \leq L \leq 2000 \tag{84}$$

is approximately

$$1.0209. \tag{85}$$

Thus the physical CMB lensing power is enhanced by approximately

$$2.09\%. \tag{86}$$

This occurs despite the lower present-day value of S_8.

The result follows from the altered line-of-sight geometry and potential history.

14.16 Official Planck lensing result

The final model was evaluated against the official Planck 2018 nine-bin lensing bandpowers, windows, and covariance matrix.

The locked result is

$$\chi^2_{\text{locked}} = 10.7379 \tag{87}$$

for nine bins.

The matched control gives

$$\chi^2_{\text{control}} = 12.0022. \tag{88}$$

Therefore

$$\Delta\chi^2 = -1.2643. \tag{89}$$

The locked p-value is

$$p_{\text{locked}} = 0.294. \tag{90}$$

No individual band exceeds approximately

$$1.6\sigma \tag{91}$$

under the diagonal-pull diagnostic.

The locked model is compatible with the measured Planck lensing bandpowers and gives a modestly lower statistic than the matched control.

14.17 PlanckLens reconstruction

The PlanckLens TT quadratic-estimator forecast gives

$$\frac{N^{(0)}_{L,\text{locked}}}{N^{(0)}_{L,\text{control}}} \simeq 1.21. \tag{92}$$

The locked primary spectrum therefore produces approximately

$$21\% \tag{93}$$

higher TT reconstruction noise under the common filtering assumptions.

This does not contradict the enhancement in the true lensing spectrum.

The true signal and reconstruction noise measure different properties.

The result confirms that the altered TT transfer function changes reconstruction efficiency even when the physical lensing power is slightly larger.

14.18 Failure of phenomenological substitutes

The tested effective modifications included:

$$f_{\text{sat}}(a,k), \qquad \mu(a,k), \tag{94}$$

and the slip closures

$$\eta = 1, \qquad \eta = \frac{1}{\mu}, \qquad \eta = 2 - \mu. \tag{95}$$

These models increased σ_8 and distorted the odd-even TT peak structure.

None reproduced the canonical-field result.

The numerical failures demonstrate that gravitational saturation is not equivalent to

$$G \to \mu G. \tag{96}$$

The model requires the actual locked background and canonical stress-energy.

14.19 Robustness

The locked center was perturbed by approximately

$$\pm 0.5\% \qquad \text{and} \qquad \pm\, 1\% \tag{97}$$

in each principal parameter direction.

Of 28 nearby points,

$$25 \tag{98}$$

retained the required behavior.

The local robustness classification is therefore

$$\boxed{\text{ROBUST}}. \tag{99}$$

The most sensitive physical direction is

$$V_\infty. \tag{100}$$

The acoustic scale remains sensitive to the early transition endpoint and width, as expected for a precision integrated observable.

14.20 What the calculation establishes

The completed numerical work establishes that the gravitational-saturation equations admit a continuous cosmological solution with:

$$\begin{aligned}
&a_{\min} > 0, \\
&\rho \le \rho_s, \\
&K_{\text{bounce}} < \infty, \\
&\Gamma_{\text{tr}} = 0 \quad \text{for} \quad H \ge 0, \\
&\rho_r \gg \rho_C \quad \text{during the post-bounce radiation era}, \\
&C \to C_s, \\
&\phi \to 0, \\
&w_C \to -1.
\end{aligned} \tag{101}$$

The same solution produces the completed acoustic, BAO, growth, CMB, and lensing results.

14.21 Physical interpretation

The numerical solution supports the interpretation that gravity can reach limiting behavior in two connected senses.

At high density,

$$\rho \to \rho_s, \tag{102}$$

and the gravitational expansion or contraction rate vanishes.

This produces the bounce.

At late time,

$$C \to C_s, \qquad \phi \to 0, \qquad K^2 \to 0. \tag{103}$$

This produces the locked residual.

The early and late cosmological limits are therefore two expressions of gravitational saturation.

14.22 The wave basis of the theory

The theory begins from the established fact that gravity carries propagating wave degrees of freedom.

A wave-bearing system can possess:

$$\text{phase}, \quad \text{interference}, \quad \text{restoration}, \quad \text{damping}, \quad \text{ringdown}, \quad \text{limiting amplitude}. \tag{104}$$

The field ϕ carries the phase structure.

The field C carries the organizational approach to saturation.

The high-density constraint limits the background gravitational response.

The potential minimum supplies the final lock.

The theory therefore gives mathematical form to the physical proposition that gravity can organize and saturate.

14.23 Dark-energy-like behavior

The late residual satisfies

$$p_C \simeq -\rho_C. \tag{105}$$

This produces accelerated expansion.

Within the theory, the negative-pressure behavior is not assigned to a separate dark-energy substance.

It follows from potential domination:

$$K^2 \ll V. \tag{106}$$

The residual is the stored energy of the saturated configuration.

Thus the phenomenon conventionally represented by dark energy is interpreted as gravitational residual saturation.

14.24 Dark-matter-like behavior

At galactic scales, the organizational response is represented by

$$g\mu\left(\frac{g}{g_*}\right) = g_N, \tag{107}$$

with

$$\mu(x) = \frac{x}{\sqrt{1+x^2}}. \tag{108}$$

In the low-acceleration limit,

$$g \simeq \sqrt{g_N g_*}. \tag{109}$$

This gives

$$v^4 \simeq GM_b g_*. \tag{110}$$

The galactic response and cosmic residual are interpreted as different regimes of one gravitational-organizational principle.

The cosmological model establishes the global locked background.

The galactic relation expresses the weak-acceleration response of incomplete local organization.

14.25 Local gravity

The locked field is near a positive-curvature minimum.

Its background velocity and perturbative sources are small.

Therefore the model does not require a strong unscreened late-time scalar force.

Local high-acceleration gravity can remain close to its standard form while the global background retains the integrated consequences of the saturation history.

This separation between local response and global history is one of the theory's central features.

14.26 The final cosmological picture

The complete physical sequence is

$$\boxed{\begin{array}{l}\text{contraction,}\\ \text{organizational kinetic growth,}\\ \text{regulated transfer into radiation,}\\ \text{finite-density gravitational saturation,}\\ \text{bounce,}\\ \text{radiation domination,}\\ \text{matter domination,}\\ \text{field damping,}\\ \text{gravitational lock,}\\ \text{late residual acceleration.}\end{array}} \tag{111}$$

The same fields exist throughout the entire sequence.

No unrelated late-time component is attached to the early model.

No singular bridge is inserted at the bounce.

No phenomenological gravitational rescaling is retained.

The Universe evolves through one continuous saturation system.

14.27 Final numerical statement

The locked model yields

$$\boxed{100\theta_* = 1.041033} \tag{112}$$

$$\boxed{r_{\mathrm{drag}} = 149.960 \text{ Mpc}} \tag{113}$$

$$\boxed{S_8 = 0.79781} \tag{114}$$

$$\boxed{\max|\delta C| = 0.03514} \tag{115}$$

$$\boxed{\max|\delta\phi| = 0.04860} \tag{116}$$

$$\boxed{\chi^2_{\mathrm{BAO}} \simeq 13.4} \tag{117}$$

$$\boxed{\chi^2_{\text{Planck lensing}} = 10.7379} \tag{118}$$

for nine bins, with

$$\boxed{\Delta\chi^2_{\mathrm{locked-control}} = -1.2643}. \tag{119}$$

These results arise from one locked background and one canonical perturbation system.

14.28 Final theoretical statement

The theory presented here proposes that gravity does more than attract and curve.

It propagates.

It carries phase.

It organizes.

It reaches dynamic balance.

It saturates.

The saturation of the high-density response prevents unrestricted collapse of the cosmological background.

The saturation of the organizational fields produces a stable residual state.

The resulting expansion history remains compatible with the completed cosmological calculations.

The final theoretical statement is therefore:

$$\boxed{\begin{array}{l}\text{a wave-bearing gravitational field can reach a finite organizational limit,}\\ \text{and that limit can govern both the nonsingular origin and late expansion of the Universe.}\end{array}} \tag{120}$$

14.29 Closing conclusion

The gravitational-saturation framework has been carried from physical hypothesis to explicit mathematics, from mathematics to numerical integration, and from numerical integration to cosmological observables.

It produces a finite-density bounce.

It preserves energy conservation.

It retains positive canonical kinetic terms.

It contains stable field-space minima.

It produces a radiation-dominated post-bounce branch.

It evolves toward a nearly constant late residual.

It reproduces the acoustic scale.

It produces a coherent BAO ruler.

It lowers the growth amplitude.

It maintains bounded linear perturbations.

It generates complete primary CMB spectra.

It remains compatible with the Planck 2018 lensing bandpowers.

The completed result is a concrete cosmological theory in which the wave nature of gravity permits organization, phase response, saturation, rebound, and lock.

The Universe described by this theory is neither born from an infinite gravitational rate nor driven at late time by an unrelated added substance.

It is a continuous gravitational system whose own dynamics carry it from contraction through saturation into the expanding state observed today.

End of Chapter 14

End Matter

The substantive theory, calculations, conclusions, and technical appendices are complete. The remaining material is the formal end matter required to close the manuscript.

Data and Code Availability

The gravitational-saturation model was evaluated through a numerical workflow containing:

$$\begin{aligned}
&\text{nonsingular background integration,}\\
&\text{dual-solver verification,}\\
&\text{locked-background construction,}\\
&\text{modified CAMB evolution,}\\
&\text{canonical } C,\phi \text{ perturbations,}\\
&\text{primary-CMB calculations,}\\
&\text{BAO and growth diagnostics,}\\
&\text{PlanckLens reconstruction forecasts,}\\
&\text{Planck 2018 lensing-bandpower evaluation.}
\end{aligned} \tag{1}$$

The frozen parameter set required to reproduce the retained model is:

$$\Omega_{m0} = 0.314114, \tag{2}$$

$$\Omega_{r0} = 9 \times 10^{-5}, \tag{3}$$

$$H_0 = 67.4 \text{ km}\,\text{s}^{-1}\,\text{Mpc}^{-1}, \tag{4}$$

$$V_\infty = 2.187987, \tag{5}$$

$$\rho_s = 1.58199 \times 10^{15}, \tag{6}$$

$$m_C^2 = 1.827221, \tag{7}$$

$$\lambda = 0.802722, \tag{8}$$

$$C_s = 0.95, \tag{9}$$

$$C_{\text{initial}} = 0.918318, \tag{10}$$

$$\Pi_{C,\text{initial}} = 0.055327, \tag{11}$$

$$a_{\text{lock}} = 2.05 \times 10^{-4}, \tag{12}$$

$$W = 0.67W_0. \tag{13}$$

The primordial inputs are:

$$\omega_b = 0.022, \tag{14}$$
$$\omega_c \simeq 0.12005, \tag{15}$$
$$A_s = 2.0 \times 10^{-9}, \tag{16}$$
$$n_s = 0.965, \tag{17}$$
$$\tau_{\rm reio} = 0.054. \tag{18}$$

The numerical equations, parameter definitions, transition rules, perturbative stress-energy terms, statistical diagnostics, and reproduction gates are provided in Appendices A through F.

The intended reproduction package contains:

`locked_parameters.json`,

`background_dop853.csv`,

`background_radau.csv`,

`background_solver_comparison.csv`,

`locked_background_table.csv`,

`cmb_locked.csv`,

`cmb_control.csv`,

`lensing_locked.csv`,

`lensing_control.csv`,

`bao_comparison.csv`,

`planck_lensing_comparison.csv`,

`plancklens_noise_comparison.csv`,

`final_locked_summary.json`.

The source archive should additionally include the modified CAMB files, the background integrator, the primary-CMB likelihood workflow, the BAO evaluation script, and the Planck-lensing window and covariance interfaces.

The archived calculation is defined by a no-retuning rule. Reproduction consists of rerunning the frozen model rather than altering its parameters to improve later observables.

Reproduction Targets

An independent implementation should recover approximately:

$$a_{\min} \simeq 2.068 \times 10^{-5}, \tag{19}$$

$$100\theta_* = 1.041033, \tag{20}$$

$$r_{\text{drag}} = 149.960 \text{ Mpc}, \tag{21}$$

$$z_* = 1089.691, \tag{22}$$

$$z_{\text{drag}} = 1058.511, \tag{23}$$

$$D_M(z_*) = 14131.86 \text{ Mpc}, \tag{24}$$

$$\sigma_8 = 0.779680, \tag{25}$$

$$S_8 = 0.79781, \tag{26}$$

$$w_C(0) \simeq -0.99988, \tag{27}$$

$$\max |\delta C| = 0.03514, \tag{28}$$

$$\max |\delta \phi| = 0.04860, \tag{29}$$

$$\chi^2_{\text{BAO}} \simeq 13.4, \tag{30}$$

$$\chi^2_{\text{Planck lensing}} = 10.7379. \tag{31}$$

The matched-control Planck-lensing statistic is

$$\chi^2_{\text{control}} = 12.0022, \tag{32}$$

giving

$$\Delta\chi^2 = -1.2643. \tag{33}$$

Small differences caused by compiler behavior, interpolation, floating-point precision, and library versions are expected. Material differences require investigation of the model normalization, transition implementation, density units, perturbative source conventions, or observational window functions.

Author Contributions

James Lawrence

James Lawrence originated the gravitational-saturation hypothesis and its mechanical interpretation.

His contributions include:

the concept of gravity as cumulative organization,
the proposition that gravitational response can saturate,
the phase-cancellation and dynamic-tension interpretation,
the connection between saturation, nonsingular rebound, and late residual energy,
the hierarchical interpretation of local and cosmological gravity,
the direction of the numerical testing program. (34)

He selected the physical questions to be investigated, interpreted the numerical outcomes, rejected phenomenological closures that failed the intended behavior, and directed the construction of the final theory and manuscript.

Computational Intelligence Assistance

Computational intelligence was used as a mathematical formalization, coding, numerical-analysis, comparison, and editorial tool.

Its contributions included assistance with:

translating the physical premise into canonical field equations,
deriving conservation and perturbation relations,
constructing numerical integration procedures,
developing CAMB modifications,
organizing robustness tests,
evaluating observational diagnostics,
preparing the manuscript. (35)

The physical hypothesis, its interpretive framework, and the decision to pursue gravitational saturation as a unified cosmological mechanism belong to the author.

The calculations should be regarded as computationally assisted work directed and interpreted by the author.

Author Position and Affiliation

Author: James Lawrence
Affiliation: Independent Researcher

The author holds no institutional appointment in cosmology, theoretical physics, or astronomy.

The work was developed independently through mechanical reasoning, numerical experimentation, study of published cosmological observations, and computational assistance.

The absence of institutional affiliation does not alter the mathematical content of the model or the reproducibility of its numerical claims.

The theory is presented for examination on the basis of its equations, calculations, observational consequences, and capacity for independent replication.

Acknowledgments

The author acknowledges the researchers and collaborations whose observational and theoretical work made the present calculation possible.

The model relies upon the accumulated scientific foundations of:

general relativity,
gravitational-wave astronomy,
cosmic microwave background physics,
Boltzmann-code development,
baryon acoustic oscillation measurements, (36)
weak-lensing surveys,
galactic scaling-relation studies,
nonsingular cosmological research.

Particular acknowledgment is due to the developers of CAMB and PlanckLens and to the Planck collaboration for providing the spectra, likelihood products, bandpower windows, and covariance information required for direct numerical testing.

The author also acknowledges the researchers who investigated:

black-hole and bulge scaling relations,
Type Ia supernova progenitor-age effects,
inhomogeneous cosmology, (37)
loop-quantum-cosmological bounce dynamics,
galactic acceleration relations,
gravitational inspiral, merger, and ringdown.

Their observations and calculations are not claimed as inventions of the present theory. They form part of the scientific record from which the gravitational-saturation hypothesis was developed and against which it must be judged.

Competing Interests

The author declares no institutional, corporate, or financial competing interest affecting the presentation of the gravitational-saturation model.

The work was not commissioned by a university, observatory, government laboratory, political organization, or commercial cosmology program.

Funding Statement

The research was conducted independently.

No institutional research grant or external scientific funding supported the development of the model.

Computational work was performed using personally available computing resources and commercially available computational-intelligence systems.

Ethical Statement

The research uses cosmological simulations, published observational products, and theoretical calculations.

It involves no human participants, animal subjects, clinical intervention, or personally identifiable research data.

No institutional ethics approval was required.

Statement on Computational Assistance

Computational intelligence assisted in:

$$\begin{aligned}
&\text{symbolic derivation,}\\
&\text{equation checking,}\\
&\text{numerical-code construction,}\\
&\text{data organization,}\\
&\text{statistical evaluation,}\\
&\text{manuscript preparation.}
\end{aligned} \tag{38}$$

Computational outputs were used as tools within an author-directed research process.

They do not possess authorship, legal responsibility, or independent scientific intent.

The author accepts responsibility for the physical interpretation, presentation, and claims of the manuscript.

Submission Classification

The manuscript may be classified under subjects including:

$$\begin{aligned}
&\text{cosmology and nongalactic astrophysics,}\\
&\text{general relativity and quantum cosmology,}\\
&\text{cosmic microwave background,}\\
&\text{modified cosmological backgrounds,}\\
&\text{nonsingular cosmology,}\\
&\text{gravitational theory.}
\end{aligned} \tag{39}$$

Suggested subject descriptors include:

$$\begin{aligned}&\text{gravitational saturation,}\\&\text{canonical scalar fields,}\\&\text{cosmic bounce,}\\&\text{Boltzmann evolution,}\\&\text{CMB acoustic structure,}\\&\text{BAO,}\\&\text{structure growth,}\\&\text{CMB lensing.}\end{aligned}\tag{40}$$

Suggested Keywords

Gravitational saturation; gravitational organization; gravitational waves; phase dynamics; non-singular cosmology; cosmic bounce; canonical scalar fields; locked expansion history; cosmic microwave background; Boltzmann code; CAMB; baryon acoustic oscillations; structure growth; S_8; CMB lensing; PlanckLens; Planck 2018.

Final Manuscript Statement

The manuscript presents a complete effective cosmological realization of gravitational saturation.

Its mathematical core is:

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right), \tag{41}$$

together with canonical organizational and phase fields governed by

$$V(C,\phi) = V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4 + A_0(1 - \cos\phi). \tag{42}$$

The full evolution includes:

$$\begin{aligned}&\text{regulated contraction,}\\&\text{a finite-density bounce,}\\&\text{radiation domination,}\\&\text{matter domination,}\\&\text{field locking,}\\&\text{late residual acceleration.}\end{aligned}\tag{43}$$

The completed numerical implementation produces:

$$100\theta_* = 1.041033, \tag{44}$$

$$r_{\rm drag} = 149.960 \text{ Mpc}, \tag{45}$$

$$S_8 = 0.79781, \tag{46}$$

bounded canonical perturbations, complete primary CMB spectra, and a Planck-lensing statistic of

$$\chi^2 = 10.7379 \tag{47}$$

for nine bins.

The theory's dominant observable influence is carried by the locked expansion history.

Its physical unity is carried by the proposition that gravity can organize, reach a finite dynamical limit, rebound, and settle into a stable residual state.

End of Manuscript

Appendix A

Complete Locked Equation Set and Parameter Card

A.1 Purpose of the appendix

This appendix consolidates the complete locked gravitational-saturation model into one reference section.

It contains:

$$\begin{gathered}\text{the canonical field potential,}\\ \text{the background field equations,}\\ \text{the contraction-era transfer law,}\\ \text{the high-density saturation constraint,}\\ \text{the affine-time formulation,}\\ \text{the linear perturbation equations,}\\ \text{the cosmological distance relations,}\\ \text{the growth and lensing definitions,}\\ \text{the final locked numerical values.}\end{gathered} \tag{1}$$

No parameter is retuned in this appendix.

The equations and values presented here define the retained unmodulated baseline used in the completed numerical calculation.

A.2 Units and conventions

The cosmological background equations are written in units satisfying

$$c = 1, \tag{2}$$

and, where stated,

$$8\pi G = 1. \tag{3}$$

The spacetime signature is

$$(-, +, +, +). \tag{4}$$

Cosmic time is denoted by

$$t. \tag{5}$$

Conformal time is denoted by

$$\tau, \tag{6}$$

with

$$d\tau = \frac{dt}{a}. \tag{7}$$

A dot denotes differentiation with respect to cosmic time:

$$\dot{X} = \frac{dX}{dt}. \tag{8}$$

A prime denotes differentiation with respect to conformal time:

$$X' = \frac{dX}{d\tau}. \tag{9}$$

The Hubble parameter is

$$H = \frac{\dot{a}}{a}. \tag{10}$$

The conformal Hubble parameter is

$$\mathcal{H} = \frac{a'}{a} = aH. \tag{11}$$

The present scale factor is normalized to

$$a_0 = 1. \tag{12}$$

Redshift is therefore

$$1 + z = \frac{1}{a}. \tag{13}$$

A.3 Gravitational and field action

The canonical gravitational and saturation-field action is

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \frac{1}{2}g^{\mu\nu}\partial_\mu C\partial_\nu C - \frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(C,\phi) + \mathcal{L}_m + \mathcal{L}_r\right]. \tag{14}$$

The fields are

$$C(x^\mu) \qquad \text{and} \qquad \phi(x^\mu). \tag{15}$$

The variable C represents the organizational or saturation state.

The variable ϕ represents the phase sector.

The high-density cosmological response is supplied by the modified background constraint stated below.

A.4 Locked potential

The retained potential is

$$\boxed{V(C,\phi) = V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4 + A_0(1 - \cos\phi)} \tag{16}$$

with

$$C_s = 0.95, \qquad V_\infty = 2.187987, \qquad m_C^2 = 1.827221, \qquad \lambda = 0.802722. \tag{17}$$

The C-derivative is

$$\boxed{V_{,C} = m_C^2(C - C_s) + \lambda(C - C_s)^3} \tag{18}$$

and the phase derivative is

$$\boxed{V_{,\phi} = A_0 \sin\phi}. \tag{19}$$

The second derivatives are

$$\boxed{V_{,CC} = m_C^2 + 3\lambda(C - C_s)^2} \qquad \text{and} \qquad \boxed{V_{,\phi\phi} = A_0 \cos\phi}. \tag{20}$$

The mixed derivative vanishes:

$$\boxed{V_{,C\phi} = 0}. \tag{21}$$

At the joint minimum,

$$C = C_s \qquad \text{and} \qquad \phi = 0, \tag{22}$$

one has

$$V_{,C} = 0, \qquad V_{,\phi} = 0, \qquad V_{,CC} = m_C^2 > 0, \qquad V_{,\phi\phi} = A_0 > 0. \tag{23}$$

The minimum energy is

$$\boxed{V(C_s, 0) = V_\infty}. \tag{24}$$

A.5 Homogeneous canonical variables

For a homogeneous background,

$$C = C(t) \qquad \text{and} \qquad \phi = \phi(t). \tag{25}$$

Define

$$\boxed{\Pi_C = \dot{C}} \qquad \text{and} \qquad \boxed{\Pi_\phi = \dot{\phi}}. \tag{26}$$

The combined kinetic quantity is

$$\boxed{K^2 = \Pi_C^2 + \Pi_\phi^2}. \tag{27}$$

The saturation-sector density is

$$\boxed{\rho_C = \frac{1}{2}\Pi_C^2 + \frac{1}{2}\Pi_\phi^2 + V(C, \phi)} \tag{28}$$

or

$$\rho_C = \frac{1}{2}K^2 + V. \tag{29}$$

The pressure is

$$\boxed{p_C = \frac{1}{2}\Pi_C^2 + \frac{1}{2}\Pi_\phi^2 - V(C, \phi)} \tag{30}$$

or

$$p_C = \frac{1}{2}K^2 - V. \tag{31}$$

The equation-of-state parameter is

$$\boxed{w_C = \frac{p_C}{\rho_C} = \frac{\frac{1}{2}K^2 - V}{\frac{1}{2}K^2 + V}}. \tag{32}$$

The identity

$$\boxed{\rho_C + p_C = K^2} \tag{33}$$

will be used repeatedly.

A.6 Matter and radiation

Pressureless matter satisfies

$$p_m = 0. \tag{34}$$

Its continuity equation is

$$\boxed{\dot{\rho}_m = -3H\rho_m}. \tag{35}$$

The solution is

$$\boxed{\rho_m = \rho_{m0}a^{-3}}. \tag{36}$$

Radiation satisfies

$$p_r = \frac{1}{3}\rho_r. \tag{37}$$

With transfer included,

$$\boxed{\dot{\rho}_r = -4H\rho_r + Q}. \tag{38}$$

After transfer shuts off,

$$Q = 0, \tag{39}$$

and therefore

$$\boxed{\rho_r = \rho_{r0}a^{-4}}. \tag{40}$$

A.7 Contraction selector

The transfer acts only on the contracting branch.

Define the one-sided contraction selector

$$\boxed{W_-(H) = \begin{cases} W_{\text{smooth}}(H), & H < 0, \\ 0, & H \geq 0, \end{cases}} \tag{41}$$

where W_{smooth} approaches unity during sufficiently deep contraction and approaches zero smoothly as

$$H \to 0^-. \tag{42}$$

A representative smooth form is

$$W_{\text{smooth}}(H) = \frac{1}{1+\exp(H/H_c)} \tag{43}$$

on the contracting branch.

The defining locked condition is

$$\boxed{W_-(H) = 0 \qquad \text{for} \qquad H \geq 0}. \tag{44}$$

A.8 Density activation

The target density is

$$\boxed{\rho_{\text{tar}} = \rho_m + \rho_r}. \tag{45}$$

The transfer activation function is

$$\boxed{S_\rho = \frac{(\rho_{\text{tar}}/\rho_\Gamma)^m}{1+(\rho_{\text{tar}}/\rho_\Gamma)^m}}. \tag{46}$$

At low target density,

$$\rho_{\text{tar}} \ll \rho_\Gamma, \tag{47}$$

one has

$$S_\rho \to 0. \tag{48}$$

At high target density,

$$\rho_{\text{tar}} \gg \rho_\Gamma, \tag{49}$$

one has

$$S_\rho \to 1. \tag{50}$$

A.9 Transfer rate

The contraction-era transfer rate is

$$\boxed{\Gamma_{\text{tr}} = W_-(H)\big[-3H + \Gamma_0 S_\rho\big]}. \tag{51}$$

Deep in contraction,

$$W_- \simeq 1, \tag{52}$$

so

$$3H + \Gamma_{\text{tr}} \simeq \Gamma_0 S_\rho. \tag{53}$$

The negative Hubble-friction coefficient is thereby replaced by nonnegative damping.

At the bounce and during expansion,

$$H \geq 0, \tag{54}$$

so

$$\boxed{\Gamma_{\mathrm{tr}} = 0}. \tag{55}$$

A.10 Energy transfer

The transfer power is

$$\boxed{Q = \Gamma_{\mathrm{tr}} K^2} \tag{56}$$

with

$$K^2 = \Pi_C^2 + \Pi_\phi^2. \tag{57}$$

The field sector obeys

$$\boxed{\dot{\rho}_C + 3H(\rho_C + p_C) = -Q}. \tag{58}$$

Radiation obeys

$$\boxed{\dot{\rho}_r + 4H\rho_r = +Q}. \tag{59}$$

Adding the two gives

$$\dot{\rho}_C + \dot{\rho}_r + 3H(\rho_C + p_C) + 4H\rho_r = 0. \tag{60}$$

Including matter,

$$\dot{\rho}_m + 3H\rho_m = 0, \tag{61}$$

the total conservation equation is

$$\boxed{\dot{\rho} + 3H(\rho + p) = 0}. \tag{62}$$

A.11 Total density and pressure

The total density is

$$\boxed{\rho = \rho_m + \rho_r + \rho_C}. \tag{63}$$

The total pressure is

$$\boxed{p = \frac{1}{3}\rho_r + p_C}. \tag{64}$$

Explicitly,

$$\rho = \rho_m + \rho_r + \frac{1}{2}K^2 + V, \tag{65}$$

and

$$p = \frac{1}{3}\rho_r + \frac{1}{2}K^2 - V. \tag{66}$$

Therefore

$$\rho + p = \rho_m + \frac{4}{3}\rho_r + K^2. \tag{67}$$

This quantity is nonnegative for the canonical matter, radiation, and field components used in the locked solution.

A.12 High-density saturation constraint

The generalized homogeneous constraint is

$$\boxed{H^2 = \frac{\rho}{3}\left[1 - \left(\frac{\rho}{\rho_s}\right)^q\right]}. \tag{68}$$

The locked implementation uses

$$\boxed{q = 1}. \tag{69}$$

Therefore

$$\boxed{H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right)}. \tag{70}$$

The locked saturation density is

$$\boxed{\rho_s = 1.58199 \times 10^{15}} \tag{71}$$

in the adopted background normalization.

At low density,

$$\rho \ll \rho_s, \tag{72}$$

the equation reduces to

$$H^2 \simeq \frac{\rho}{3}. \tag{73}$$

At

$$\rho = \rho_s, \tag{74}$$

one has

$$\boxed{H = 0}. \tag{75}$$

A.13 Raychaudhuri equation

Differentiating the generalized constraint and using total conservation gives

$$\boxed{\dot{H} = -\frac{1}{2}(\rho + p)\left[1 - (q+1)\left(\frac{\rho}{\rho_s}\right)^q\right]}. \tag{76}$$

For

$$q = 1, \tag{77}$$

this becomes

$$\boxed{\dot{H} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right)}. \tag{78}$$

At the bounce,

$$\rho = \rho_s, \tag{79}$$

so

$$\boxed{\dot{H}_{\mathrm{b}} = \frac{1}{2}(\rho_{\mathrm{b}} + p_{\mathrm{b}})}. \tag{80}$$

Since the locked components satisfy

$$\rho_{\mathrm{b}} + p_{\mathrm{b}} > 0, \tag{81}$$

one has

$$\dot{H}_{\mathrm{b}} > 0. \tag{82}$$

A.14 Acceleration equation

The scale factor satisfies

$$\dot{a} = aH. \tag{83}$$

Therefore

$$\frac{\ddot{a}}{a} = \dot{H} + H^2. \tag{84}$$

For $q = 1$,

$$\boxed{\frac{\ddot{a}}{a} = -\frac{1}{6}(\rho + 3p) + \frac{\rho}{\rho_s}\left(\frac{2}{3}\rho + p\right)}. \tag{85}$$

At low density,

$$\rho/\rho_s \to 0, \tag{86}$$

this reduces to

$$\frac{\ddot{a}}{a} \simeq -\frac{1}{6}(\rho + 3p). \tag{87}$$

At the bounce,

$$H = 0, \tag{88}$$

and

$$\boxed{\frac{\ddot{a}_{\mathrm{b}}}{a_{\mathrm{b}}} = \frac{1}{2}(\rho_{\mathrm{b}} + p_{\mathrm{b}}) > 0}. \tag{89}$$

A.15 Maximum gravitational rate

Define

$$x = \frac{\rho}{\rho_s}. \tag{90}$$

Then

$$H^2 = \frac{\rho_s}{3}x(1 - x). \tag{91}$$

The maximum occurs at

$$x = \frac{1}{2}. \tag{92}$$

Therefore

$$\boxed{H^2_{\max} = \frac{\rho_s}{12}} \qquad \text{and} \qquad \boxed{|H|_{\max} = \sqrt{\frac{\rho_s}{12}}}. \tag{93}$$

The expansion and contraction rates are finite.

A.16 Complete cosmic-time system

The retained homogeneous state vector is

$$\boxed{\mathbf{y} = \big(a, H, C, \Pi_C, \phi, \Pi_\phi, \rho_m, \rho_r\big)}. \tag{94}$$

The complete first-order system is

$$\boxed{\dot{a} = aH}, \tag{95}$$

$$\boxed{\dot{H} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right)}, \tag{96}$$

$$\boxed{\dot{C} = \Pi_C}, \tag{97}$$

$$\boxed{\dot{\Pi}_C = -(3H + \Gamma_{\rm tr})\Pi_C - V_{,C}}, \tag{98}$$

$$\boxed{\dot{\phi} = \Pi_\phi}, \tag{99}$$

$$\boxed{\dot{\Pi}_\phi = -(3H + \Gamma_{\rm tr})\Pi_\phi - V_{,\phi}}, \tag{100}$$

$$\boxed{\dot{\rho}_m = -3H\rho_m}, \tag{101}$$

$$\boxed{\dot{\rho}_r = -4H\rho_r + \Gamma_{\rm tr}(\Pi_C^2 + \Pi_\phi^2)}. \tag{102}$$

The algebraic definitions are

$$K^2 = \Pi_C^2 + \Pi_\phi^2, \tag{103}$$

$$\rho_C = \frac{1}{2}K^2 + V, \tag{104}$$

$$p_C = \frac{1}{2}K^2 - V, \tag{105}$$

$$\rho = \rho_m + \rho_r + \rho_C, \tag{106}$$

$$p = \frac{1}{3}\rho_r + p_C. \tag{107}$$

A.17 Constraint residual

The numerical Friedmann residual is

$$\boxed{\mathcal{C}_H = H^2 - \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right)}. \tag{108}$$

An exact solution satisfies

$$\mathcal{C}_H = 0. \tag{109}$$

A normalized diagnostic is

$$\boxed{\epsilon_H = \frac{\left|H^2 - \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right)\right|}{\max\left[H^2, \frac{\rho}{3}, \epsilon_{\rm floor}\right]}}. \tag{110}$$

The nonzero floor prevents an artificial division by zero near the bounce.

A.18 Constraint projection

At numerical checkpoints, the magnitude of H may be projected onto the constraint:

$$\boxed{H \to \operatorname{sgn}(H)\sqrt{\frac{\rho}{3}\left(1-\frac{\rho}{\rho_s}\right)}}. \tag{111}$$

The projection controls drift but does not replace the evolution equation for H.

The sign is negative before the bounce and positive after the bounce.

A.19 Nonsingular affine variables

Define the scaled variables

$$\boxed{h = \frac{H}{\sqrt{\rho_s}}}, \tag{112}$$

$$\boxed{u_C = \frac{\Pi_C}{\sqrt{\rho_s}}}, \tag{113}$$

$$\boxed{u_\phi = \frac{\Pi_\phi}{\sqrt{\rho_s}}}, \tag{114}$$

$$\boxed{\bar{\rho} = \frac{\rho}{\rho_s}}, \tag{115}$$

$$\boxed{\bar{p} = \frac{p}{\rho_s}}. \tag{116}$$

Define affine time by

$$\boxed{d\tau_{\rm aff} = \sqrt{\rho_s}\, dt}. \tag{117}$$

The locked constraint becomes

$$\boxed{h^2 = \frac{\bar{\rho}}{3}(1-\bar{\rho})}. \tag{118}$$

The scale-factor equation is

$$\boxed{\frac{da}{d\tau_{\rm aff}} = ah}. \tag{119}$$

The field-position equations are

$$\boxed{\frac{dC}{d\tau_{\rm aff}} = u_C} \quad \text{and} \quad \boxed{\frac{d\phi}{d\tau_{\rm aff}} = u_\phi}. \tag{120}$$

The scaled Raychaudhuri equation is

$$\boxed{\frac{dh}{d\tau_{\rm aff}} = -\frac{1}{2}(\bar{\rho}+\bar{p})(1-2\bar{\rho})}. \tag{121}$$

No equation is divided by H or h.

A.20 Prohibited singular variable through the bounce

The variable

$$N = \ln a \tag{122}$$

satisfies

$$\frac{d}{dN} = \frac{1}{H}\frac{d}{dt}. \tag{123}$$

Therefore

$$\frac{dC}{dN} = \frac{\Pi_C}{H}, \qquad \frac{d\phi}{dN} = \frac{\Pi_\phi}{H}, \qquad \frac{d\rho_r}{dN} = -4\rho_r + \frac{Q}{H}. \tag{124}$$

These expressions are singular at

$$H = 0. \tag{125}$$

The locked integration rule is therefore

$$\boxed{\text{No } N = \ln a \text{ integration through the bounce}}. \tag{126}$$

The bounce is evolved only with nonsingular cosmic-time or affine-time equations.

A.21 Piecewise integration rule

The trajectory is integrated in multiple nonsingular segments.

A representative division is

$$\text{contraction} \to \text{bounce} \to a \sim 10^{-3}, \qquad a \sim 10^{-3} \to a \sim 10^{-2}, \qquad a \sim 10^{-2} \to a = 1. \tag{127}$$

The complete state vector is passed from one segment to the next.

The physical equations do not change between segments.

The locked geometry rule is

$$\boxed{\text{Piecewise nonsingular integration only}}. \tag{128}$$

A.22 Dual-solver rule

Every retained background geometry result is verified with

$$\boxed{\text{DOP853}} \qquad \text{and} \qquad \boxed{\text{Radau}}. \tag{129}$$

The compared quantities include

$$a_{\min}, \quad K_{\text{bounce}}, \quad \rho_C(0), \quad C(0), \quad E(z), \quad r_{\text{drag}}, \quad 100\theta_*, \tag{130}$$

and the BAO observables.

The accepted geometry requires solver agreement much smaller than one percent.

A.23 Expanding-branch conversion

After the bounce, the scale factor is monotonic.

The redshift relation is

$$\boxed{z = \frac{1}{a} - 1}. \tag{131}$$

The expansion function is

$$\boxed{E(z) = \frac{H(z)}{H_0}}. \tag{132}$$

The physical Hubble rate is

$$\boxed{H(z) = H_0 E(z)}. \tag{133}$$

The final Hubble constant is

$$\boxed{H_0 = 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}}. \tag{134}$$

The retained normalization satisfies

$$\boxed{E(0) = 1}. \tag{135}$$

Equivalently,

$$E(a = 1) = 1. \tag{136}$$

A.24 Locked transition

The expanding background is passed into CAMB through a smooth transition ending at

$$\boxed{a_{\mathrm{lock}} = 2.05 \times 10^{-4}}. \tag{137}$$

The final transition width is

$$\boxed{W = 0.67 W_0}. \tag{138}$$

A generic smoothstep variable is

$$x = \frac{a - a_{\mathrm{start}}}{a_{\mathrm{lock}} - a_{\mathrm{start}}}. \tag{139}$$

The cubic smoothstep is

$$\boxed{S(x) = 3x^2 - 2x^3}. \tag{140}$$

It satisfies

$$S(0) = 0, \qquad S(1) = 1, \qquad S'(0) = 0, \qquad S'(1) = 0. \tag{141}$$

A schematic background blend is

$$\boxed{E^2_{\mathrm{blend}}(a) = [1 - S(a)]E^2_{\mathrm{early}}(a) + S(a)E^2_{\mathrm{locked}}(a)}. \tag{142}$$

The final implementation preserves a smooth background and first derivative through the transition.

A.25 Cosmological density parameters

The locked matter fraction is

$$\boxed{\Omega_{m0} = 0.314114}. \tag{143}$$

The locked radiation fraction is

$$\boxed{\Omega_{r0} = 9 \times 10^{-5}}. \tag{144}$$

The physical baryon density is

$$\boxed{\omega_b = \Omega_b h^2 = 0.022}. \tag{145}$$

The physical cold-matter input is approximately

$$\boxed{\omega_c \simeq 0.12005}. \tag{146}$$

The primordial scalar index is

$$\boxed{n_s = 0.965}. \tag{147}$$

The primordial scalar amplitude is

$$\boxed{A_s = 2.0 \times 10^{-9}}. \tag{148}$$

The reionization optical depth is

$$\boxed{\tau_{\text{reio}} = 0.054}. \tag{149}$$

A.26 Locked initial field values

The retained organizational initial value is

$$\boxed{C_{\text{initial}} = 0.918318}. \tag{150}$$

The retained initial momentum is

$$\boxed{\Pi_{C,\text{initial}} = 0.055327}. \tag{151}$$

The saturation value is

$$\boxed{C_s = 0.95}. \tag{152}$$

The initial displacement from saturation is

$$C_{\text{initial}} - C_s = -0.031682. \tag{153}$$

Thus the initial organizational field begins below the stable saturation minimum and evolves toward it.

A.27 Bounce values

The minimum scale factor is approximately

$$\boxed{a_{\min} \simeq 2.068 \times 10^{-5}}. \tag{154}$$

The corresponding redshift is

$$\boxed{z_{\text{bounce}} \simeq 4.84 \times 10^{4}}. \tag{155}$$

At the bounce,

$$\boxed{H_{\text{b}} = 0} \qquad \text{and} \qquad \boxed{\rho_{\text{b}} = \rho_s}. \tag{156}$$

The field kinetic quantity remains finite:

$$\boxed{K^2_{\text{bounce}} = \Pi^2_{C,\text{b}} + \Pi^2_{\phi,\text{b}} < \infty}. \tag{157}$$

The bounce satisfies

$$\boxed{\dot{H}_{\text{b}} > 0}. \tag{158}$$

A.28 Matter-radiation equality

Equality is defined by

$$\rho_m(a_{\text{eq}}) = \rho_r(a_{\text{eq}}). \tag{159}$$

Therefore

$$\boxed{a_{\text{eq}} = \frac{\Omega_{r0}}{\Omega_{m0}}}. \tag{160}$$

Using the locked values,

$$\boxed{a_{\text{eq}} \simeq 2.865 \times 10^{-4}}. \tag{161}$$

The equality redshift is

$$\boxed{z_{\text{eq}} \simeq 3489}. \tag{162}$$

The bounce-to-equality ratio is

$$\boxed{\frac{a_{\min}}{a_{\text{eq}}} \simeq 0.0722}. \tag{163}$$

The bounce therefore occurs before equality and leaves a clear radiation-dominated expanding interval.

A.29 Late saturation state

At late time,

$$C \to C_s, \qquad \phi \to 0, \qquad \Pi_C \to 0, \qquad \Pi_\phi \to 0. \tag{164}$$

Therefore

$$\boxed{\rho_C \to V_\infty}, \qquad \boxed{p_C \to -V_\infty}, \qquad \boxed{w_C \to -1}. \tag{165}$$

The present equation of state is

$$\boxed{w_C(0) \simeq -0.99988}. \tag{166}$$

The present residual density is approximately

$$\boxed{\rho_C(0) \simeq 2.192} \tag{167}$$

in the adopted normalization.

A.30 Cosmological distances

The Hubble distance is

$$\boxed{D_H(z) = \frac{c}{H(z)}}. \tag{168}$$

For a spatially flat universe, the transverse comoving distance is

$$\boxed{D_M(z) = \int_0^z \frac{c\,dz'}{H(z')}}. \tag{169}$$

The angular-diameter distance is

$$\boxed{D_A(z) = \frac{D_M(z)}{1+z}}. \tag{170}$$

The luminosity distance is

$$\boxed{D_L(z) = (1+z)D_M(z)}. \tag{171}$$

The isotropic BAO distance is

$$\boxed{D_V(z) = \left[zD_M^2(z)D_H(z)\right]^{1/3}}. \tag{172}$$

A.31 Photon-baryon sound speed

The baryon-loading ratio is

$$\boxed{R(z) = \frac{3\rho_b}{4\rho_\gamma}}. \tag{173}$$

The sound speed is

$$\boxed{c_s(z) = \frac{c}{\sqrt{3[1+R(z)]}}}. \tag{174}$$

The recombination sound horizon is

$$\boxed{r_s(z_*) = \int_{z_*}^{\infty} \frac{c_s(z)}{H(z)}\,dz}. \tag{175}$$

The drag horizon is

$$\boxed{r_{\rm drag} = \int_{z_{\rm drag}}^{\infty} \frac{c_s(z)}{H(z)}\,dz}. \tag{176}$$

A.32 Acoustic angle

The angular acoustic scale is

$$\boxed{\theta_* = \frac{r_s(z_*)}{D_M(z_*)}}. \tag{177}$$

CAMB reports

$$\boxed{100\theta_*}. \tag{178}$$

The final locked value is

$$\boxed{100\theta_* = 1.041033}. \tag{179}$$

The reference comparison value is

$$1.0411. \tag{180}$$

The fractional difference is

$$\boxed{\frac{1.041033 - 1.0411}{1.0411} \times 100\% \simeq -0.0064\%}. \tag{181}$$

A.33 Final acoustic quantities

The final drag horizon is

$$\boxed{r_{\text{drag}} = 149.960 \text{ Mpc}}. \tag{182}$$

The recombination redshift is

$$\boxed{z_* = 1089.691}. \tag{183}$$

The drag redshift is

$$\boxed{z_{\text{drag}} = 1058.511}. \tag{184}$$

The comoving distance to last scattering is

$$\boxed{D_M(z_*) = 14131.86 \text{ Mpc}}. \tag{185}$$

The physical angular-diameter distance is

$$\boxed{D_A(z_*) = 12.9568 \text{ Mpc}}. \tag{186}$$

A.34 BAO observables

The anisotropic BAO observables are

$$\boxed{\frac{D_M(z)}{r_{\text{drag}}}} \quad \text{and} \quad \boxed{\frac{D_H(z)}{r_{\text{drag}}}}. \tag{187}$$

The isotropic observable is

$$\boxed{\frac{D_V(z)}{r_{\text{drag}}}}. \tag{188}$$

The final covariance-weighted diagnostic value is

$$\boxed{\chi^2_{\rm BAO} \simeq 13.4}. \tag{189}$$

The same locked background is used for both the distances and

$$r_{\rm drag}. \tag{190}$$

A.35 Linear growth equation

On the expanding branch, define

$$N = \ln a. \tag{191}$$

The standard linear matter-growth equation is

$$\boxed{\frac{d^2\delta_m}{dN^2} + \left(2 + \frac{d\ln H}{dN}\right)\frac{d\delta_m}{dN} - \frac{3}{2}\Omega_m(N)\delta_m = 0}. \tag{192}$$

The matter fraction is

$$\boxed{\Omega_m(a) = \frac{\Omega_{m0}a^{-3}}{E^2(a)}}. \tag{193}$$

The growth rate is

$$\boxed{f(a) = \frac{d\ln\delta_m}{d\ln a}}. \tag{194}$$

The present locked growth rate is approximately

$$\boxed{f(0) \simeq 0.527}. \tag{195}$$

A.36 Fluctuation amplitude

The present matter amplitude is

$$\boxed{\sigma_8 = 0.779680}. \tag{196}$$

The weak-lensing combination is

$$\boxed{S_8 = \sigma_8\sqrt{\frac{\Omega_{m0}}{0.3}}}. \tag{197}$$

The final value is

$$\boxed{S_8 = 0.79781}. \tag{198}$$

The matched control values are approximately

$$\sigma_{8,\rm control} = 0.794561, \qquad S_{8,\rm control} = 0.813037. \tag{199}$$

A.37 Linear saturation perturbations

On the expanding branch,

$$C(\tau,\mathbf{x}) = C_{\rm bg}(\tau) + \delta C(\tau,\mathbf{x}), \qquad \phi(\tau,\mathbf{x}) = \phi_{\rm bg}(\tau) + \delta\phi(\tau,\mathbf{x}). \tag{200}$$

In Fourier space, define

$$v_C = \delta C', \qquad v_\phi = \delta\phi'. \tag{201}$$

The first-order system is

$$\boxed{\delta C' = v_C}, \tag{202}$$

$$\boxed{v_C' = -2\mathcal{H}v_C - \big(k^2 + a^2 V_{,CC}\big)\delta C - \frac{1}{2}C_{\rm bg}'h'}, \tag{203}$$

$$\boxed{\delta\phi' = v_\phi}, \tag{204}$$

$$\boxed{v_\phi' = -2\mathcal{H}v_\phi - \big(k^2 + a^2 V_{,\phi\phi}\big)\delta\phi - \frac{1}{2}\phi_{\rm bg}'h'}. \tag{205}$$

Equivalently, the second-order equations are

$$\boxed{\delta C'' + 2\mathcal{H}\delta C' + \big(k^2 + a^2 V_{,CC}\big)\delta C + \frac{1}{2}C_{\rm bg}'h' = 0} \tag{206}$$

and

$$\boxed{\delta\phi'' + 2\mathcal{H}\delta\phi' + \big(k^2 + a^2 V_{,\phi\phi}\big)\delta\phi + \frac{1}{2}\phi_{\rm bg}'h' = 0}. \tag{207}$$

A.38 CAMB internal metric form

The implemented CAMB equations use an internal synchronous-gauge metric variable kz.

The forcing terms are represented in code as

$$-kz\,C_{\rm bg}' \qquad \text{and} \qquad -kz\,\phi_{\rm bg}'. \tag{208}$$

These correspond to the synchronous-gauge metric forcing written above as

$$-\frac{1}{2}h'C_{\rm bg}' \qquad \text{and} \qquad -\frac{1}{2}h'\phi_{\rm bg}'. \tag{209}$$

The implementation retains the same canonical perturbation structure.

A.39 Linear field stress-energy

The density perturbation is

$$\boxed{\delta\rho_C = \frac{1}{a^2}\left[C_{\rm bg}'v_C + \phi_{\rm bg}'v_\phi\right] + V_{,C}\delta C + V_{,\phi}\delta\phi}. \tag{210}$$

The pressure perturbation is

$$\boxed{\delta p_C = \frac{1}{a^2}\left[C'_{\rm bg} v_C + \phi'_{\rm bg} v_\phi\right] - V_{,C}\delta C - V_{,\phi}\delta\phi}. \tag{211}$$

The momentum perturbation is

$$\boxed{\delta q_C = \frac{k}{a^2}\left[C'_{\rm bg}\delta C + \phi'_{\rm bg}\delta\phi\right]}. \tag{212}$$

The intrinsic anisotropic stress is

$$\boxed{\pi_C = 0}. \tag{213}$$

The CAMB-normalized source forms are

$$a^2\delta\rho_C = C'_{\rm bg} v_C + \phi'_{\rm bg} v_\phi + a^2\big(V_{,C}\delta C + V_{,\phi}\delta\phi\big), \tag{214}$$
$$a^2\delta p_C = C'_{\rm bg} v_C + \phi'_{\rm bg} v_\phi - a^2\big(V_{,C}\delta C + V_{,\phi}\delta\phi\big), \tag{215}$$
$$a^2\delta q_C = k\big(C'_{\rm bg}\delta C + \phi'_{\rm bg}\delta\phi\big). \tag{216}$$

A.40 Kinetic and gradient stability

The field-space kinetic matrix is

$$\boxed{K_{IJ} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}. \tag{217}$$

Its eigenvalues are

$$\boxed{\kappa_1 = 1, \qquad \kappa_2 = 1}. \tag{218}$$

The sound speeds are

$$\boxed{c^2_{s,C} = 1} \qquad \text{and} \qquad \boxed{c^2_{s,\phi} = 1}. \tag{219}$$

The mass matrix is

$$\boxed{M^2_{IJ} = \begin{pmatrix} V_{,CC} & 0 \\ 0 & V_{,\phi\phi} \end{pmatrix}}. \tag{220}$$

Near the locked state,

$$M^2_{IJ} \simeq \begin{pmatrix} m_C^2 & 0 \\ 0 & A_0 \end{pmatrix}. \tag{221}$$

For

$$A_0 > 0, \tag{222}$$

both mass eigenvalues are positive.

A.41 Final field amplitudes

The maximum organizational-field perturbation is

$$\boxed{\max|\delta C| = 0.03514}. \tag{223}$$

The maximum phase-field perturbation is

$$\boxed{\max|\delta\phi| = 0.04860}. \tag{224}$$

The field modes remain bounded.

The direct fractional change in σ_8 due to the canonical perturbations on the already locked background is approximately

$$\boxed{\mathcal{O}(5\times10^{-6})}. \tag{225}$$

The direct peak-height changes are approximately

$$\boxed{10^{-6}\text{–}10^{-7}}. \tag{226}$$

A.42 Primary CMB spectra

The Boltzmann calculation produces

$$\boxed{C_\ell^{TT}}, \qquad \boxed{C_\ell^{EE}}, \qquad \boxed{C_\ell^{BB}}, \qquad \boxed{C_\ell^{TE}}. \tag{227}$$

The plotted spectra are

$$\boxed{D_\ell^{XY} = \frac{\ell(\ell+1)}{2\pi}C_\ell^{XY}}. \tag{228}$$

The final calculation extends to approximately

$$\boxed{\ell_{\max} = 2500}. \tag{229}$$

For the TE cross-spectrum, the stable residual measure is

$$\boxed{\frac{\Delta TE}{\sqrt{TT_{\text{control}}\,EE_{\text{control}}}}}. \tag{230}$$

A.43 Primary spectral residuals

For TT, the RMS residuals relative to the matched control are approximately

$$\boxed{28.20\% \quad (2\le\ell\le29)}, \tag{231}$$

$$\boxed{10.41\% \quad (30\le\ell\le600)}, \tag{232}$$

$$\boxed{2.98\% \quad (601\le\ell\le1500)}, \tag{233}$$

$$\boxed{1.73\% \quad (1501\le\ell\le2500)}. \tag{234}$$

For normalized TE, the RMS residuals are approximately

$$\boxed{12.03\%}, \qquad \boxed{8.16\%}, \qquad \boxed{2.37\%}, \qquad \boxed{1.07\%} \tag{235}$$

over the same bands.

For EE, they are approximately

$$\boxed{6.93\%}, \quad \boxed{5.93\%}, \quad \boxed{4.23\%}, \quad \boxed{2.25\%}. \tag{236}$$

A.44 Amplitude and shape decomposition

The best locked/control TT amplitude is

$$\boxed{A_{TT} = 0.9459}. \tag{237}$$

The remaining TT shape RMS is

$$\boxed{7.90\%}. \tag{238}$$

For EE,

$$\boxed{A_{EE} = 1.0233} \tag{239}$$

with shape RMS

$$\boxed{4.37\%}. \tag{240}$$

For TE,

$$\boxed{A_{TE} = 0.9802} \tag{241}$$

with shape RMS

$$\boxed{4.10\%}. \tag{242}$$

The differences are therefore not removable through one common amplitude rescaling.

A.45 CMB lensing potential

The CMB lensing potential is

$$\boxed{\varphi_{\rm lens}(\hat{\mathbf{n}}) = -2\int_0^{\chi_*} d\chi \, \frac{\chi_* - \chi}{\chi_* \chi} \Phi_W}. \tag{243}$$

The Weyl potential is

$$\boxed{\Phi_W = \frac{\Phi + \Psi}{2}}. \tag{244}$$

The angular lensing spectrum is

$$\boxed{C_L^{\varphi\varphi}}. \tag{245}$$

The weighted locked/control ratio over

$$40 \le L \le 2000 \tag{246}$$

is approximately

$$\boxed{1.0209}. \tag{247}$$

The physical lensing potential is therefore enhanced by approximately

$$\boxed{2.09\%} \tag{248}$$

relative to the matched control over that weighted range.

A.46 Official Planck lensing statistic

The nine-bin model prediction is

$$\boxed{C_b^{\varphi\varphi} = \sum_L W_{bL} C_L^{\varphi\varphi}}. \tag{249}$$

The residual vector is

$$\boxed{\Delta_b = C_{b,\mathrm{model}}^{\varphi\varphi} - C_{b,\mathrm{Planck}}^{\varphi\varphi}}. \tag{250}$$

The covariance statistic is

$$\boxed{\chi^2 = \Delta^{\mathrm{T}} \Sigma^{-1} \Delta}. \tag{251}$$

The locked model gives

$$\boxed{\chi^2_{\mathrm{locked}} = 10.7379} \tag{252}$$

for nine bins.

The matched control gives

$$\boxed{\chi^2_{\mathrm{control}} = 12.0022}. \tag{253}$$

Therefore

$$\boxed{\Delta\chi^2 = -1.2643}. \tag{254}$$

The locked p-value is

$$\boxed{p_{\mathrm{locked}} = 0.294}. \tag{255}$$

A.47 Profiled lensing amplitude

Applying a global lensing amplitude,

$$C_L^{\varphi\varphi} \to A_\varphi C_L^{\varphi\varphi}, \tag{256}$$

the locked best-fit amplitude is approximately

$$\boxed{A_{\varphi,\mathrm{locked}} = 1.0436}. \tag{257}$$

The matched control gives

$$\boxed{A_{\varphi,\mathrm{control}} = 1.0505}. \tag{258}$$

A.48 PlanckLens reconstruction noise

The TT quadratic-estimator Gaussian noise is

$$\boxed{N_L^{(0)}}. \tag{259}$$

The final forecast uses

$$\boxed{\ell_{\max}^{\mathrm{CMB}} = 512}, \tag{260}$$
$$\boxed{L_{\max} = 400}, \tag{261}$$
$$\boxed{\ell_{\min}^{\mathrm{CMB}} = 30}, \tag{262}$$
$$\boxed{\theta_{\mathrm{beam}} = 6\ \mathrm{arcmin}}, \tag{263}$$
$$\boxed{\Delta_T = 35\ \mu\mathrm{K\,arcmin}}. \tag{264}$$

The mean locked/control noise ratio over

$$8 \le L \le 400 \tag{265}$$

is

$$\boxed{\frac{N_{L,\mathrm{locked}}^{(0)}}{N_{L,\mathrm{control}}^{(0)}} \simeq 1.213}. \tag{266}$$

The locked TT reconstruction noise is therefore approximately

$$\boxed{21.3\%} \tag{267}$$

higher under the common forecast assumptions.

A.49 Robustness classification

The locked center was perturbed one parameter at a time by approximately

$$\pm 0.5\% \qquad \text{and} \qquad \pm\, 1\%. \tag{268}$$

The tested parameters were

$$\Omega_{m0}, \quad V_\infty, \quad \rho_s, \quad m_C^2, \quad \lambda, \quad C_{\mathrm{initial}}, \quad \Pi_{C,\mathrm{initial}}. \tag{269}$$

The number of passing nearby points was

$$\boxed{25/28}. \tag{270}$$

The locked model is therefore classified as

$$\boxed{\mathrm{ROBUST}}. \tag{271}$$

The most sensitive direction is

$$\boxed{V_\infty}. \tag{272}$$

A.50 Matched-control card

The matched control uses the same

$$H_0, \quad \Omega_{m0}, \quad \Omega_{r0}, \quad \omega_b, \quad \omega_c, \quad A_s, \quad n_s, \quad \tau_{\rm reio}, \tag{273}$$

and neutrino content.

Its approximate derived values are

$$\boxed{100\theta_{*,\rm control} = 1.042322}, \tag{274}$$

$$\boxed{r_{\rm drag,control} = 147.503 \text{ Mpc}}, \tag{275}$$

$$\boxed{D_{M,\rm control}(z_*) = 13883.67 \text{ Mpc}}, \tag{276}$$

$$\boxed{\sigma_{8,\rm control} = 0.794561}, \tag{277}$$

$$\boxed{S_{8,\rm control} = 0.813037}. \tag{278}$$

A.51 Complete locked parameter card

$$\boxed{\begin{aligned} \Omega_{m0} &= 0.314114, \\ \Omega_{r0} &= 9 \times 10^{-5}, \\ H_0 &= 67.4 \text{ km}\,\text{s}^{-1}\,\text{Mpc}^{-1}, \\ V_\infty &= 2.187987, \\ \rho_s &= 1.58199 \times 10^{15}, \\ m_C^2 &= 1.827221, \\ \lambda &= 0.802722, \\ C_s &= 0.95, \\ C_{\rm initial} &= 0.918318, \\ \Pi_{C,\rm initial} &= 0.055327, \\ a_{\rm min} &\simeq 2.068 \times 10^{-5}, \\ a_{\rm lock} &= 2.05 \times 10^{-4}, \\ W &= 0.67 W_0. \end{aligned}} \tag{279}$$

A.52 Complete locked result card

$$\boxed{\begin{aligned} 100\theta_* &= 1.041033, \\ r_{\rm drag} &= 149.960 \text{ Mpc}, \\ z_* &= 1089.691, \\ z_{\rm drag} &= 1058.511, \\ D_M(z_*) &= 14131.86 \text{ Mpc}, \\ D_A(z_*) &= 12.9568 \text{ Mpc}, \\ \rho_C(0) &\simeq 2.192, \\ w_C(0) &\simeq -0.99988, \\ f(0) &\simeq 0.527, \\ \sigma_8 &= 0.779680, \\ S_8 &= 0.79781, \\ \max|\delta C| &= 0.03514, \\ \max|\delta\phi| &= 0.04860, \\ \chi^2_{\rm BAO} &\simeq 13.4, \\ \chi^2_{\rm Planck\ lensing} &= 10.7379, \\ \Delta\chi^2_{\rm lensing} &= -1.2643. \end{aligned}} \tag{280}$$

A.53 Locked implementation rules

The retained calculation obeys the following implementation rules:

$$\boxed{\text{Piecewise nonsingular integration only}}, \tag{281}$$

$$\boxed{\text{No division by } H \text{ or } h \text{ through the bounce}}, \tag{282}$$

$$\boxed{\text{No } N = \ln a \text{ integration through } H = 0}, \tag{283}$$

$$\boxed{\text{DOP853 and Radau verification for geometry}}, \tag{284}$$

$$\boxed{\Gamma_{\rm tr} = 0 \quad \text{for} \quad H \geq 0}, \tag{285}$$

$$\boxed{\text{Second-order canonical } C, \phi \text{ fields}}, \tag{286}$$

$$\boxed{\text{No separate dark-energy fluid}}, \tag{287}$$

$$\boxed{\text{No retained phenomenological } \mu \text{ or } \eta \text{ closure}}, \tag{288}$$

$$\boxed{\text{The same locked background determines CMB, BAO, growth, and lensing}}. \tag{289}$$

A.54 Final compact equation set

The complete theory may be compressed into the following locked system:

$$\boxed{V = V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4 + A_0(1 - \cos\phi)}, \tag{290}$$

$$\boxed{H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right)}, \tag{291}$$

$$\boxed{\dot{H} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right)}, \tag{292}$$

$$\boxed{\dot{C} = \Pi_C}, \tag{293}$$

$$\boxed{\dot{\Pi}_C = -(3H + \Gamma_{\rm tr})\Pi_C - V_{,C}}, \tag{294}$$

$$\boxed{\dot{\phi} = \Pi_\phi}, \tag{295}$$

$$\boxed{\dot{\Pi}_\phi = -(3H + \Gamma_{\rm tr})\Pi_\phi - V_{,\phi}}, \tag{296}$$

$$\boxed{\dot{\rho}_m = -3H\rho_m}, \tag{297}$$

$$\boxed{\dot{\rho}_r = -4H\rho_r + \Gamma_{\rm tr}(\Pi_C^2 + \Pi_\phi^2)}, \tag{298}$$

$$\boxed{\Gamma_{\rm tr} = W_-(H)[-3H + \Gamma_0 S_\rho]}, \tag{299}$$

$$\boxed{\Gamma_{\rm tr} = 0 \quad \text{for} \quad H \geq 0}, \tag{300}$$

$$\boxed{\rho_C = \frac{1}{2}(\Pi_C^2 + \Pi_\phi^2) + V}, \tag{301}$$

$$\boxed{p_C = \frac{1}{2}(\Pi_C^2 + \Pi_\phi^2) - V}, \tag{302}$$

$$\boxed{\rho = \rho_m + \rho_r + \rho_C}, \tag{303}$$

$$\boxed{p = \frac{1}{3}\rho_r + p_C}. \tag{304}$$

The expanding-branch perturbations are

$$\boxed{\delta C'' + 2\mathcal{H}\delta C' + (k^2 + a^2 V_{,CC})\delta C + \frac{1}{2}C'_{\rm bg}h' = 0} \tag{305}$$

and

$$\boxed{\delta\phi'' + 2\mathcal{H}\delta\phi' + (k^2 + a^2 V_{,\phi\phi})\delta\phi + \frac{1}{2}\phi'_{\rm bg}h' = 0}. \tag{306}$$

This equation set, together with the locked parameter card, defines the completed gravitational-saturation cosmology used throughout the manuscript.

End of Appendix A

Appendix B

Numerical Integration, CAMB Modifications, and Reproduction Protocol

B.1 Purpose of the appendix

This appendix records the numerical procedure used to construct, verify, and reproduce the locked gravitational-saturation cosmology.

The workflow contains five linked stages:

$$\begin{aligned}
&\text{nonsingular background integration,}\\
&\text{locked expanding-branch construction,}\\
&\text{CAMB background modification,}\\
&\text{canonical } C, \phi \text{ perturbation evolution,}\\
&\text{CMB, BAO, growth, and lensing evaluation.}
\end{aligned} \tag{1}$$

The essential rule is that all final observables are generated from the same retained background and parameter set.

No parameter is retuned between the final acoustic, BAO, growth, primary-CMB, and lensing calculations.

B.2 Reproduction environment

The numerical cosmology is implemented through a modified CAMB Python and Fortran workflow.

The recommended software environment is:

$$\boxed{\text{Python 3.10 or later}} \tag{2}$$

with:

$$\boxed{\text{CAMB}}, \quad \boxed{\text{NumPy}}, \quad \boxed{\text{SciPy}}, \quad \boxed{\text{pandas}}, \quad \boxed{\text{Matplotlib}} \tag{3}$$

and, for reconstruction calculations,

$$\boxed{\text{PlanckLens}}. \tag{4}$$

A modern Fortran compiler is required for rebuilding CAMB.

The tested compiler family is

$$\boxed{\text{gfortran}}. \tag{5}$$

An editable CAMB installation is created through

```
pip install -e .
```

from the CAMB source directory.

After Fortran modifications, the code is rebuilt using

```
python setup.py make
```

If stale module files appear after changing wrapped Fortran types, a clean rebuild is performed with

```
python setup.py clean
```

followed by

```
python setup.py make
```

B.3 Code-formatting and validation rules

The Python and Fortran code follow the CAMB source conventions.

Python formatting and static checks are run through

```
pre-commit run --all-files
```

The primary CAMB test is

```
python -m unittest camb.tests.camb_test
```

Numerical-accuracy checks may be performed with

```
python -m camb.check_accuracy
```

The modified calculation is accepted only when:

$$\begin{aligned}&\text{CAMB builds successfully,}\\&\text{the background tables are finite,}\\&\text{the canonical perturbations remain bounded,}\\&\text{the requested spectra and derived quantities are produced.}\end{aligned} \tag{6}$$

B.4 Locked input parameters

The numerical reconstruction begins with the locked values

$$\Omega_{m0} = 0.314114, \tag{7}$$

$$\Omega_{r0} = 9 \times 10^{-5}, \tag{8}$$

$$H_0 = 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{9}$$

$$V_\infty = 2.187987, \tag{10}$$

$$\rho_s = 1.58199 \times 10^{15}, \tag{11}$$

$$m_C^2 = 1.827221, \tag{12}$$

$$\lambda = 0.802722, \tag{13}$$

$$C_s = 0.95, \tag{14}$$

$$C_{\text{initial}} = 0.918318, \tag{15}$$

$$\Pi_{C,\text{initial}} = 0.055327. \tag{16}$$

The final transition values are

$$a_{\text{lock}} = 2.05 \times 10^{-4}, \qquad W = 0.67 W_0. \tag{17}$$

The primordial inputs are

$$\begin{aligned} \omega_b &= 0.022, && (18)\\ \omega_c &\simeq 0.12005, && (19)\\ A_s &= 2.0 \times 10^{-9}, && (20)\\ n_s &= 0.965, && (21)\\ \tau_{\text{reio}} &= 0.054. && (22) \end{aligned}$$

These values are treated as frozen during reproduction of the final model.

B.5 Background state vector

The nonsingular background state is

$$\mathbf{y} = \big(a, H, C, \Pi_C, \phi, \Pi_\phi, \rho_m, \rho_r\big). \tag{23}$$

At every integration step, the auxiliary quantities are reconstructed:

$$\begin{aligned} K^2 &= \Pi_C^2 + \Pi_\phi^2, && (24)\\ V &= V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4 + A_0(1 - \cos\phi), && (25)\\ \rho_C &= \frac{1}{2}K^2 + V, && (26)\\ p_C &= \frac{1}{2}K^2 - V, && (27)\\ \rho &= \rho_m + \rho_r + \rho_C, && (28)\\ p &= \frac{1}{3}\rho_r + p_C. && (29) \end{aligned}$$

The Friedmann residual is evaluated continuously.

B.6 Background derivative function

The cosmic-time derivative function is

$$\frac{d\mathbf{y}}{dt} = \mathbf{F}(t, \mathbf{y}). \tag{30}$$

Its components are

$$\dot{a} = aH, \tag{31}$$

$$\dot{H} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right), \tag{32}$$

$$\dot{C} = \Pi_C, \tag{33}$$

$$\dot{\Pi}_C = -(3H + \Gamma_{\rm tr})\Pi_C - m_C^2(C - C_s) - \lambda(C - C_s)^3, \tag{34}$$

$$\dot{\phi} = \Pi_\phi, \tag{35}$$

$$\dot{\Pi}_\phi = -(3H + \Gamma_{\rm tr})\Pi_\phi - A_0 \sin\phi, \tag{36}$$

$$\dot{\rho}_m = -3H\rho_m, \tag{37}$$

$$\dot{\rho}_r = -4H\rho_r + \Gamma_{\rm tr}\big(\Pi_C^2 + \Pi_\phi^2\big). \tag{38}$$

The derivative function never divides by H.

B.7 Contraction transfer implementation

The contraction selector is evaluated from the sign of H.

For

$$H \geq 0, \tag{39}$$

the implementation returns

$$W_- = 0, \tag{40}$$

and therefore

$$\Gamma_{\rm tr} = 0. \tag{41}$$

For

$$H < 0, \tag{42}$$

a smooth one-sided function is used.

The transfer rate is

$$\Gamma_{\rm tr} = W_-(H)\big[-3H + \Gamma_0 S_\rho\big], \tag{43}$$

where

$$S_\rho = \frac{(\rho_{\rm tar}/\rho_\Gamma)^m}{1 + (\rho_{\rm tar}/\rho_\Gamma)^m}, \tag{44}$$

and

$$\rho_{\rm tar} = \rho_m + \rho_r. \tag{45}$$

The transfer source added to radiation is

$$Q = \Gamma_{\rm tr}\big(\Pi_C^2 + \Pi_\phi^2\big). \tag{46}$$

The same quantity is removed from the field sector through its damping terms.

B.8 Bounce event detection

The bounce is identified by the event

$$H = 0 \tag{47}$$

on a trajectory for which

$$H < 0 \tag{48}$$

before the event and

$$\dot{H} > 0 \tag{49}$$

at the crossing.

The event function is

$$\mathcal{E}_{\mathrm{bounce}}(t, \mathbf{y}) = H. \tag{50}$$

The required direction is

$$-1 \to +1. \tag{51}$$

At the event, the following quantities are stored:

$$a_{\min}, \quad C_{\mathrm{b}}, \quad \Pi_{C,\mathrm{b}}, \quad \phi_{\mathrm{b}}, \quad \Pi_{\phi,\mathrm{b}}, \quad \rho_{\mathrm{b}}, \quad K_{\mathrm{b}}^2, \quad \dot{H}_{\mathrm{b}}. \tag{52}$$

The bounce passes only if

$$a_{\min} > 0, \qquad \rho_{\mathrm{b}} \simeq \rho_s, \qquad K_{\mathrm{b}}^2 < \infty, \qquad \dot{H}_{\mathrm{b}} > 0. \tag{53}$$

B.9 Constraint monitoring

The Friedmann constraint is

$$H_{\mathrm{con}}^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{54}$$

The absolute residual is

$$\mathcal{C}_H = H^2 - H_{\mathrm{con}}^2. \tag{55}$$

A normalized error is

$$\epsilon_H = \frac{|\mathcal{C}_H|}{\max\bigl(H^2, \rho/3, \epsilon_{\mathrm{floor}}\bigr)}. \tag{56}$$

The constraint is checked:

$$\begin{aligned} &\text{before the bounce,} \\ &\text{at the bounce,} \\ &\text{after the bounce,} \\ &\text{at every piecewise checkpoint.} \end{aligned} \tag{57}$$

A sign-preserving projection may be applied at checkpoints:

$$H \leftarrow \mathrm{sgn}(H)\sqrt{\max\left[0, \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right)\right]}. \tag{58}$$

The square-root argument is clipped only against small negative roundoff error.

A genuinely negative physical argument is treated as a failed integration.

B.10 Piecewise integration protocol

The integration is divided into intervals because the system changes stiffness across the bounce and through many decades in scale factor.

A practical protocol is:

Segment 1. Integrate from the contracting initial state through

$$H = 0 \tag{59}$$

and onward until

$$a \sim 10^{-3}. \tag{60}$$

Segment 2. Restart from the full stored state and integrate to

$$a \sim 10^{-2}. \tag{61}$$

Segment 3. Restart again and integrate to

$$a = 1. \tag{62}$$

Additional segments may be inserted if the adaptive solver requires them.

No equation or parameter changes between segments.

The segmentation is numerical only.

B.11 DOP853 run

The first background realization uses the DOP853 solver.

The integration call has the schematic form

```
solve_ivp(F, [t_i, t_f], y_i, method="DOP853")
```

Dense output is enabled when constructing smooth background tables.

The required settings include:

$$\begin{aligned}&\text{tight relative tolerance,}\\&\text{tight absolute tolerance,}\\&\text{restricted maximum step near the bounce.}\end{aligned} \tag{63}$$

A representative precision target is

$$\mathrm{rtol} \le 10^{-10}. \tag{64}$$

The absolute tolerances may be component dependent because

$$a,\ H,\ \rho_i,\ C, \tag{65}$$

and the momenta have different scales.

B.12 Radau run

The same initial state and parameters are integrated with the implicit Radau method:

```
solve_ivp(F, [t_i, t_f], y_i, method="Radau")
```

Radau is used as an independent stiff-system check.

The accepted solution requires agreement with DOP853 in:

$$a_{\min}, \quad K_{\mathrm{bounce}}, \quad C(a), \quad \rho_C(a), \quad E(z), \quad r_{\mathrm{drag}}, \quad 100\theta_*. \tag{66}$$

The final geometry is not accepted from only one solver.

B.13 Solver-comparison table

For every retained run, a comparison table is produced with columns:

$$\text{quantity}, \quad \text{DOP853 result}, \quad \text{Radau result}, \quad \text{absolute difference}, \quad \text{fractional difference}. \tag{67}$$

The table includes at minimum:

$$\begin{aligned} &a_{\min},\ \rho_C(0),\ C(0),\ w_C(0),\\ &E(0.5),\ E(1),\ E(10),\ E(100),\ E(1000),\\ &r_{\mathrm{drag}},\ 100\theta_*. \end{aligned} \tag{68}$$

The dual-solver mean is used for archived geometry values when the difference is negligible.

B.14 Conversion to an expanding background table

Only the expanding branch is exported to CAMB.

The branch is selected through

$$H > 0. \tag{69}$$

The stored columns include:

$$a,\ z,\ H,\ E = H/H_0,\ C,\ \Pi_C,\ \phi,\ \Pi_\phi,\ \rho_C,\ p_C,\ w_C. \tag{70}$$

The conformal derivatives are constructed as

$$C' = a\Pi_C, \qquad \phi' = a\Pi_\phi. \tag{71}$$

The table is sorted by increasing a.

Duplicate scale-factor entries are removed before interpolation.

B.15 Background interpolation requirements

The interpolator must preserve:

$$E(a) > 0, \qquad \rho_C(a) > 0, \tag{72}$$

and smooth first derivatives.

A shape-preserving cubic interpolator is preferred.

Ordinary high-degree polynomial interpolation is avoided because it may introduce oscillations.

The following functions are interpolated independently:

$$\begin{gathered} E(a),\ C(a),\ C'(a),\ \phi(a),\ \phi'(a), \\ V_{,C}(a),\ V_{,\phi}(a),\ V_{,CC}(a),\ V_{,\phi\phi}(a). \end{gathered} \tag{73}$$

The interpolation is checked against the raw integration grid.

B.16 Locked transition construction

The CAMB background uses the retained transition ending at

$$a_{\text{lock}} = 2.05 \times 10^{-4}. \tag{74}$$

The narrowed width is

$$W = 0.67 W_0. \tag{75}$$

Define

$$a_{\text{start}} = a_{\text{lock}} - W. \tag{76}$$

The normalized transition coordinate is

$$x = \frac{a - a_{\text{start}}}{W}. \tag{77}$$

The smoothstep function is

$$S(x) = \begin{cases} 0, & x \le 0, \\ 3x^2 - 2x^3, & 0 < x < 1, \\ 1, & x \ge 1. \end{cases} \tag{78}$$

The background is blended as

$$E^2_{\text{CAMB}}(a) = [1 - S(a)] E^2_{\text{early}}(a) + S(a) E^2_{\text{lock}}(a). \tag{79}$$

The same transition convention is applied consistently to any required background derivative.

B.17 Background derivative through the transition

CAMB requires smooth derivatives of the background.

Differentiating the blend gives

$$\frac{dE^2_{\rm CAMB}}{da} = -[S'(a)]E^2_{\rm early} + [1 - S(a)]\frac{dE^2_{\rm early}}{da} + [S'(a)]E^2_{\rm lock} + S(a)\frac{dE^2_{\rm lock}}{da}. \tag{80}$$

Equivalently,

$$\frac{dE^2_{\rm CAMB}}{da} = [1 - S]\frac{dE^2_{\rm early}}{da} + S\frac{dE^2_{\rm lock}}{da} + S'\big(E^2_{\rm lock} - E^2_{\rm early}\big). \tag{81}$$

The logarithmic derivative is

$$\frac{d\ln H}{d\ln a} = \frac{a}{2E^2_{\rm CAMB}}\frac{dE^2_{\rm CAMB}}{da}. \tag{82}$$

This expression is used wherever the Boltzmann hierarchy requires the background derivative.

B.18 Present normalization

After constructing the blended function, define

$$\widetilde{E}(a) = E_{\rm CAMB}(a). \tag{83}$$

The normalized function is

$$E(a) = \frac{\widetilde{E}(a)}{\widetilde{E}(1)}. \tag{84}$$

This guarantees

$$E(1) = 1. \tag{85}$$

The same normalization factor must be applied consistently to derivatives and any dimensional Hubble-rate output.

B.19 CAMB parameter setup

The Python CAMB driver initializes

```
camb.CAMBparams()
```

The cosmological settings are passed schematically as

```
pars.set_cosmology
```

with:

$$H_0 = 67.4, \quad \omega_b = 0.022, \quad \omega_c \simeq 0.12005, \tag{86}$$

the retained neutrino mass,

$$\Omega_k = 0, \tag{87}$$

and the chosen optical depth.

The primordial spectrum is set through

```
pars.InitPower.set_params
```

with

$$A_s = 2.0 \times 10^{-9}, \qquad n_s = 0.965. \tag{88}$$

The spectrum calculation is extended to

$$\ell_{\max} = 2500. \tag{89}$$

Lensing is enabled.

B.20 CAMB background modification

The standard CAMB background call is replaced or supplemented by a locked-background function returning

$$H(a), \qquad \mathcal{H}(a) = aH(a), \tag{90}$$

and the required derivatives.

The implementation must ensure that all CAMB modules use the same modified background.

The locked expansion must affect:

$$\begin{aligned} &\text{conformal-time integration,}\\ &\text{recombination distances,}\\ &\text{sound-horizon integrals,}\\ &\text{perturbation timing,}\\ &\text{growth,}\\ &\text{lensing projection.} \end{aligned} \tag{91}$$

A partial replacement in only the distance calculator is not sufficient.

B.21 Canonical perturbation variables in CAMB

Four new linear state variables are added:

$$\delta C, \qquad v_C = \delta C', \qquad \delta\phi, \qquad v_\phi = \delta\phi'. \tag{92}$$

Their positions in the CAMB perturbation vector are fixed and passed consistently through:

$$\begin{aligned} &\text{initialization,}\\ &\text{derivative evaluation,}\\ &\text{source construction,}\\ &\text{output diagnostics.} \end{aligned} \tag{93}$$

Changing the Fortran state layout requires a clean rebuild.

B.22 Canonical perturbation derivatives

At each wave number k and conformal time, the code evaluates

$$\delta C' = v_C, \tag{94}$$

$$v_C' = -2\mathcal{H}v_C - \left(k^2 + a^2 V_{,CC}\right)\delta C - \frac{1}{2}C'_{\rm bg}h', \tag{95}$$

$$\delta\phi' = v_\phi, \tag{96}$$

$$v_\phi' = -2\mathcal{H}v_\phi - \left(k^2 + a^2 V_{,\phi\phi}\right)\delta\phi - \frac{1}{2}\phi'_{\rm bg}h'. \tag{97}$$

In CAMB internal notation, the metric source is evaluated through the corresponding synchronous variable kz.

The code must not confuse the phase field ϕ with the CMB lensing potential.

Separate variable names are used internally.

B.23 Field stress-energy sources

At each perturbation step, the field sources are

$$a^2\delta\rho_C = C'_{\rm bg}v_C + \phi'_{\rm bg}v_\phi + a^2\left(V_{,C}\delta C + V_{,\phi}\delta\phi\right), \tag{98}$$

$$a^2\delta p_C = C'_{\rm bg}v_C + \phi'_{\rm bg}v_\phi - a^2\left(V_{,C}\delta C + V_{,\phi}\delta\phi\right), \tag{99}$$

$$a^2\delta q_C = k\left(C'_{\rm bg}\delta C + \phi'_{\rm bg}\delta\phi\right). \tag{100}$$

The anisotropic stress is

$$\pi_C = 0. \tag{101}$$

These terms are added to the total Einstein sources.

They do not replace the ordinary matter or radiation terms.

B.24 Initial perturbation conditions

The field perturbations are initialized at small amplitudes compatible with regular early evolution.

The initial state satisfies

$$|\delta C_{\rm ini}| \ll 1, \qquad |v_{C,\rm ini}| \ll 1, \qquad |\delta\phi_{\rm ini}| \ll 1, \qquad |v_{\phi,\rm ini}| \ll 1. \tag{102}$$

The same initialization convention is retained for all final comparisons.

The amplitudes are then evolved dynamically through the metric forcing and mass terms.

They are not reset at later times.

B.25 Perturbation safety checks

During every CAMB run, the following quantities are monitored:

$$\max|\delta C|, \quad \max|v_C|, \quad \max|\delta\phi|, \quad \max|v_\phi|, \quad \max|\delta\rho_C|, \quad \max|\delta p_C|. \tag{103}$$

The final run gives

$$\max|\delta C| = 0.03514, \qquad \max|\delta\phi| = 0.04860. \tag{104}$$

A run fails if any mode exhibits:

$$\text{nonfinite values}, \qquad \text{rapid exponential growth}, \qquad \text{loss of linearity}. \tag{105}$$

B.26 Background-only control inside the locked model

To isolate the direct field-perturbation contribution, a second locked-background run is performed with the same

$$H(a) \tag{106}$$

but with

$$\delta\rho_C = \delta p_C = \delta q_C = 0. \tag{107}$$

The difference between this run and the full canonical-field run measures the direct linear back-reaction.

The observed change is approximately

$$\mathcal{O}(10^{-6}) \tag{108}$$

or smaller in the principal derived quantities.

This confirms that the background dominates the final signal.

B.27 Matched reference control

A matched flat control is generated using the same:

$$H_0, \quad \Omega_{m0}, \quad \Omega_{r0}, \quad \omega_b, \quad \omega_c, \quad A_s, \quad n_s, \quad \tau_{\rm reio}, \tag{109}$$

and neutrino settings.

Its expansion is

$$E^2_{\Lambda\rm CDM}(a) = \Omega_{r0}a^{-4} + \Omega_{m0}a^{-3} + \Omega_{\Lambda 0}, \tag{110}$$

where

$$\Omega_{\Lambda 0} = 1 - \Omega_{m0} - \Omega_{r0}. \tag{111}$$

No parameter is separately optimized for the control.

B.28 Phenomenological closure tests

The failed effective-gravity tests are reproduced by modifying the scalar Einstein constraints while holding the locked background fixed.

A Poisson modification is written as

$$k^2\Psi = -4\pi G a^2 \mu(a,k)\rho\Delta. \tag{112}$$

The tested slip relations are

$$\eta = 1, \qquad \eta = \frac{1}{\mu}, \qquad \eta = 2 - \mu. \tag{113}$$

A single shared saturation factor

$$f_{\mathrm{sat}}(a, k) \tag{114}$$

is also tested in the Hamiltonian and momentum constraints.

Each closure is run against the same locked background and primordial inputs.

The resulting changes in σ_8 and acoustic-peak structure are archived.

B.29 CAMB output extraction

The final CAMB result object is obtained through

```
camb.get_results(pars)
```

Derived background parameters are read from the result object.

The CMB spectra are extracted through

```
results.get_cmb_power_spectra
```

The output arrays contain

$$TT, \quad EE, \quad BB, \quad TE. \tag{115}$$

The lensing-potential spectrum is extracted from the corresponding CAMB lensing output.

All spectra are written to plain-text or CSV files with the multipole column included.

B.30 Required CAMB output table

The principal derived-output table contains:

$$\begin{aligned} &100\theta_*,\ r_s(z_*),\ r_{\mathrm{drag}},\ z_*,\ z_{\mathrm{drag}}, \\ &D_M(z_*),\ D_A(z_*),\ \sigma_8,\ S_8, \end{aligned} \tag{116}$$

and the first five acoustic peak locations and amplitudes.

The same table is produced for:

$$\text{locked full fields}, \qquad \text{locked background only}, \qquad \text{matched control}. \tag{117}$$

B.31 Acoustic peak extraction

The TT peak locations are obtained from the local maxima of

$$D_\ell^{TT} = \frac{\ell(\ell+1)}{2\pi} C_\ell^{TT}. \tag{118}$$

The extraction excludes the very low multipoles and searches within expected acoustic intervals.

For peak number n, store

$$\ell_n \qquad \text{and} \qquad D^{TT}_{\ell_n}. \tag{119}$$

A smoothing operation may be used only to locate candidate maxima.

The reported amplitudes are taken from the unsmoothed spectrum.

B.32 Spectrum residuals

For TT and EE, the fractional residual is

$$R^{XX}_\ell = \frac{D^{XX}_{\ell,\text{locked}} - D^{XX}_{\ell,\text{control}}}{D^{XX}_{\ell,\text{control}}}, \tag{120}$$

where

$$XX \in \{TT, EE\}. \tag{121}$$

For TE, use

$$R^{TE}_\ell = \frac{D^{TE}_{\ell,\text{locked}} - D^{TE}_{\ell,\text{control}}}{\sqrt{D^{TT}_{\ell,\text{control}} D^{EE}_{\ell,\text{control}}}}. \tag{122}$$

The RMS in a multipole band $\mathcal{B}$ is

$$\text{RMS}_{\mathcal{B}} = \sqrt{\frac{1}{N_{\mathcal{B}}} \sum_{\ell\in\mathcal{B}} R^2_\ell}. \tag{123}$$

The final bands are

$$2 \le \ell \le 29, \qquad 30 \le \ell \le 600, \qquad 601 \le \ell \le 1500, \qquad 1501 \le \ell \le 2500. \tag{124}$$

B.33 Amplitude profiling

For an auto-spectrum, a best-fit amplitude A is obtained by minimizing

$$\sum_\ell w_\ell \big[D_{\ell,\text{locked}} - A D_{\ell,\text{control}}\big]^2. \tag{125}$$

The solution is

$$A = \frac{\sum_\ell w_\ell D_{\ell,\text{locked}} D_{\ell,\text{control}}}{\sum_\ell w_\ell (D_{\ell,\text{control}})^2}. \tag{126}$$

After profiling, the shape residual is

$$R_{\ell,\text{shape}} = \frac{D_{\ell,\text{locked}} - A D_{\ell,\text{control}}}{D_{\ell,\text{control}}}. \tag{127}$$

The final profiled amplitudes are

$$A_{TT} = 0.9459, \qquad A_{EE} = 1.0233, \qquad A_{TE} = 0.9802. \tag{128}$$

B.34 BAO calculation

The locked expansion table is used to compute

$$D_H(z) = \frac{c}{H(z)}, \tag{129}$$

$$D_M(z) = \int_0^z \frac{c\,dz'}{H(z')}, \tag{130}$$

$$D_V(z) = \left[z D_M^2(z) D_H(z)\right]^{1/3}. \tag{131}$$

The same CAMB run supplies

$$r_{\rm drag} = 149.960 \text{ Mpc}. \tag{132}$$

The predicted BAO vector contains the required combinations

$$D_M/r_{\rm drag}, \qquad D_H/r_{\rm drag}, \qquad D_V/r_{\rm drag}. \tag{133}$$

For observed vector $\mathbf{d}$, model vector $\mathbf{m}$, and covariance $\mathbf{C}$,

$$\chi^2_{\rm BAO} = (\mathbf{m} - \mathbf{d})^{\rm T} \mathbf{C}^{-1} (\mathbf{m} - \mathbf{d}). \tag{134}$$

The final diagnostic value is

$$\chi^2_{\rm BAO} \simeq 13.4. \tag{135}$$

B.35 Growth calculation

CAMB supplies the matter transfer functions and σ_8.

The derived value is

$$S_8 = \sigma_8 \sqrt{\frac{\Omega_{m0}}{0.3}}. \tag{136}$$

For the locked model,

$$\sigma_8 = 0.779680, \qquad S_8 = 0.79781. \tag{137}$$

The matched control gives

$$\sigma_8 = 0.794561, \qquad S_8 = 0.813037. \tag{138}$$

The same primordial amplitude is used in both models.

B.36 Lensing-spectrum output

The CMB lensing potential is exported as

$$C_L^{\varphi\varphi} \tag{139}$$

for the locked and control models.

The ratio is

$$R_L^{\varphi\varphi} = \frac{C_{L,\text{locked}}^{\varphi\varphi}}{C_{L,\text{control}}^{\varphi\varphi}}. \tag{140}$$

A weighted mean is computed over

$$40 \leq L \leq 2000. \tag{141}$$

The final mean ratio is

$$1.0209. \tag{142}$$

The spectrum files must identify clearly that φ denotes the CMB lensing potential, not the saturation phase field.

B.37 Planck 2018 bandpower calculation

The official conservative Planck lensing dataset contains nine bandpower windows

$$W_{bL}. \tag{143}$$

For each bin,

$$C_b^{\varphi\varphi} = \sum_L W_{bL} C_L^{\varphi\varphi}. \tag{144}$$

The observed vector is

$$\mathbf{C}_{\text{obs}}. \tag{145}$$

The residual is

$$\Delta\mathbf{C} = \mathbf{C}_{\text{model}} - \mathbf{C}_{\text{obs}}. \tag{146}$$

Using the official 9×9 covariance,

$$\chi^2 = \Delta\mathbf{C}^{\mathrm{T}} \Sigma^{-1} \Delta\mathbf{C}. \tag{147}$$

The final results are

$$\chi^2_{\text{locked}} = 10.7379, \qquad \chi^2_{\text{control}} = 12.0022. \tag{148}$$

B.38 Planck bandpower table

The archived comparison table contains:

$$\begin{aligned}
&\text{bin number},\ L_{\min},\ L_{\max},\ L_{\text{eff}},\\
&C_{b,\text{Planck}}^{\varphi\varphi},\ C_{b,\text{locked}}^{\varphi\varphi},\ C_{b,\text{control}}^{\varphi\varphi},\\
&\text{locked diagonal pull},\ \text{control diagonal pull},\\
&\text{locked/control ratio}.
\end{aligned} \tag{149}$$

The diagonal pull is used only as an individual-bin visual diagnostic.

The final statistical result uses the full covariance.

B.39 Profiled lensing amplitude

The model spectrum may be multiplied by a global amplitude A.

The covariance-weighted best fit is

$$A = \frac{\mathbf{p}^{\mathrm{T}}\Sigma^{-1}\mathbf{d}}{\mathbf{p}^{\mathrm{T}}\Sigma^{-1}\mathbf{p}}, \tag{150}$$

where $\mathbf{p}$ is the model prediction and $\mathbf{d}$ the observed bandpower vector.

The profiled residual is

$$\Delta_A = A\mathbf{p} - \mathbf{d}. \tag{151}$$

The locked best-fit amplitude is

$$A_{\mathrm{locked}} = 1.0436. \tag{152}$$

The matched-control value is

$$A_{\mathrm{control}} = 1.0505. \tag{153}$$

B.40 PlanckLens forecast

The CAMB temperature spectra are passed into PlanckLens.

The retained forecast uses:

$$\ell_{\mathrm{min}}^{\mathrm{CMB}} = 30, \tag{154}$$

$$\ell_{\mathrm{max}}^{\mathrm{CMB}} = 512, \tag{155}$$

$$L_{\mathrm{max}} = 400, \tag{156}$$

$$\theta_{\mathrm{FWHM}} = 6 \text{ arcmin}, \tag{157}$$

$$\Delta_T = 35\ \mu\mathrm{K}\,\mathrm{arcmin}. \tag{158}$$

The beam transfer function is

$$B_\ell = \exp\left[-\frac{\ell(\ell+1)\sigma_b^2}{2}\right], \tag{159}$$

where

$$\sigma_b = \frac{\theta_{\mathrm{FWHM}}}{\sqrt{8\ln 2}}. \tag{160}$$

The temperature-noise spectrum is

$$N_\ell^{TT} = (\Delta_T)^2 B_\ell^{-2} \tag{161}$$

after conversion to radians.

The quadratic estimator returns

$$N_L^{(0)}. \tag{162}$$

B.41 PlanckLens comparison

For locked and control runs, compute

$$R_L^{N0} = \frac{N_{L,\mathrm{locked}}^{(0)}}{N_{L,\mathrm{control}}^{(0)}}. \tag{163}$$

The mean over the tested range is approximately

$$1.213. \tag{164}$$

The ratio is archived by L bin.

The PlanckLens calculation must use the same beam, noise, filtering range, and estimator settings for both cosmologies.

B.42 Robustness protocol

Each locked parameter is varied independently by

$$-1\%, \quad -0.5\%, \quad +0.5\%, \quad +1\%. \tag{165}$$

The tested parameters are

$$\Omega_{m0}, \quad V_\infty, \quad \rho_s, \quad m_C^2, \quad \lambda, \quad C_{\mathrm{initial}}, \quad \Pi_{C,\mathrm{initial}}. \tag{166}$$

All other parameters remain fixed.

No global optimizer is launched.

B.43 Robustness gates

For each perturbed point, record:

$$\begin{gathered} \text{DOP853 success, Radau success,} \\ a_{\min},\ \rho_C(0),\ E(0.5),\ E(1), \\ 100\theta_*,\ r_{\mathrm{drag}},\ \chi^2_{\mathrm{BAO}}, \\ \text{all gate failures.} \end{gathered} \tag{167}$$

A point passes when:

$$\begin{gathered} \text{both solvers complete,} \\ a_{\min} > 0, \\ K_{\mathrm{bounce}} < \infty, \\ E(1) = 1, \\ \chi^2_{\mathrm{BAO}} < 30, \\ \text{the late residual remains stable.} \end{gathered} \tag{168}$$

The result is

$$25/28 \tag{169}$$

passing perturbations.

B.44 Transition sensitivity protocol

The transition width is tested at:

$$W = W_0, \qquad W = 1.5W_0, \qquad W = 0.67W_0. \tag{170}$$

The lock endpoint is tested near

$$2.10 \times 10^{-4}, \quad 2.05 \times 10^{-4}, \quad 2.00 \times 10^{-4}, \quad 1.95 \times 10^{-4}. \tag{171}$$

For each case, record:

$$100\theta_*, \quad r_{\rm drag}, \quad S_8, \quad \chi^2_{\rm BAO}, \quad \text{the field amplitudes.} \tag{172}$$

The retained endpoint is

$$2.05 \times 10^{-4}. \tag{173}$$

B.45 High-redshift integration grid

The sound-horizon and distance calculations are checked using progressively finer grids.

Representative high-redshift grid sizes are

$$12{,}000, \quad 24{,}000, \quad 48{,}000, \quad 96{,}000 \tag{174}$$

points.

The accepted integral must converge under refinement.

The grid is constructed to resolve:

$$\text{the transition}, \quad \text{recombination}, \quad \text{drag}, \quad \text{the high-redshift tail.} \tag{175}$$

B.46 Numerical quadrature

For a sampled function $F(z)$, the integral

$$I = \int_{z_1}^{z_2} F(z)\, dz \tag{176}$$

is evaluated with a stable composite quadrature.

The calculation may use:

$$\text{Simpson integration} \qquad \text{or} \qquad \text{high-resolution trapezoidal integration.} \tag{177}$$

Adaptive quadrature is used only when its endpoint behavior is verified against the fixed-grid result.

The final acoustic quantities are accepted only when independent quadrature methods agree.

B.47 Required archive files

A complete reproduction archive contains:

`locked_parameters.json`,
`background_dop853.csv`,
`background_radau.csv`,
`background_solver_comparison.csv`,
`locked_background_table.csv`,
`cmb_locked.csv`,
`cmb_control.csv`,
`lensing_locked.csv`,
`lensing_control.csv`,
`bao_comparison.csv`,
`planck_lensing_comparison.csv`,
`plancklens_noise_comparison.csv`,
`final_locked_summary.json`.

The archive also contains the modified source files and exact run scripts.

B.48 Locked result validation

A successful reproduction should recover approximately:

$$a_{\min} \simeq 2.068 \times 10^{-5}, \tag{178}$$
$$100\theta_* = 1.041033, \tag{179}$$
$$r_{\rm drag} = 149.960 \text{ Mpc}, \tag{180}$$
$$z_* = 1089.691, \tag{181}$$
$$z_{\rm drag} = 1058.511, \tag{182}$$
$$\sigma_8 = 0.779680, \tag{183}$$
$$S_8 = 0.79781, \tag{184}$$
$$\max|\delta C| = 0.03514, \tag{185}$$
$$\max|\delta\phi| = 0.04860, \tag{186}$$
$$\chi^2_{\rm BAO} \simeq 13.4, \tag{187}$$
$$\chi^2_{\rm Planck\ lensing} = 10.7379. \tag{188}$$

Small floating-point differences are expected.

A materially different result requires investigation of the background normalization, transition, parameter units, or source conventions.

B.49 Common implementation errors

Dividing by H at the bounce. Using

$$N = \ln a \tag{189}$$

through

$$H = 0 \tag{190}$$

creates a numerical singularity.

Mixing physical and normalized densities. The ratio

$$\rho/\rho_s \tag{191}$$

must use consistent units.

Leaving transfer active during expansion. The locked rule is

$$\Gamma_{\rm tr} = 0 \tag{192}$$

for

$$H \geq 0. \tag{193}$$

Using an inconsistent sound horizon. The BAO ruler must come from the same CAMB background as the distances.

Adding field stress-energy twice. The locked expansion already includes the background field energy.

The perturbation module adds only the linear field perturbations to the Einstein sources.

Confusing the phase field with the lensing potential. The saturation phase ϕ and CMB lensing potential $\varphi_{\rm lens}$ require distinct internal names.

Applying a phenomenological μ factor together with canonical fields. The final model retains the canonical fields and does not retain the failed effective-gravity closure.

B.50 Minimal Python CAMB driver

A minimal driver has the following structure:

```
import camb
pars = camb.CAMBparams()
pars.set_cosmology(
    H0=67.4,
    ombh2=0.022,
    omch2=0.12005,
```

```
    mnu=0.06,
    omk=0.0,
    tau=0.054,
)
pars.InitPower.set_params(
    As=2.0e-9,
    ns=0.965,
)
pars.set_for_lmax(
    2500,
    lens_potential_lmax=2500,
)
results = camb.get_results(pars)
powers = results.get_cmb_power_spectra(
    pars,
    CMB_unit="muK",
)
total_cls = powers["total"]
```

In the modified branch, the locked-background and saturation-field options are enabled before calling

```
camb.get_results
```

The standard driver alone does not activate the locked model unless the modified Fortran and Python interfaces are present.

B.51 Required model switch

The modified CAMB parameter structure includes an explicit switch such as

```
use_locked_saturation
```

When false:

$$\text{standard matched control is run.} \tag{194}$$

When true:

$$\text{locked } H(a) \tag{195}$$

and the canonical

$$C, \phi \tag{196}$$

sector are activated.

A second diagnostic switch may disable only the field perturbations while leaving the locked background active.

This produces the background-only spectator test.

B.52 Reproduction sequence

The complete run order is:

1. Load the locked parameter file.
2. Integrate the nonsingular background with DOP853.
3. Integrate the same background with Radau.
4. Compare both trajectories and derived geometry.
5. Export the accepted expanding-branch table.
6. Construct the smooth locked CAMB transition.
7. Build the modified CAMB source.
8. Run the matched control.
9. Run the locked background without field perturbations.
10. Run the locked background with canonical field perturbations.
11. Extract derived parameters and TT, TE, EE, and lensing spectra.
12. Compute BAO distances using the same r_{drag}.
13. Evaluate the primary-CMB workflow.
14. Run the PlanckLens reconstruction forecast.
15. Apply the official Planck 2018 lensing windows and covariance.
16. Produce the final summary and comparison tables.

B.53 Reproduction gates

The final pipeline passes only if all of the following hold:

$$\boxed{a_{\min} > 0}, \tag{197}$$

$$\boxed{K_{\text{bounce}} < \infty}, \tag{198}$$

$$\boxed{E(1) = 1}, \tag{199}$$

$$\boxed{\text{DOP853 and Radau agree}}, \tag{200}$$

$$\boxed{V_{,CC} > 0}, \tag{201}$$

$$\boxed{c_{s,C}^2 = c_{s,\phi}^2 = 1}, \tag{202}$$

$$\boxed{\max |\delta C| < 1}, \tag{203}$$

$$\boxed{\max |\delta \phi| < 1}, \tag{204}$$

$$\boxed{100\theta_* \simeq 1.041033}, \tag{205}$$

$$\boxed{\chi^2_{\text{BAO}} \simeq 13.4}, \tag{206}$$

$$\boxed{\chi^2_{\text{Planck lensing}} \simeq 10.7379}. \tag{207}$$

A failed gate is reported rather than corrected by automatic retuning.

B.54 No-retuning rule

The final reproduction must not:

$$\begin{aligned}
&\text{change } V_\infty,\\
&\text{change } \rho_s,\\
&\text{change } m_C^2,\\
&\text{change } \lambda,\\
&\text{change } C_{\text{initial}},\\
&\text{change } \Pi_{C,\text{initial}},\\
&\text{change } a_{\text{lock}},\\
&\text{change the transition width}
\end{aligned} \tag{208}$$

to improve a later observable.

The purpose of reproduction is to recover the archived locked result, not to generate a new best fit.

B.55 Final numerical protocol statement

The gravitational-saturation calculation is reproduced through one continuous numerical chain:

$$\boxed{\begin{array}{l}\text{canonical background equations,}\\ \text{nonsingular cosmic-time integration,}\\ \text{dual-solver verification,}\\ \text{expanding-branch interpolation,}\\ \text{smooth CAMB transition,}\\ \text{canonical field perturbations,}\\ \text{standard Einstein-Boltzmann evolution,}\\ \text{CMB, BAO, growth, and lensing evaluation.}\end{array}} \tag{209}$$

The calculation is considered reproduced when the same frozen parameters recover the archived background, perturbative amplitudes, acoustic quantities, growth amplitude, and Planck lensing statistic without further tuning.

End of Appendix B

Appendix C

Observational Diagnostics, Statistical Definitions, and Final Comparison Tables

C.1 Purpose of the appendix

This appendix records the observational diagnostics used to evaluate the locked gravitational-saturation cosmology.

The calculations include

$$\begin{aligned}
&\text{background expansion,}\\
&\text{acoustic geometry,}\\
&\text{baryon acoustic oscillations,}\\
&\text{linear matter growth,}\\
&\text{primary CMB temperature and polarization,}\\
&\text{CMB lensing,}\\
&\text{lensing reconstruction noise.}
\end{aligned} \tag{1}$$

Every diagnostic is evaluated from the same frozen parameter set and the same locked expansion history.

The statistical definitions in this appendix are intended to make the final reported quantities reproducible and unambiguous.

C.2 Final locked cosmology

The retained cosmological parameters are

$$\Omega_{m0} = 0.314114, \tag{2}$$

$$\Omega_{r0} = 9 \times 10^{-5}, \tag{3}$$

$$H_0 = 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{4}$$

$$\omega_b = 0.022, \tag{5}$$

$$\omega_c \simeq 0.12005, \tag{6}$$

$$A_s = 2.0 \times 10^{-9}, \tag{7}$$

$$n_s = 0.965, \tag{8}$$

$$\tau_{\mathrm{reio}} = 0.054. \tag{9}$$

The locked saturation parameters are

$$V_\infty = 2.187987, \tag{10}$$
$$\rho_s = 1.58199 \times 10^{15}, \tag{11}$$
$$m_C^2 = 1.827221, \tag{12}$$
$$\lambda = 0.802722, \tag{13}$$
$$C_s = 0.95, \tag{14}$$
$$C_{\text{initial}} = 0.918318, \tag{15}$$
$$\Pi_{C,\text{initial}} = 0.055327. \tag{16}$$

The retained numerical transition is

$$a_{\text{lock}} = 2.05 \times 10^{-4}, \tag{17}$$

with width

$$W = 0.67W_0. \tag{18}$$

C.3 Matched reference cosmology

The matched reference uses the same

$$H_0, \quad \Omega_{m0}, \quad \Omega_{r0}, \quad \omega_b, \quad \omega_c, \quad A_s, \quad n_s, \quad \tau_{\text{reio}}, \tag{19}$$

and neutrino inputs.

Its flat expansion function is

$$E^2_{\text{control}}(z) = \Omega_{r0}(1+z)^4 + \Omega_{m0}(1+z)^3 + \Omega_{\Lambda 0}, \tag{20}$$

where

$$\Omega_{\Lambda 0} = 1 - \Omega_{m0} - \Omega_{r0}. \tag{21}$$

The control is used to identify the consequences of the locked expansion history.

It is not independently retuned.

C.4 Expansion-history comparison

The relative expansion residual is

$$\boxed{\Delta_E(z) = \frac{E_{\text{locked}}(z) - E_{\text{control}}(z)}{E_{\text{control}}(z)}}. \tag{22}$$

At the present epoch,

$$E_{\text{locked}}(0) = E_{\text{control}}(0) = 1. \tag{23}$$

At intermediate and higher redshift, the locked expansion is approximately

$$1\%–2\% \tag{24}$$

below the matched control over the previously evaluated interval.

A negative value,

$$\Delta_E(z) < 0, \tag{25}$$

implies

$$D_H^{\text{locked}}(z) > D_H^{\text{control}}(z), \tag{26}$$

because

$$D_H = \frac{c}{H}. \tag{27}$$

It also tends to increase

$$D_M(z) = \int_0^z \frac{c\,dz'}{H(z')}. \tag{28}$$

C.5 Acoustic-scale definitions

The comoving sound horizon is

$$r_s(z) = \int_z^\infty \frac{c_s(z')}{H(z')}\,dz'. \tag{29}$$

The sound speed is

$$c_s(z) = \frac{c}{\sqrt{3[1+R(z)]}}, \tag{30}$$

where

$$R(z) = \frac{3\rho_b}{4\rho_\gamma}. \tag{31}$$

The acoustic angle is

$$\boxed{\theta_* = \frac{r_s(z_*)}{D_M(z_*)}}. \tag{32}$$

CAMB reports

$$100\theta_*. \tag{33}$$

The locked result is

$$\boxed{100\theta_* = 1.041033}. \tag{34}$$

The comparison value is

$$1.0411. \tag{35}$$

The fractional residual is

$$\boxed{\delta_{\theta_*} = \frac{1.041033 - 1.0411}{1.0411} = -6.44 \times 10^{-5}}. \tag{36}$$

Expressed as a percentage,

$$\boxed{100\delta_{\theta_*} \simeq -0.0064\%}. \tag{37}$$

C.6 Final acoustic table

Quantity	Locked model	Matched control
$100\theta_*$	1.041033	1.042322
$r_{\rm drag}$	149.960 Mpc	147.503 Mpc
z_*	1089.691	matched-control output
$z_{\rm drag}$	1058.511	matched-control output
$D_M(z_*)$	14131.86 Mpc	13883.67 Mpc
$D_A(z_*)$	12.9568 Mpc	derived from control

The locked sound horizon and comoving distance are both larger than the corresponding matched-control values.

Their ratio produces an acoustic angle close to the retained reference.

C.7 Drag-horizon comparison

The locked drag horizon is

$$r_{{\rm drag},L} = 149.960 \text{ Mpc}. \tag{38}$$

The matched-control drag horizon is

$$r_{{\rm drag},C} = 147.503 \text{ Mpc}. \tag{39}$$

The absolute difference is

$$\Delta r_{\rm drag} = 2.457 \text{ Mpc}. \tag{40}$$

The fractional difference is

$$\frac{\Delta r_{\rm drag}}{r_{{\rm drag},C}} = \frac{2.457}{147.503} \simeq 0.01666. \tag{41}$$

Therefore

$$\boxed{r_{{\rm drag},L} \text{ is approximately 1.67\% larger than the control ruler}}. \tag{42}$$

C.8 Last-scattering-distance comparison

The locked comoving distance is

$$D_{M,L}(z_*) = 14131.86 \text{ Mpc}. \tag{43}$$

The matched-control value is

$$D_{M,C}(z_*) = 13883.67 \text{ Mpc}. \tag{44}$$

The difference is

$$\Delta D_M = 248.19 \text{ Mpc}. \tag{45}$$

The fractional difference is

$$\frac{248.19}{13883.67} \simeq 0.01788. \tag{46}$$

Therefore

$$\boxed{D_{M,L}(z_*) \text{ is approximately } 1.79\% \text{ larger}}. \tag{47}$$

The simultaneous increases in

$$r_s \tag{48}$$

and

$$D_M \tag{49}$$

produce the near-reference value of θ_*.

C.9 BAO distance definitions

The Hubble distance is

$$D_H(z) = \frac{c}{H(z)}. \tag{50}$$

The transverse comoving distance is

$$D_M(z) = \int_0^z \frac{c\,dz'}{H(z')}. \tag{51}$$

The isotropic volume distance is

$$D_V(z) = \left[z D_M^2(z) D_H(z)\right]^{1/3}. \tag{52}$$

The BAO observables are

$$\boxed{\frac{D_M(z)}{r_{\rm drag}}}, \qquad \boxed{\frac{D_H(z)}{r_{\rm drag}}}, \qquad \boxed{\frac{D_V(z)}{r_{\rm drag}}}. \tag{53}$$

The same locked value

$$r_{\rm drag} = 149.960 \text{ Mpc} \tag{54}$$

is used for every locked-model BAO ratio.

C.10 BAO residual vector

Let the observed BAO vector be

$$\mathbf{d}_{\rm BAO}. \tag{55}$$

Let the model prediction be

$$\mathbf{m}_{\rm BAO}. \tag{56}$$

The residual vector is

$$\boxed{\Delta_{\rm BAO} = \mathbf{m}_{\rm BAO} - \mathbf{d}_{\rm BAO}}. \tag{57}$$

With covariance matrix

$$\mathbf{C}_{\rm BAO}, \tag{58}$$

the statistic is

$$\boxed{\chi^2_{\mathrm{BAO}} = \Delta^{\mathrm{T}}_{\mathrm{BAO}} \mathbf{C}^{-1}_{\mathrm{BAO}} \Delta_{\mathrm{BAO}}}. \tag{59}$$

The final locked diagnostic result is

$$\boxed{\chi^2_{\mathrm{BAO}} \simeq 13.4}. \tag{60}$$

The earlier value near

$$27 \tag{61}$$

used an inconsistent external ruler.

The final value uses the CAMB-derived locked ruler.

C.11 Meaning of the BAO result

The locked background is slightly slower than the matched control over part of the distance integral.

This tends to increase

$$D_M \tag{62}$$

and

$$D_H. \tag{63}$$

The same slower history increases

$$r_{\mathrm{drag}}. \tag{64}$$

Because the BAO observables are ratios, these changes partially cancel.

The final statistic therefore tests the complete locked history rather than only the late-time distances.

C.12 Linear growth definitions

The linear growth factor is denoted

$$D(a). \tag{65}$$

The growth rate is

$$\boxed{f(a) = \frac{d \ln D}{d \ln a}}. \tag{66}$$

The subhorizon growth equation is

$$\frac{d^2 D}{dN^2} + \left[2 + \frac{d \ln H}{dN}\right] \frac{dD}{dN} - \frac{3}{2}\Omega_m(a) D = 0, \tag{67}$$

where

$$N = \ln a \tag{68}$$

is used only on the expanding branch.

The matter fraction is

$$\Omega_m(a) = \frac{\Omega_{m0} a^{-3}}{E^2(a)}. \tag{69}$$

The present growth rate is approximately

$$\boxed{f(0) \simeq 0.527}. \tag{70}$$

C.13 σ_8 definition

The linear matter variance within a top-hat radius

$$R_8 = 8h^{-1}\mathrm{Mpc} \tag{71}$$

is

$$\boxed{\sigma_8^2 = \int_0^\infty \frac{dk}{k}\Delta_m^2(k)W^2(kR_8)}. \tag{72}$$

The Fourier-space top-hat window is

$$W(x) = 3\frac{\sin x - x\cos x}{x^3}. \tag{73}$$

The locked result is

$$\boxed{\sigma_{8,L} = 0.779680}. \tag{74}$$

The control result is

$$\boxed{\sigma_{8,C} = 0.794561}. \tag{75}$$

The difference is

$$\Delta\sigma_8 = -0.014881. \tag{76}$$

The fractional change is

$$\boxed{\frac{\sigma_{8,L} - \sigma_{8,C}}{\sigma_{8,C}} \simeq -1.87\%}. \tag{77}$$

C.14 S_8 definition

The weak-lensing compression parameter is

$$\boxed{S_8 = \sigma_8\sqrt{\frac{\Omega_{m0}}{0.3}}}. \tag{78}$$

For the locked model,

$$\boxed{S_{8,L} = 0.79781}. \tag{79}$$

For the matched control,

$$\boxed{S_{8,C} = 0.813037}. \tag{80}$$

Because both runs use the same Ω_{m0}, the fractional difference equals the fractional σ_8 difference:

$$\boxed{\frac{S_{8,L} - S_{8,C}}{S_{8,C}} \simeq -1.87\%}. \tag{81}$$

C.15 Growth comparison table

Quantity	Locked model	Matched control	Locked/control
σ_8	0.779680	0.794561	0.98127
S_8	0.79781	0.813037	0.98127
$f(0)$	$\simeq 0.527$	control output	—

The reduced locked growth amplitude is produced by the expansion history. It is not produced by a retained reduction of Newton's constant.

C.16 Primary CMB spectra

The angular spectra are

$$C_\ell^{TT}, \quad C_\ell^{EE}, \quad C_\ell^{BB}, \quad C_\ell^{TE}. \tag{82}$$

The plotted spectra are

$$\boxed{D_\ell^{XY} = \frac{\ell(\ell+1)}{2\pi} C_\ell^{XY}}. \tag{83}$$

The final calculation extends through approximately

$$\ell_{\max} = 2500. \tag{84}$$

The primary-CMB evaluation uses the full computed spectra rather than only the acoustic-angle summary.

C.17 TT fractional residual

The TT fractional residual is

$$\boxed{R_\ell^{TT} = \frac{D_{\ell,L}^{TT} - D_{\ell,C}^{TT}}{D_{\ell,C}^{TT}}}. \tag{85}$$

For a multipole band

$$\ell_{\min} \le \ell \le \ell_{\max}, \tag{86}$$

the RMS residual is

$$\boxed{\mathrm{RMS}_{TT} = \sqrt{\frac{1}{N} \sum_{\ell=\ell_{\min}}^{\ell_{\max}} \left(R_\ell^{TT}\right)^2}}. \tag{87}$$

The locked TT residuals are approximately:

Multipole range	TT RMS residual
$2 \le \ell \le 29$	28.20%
$30 \le \ell \le 600$	10.41%
$601 \le \ell \le 1500$	2.98%
$1501 \le \ell \le 2500$	1.73%

C.18 EE fractional residual

The EE residual is

$$\boxed{R_\ell^{EE} = \frac{D_{\ell,L}^{EE} - D_{\ell,C}^{EE}}{D_{\ell,C}^{EE}}}. \tag{88}$$

The final RMS values are:

Multipole range	EE RMS residual
$2 \leq \ell \leq 29$	6.93%
$30 \leq \ell \leq 600$	5.93%
$601 \leq \ell \leq 1500$	4.23%
$1501 \leq \ell \leq 2500$	2.25%

The EE residual decreases toward the damping tail but remains scale dependent.

C.19 TE normalized residual

Because

$$C_\ell^{TE} \tag{89}$$

crosses zero, a direct fractional residual can diverge.

The stable residual is

$$\boxed{R_\ell^{TE} = \frac{D_{\ell,L}^{TE} - D_{\ell,C}^{TE}}{\sqrt{D_{\ell,C}^{TT} D_{\ell,C}^{EE}}}}. \tag{90}$$

The final RMS values are:

Multipole range	Normalized TE RMS
$2 \leq \ell \leq 29$	12.03%
$30 \leq \ell \leq 600$	8.16%
$601 \leq \ell \leq 1500$	2.37%
$1501 \leq \ell \leq 2500$	1.07%

C.20 Primary residual summary

The primary residuals are largest in

$$\text{the low-multipole region} \tag{91}$$

and

$$\text{the first acoustic region.} \tag{92}$$

They decrease toward high multipoles.

The pattern is therefore

$$\boxed{\text{scale dependent}} \tag{93}$$

rather than a uniform multiplicative shift.

The residuals reflect changes in

$$\begin{aligned}&\text{potential evolution,}\\&\text{radiation driving,}\\&\text{projection geometry,}\\&\text{lensing smoothing.}\end{aligned} \tag{94}$$

C.21 Amplitude profiling

For a model spectrum $\mathbf{p}$ and comparison spectrum $\mathbf{d}$, the weighted best-fit amplitude is

$$\boxed{A = \frac{\mathbf{p}^{\mathrm{T}}\mathbf{W}\mathbf{d}}{\mathbf{d}^{\mathrm{T}}\mathbf{W}\mathbf{d}}} \tag{95}$$

when the control is treated as the comparison template.

Equivalently, for diagonal weights,

$$A = \frac{\sum_\ell w_\ell D_{\ell,L} D_{\ell,C}}{\sum_\ell w_\ell (D_{\ell,C})^2}. \tag{96}$$

The profiled shape residual is

$$R_{\ell,\text{shape}} = \frac{D_{\ell,L} - A D_{\ell,C}}{D_{\ell,C}}. \tag{97}$$

C.22 Amplitude-profile results

The final amplitudes are

$$\boxed{A_{TT} = 0.9459}, \qquad \boxed{A_{EE} = 1.0233}, \qquad \boxed{A_{TE} = 0.9802}. \tag{98}$$

The remaining shape RMS values are

$$\boxed{\mathrm{RMS}_{TT,\text{shape}} = 7.90\%}, \qquad \boxed{\mathrm{RMS}_{EE,\text{shape}} = 4.37\%}, \qquad \boxed{\mathrm{RMS}_{TE,\text{shape}} = 4.10\%}. \tag{99}$$

A single amplitude correction therefore does not remove the locked-control differences.

C.23 Primary-CMB likelihood structure

Let the full primary-CMB data vector be

$$\mathbf{d}_{\text{CMB}}. \tag{100}$$

Let the locked-model prediction be

$$\mathbf{m}_{\text{CMB}}. \tag{101}$$

Let the complete covariance or likelihood operator be represented by

$$\mathbf{C}_{\mathrm{CMB}}. \tag{102}$$

The Gaussian form is

$$\boxed{\chi^2_{\mathrm{CMB}} = \left(\mathbf{m}_{\mathrm{CMB}} - \mathbf{d}_{\mathrm{CMB}}\right)^{\mathrm{T}} \mathbf{C}^{-1}_{\mathrm{CMB}} \left(\mathbf{m}_{\mathrm{CMB}} - \mathbf{d}_{\mathrm{CMB}}\right)}. \tag{103}$$

The completed workflow evaluates the primary

$$TT, \qquad TE, \qquad EE \tag{104}$$

spectra through the retained likelihood structure.

The acoustic geometry, polarization response, damping tail, and scale-dependent residuals are therefore included in the primary-CMB evaluation.

C.24 Canonical-field spectator comparison

Three principal CAMB runs are compared:

$$\text{matched control}, \qquad \text{locked background only}, \qquad \text{locked background plus canonical fields}. \tag{105}$$

Define the direct canonical-field change as

$$\Delta X_{\mathrm{fields}} = X_{\mathrm{locked+fields}} - X_{\mathrm{locked-background}}. \tag{106}$$

The fractional σ_8 change is approximately

$$\boxed{\frac{\Delta\sigma_8}{\sigma_8} \sim -5 \times 10^{-6}}. \tag{107}$$

The primary peak-height changes are approximately

$$\boxed{10^{-6}\text{–}10^{-7}}. \tag{108}$$

The canonical fields are therefore linear spectators around the locked background.

C.25 Final field-amplitude table

Field diagnostic	Final value
$\max\lvert\delta C\rvert$	0.03514
$\max\lvert\delta\phi\rvert$	0.04860
Direct fractional σ_8 shift	$\mathcal{O}(5\times 10^{-6})$
Direct peak-height shifts	$\mathcal{O}(10^{-6}$–$10^{-7})$
Intrinsic anisotropic stress	0
$c^2_{s,C}$	1
$c^2_{s,\phi}$	1

C.26 CMB lensing-potential definition

The projected CMB lensing potential is

$$\varphi_{\rm lens}(\hat{\mathbf{n}}) = -2\int_0^{\chi_*} d\chi \frac{\chi_* - \chi}{\chi_* \chi}\Phi_W\big(\chi\hat{\mathbf{n}}, \eta_0 - \chi\big). \tag{109}$$

The Weyl potential is

$$\Phi_W = \frac{\Phi + \Psi}{2}. \tag{110}$$

The corresponding angular spectrum is

$$\boxed{C_L^{\varphi\varphi}}. \tag{111}$$

The phase field of the saturation sector is not the same quantity as the CMB lensing potential.

C.27 Locked-control lensing ratio

Define

$$\boxed{R_L^{\varphi\varphi} = \frac{C_{L,L}^{\varphi\varphi}}{C_{L,C}^{\varphi\varphi}}}. \tag{112}$$

The approximate mean ratios are:

Lensing multipole range	Locked/control ratio
$40 \le L \le 100$	1.0098
$100 \le L \le 400$	1.0049
$400 \le L \le 1000$	1.0182
$1000 \le L \le 2000$	1.0223

The weighted ratio over

$$40 \le L \le 2000 \tag{113}$$

is

$$\boxed{\big\langle R_L^{\varphi\varphi}\big\rangle \simeq 1.0209}. \tag{114}$$

The true CMB lensing power is therefore enhanced by approximately

$$2.09\%. \tag{115}$$

C.28 Lensing shape residual

Let

$$A_{\varphi,LC} \tag{116}$$

be the best common locked/control amplitude over the chosen L interval.

The residual shape is

$$R^{\varphi\varphi}_{L,\mathrm{shape}} = \frac{C^{\varphi\varphi}_{L,L} - A_{\varphi,LC} C^{\varphi\varphi}_{L,C}}{C^{\varphi\varphi}_{L,C}}. \tag{117}$$

The lensing shape scatter is approximately

$$\boxed{0.65\%}. \tag{118}$$

The difference is therefore mostly a smooth amplitude-like enhancement with a smaller residual scale dependence.

C.29 Planck lensing bandpower prediction

The Planck window matrix is

$$W_{bL}. \tag{119}$$

The binned model prediction is

$$\boxed{P_b = \sum_L W_{bL} C^{\varphi\varphi}_L}. \tag{120}$$

Let the observed nine-bin vector be

$$\mathbf{D}. \tag{121}$$

The residual vector is

$$\boxed{\Delta = \mathbf{P} - \mathbf{D}}. \tag{122}$$

The full covariance is

$$\mathbf{\Sigma}. \tag{123}$$

The statistic is

$$\boxed{\chi^2_{\mathrm{lens}} = \Delta^{\mathrm{T}} \mathbf{\Sigma}^{-1} \Delta}. \tag{124}$$

C.30 Official Planck lensing results

For the locked model,

$$\boxed{\chi^2_{\mathrm{locked}} = 10.7379} \tag{125}$$

for nine bins.

For the matched control,

$$\boxed{\chi^2_{\text{control}} = 12.0022}. \tag{126}$$

The difference is

$$\boxed{\Delta\chi^2 = \chi^2_{\text{locked}} - \chi^2_{\text{control}} = -1.2643}. \tag{127}$$

The locked p-value is

$$\boxed{p_{\text{locked}} = 0.294}. \tag{128}$$

The control p-value is approximately

$$\boxed{p_{\text{control}} = 0.213}. \tag{129}$$

C.31 Interpretation of the lensing statistic

Both the locked model and matched control are compatible with the nine-bin lensing dataset.

The locked model gives the lower statistic.

The result may be stated as

$$\boxed{\text{the locked model gives a modestly lower Planck-lensing } \chi^2 \text{ than the matched control}}. \tag{130}$$

The result is not interpreted as an independent proof of the saturation mechanism.

It shows that the locked growth and geometry do not create a conflict with the measured lensing bandpowers.

C.32 Individual lensing pulls

For visual diagnostics, a diagonal pull may be written as

$$\boxed{p_b = \frac{P_b - D_b}{\sqrt{\Sigma_{bb}}}}. \tag{131}$$

The full statistic does not equal the sum of squared diagonal pulls when the covariance contains off-diagonal terms.

No locked-model bin exceeds approximately

$$1.6\sigma \tag{132}$$

in the diagonal-pull display.

The largest negative pull occurs in the high-L portion of the conservative bin range.

C.33 Profiled Planck lensing amplitude

Let the model bandpower vector be

$$\mathbf{P}. \tag{133}$$

Apply an amplitude

$$A_\varphi : \qquad \mathbf{P} \to A_\varphi \mathbf{P}. \tag{134}$$

The covariance-weighted best amplitude is

$$\boxed{A_\varphi = \frac{\mathbf{P}^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}\mathbf{D}}{\mathbf{P}^{\mathrm{T}}\boldsymbol{\Sigma}^{-1}\mathbf{P}}}. \tag{135}$$

The final values are

$$\boxed{A_{\varphi,L} = 1.0436} \qquad \text{and} \qquad \boxed{A_{\varphi,C} = 1.0505}. \tag{136}$$

The locked model requires a slightly smaller upward amplitude adjustment than the control.

C.34 Quadratic-estimator reconstruction noise

For a quadratic lensing estimator,

$$\widehat{\varphi}_{LM} = A_L \sum_{\ell_1 m_1 \ell_2 m_2} W^{LM}_{\ell_1 m_1 \ell_2 m_2} X_{\ell_1 m_1} Y_{\ell_2 m_2}. \tag{137}$$

The disconnected Gaussian reconstruction noise is

$$\boxed{N_L^{(0)}}. \tag{138}$$

The locked and control forecasts use identical

$$\text{beam}, \quad \text{noise}, \quad \ell\text{-range}, \quad \text{estimator settings}. \tag{139}$$

C.35 PlanckLens settings

The retained TT-estimator settings are

$$\ell_{\min}^{\mathrm{CMB}} = 30, \tag{140}$$
$$\ell_{\max}^{\mathrm{CMB}} = 512, \tag{141}$$
$$L_{\max} = 400, \tag{142}$$
$$\theta_{\mathrm{FWHM}} = 6 \text{ arcmin}, \tag{143}$$
$$\Delta_T = 35\ \mu\mathrm{K\,arcmin}. \tag{144}$$

The beam width is

$$\sigma_b = \frac{\theta_{\mathrm{FWHM}}}{\sqrt{8\ln 2}}. \tag{145}$$

The beam transfer function is

$$B_\ell = \exp\Big[-\frac{1}{2}\ell(\ell+1)\sigma_b^2\Big]. \tag{146}$$

The temperature-noise spectrum is

$$N_\ell^{TT} = \Delta_T^2 B_\ell^{-2} \tag{147}$$

after conversion of Δ_T to radians.

C.36 Reconstruction-noise comparison

Define

$$\boxed{R_L^{N0} = \frac{N_{L,L}^{(0)}}{N_{L,C}^{(0)}}}. \tag{148}$$

The approximate ratios are:

Reconstruction range	Locked/control $N_L^{(0)}$
$8 \leq L \leq 40$	1.223
$41 \leq L \leq 100$	1.236
$101 \leq L \leq 200$	1.280
$201 \leq L \leq 400$	1.171

The average ratio is

$$\boxed{\left\langle R_L^{N0} \right\rangle \simeq 1.213}. \tag{149}$$

The locked TT reconstruction noise is therefore approximately

$$21.3\% \tag{150}$$

higher under the common forecast assumptions.

C.37 Signal-versus-noise distinction

The physical lensing signal is

$$C_L^{\varphi\varphi}. \tag{151}$$

The estimator noise is

$$N_L^{(0)}. \tag{152}$$

The locked model produces

$$C_{L,L}^{\varphi\varphi} > C_{L,C}^{\varphi\varphi} \tag{153}$$

over much of the tested range, while also producing

$$N_{L,L}^{(0)} > N_{L,C}^{(0)}. \tag{154}$$

These results are not contradictory.

The first concerns the actual projected gravitational potential.

The second concerns how efficiently the altered primary TT spectrum can reconstruct that potential.

C.38 Phenomenological-closure comparison

The tested phenomenological scalar constraints used

$$k^2\Psi = -4\pi G a^2 \mu(a,k)\rho\Delta. \tag{155}$$

The tested slip relations were

$$\eta = 1, \qquad \eta = \frac{1}{\mu}, \qquad \eta = 2 - \mu. \tag{156}$$

A shared multiplicative factor

$$f_{\text{sat}}(a,k) \tag{157}$$

was also applied to the Hamiltonian and momentum constraints.

These tests produced

$$\text{increased } \sigma_8 \tag{158}$$

and

$$\text{distorted odd-even acoustic peaks.} \tag{159}$$

The final canonical-field model does not retain these closures.

C.39 Statistical comparison rules

Every comparison follows these rules:

1. The locked and control runs use the same primordial parameters.
2. The locked BAO ratios use the locked r_{drag}.
3. The control BAO ratios use the control r_{drag}.
4. The Planck lensing statistic uses the full supplied covariance.
5. Cross-spectrum residuals are normalized without division by TE zero crossings.
6. Amplitude-profiled residuals are distinguished from unprofiled residuals.
7. Background-only and canonical-field locked runs are compared separately.
8. No post hoc parameter retuning is performed between diagnostics.

C.40 Robustness statistic

The local robustness fraction is

$$\boxed{R_{\text{robust}} = \frac{N_{\text{pass}}}{N_{\text{total}}}}. \tag{160}$$

For the retained scan,

$$N_{\text{pass}} = 25, \qquad N_{\text{total}} = 28. \tag{161}$$

Therefore

$$\boxed{R_{\rm robust} = \frac{25}{28} \simeq 0.893}. \tag{162}$$

Approximately

$$89.3\% \tag{163}$$

of the local one-parameter perturbations pass the retained gates.

C.41 Final comparison table

Diagnostic	Locked model	Matched control	Locked-control interpretation
$100\theta_*$	1.041033	1.042322	Locked closer to 1.0411 reference
$r_{\rm drag}$	149.960 Mpc	147.503 Mpc	Locked ruler 1.67% larger
$D_M(z_*)$	14131.86 Mpc	13883.67 Mpc	Locked distance 1.79% larger
σ_8	0.779680	0.794561	Locked 1.87% lower
S_8	0.79781	0.813037	Locked 1.87% lower
Mean $C_L^{\varphi\varphi}$ ratio	1.0209	1	Locked lensing $\sim$ 2.09% higher
Planck lensing χ^2	10.7379	12.0022	$\Delta\chi^2 = -1.2643$
Mean TT $N_L^{(0)}$ ratio	1.213	1	Locked reconstruction noise 21.3% higher

C.42 Final primary-spectrum table

Multipole range	TT RMS	Normalized TE RMS	EE RMS
$2 \leq \ell \leq 29$	28.20%	12.03%	6.93%
$30 \leq \ell \leq 600$	10.41%	8.16%	5.93%
$601 \leq \ell \leq 1500$	2.98%	2.37%	4.23%
$1501 \leq \ell \leq 2500$	1.73%	1.07%	2.25%

The principal residual structure is concentrated at low multipoles and through the first acoustic region.

C.43 Final amplitude-profile table

Spectrum	Best amplitude	Residual shape RMS
TT	0.9459	7.90%
EE	1.0233	4.37%
TE	0.9802	4.10%

The primary differences are not explained by a single common amplitude change.

C.44 Final locked result card

$$\boxed{\begin{aligned} a_{\min} &\simeq 2.068 \times 10^{-5}, \\ 100\theta_* &= 1.041033, \\ r_{\rm drag} &= 149.960 \text{ Mpc}, \\ z_* &= 1089.691, \\ z_{\rm drag} &= 1058.511, \\ D_M(z_*) &= 14131.86 \text{ Mpc}, \\ \sigma_8 &= 0.779680, \\ S_8 &= 0.79781, \\ w_C(0) &\simeq -0.99988, \\ \max|\delta C| &= 0.03514, \\ \max|\delta\phi| &= 0.04860, \\ \chi^2_{\rm BAO} &\simeq 13.4, \\ \chi^2_{\rm lens} &= 10.7379, \\ \Delta\chi^2_{\rm lens} &= -1.2643, \\ \left\langle C^{\varphi\varphi}_{L,L}/C^{\varphi\varphi}_{L,C} \right\rangle &= 1.0209, \\ \left\langle N^{(0)}_{L,L}/N^{(0)}_{L,C} \right\rangle &= 1.213. \end{aligned}} \tag{164}$$

C.45 Interpretation of the combined diagnostics

The completed observational calculations show one coherent pattern.

The locked model has

$$\begin{aligned} &\text{a larger acoustic ruler,} \\ &\text{a larger distance to last scattering,} \\ &\text{a near-reference acoustic angle,} \\ &\text{a coherent BAO ladder,} \\ &\text{a lower present growth amplitude,} \\ &\text{small canonical field perturbations,} \\ &\text{scale-dependent primary CMB residuals,} \\ &\text{a mildly enhanced physical lensing spectrum,} \\ &\text{higher TT reconstruction noise.} \end{aligned} \tag{165}$$

These results arise from one locked expansion history.

They are not independent phenomenological adjustments.

C.46 Final statistical statement

The gravitational-saturation cosmology is evaluated through correlated background and perturbative observables.

The final numbers establish that the retained locked model:

$$\boxed{\text{reproduces the acoustic angular scale}}, \tag{166}$$

$$\boxed{\text{maintains a coherent BAO ruler and distance ladder}}, \tag{167}$$

$$\boxed{\text{reduces the late growth amplitude}}, \tag{168}$$

$$\boxed{\text{keeps the canonical saturation fields perturbative}}, \tag{169}$$

$$\boxed{\text{produces complete primary CMB spectra}}, \tag{170}$$

$$\boxed{\text{remains compatible with the official Planck lensing bandpowers}}. \tag{171}$$

The statistical identity of the theory is therefore defined by the complete correlated pattern rather than by any single fitted observable.

End of Appendix C

Appendix D

Symbol Glossary, Normalization Conventions, and Mathematical Consistency Checks

D.1 Purpose of the appendix

This appendix defines the principal symbols, units, normalization conventions, and internal mathematical checks used throughout the gravitational-saturation theory.

The completed framework contains quantities written in several related forms:

$$\begin{aligned}&\text{cosmic-time variables,}\\&\text{conformal-time variables,}\\&\text{dimensionless background variables,}\\&\text{CAMB-normalized perturbation sources,}\\&\text{physical observational units.}\end{aligned} \tag{1}$$

The purpose of this section is to keep those forms distinct and to show explicitly how they are related.

The appendix also records the algebraic checks connecting:

$$\begin{aligned}&\text{the modified Friedmann constraint,}\\&\text{the Raychaudhuri equation,}\\&\text{total energy conservation,}\\&\text{the contraction-era transfer,}\\&\text{the canonical perturbation system.}\end{aligned} \tag{2}$$

D.2 Spacetime conventions

The metric signature is

$$\boxed{(-,+,+,+)}. \tag{3}$$

The flat homogeneous and isotropic line element is

$$\boxed{ds^2 = -dt^2 + a^2(t)\delta_{ij}dx^i dx^j}. \tag{4}$$

The present scale factor is normalized to

$$\boxed{a_0 = 1}. \tag{5}$$

Redshift is therefore

$$\boxed{1 + z = \frac{1}{a}} \qquad \text{or} \qquad \boxed{z = \frac{1}{a} - 1}. \tag{6}$$

Cosmic time is denoted by

$$t. \tag{7}$$

Conformal time is denoted by

$$\tau, \tag{8}$$

with

$$\boxed{d\tau = \frac{dt}{a}}. \tag{9}$$

The affine time used in the scaled bounce integration is denoted separately by

$$\tau_{\rm aff}. \tag{10}$$

It satisfies

$$\boxed{d\tau_{\rm aff} = \sqrt{\rho_s}\, dt}. \tag{11}$$

D.3 Derivative conventions

A dot denotes a cosmic-time derivative:

$$\boxed{\dot{X} = \frac{dX}{dt}}. \tag{12}$$

A prime denotes a conformal-time derivative:

$$\boxed{X' = \frac{dX}{d\tau}}. \tag{13}$$

An affine-time derivative may be written as

$$\boxed{\frac{dX}{d\tau_{\rm aff}}}. \tag{14}$$

The cosmic-time and conformal-time derivatives are related by

$$\boxed{X' = a\dot{X}}. \tag{15}$$

Therefore

$$C' = a\Pi_C, \qquad \phi' = a\Pi_\phi. \tag{16}$$

For the scale factor,

$$a' = a\dot{a}. \tag{17}$$

Since

$$\dot{a} = aH, \tag{18}$$

one obtains

$$\boxed{a' = a^2 H}. \tag{19}$$

D.4 Hubble quantities

The physical Hubble parameter is

$$\boxed{H = \frac{\dot{a}}{a}}. \tag{20}$$

The conformal Hubble parameter is

$$\boxed{\mathcal{H} = \frac{a'}{a} = aH}. \tag{21}$$

The dimensionless expansion function is

$$\boxed{E(a) = \frac{H(a)}{H_0}}. \tag{22}$$

In redshift form,

$$\boxed{E(z) = \frac{H(z)}{H_0}}. \tag{23}$$

The final Hubble constant is

$$\boxed{H_0 = 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}}. \tag{24}$$

The reduced Hubble parameter is

$$\boxed{h = \frac{H_0}{100\ \mathrm{km\,s^{-1}\,Mpc^{-1}}} = 0.674}. \tag{25}$$

The symbol h is also commonly used for the synchronous-gauge metric trace perturbation. The meaning is determined by context.

To avoid ambiguity:

$$h_{\mathrm{Hubble}} = 0.674 \tag{26}$$

may be used for the reduced Hubble constant, while

$$h_{\mathrm{syn}} \tag{27}$$

may be used for the synchronous metric perturbation.

D.5 Gravitational units

The theoretical background derivations use

$$\boxed{c = 1}. \tag{28}$$

Where stated, they also use

$$\boxed{8\pi G = 1}. \tag{29}$$

In this normalization, the ordinary flat Friedmann equation is

$$H^2 = \frac{\rho}{3}. \tag{30}$$

Restoring $8\pi G$, the standard low-density limit is

$$\boxed{H^2 = \frac{8\pi G}{3}\rho}. \tag{31}$$

The saturated form may then be written as

$$\boxed{H^2 = \frac{8\pi G}{3}\rho\left(1 - \frac{\rho}{\rho_s}\right)}. \tag{32}$$

The numerical CAMB calculation restores the physical Hubble normalization through H_0 and the density parameters.

D.6 Density dimensions

In units with

$$8\pi G = 1, \tag{33}$$

energy density has the same dimensions as

$$H^2. \tag{34}$$

Therefore

$$[\rho] = [H]^2. \tag{35}$$

The saturation density satisfies

$$[\rho_s] = [H]^2. \tag{36}$$

The normalized ratio

$$\boxed{\frac{\rho}{\rho_s}} \tag{37}$$

is dimensionless.

The potential has density dimensions:

$$[V] = [\rho]. \tag{38}$$

The kinetic terms also have density dimensions:

$$[\Pi_C^2] = [\rho], \qquad [\Pi_\phi^2] = [\rho]. \tag{39}$$

D.7 Field dimensions

In the locked numerical normalization, the organizational variable

$$C \tag{40}$$

and phase variable

$$\phi \tag{41}$$

are treated as dimensionless.

Therefore

$$[C] = 1, \qquad [\phi] = 1. \tag{42}$$

The phase field is periodic:

$$\phi \sim \phi + 2\pi n. \tag{43}$$

The field momenta are

$$\Pi_C = \dot{C}, \qquad \Pi_\phi = \dot{\phi}. \tag{44}$$

They therefore have inverse-time dimensions:

$$[\Pi_C] = [H], \qquad [\Pi_\phi] = [H]. \tag{45}$$

The quantities

$$m_C^2 \tag{46}$$

and

$$A_0 \tag{47}$$

carry the dimensions required to make the potential an energy density in the adopted normalization.

D.8 Principal field symbols

The organizational field is

$$\boxed{C}. \tag{48}$$

Its stable saturation value is

$$\boxed{C_s = 0.95}. \tag{49}$$

The displacement from saturation is

$$\boxed{\Delta C = C - C_s}. \tag{50}$$

The phase field is

$$\boxed{\phi}. \tag{51}$$

Its locked minimum is

$$\boxed{\phi = 0 \pmod{2\pi}}. \tag{52}$$

The field momenta are

$$\boxed{\Pi_C = \dot{C}} \qquad \text{and} \qquad \boxed{\Pi_\phi = \dot{\phi}}. \tag{53}$$

The combined field kinetic quantity is

$$\boxed{K^2 = \Pi_C^2 + \Pi_\phi^2}. \tag{54}$$

D.9 Potential symbols

The residual potential floor is

$$\boxed{V_\infty = 2.187987}. \tag{55}$$

The organizational quadratic curvature is

$$\boxed{m_C^2 = 1.827221}. \tag{56}$$

The organizational quartic coupling is

$$\boxed{\lambda = 0.802722}. \tag{57}$$

The phase-potential amplitude is

$$\boxed{A_0}. \tag{58}$$

The complete potential is

$$V(C,\phi) = V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4 + A_0(1 - \cos\phi). \tag{59}$$

The potential gradients are

$$V_{,C} = m_C^2(C - C_s) + \lambda(C - C_s)^3, \qquad V_{,\phi} = A_0 \sin\phi. \tag{60}$$

D.10 Background-fluid symbols

The pressureless matter density is

$$\boxed{\rho_m}. \tag{61}$$

Its pressure is

$$\boxed{p_m = 0}. \tag{62}$$

The radiation density is

$$\boxed{\rho_r}. \tag{63}$$

Its pressure is

$$\boxed{p_r = \frac{1}{3}\rho_r}. \tag{64}$$

The saturation-field density is

$$\boxed{\rho_C = \frac{1}{2}K^2 + V}. \tag{65}$$

The saturation pressure is

$$\boxed{p_C = \frac{1}{2}K^2 - V}. \tag{66}$$

The total density is

$$\boxed{\rho = \rho_m + \rho_r + \rho_C}. \tag{67}$$

The total pressure is

$$\boxed{p = \frac{1}{3}\rho_r + p_C}. \tag{68}$$

D.11 Equation-of-state identities

The saturation equation of state is

$$w_C = \frac{\frac{1}{2}K^2 - V}{\frac{1}{2}K^2 + V}. \tag{69}$$

Adding one gives

$$1 + w_C = 1 + \frac{\frac{1}{2}K^2 - V}{\frac{1}{2}K^2 + V}. \tag{70}$$

Combining the numerator,

$$1 + w_C = \frac{\frac{1}{2}K^2 + V + \frac{1}{2}K^2 - V}{\frac{1}{2}K^2 + V}. \tag{71}$$

Therefore

$$\boxed{1 + w_C = \frac{K^2}{\rho_C}}. \tag{72}$$

At the present epoch,

$$w_C(0) \simeq -0.99988. \tag{73}$$

Thus

$$1 + w_C(0) \simeq 1.2 \times 10^{-4}. \tag{74}$$

Therefore the present kinetic fraction is

$$\boxed{\frac{K^2}{\rho_C} \simeq 1.2 \times 10^{-4}}. \tag{75}$$

D.12 Saturation-density symbols

The limiting density is

$$\boxed{\rho_s = 1.58199 \times 10^{15}}. \tag{76}$$

The generalized saturation exponent is

$$\boxed{q}. \tag{77}$$

The locked value is

$$\boxed{q = 1}. \tag{78}$$

Define the dimensionless density ratio

$$\boxed{x = \frac{\rho}{\rho_s}}. \tag{79}$$

The allowed homogeneous branch satisfies

$$0 \leq x \leq 1. \tag{80}$$

The locked Friedmann equation is

$$H^2 = \frac{\rho_s}{3} x(1 - x). \tag{81}$$

D.13 Transfer symbols

The contraction selector is

$$\boxed{W_-(H)}. \tag{82}$$

The target density is

$$\boxed{\rho_{\text{tar}} = \rho_m + \rho_r}. \tag{83}$$

The activation density is

$$\boxed{\rho_\Gamma}. \tag{84}$$

The activation steepness is

$$\boxed{m}. \tag{85}$$

The density switch is

$$\boxed{S_\rho = \frac{(\rho_{\text{tar}}/\rho_\Gamma)^m}{1 + (\rho_{\text{tar}}/\rho_\Gamma)^m}}. \tag{86}$$

The residual damping scale is

$$\boxed{\Gamma_0}. \tag{87}$$

The total transfer rate is

$$\boxed{\Gamma_{\text{tr}} = W_-(H)[-3H + \Gamma_0 S_\rho]}. \tag{88}$$

The transfer power is

$$\boxed{Q = \Gamma_{\text{tr}} K^2}. \tag{89}$$

D.14 Scale-factor milestones

The minimum scale factor is

$$\boxed{a_{\min} \simeq 2.068 \times 10^{-5}}. \tag{90}$$

The locked transition endpoint is

$$\boxed{a_{\text{lock}} = 2.05 \times 10^{-4}}. \tag{91}$$

Matter-radiation equality occurs at

$$\boxed{a_{\rm eq} \simeq 2.865 \times 10^{-4}}. \tag{92}$$

The ordering is

$$\boxed{a_{\rm min} < a_{\rm lock} < a_{\rm eq} < 1}. \tag{93}$$

Numerically,

$$\frac{a_{\rm lock}}{a_{\rm min}} \simeq 9.91, \qquad \frac{a_{\rm eq}}{a_{\rm lock}} \simeq 1.40. \tag{94}$$

Thus the lock transition lies after the bounce and before equality.

D.15 Redshift milestones

The bounce redshift is

$$\boxed{z_{\rm bounce} \simeq 4.84 \times 10^{4}}. \tag{95}$$

The equality redshift is

$$\boxed{z_{\rm eq} \simeq 3489}. \tag{96}$$

The recombination redshift is

$$\boxed{z_{*} = 1089.691}. \tag{97}$$

The drag redshift is

$$\boxed{z_{\rm drag} = 1058.511}. \tag{98}$$

The ordering is

$$\boxed{z_{\rm bounce} > z_{\rm eq} > z_{*} > z_{\rm drag}}. \tag{99}$$

D.16 Density parameters

The present total matter fraction is

$$\boxed{\Omega_{m0} = 0.314114}. \tag{100}$$

The present radiation fraction is

$$\boxed{\Omega_{r0} = 9 \times 10^{-5}}. \tag{101}$$

The physical baryon density is

$$\boxed{\omega_b = \Omega_b h^2 = 0.022}. \tag{102}$$

The physical cold-matter input is

$$\boxed{\omega_c \simeq 0.12005}. \tag{103}$$

The present matter-radiation equality scale follows from

$$a_{\rm eq} = \frac{\Omega_{r0}}{\Omega_{m0}}. \tag{104}$$

Substituting,

$$a_{\rm eq} = \frac{9 \times 10^{-5}}{0.314114} \simeq 2.865 \times 10^{-4}. \tag{105}$$

D.17 Perturbation symbols

The background organizational field is

$$\boxed{C_{\rm bg}(\tau)}. \tag{106}$$

Its perturbation is

$$\boxed{\delta C(\tau, \mathbf{k})}. \tag{107}$$

The background phase field is

$$\boxed{\phi_{\rm bg}(\tau)}. \tag{108}$$

Its perturbation is

$$\boxed{\delta\phi(\tau, \mathbf{k})}. \tag{109}$$

The first-order velocity variables are

$$\boxed{v_C = \delta C'} \qquad \text{and} \qquad \boxed{v_\phi = \delta\phi'}. \tag{110}$$

The Fourier wave number is

$$\boxed{k}. \tag{111}$$

The synchronous metric variables are

$$\boxed{h_{\rm syn}} \qquad \text{and} \qquad \boxed{\eta_{\rm syn}}. \tag{112}$$

D.18 Lensing-potential notation

The phase field is denoted

$$\phi. \tag{113}$$

The CMB lensing potential is a different quantity and is denoted

$$\boxed{\varphi_{\rm lens}}. \tag{114}$$

Its spectrum is

$$\boxed{C_L^{\varphi\varphi}}. \tag{115}$$

When the superscript $\varphi\varphi$ appears in a CMB-lensing expression, it refers to the projected lensing potential, not the saturation phase field.

The Weyl potential is

$$\boxed{\Phi_W = \frac{\Phi + \Psi}{2}}. \tag{116}$$

D.19 Acoustic symbols

The recombination sound horizon is

$$\boxed{r_s(z_*)}. \tag{117}$$

The drag horizon is

$$\boxed{r_{\mathrm{drag}}}. \tag{118}$$

The comoving distance is

$$\boxed{D_M(z)}. \tag{119}$$

The angular-diameter distance is

$$\boxed{D_A(z)}. \tag{120}$$

The Hubble distance is

$$\boxed{D_H(z) = \frac{c}{H(z)}}. \tag{121}$$

The acoustic angle is

$$\boxed{\theta_* = \frac{r_s(z_*)}{D_M(z_*)}}. \tag{122}$$

The reported quantity is

$$\boxed{100\theta_* = 1.041033}. \tag{123}$$

D.20 Growth symbols

The linear matter contrast is

$$\boxed{\delta_m}. \tag{124}$$

The normalized growth function is

$$\boxed{D(a)}. \tag{125}$$

The growth rate is

$$\boxed{f(a) = \frac{d\ln D}{d\ln a}}. \tag{126}$$

The present matter variance is

$$\boxed{\sigma_8 = 0.779680}. \tag{127}$$

The weak-lensing compression parameter is

$$\boxed{S_8 = \sigma_8\sqrt{\frac{\Omega_{m0}}{0.3}} = 0.79781}. \tag{128}$$

D.21 CMB-spectrum symbols

The temperature spectrum is

$$\boxed{C_\ell^{TT}}. \tag{129}$$

The electric-polarization spectrum is

$$\boxed{C_\ell^{EE}}. \tag{130}$$

The magnetic-polarization spectrum is

$$\boxed{C_\ell^{BB}}. \tag{131}$$

The temperature-polarization cross-spectrum is

$$\boxed{C_\ell^{TE}}. \tag{132}$$

The rescaled form is

$$\boxed{D_\ell^{XY} = \frac{\ell(\ell+1)}{2\pi} C_\ell^{XY}}. \tag{133}$$

For TE comparisons, the stable normalized residual is

$$\boxed{\frac{\Delta TE}{\sqrt{TT \cdot EE}}}. \tag{134}$$

D.22 Statistical symbols

The residual vector is generally denoted

$$\boxed{\Delta}. \tag{135}$$

The covariance matrix is denoted

$$\boxed{\Sigma} \qquad \text{or} \qquad \boxed{C} \tag{136}$$

depending upon the calculation.

The Gaussian statistic is

$$\boxed{\chi^2 = \Delta^{\mathrm{T}} \Sigma^{-1} \Delta}. \tag{137}$$

The number of degrees of freedom is

$$\boxed{\nu}. \tag{138}$$

The probability of obtaining a larger statistic under the model is

$$\boxed{p = P(\chi^2_\nu \geq \chi^2_{\mathrm{measured}})}. \tag{139}$$

For the nine-bin Planck lensing comparison,

$$\chi^2_{\mathrm{locked}} = 10.7379, \qquad \nu = 9, \qquad p_{\mathrm{locked}} = 0.294. \tag{140}$$

D.23 Consistency Check I: Differentiating the Saturated Constraint

The generalized constraint is

$$H^2 = \frac{\rho}{3}\left[1 - \left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{141}$$

Define

$$F(\rho) = \frac{\rho}{3}\left[1 - \left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{142}$$

Then

$$2H\dot{H} = \frac{dF}{d\rho}\dot{\rho}. \tag{143}$$

Differentiate F:

$$\frac{dF}{d\rho} = \frac{1}{3}\left[1 - \left(\frac{\rho}{\rho_s}\right)^q\right] - \frac{\rho}{3}q\left(\frac{\rho}{\rho_s}\right)^{q-1}\frac{1}{\rho_s}. \tag{144}$$

Since

$$\frac{\rho}{\rho_s}\left(\frac{\rho}{\rho_s}\right)^{q-1} = \left(\frac{\rho}{\rho_s}\right)^q, \tag{145}$$

one obtains

$$\frac{dF}{d\rho} = \frac{1}{3}\left[1 - (q+1)\left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{146}$$

Thus

$$2H\dot{H} = \frac{\dot{\rho}}{3}\left[1 - (q+1)\left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{147}$$

Using

$$\dot{\rho} = -3H(\rho + p), \tag{148}$$

gives

$$2H\dot{H} = -H(\rho + p)\left[1 - (q+1)\left(\frac{\rho}{\rho_s}\right)^q\right]. \tag{149}$$

The continuous equation is therefore

$$\boxed{\dot{H} = -\frac{1}{2}(\rho + p)\left[1 - (q+1)\left(\frac{\rho}{\rho_s}\right)^q\right]}. \tag{150}$$

For

$$q = 1, \tag{151}$$

$$\boxed{\dot{H} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right)}. \tag{152}$$

D.24 Consistency Check II: Bounce Direction

At the bounce,

$$\rho = \rho_s. \tag{153}$$

For $q = 1$,

$$\dot{H}_{\mathrm{b}} = -\frac{1}{2}(\rho_{\mathrm{b}} + p_{\mathrm{b}})(1 - 2). \tag{154}$$

Therefore

$$\boxed{\dot{H}_{\mathrm{b}} = \frac{1}{2}(\rho_{\mathrm{b}} + p_{\mathrm{b}})}. \tag{155}$$

The total enthalpy is

$$\rho + p = \rho_m + \frac{4}{3}\rho_r + K^2. \tag{156}$$

Every term is nonnegative.

For a nonempty bounce,

$$\rho + p > 0. \tag{157}$$

Therefore

$$\boxed{\dot{H}_{\rm b} > 0}. \tag{158}$$

The trajectory crosses from negative to positive H.

D.25 Consistency Check III: Scale-Factor Minimum

The scale factor satisfies

$$\dot{a} = aH. \tag{159}$$

At the bounce,

$$H_{\rm b} = 0. \tag{160}$$

Therefore

$$\dot{a}_{\rm b} = 0. \tag{161}$$

Differentiate:

$$\ddot{a} = \dot{a}H + a\dot{H}. \tag{162}$$

Since

$$\dot{a} = aH, \tag{163}$$

one obtains

$$\ddot{a} = aH^2 + a\dot{H}. \tag{164}$$

At the bounce,

$$H = 0, \tag{165}$$

so

$$\boxed{\ddot{a}_{\rm b} = a_{\rm min}\dot{H}_{\rm b}}. \tag{166}$$

Because

$$a_{\rm min} > 0 \tag{167}$$

and

$$\dot{H}_{\rm b} > 0, \tag{168}$$

one obtains

$$\boxed{\ddot{a}_{\rm b} > 0}. \tag{169}$$

Thus $a_{\rm min}$ is a true local minimum.

D.26 Consistency Check IV: Field Energy Transfer

The field density is

$$\rho_C = \frac{1}{2}\Pi_C^2 + \frac{1}{2}\Pi_\phi^2 + V. \tag{170}$$

Differentiate:

$$\dot{\rho}_C = \Pi_C \dot{\Pi}_C + \Pi_\phi \dot{\Pi}_\phi + V_{,C}\dot{C} + V_{,\phi}\dot{\phi}. \tag{171}$$

Using

$$\dot{C} = \Pi_C, \qquad \dot{\phi} = \Pi_\phi, \qquad \dot{\Pi}_C = -(3H + \Gamma_{\rm tr})\Pi_C - V_{,C}, \qquad \dot{\Pi}_\phi = -(3H + \Gamma_{\rm tr})\Pi_\phi - V_{,\phi}, \tag{172}$$

gives

$$\dot{\rho}_C = -(3H + \Gamma_{\rm tr})\big(\Pi_C^2 + \Pi_\phi^2\big). \tag{173}$$

Therefore

$$\dot{\rho}_C = -(3H + \Gamma_{\rm tr})K^2. \tag{174}$$

Since

$$\rho_C + p_C = K^2, \tag{175}$$

one obtains

$$\boxed{\dot{\rho}_C + 3H(\rho_C + p_C) = -\Gamma_{\rm tr}K^2}. \tag{176}$$

Thus

$$\boxed{Q = \Gamma_{\rm tr}K^2} \tag{177}$$

is the exact energy removed from the field sector.

D.27 Consistency Check V: Total Energy Conservation

Radiation obeys

$$\dot{\rho}_r + 4H\rho_r = Q. \tag{178}$$

The field sector obeys

$$\dot{\rho}_C + 3H(\rho_C + p_C) = -Q. \tag{179}$$

Matter obeys

$$\dot{\rho}_m + 3H\rho_m = 0. \tag{180}$$

Adding,

$$\dot{\rho}_m + \dot{\rho}_r + \dot{\rho}_C + 3H\rho_m + 4H\rho_r + 3H(\rho_C + p_C) = 0. \tag{181}$$

Since

$$4H\rho_r = 3H\big(\rho_r + \frac{1}{3}\rho_r\big), \tag{182}$$

the expression becomes

$$\dot{\rho} + 3H\Big[\rho_m + \rho_r + \rho_C + \frac{1}{3}\rho_r + p_C\Big] = 0. \tag{183}$$

The bracket equals

$$\rho + p. \tag{184}$$

Therefore

$$\boxed{\dot{\rho} + 3H(\rho + p) = 0}. \tag{185}$$

The transfer is exactly conservative.

D.28 Consistency Check VI: Anti-Friction Cancellation

During contraction,

$$H < 0. \tag{186}$$

The transfer rate is

$$\Gamma_{\rm tr} = W_{-}[-3H + \Gamma_0 S_\rho]. \tag{187}$$

Deep in contraction,

$$W_{-} \simeq 1. \tag{188}$$

Therefore

$$3H + \Gamma_{\rm tr} \simeq 3H - 3H + \Gamma_0 S_\rho. \tag{189}$$

Thus

$$\boxed{3H + \Gamma_{\rm tr} \simeq \Gamma_0 S_\rho}. \tag{190}$$

For

$$\Gamma_0 > 0 \tag{191}$$

and

$$S_\rho \geq 0, \tag{192}$$

the effective damping coefficient is nonnegative.

The contraction anti-friction is therefore cancelled.

D.29 Consistency Check VII: Transfer Shutdown

The one-sided selector satisfies

$$W_{-}(H) = 0 \tag{193}$$

for

$$H \geq 0. \tag{194}$$

Therefore

$$\Gamma_{\rm tr} = W_{-}[-3H + \Gamma_0 S_\rho] = 0. \tag{195}$$

Thus

$$\boxed{Q = 0 \qquad \text{for} \qquad H \geq 0}. \tag{196}$$

The expanding field equations reduce to

$$\ddot{C} + 3H\dot{C} + V_{,C} = 0, \qquad \ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0. \tag{197}$$

Radiation returns to

$$\dot{\rho}_r = -4H\rho_r. \tag{198}$$

D.30 Consistency Check VIII: Potential Boundedness

The organizational potential above its floor is

$$V_C - V_\infty = \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4. \tag{199}$$

For

$$m_C^2 > 0 \tag{200}$$

and

$$\lambda > 0, \tag{201}$$

both terms are nonnegative.

The phase term is

$$A_0(1 - \cos\phi). \tag{202}$$

For

$$A_0 > 0, \tag{203}$$

and since

$$1 - \cos\phi \geq 0, \tag{204}$$

the phase term is nonnegative.

Therefore

$$\boxed{V(C, \phi) \geq V_\infty}. \tag{205}$$

Since the kinetic energy is also nonnegative,

$$\boxed{\rho_C \geq V_\infty}. \tag{206}$$

The field energy is bounded below.

D.31 Consistency Check IX: Unique Organizational Minimum

Set

$$x = C - C_s. \tag{207}$$

Then

$$V_{,C} = x(m_C^2 + \lambda x^2). \tag{208}$$

For

$$m_C^2 > 0 \tag{209}$$

and

$$\lambda > 0, \tag{210}$$

the factor

$$m_C^2 + \lambda x^2 \tag{211}$$

is always positive.

Therefore

$$V_{,C} = 0 \tag{212}$$

only when

$$x = 0. \tag{213}$$

Thus

$$\boxed{C = C_s} \tag{214}$$

is the unique real stationary point in the organizational direction.

The second derivative is

$$V_{,CC} = m_C^2 + 3\lambda x^2 > 0. \tag{215}$$

Therefore the stationary point is a global minimum.

D.32 Consistency Check X: Late Residual Limit

During expansion,

$$\Gamma_{\rm tr} = 0. \tag{216}$$

The field energy satisfies

$$\dot{\rho}_C = -3HK^2. \tag{217}$$

For

$$H > 0, \tag{218}$$

one has

$$\dot{\rho}_C \le 0. \tag{219}$$

The energy decreases but is bounded below:

$$\rho_C \ge V_\infty. \tag{220}$$

Therefore the energy approaches a finite limit.

The restoring dynamics imply

$$C \to C_s, \qquad \phi \to 0. \tag{221}$$

Hubble damping gives

$$K^2 \to 0. \tag{222}$$

Therefore

$$\boxed{\rho_C \to V_\infty} \quad \text{and} \quad \boxed{p_C \to -V_\infty}. \tag{223}$$

Thus

$$\boxed{w_C \to -1}. \tag{224}$$

D.33 Consistency Check XI: Maximum $|H|$

For $q = 1$,

$$H^2 = \frac{\rho_s}{3} x(1 - x). \tag{225}$$

Differentiate with respect to x:

$$\frac{dH^2}{dx} = \frac{\rho_s}{3}(1 - 2x). \tag{226}$$

Setting the derivative to zero gives

$$x = \frac{1}{2}. \tag{227}$$

The second derivative is

$$\frac{d^2H^2}{dx^2} = -\frac{2\rho_s}{3} < 0. \tag{228}$$

Therefore the point is a maximum.

Substituting,

$$H^2_{\text{max}} = \frac{\rho_s}{3}\left(\frac{1}{2}\right)\left(\frac{1}{2}\right). \tag{229}$$

Thus

$$\boxed{H^2_{\text{max}} = \frac{\rho_s}{12}}. \tag{230}$$

The Hubble rate is bounded.

D.34 Consistency Check XII: Low-Density Recovery

For

$$\rho \ll \rho_s, \tag{231}$$

the ratio

$$\epsilon = \frac{\rho}{\rho_s} \tag{232}$$

is small.

The locked constraint is

$$H^2 = \frac{\rho}{3}(1 - \epsilon). \tag{233}$$

To leading order,

$$H^2 = \frac{\rho}{3} - \frac{\rho^2}{3\rho_s}. \tag{234}$$

The fractional correction is

$$\frac{H^2 - \rho/3}{\rho/3} = -\frac{\rho}{\rho_s}. \tag{235}$$

Therefore

$$\boxed{\frac{\Delta H^2}{H^2_{\text{standard}}} = -\frac{\rho}{\rho_s} \to 0} \tag{236}$$

as

$$\rho/\rho_s \to 0. \tag{237}$$

The ordinary Friedmann limit is recovered continuously.

D.35 Consistency Check XIII: No Background Ghost

The homogeneous kinetic contribution is

$$\frac{1}{2}\dot{C}^2 + \frac{1}{2}\dot{\phi}^2. \tag{238}$$

The field-space kinetic matrix is

$$K_{IJ} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{239}$$

Its eigenvalues are

$$1 \qquad \text{and} \qquad 1. \tag{240}$$

Both are positive.

Therefore

$$\boxed{\text{the canonical field sector contains no negative-kinetic mode}}. \tag{241}$$

The bounce is produced by the saturated background response rather than a ghost field.

D.36 Consistency Check XIV: Gradient Stability

The quadratic perturbation action contains

$$-\frac{1}{2}(\nabla\delta C)^2 \qquad \text{and} \qquad -\frac{1}{2}(\nabla\delta\phi)^2. \tag{242}$$

The mode equations contain

$$+k^2\delta C \qquad \text{and} \qquad +k^2\delta\phi. \tag{243}$$

Therefore the propagation speeds are

$$\boxed{c_{s,C}^2 = 1} \qquad \text{and} \qquad \boxed{c_{s,\phi}^2 = 1}. \tag{244}$$

No negative gradient coefficient appears.

D.37 Consistency Check XV: Mass Stability at the Lock

The Hessian is

$$M^2 = \begin{pmatrix} V_{,CC} & 0 \\ 0 & V_{,\phi\phi} \end{pmatrix}. \tag{245}$$

At

$$C = C_s, \qquad \phi = 0, \tag{246}$$

one has

$$M^2_{\text{lock}} = \begin{pmatrix} m_C^2 & 0 \\ 0 & A_0 \end{pmatrix}. \tag{247}$$

For

$$m_C^2 > 0 \tag{248}$$

and

$$A_0 > 0, \tag{249}$$

both eigenvalues are positive.

Therefore

$$\boxed{\text{the locked field configuration is mass-stable}}. \tag{250}$$

D.38 Consistency Check XVI: Suppression of Linear Field Sources

The field density perturbation is

$$\delta\rho_C = \frac{1}{a^2}\left(C'_{\rm bg}v_C + \phi'_{\rm bg}v_\phi\right) + V_{,C}\delta C + V_{,\phi}\delta\phi. \tag{251}$$

Near the lock,

$$C'_{\rm bg} \to 0, \qquad \phi'_{\rm bg} \to 0, \qquad V_{,C} \to 0, \qquad V_{,\phi} \to 0. \tag{252}$$

Therefore

$$\boxed{\delta\rho_C \to 0} \tag{253}$$

for finite field perturbations.

Similarly,

$$\boxed{\delta p_C \to 0} \qquad \text{and} \qquad \boxed{\delta q_C \to 0}. \tag{254}$$

This explains the spectator behavior of the linear saturation fields.

D.39 Consistency Check XVII: Matter-Radiation Equality

Matter evolves as

$$\rho_m = \rho_{m0}a^{-3}. \tag{255}$$

Radiation evolves after the transfer shuts off as

$$\rho_r = \rho_{r0}a^{-4}. \tag{256}$$

At equality,

$$\rho_{m0}a_{\rm eq}^{-3} = \rho_{r0}a_{\rm eq}^{-4}. \tag{257}$$

Multiplying by $a_{\rm eq}^4$,

$$\rho_{m0}a_{\rm eq} = \rho_{r0}. \tag{258}$$

Thus

$$a_{\rm eq} = \frac{\rho_{r0}}{\rho_{m0}} = \frac{\Omega_{r0}}{\Omega_{m0}}. \tag{259}$$

Using the locked values gives

$$\boxed{a_{\rm eq} \simeq 2.865 \times 10^{-4}}. \tag{260}$$

D.40 Consistency Check XVIII: S_8

The definition is

$$S_8 = \sigma_8 \sqrt{\frac{\Omega_{m0}}{0.3}}. \tag{261}$$

Substituting

$$\sigma_8 = 0.779680, \qquad \Omega_{m0} = 0.314114, \tag{262}$$

gives

$$S_8 = 0.779680 \sqrt{\frac{0.314114}{0.3}}. \tag{263}$$

The ratio is

$$\frac{0.314114}{0.3} = 1.0470467. \tag{264}$$

Its square root is approximately

$$1.02325. \tag{265}$$

Therefore

$$S_8 \simeq 0.779680 \times 1.02325. \tag{266}$$

Hence

$$\boxed{S_8 \simeq 0.79781}. \tag{267}$$

D.41 Consistency Check XIX: Acoustic-Angle Difference

The locked value is

$$100\theta_{*,L} = 1.041033. \tag{268}$$

The reference value is

$$100\theta_{*,R} = 1.0411. \tag{269}$$

The difference is

$$\Delta(100\theta_*) = -0.000067. \tag{270}$$

The fractional difference is

$$\frac{-0.000067}{1.0411} \simeq -6.44 \times 10^{-5}. \tag{271}$$

Multiplying by 100 gives

$$\boxed{-0.00644\%}. \tag{272}$$

Rounded,

$$\boxed{-0.0064\%}. \tag{273}$$

D.42 Consistency Check XX: Lensing $\Delta\chi^2$

The locked statistic is

$$\chi^2_L = 10.7379. \tag{274}$$

The control statistic is

$$\chi_C^2 = 12.0022. \tag{275}$$

Therefore

$$\Delta\chi^2 = \chi_L^2 - \chi_C^2. \tag{276}$$

Substituting,

$$\Delta\chi^2 = 10.7379 - 12.0022. \tag{277}$$

Thus

$$\boxed{\Delta\chi^2 = -1.2643}. \tag{278}$$

A negative value means the locked model has the lower statistic in this matched comparison.

D.43 Consistency Check XXI: Reconstruction-Noise Increase

The mean noise ratio is

$$R_{N0} = 1.213. \tag{279}$$

The fractional increase is

$$R_{N0} - 1 = 0.213. \tag{280}$$

Therefore the percentage increase is

$$\boxed{21.3\%}. \tag{281}$$

This statistic concerns estimator noise and is separate from the physical lensing-spectrum ratio.

D.44 Consistency Check XXII: Lensing-Power Increase

The mean lensing-potential ratio is

$$R_\varphi = 1.0209. \tag{282}$$

The fractional increase is

$$R_\varphi - 1 = 0.0209. \tag{283}$$

Therefore the percentage increase is

$$\boxed{2.09\%}. \tag{284}$$

Thus the completed model predicts both

$$2.09\% \tag{285}$$

greater mean physical lensing power over the stated weighted interval and

$$21.3\% \tag{286}$$

greater mean TT reconstruction noise over the forecast interval.

D.45 Symbol glossary: geometry

Symbol	Meaning
$g_{\mu\nu}$	Spacetime metric
$R_{\mu\nu}$	Ricci tensor
R	Ricci scalar
$G_{\mu\nu}$	Einstein tensor
$a(t)$	Scale factor
H	Cosmic-time Hubble parameter
$\mathcal{H}$	Conformal Hubble parameter
H_0	Present physical Hubble constant
$E(a)$	Dimensionless expansion function
t	Cosmic time
τ	Conformal time
$\tau_{\rm aff}$	Scaled affine time
z	Redshift
k	Fourier wave number

D.46 Symbol glossary: saturation sector

Symbol	Meaning
C	Gravitational organizational field
C_s	Stable saturation value
ϕ	Saturation phase field
Π_C	Cosmic-time momentum $\dot{C}$
Π_ϕ	Cosmic-time momentum $\dot{\phi}$
K^2	Combined kinetic quantity
V_∞	Residual potential floor
m_C^2	Organizational potential curvature
λ	Organizational quartic coupling
A_0	Phase-potential amplitude
ρ_C	Saturation-field density
p_C	Saturation-field pressure
w_C	Saturation-field equation of state
ρ_s	High-density saturation scale
q	Saturation exponent

D.47 Symbol glossary: transfer sector

Symbol	Meaning
$W_-(H)$	Contracting-branch selector
ρ_{tar}	Matter-plus-radiation target density
ρ_Γ	Transfer activation density
m	Transfer-switch steepness
S_ρ	Density activation function
Γ_0	Positive residual damping scale
Γ_{tr}	Total contraction transfer rate
Q	Field-to-radiation energy-transfer power

D.48 Symbol glossary: cosmological fluids

Symbol	Meaning
ρ_m	Pressureless matter density
p_m	Matter pressure
ρ_r	Radiation density
p_r	Radiation pressure
ρ	Total background density
p	Total background pressure
Ω_{m0}	Present matter fraction
Ω_{r0}	Present radiation fraction
ω_b	Physical baryon density
ω_c	Physical cold-matter density

D.49 Symbol glossary: perturbations

Symbol	Meaning
δC	Organizational-field perturbation
v_C	$\delta C'$
$\delta\phi$	Phase-field perturbation
v_ϕ	$\delta\phi'$
h_{syn}	Synchronous-gauge trace perturbation
η_{syn}	Synchronous-gauge curvature perturbation
$\delta\rho_C$	Saturation density perturbation
δp_C	Saturation pressure perturbation
δq_C	Saturation momentum perturbation
π_C	Saturation anisotropic stress
c_s^2	Perturbative propagation-speed squared

D.50 Symbol glossary: observations

Symbol	Meaning
$r_s(z_*)$	Sound horizon at recombination
$r_{\rm drag}$	Sound horizon at baryon drag
z_*	Recombination redshift
$z_{\rm drag}$	Drag redshift
D_M	Transverse comoving distance
D_A	Angular-diameter distance
D_H	Hubble distance
D_V	Isotropic BAO distance
θ_*	CMB angular acoustic scale
σ_8	Matter variance at $8h^{-1}$ Mpc
S_8	Weak-lensing compression parameter
f	Linear growth rate
C_ℓ^{TT}	Temperature spectrum
C_ℓ^{TE}	Temperature-polarization spectrum
C_ℓ^{EE}	Electric-polarization spectrum
$C_L^{\varphi\varphi}$	CMB lensing-potential spectrum
$N_L^{(0)}$	Gaussian lensing-reconstruction noise

D.51 Final dimensional hierarchy

The complete numerical hierarchy is

$$\boxed{\text{dimensionless fields} \to \text{normalized densities} \to E(a) \to H_0E(a) \to \text{physical distances and spectra}}. \tag{287}$$

The nonsingular field evolution is carried out in the model normalization.

The final expansion is normalized through

$$E(1) = 1. \tag{288}$$

Physical distances are restored through

$$H_0 \tag{289}$$

and

$$c. \tag{290}$$

CAMB then calculates the recombination, sound-horizon, growth, CMB, and lensing quantities in their standard physical units.

D.52 Final consistency statement

The locked gravitational-saturation system is internally connected by the following identities:

$$\boxed{\text{constraint differentiation} + \text{total conservation} \Rightarrow \text{Raychaudhuri equation}}, \tag{291}$$

$$\boxed{\text{field equations} \Rightarrow \text{field continuity with transfer}}, \tag{292}$$

$$\boxed{\text{field loss} + \text{radiation gain} \Rightarrow \text{total conservation}}, \tag{293}$$

$$\boxed{\rho = \rho_s \Rightarrow H = 0 \text{ and } \dot{H} > 0}, \tag{294}$$

$$\boxed{H > 0 \Rightarrow \Gamma_{\rm tr} = 0 \text{ and } \dot{\rho}_C = -3HK^2}, \tag{295}$$

$$\boxed{V \geq V_\infty \text{ and } K^2 \to 0 \Rightarrow \rho_C \to V_\infty}, \tag{296}$$

$$\boxed{C \to C_s,\ \phi \to 0 \Rightarrow w_C \to -1}, \tag{297}$$

$$\boxed{C'_{\rm bg}, \phi'_{\rm bg}, V_{,C}, V_{,\phi} \to 0 \Rightarrow \delta T^{C\phi}_{\mu\nu} \text{ is suppressed}}. \tag{298}$$

These relations verify that the nonsingular background, conservative transfer system, stable field minimum, locked residual, and spectator perturbations form one mathematically connected model.

D.53 Final glossary card

$$\boxed{\begin{aligned}
C &= \text{organizational field}, \\
\phi &= \text{phase field}, \\
C_s &= 0.95, \\
V_\infty &= 2.187987, \\
m_C^2 &= 1.827221, \\
\lambda &= 0.802722, \\
\rho_s &= 1.58199 \times 10^{15}, \\
a_{\min} &\simeq 2.068 \times 10^{-5}, \\
a_{\rm lock} &= 2.05 \times 10^{-4}, \\
100\theta_* &= 1.041033, \\
r_{\rm drag} &= 149.960\ \text{Mpc}, \\
\sigma_8 &= 0.779680, \\
S_8 &= 0.79781, \\
\chi^2_{\rm BAO} &\simeq 13.4, \\
\chi^2_{\rm lens} &= 10.7379.
\end{aligned}} \tag{299}$$

This notation and consistency framework completes the mathematical reference material for the locked gravitational-saturation cosmology.

End of Appendix D

Appendix E

Scope, Assumptions, Claim Boundaries, and Formal Statement of Results

E.1 Purpose of the appendix

This appendix defines the precise scope of the gravitational-saturation theory and the completed numerical calculation.

Its purpose is to distinguish among:

$$\begin{aligned}
&\text{the physical hypothesis,}\\
&\text{the mathematical model,}\\
&\text{the numerical implementation,}\\
&\text{the calculated observational results,}\\
&\text{the broader physical interpretation.}
\end{aligned} \tag{1}$$

The theory is evaluated as one connected system.

Its central conclusion does not depend upon any single equation or observable.

The retained result is the combined demonstration that a gravitational-saturation background can:

$$\begin{aligned}
&\text{cross a finite-density bounce,}\\
&\text{produce a stable expanding branch,}\\
&\text{enter radiation and matter domination,}\\
&\text{settle into a late residual state,}\\
&\text{evolve canonical linear perturbations,}\\
&\text{produce the completed CMB, BAO, growth, and lensing calculations.}
\end{aligned} \tag{2}$$

E.2 The physical hypothesis

The physical hypothesis is:

$$\boxed{\text{gravity is a dynamical, wave-bearing and organizational phenomenon whose cumulative response can saturat}} \tag{3}$$

The wave character supplies the physical basis for:

$$\begin{aligned}
&\text{propagation,}\\
&\text{phase,}\\
&\text{interference,}\\
&\text{damping,}\\
&\text{restoration,}\\
&\text{limiting behavior.}
\end{aligned} \tag{4}$$

The organizational field C represents the state of gravitational organization.

The phase field ϕ represents the periodic response of that sector.

The hypothesis does not state that gravity is ordinary sound.

It states that gravity possesses dynamical wave properties capable of supporting saturation-like behavior.

E.3 The mathematical realization

The physical premise is represented through three connected mathematical structures.

Canonical field sector

$$C(x^\mu), \qquad \phi(x^\mu), \tag{5}$$

with potential

$$V(C,\phi) = V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4 + A_0(1 - \cos\phi). \tag{6}$$

High-density saturation response

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{7}$$

Contraction-era energy transfer

$$\Gamma_{\rm tr} = W_-(H)\big[-3H + \Gamma_0 S_\rho\big], \tag{8}$$

with

$$Q = \Gamma_{\rm tr}\big(\Pi_C^2 + \Pi_\phi^2\big). \tag{9}$$

These structures are evolved together rather than treated as independent phenomenological corrections.

E.4 The retained model is canonical in its field sector

The field kinetic matrix is

$$K_{IJ} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{10}$$

The kinetic energy is

$$\frac{1}{2}\Pi_C^2 + \frac{1}{2}\Pi_\phi^2. \tag{11}$$

Both kinetic coefficients are positive.

The gradient terms are standard:

$$-\frac{1}{2}(\nabla C)^2, \qquad -\frac{1}{2}(\nabla\phi)^2. \tag{12}$$

Therefore

$$c_{s,C}^2 = 1, \qquad c_{s,\phi}^2 = 1. \tag{13}$$

The retained cosmological solution does not use a negative-kinetic-energy field to generate the bounce.

The reversal is produced by the high-density saturation law.

E.5 The retained model preserves total energy conservation

Matter satisfies

$$\dot{\rho}_m + 3H\rho_m = 0. \tag{14}$$

Radiation satisfies

$$\dot{\rho}_r + 4H\rho_r = Q. \tag{15}$$

The field sector satisfies

$$\dot{\rho}_C + 3H(\rho_C + p_C) = -Q. \tag{16}$$

Adding the equations gives

$$\boxed{\dot{\rho} + 3H(\rho + p) = 0}. \tag{17}$$

The contraction transfer redistributes energy.

It does not destroy energy or create energy outside the total conserved system.

E.6 The bounce claim

The completed numerical trajectory satisfies

$$a_{\min} \simeq 2.068 \times 10^{-5}, \qquad H_{\mathrm{b}} = 0, \qquad \rho_{\mathrm{b}} = \rho_s. \tag{18}$$

The field variables remain finite.

The kinetic quantity remains finite:

$$K^2_{\mathrm{bounce}} = \Pi^2_{C,\mathrm{b}} + \Pi^2_{\phi,\mathrm{b}} < \infty. \tag{19}$$

The derivative of the Hubble parameter is

$$\dot{H}_{\mathrm{b}} = \frac{1}{2}(\rho_{\mathrm{b}} + p_{\mathrm{b}}). \tag{20}$$

Therefore

$$\ddot{a}_{\mathrm{b}} > 0. \tag{21}$$

The retained solution contains a genuine local minimum of the scale factor.

The formal bounce claim is:

$$\boxed{\text{the locked equations admit a continuous finite-density transition from contraction to expansion}} \tag{22}$$

E.7 The bounded-response claim

The saturated constraint may be written as

$$H^2 = \frac{\rho_s}{3} x(1-x), \tag{23}$$

where

$$x = \frac{\rho}{\rho_s}. \tag{24}$$

The maximum occurs at

$$x = \frac{1}{2}. \tag{25}$$

Therefore

$$|H|_{\max} = \sqrt{\frac{\rho_s}{12}}. \tag{26}$$

The Hubble rate is bounded.

The formal saturation claim is:

$$\boxed{\text{the homogeneous gravitational response does not grow without limit as } \rho \to \rho_s} \tag{27}$$

Instead,

$$H^2 \to 0 \tag{28}$$

at finite density.

E.8 The radiation-era claim

Matter-radiation equality occurs at

$$a_{\rm eq} = \frac{\Omega_{r0}}{\Omega_{m0}} \simeq 2.865 \times 10^{-4}. \tag{29}$$

Since

$$a_{\min} \simeq 2.068 \times 10^{-5}, \tag{30}$$

one has

$$a_{\min} < a_{\rm eq}. \tag{31}$$

The expanding branch therefore contains an interval satisfying

$$\rho_r > \rho_m. \tag{32}$$

The transfer is shut off for

$$H \geq 0. \tag{33}$$

Radiation then evolves as

$$\rho_r \propto a^{-4}. \tag{34}$$

The formal radiation-era claim is:

$$\boxed{\text{the nonsingular expanding branch enters an ordinary radiation-dominated interval before equality}} \tag{35}$$

E.9 The matter-era claim

After equality,

$$\rho_m > \rho_r. \tag{36}$$

Matter evolves as

$$\rho_m \propto a^{-3}. \tag{37}$$

The high-density correction becomes negligible because

$$\rho \ll \rho_s. \tag{38}$$

The background consequently approaches the ordinary low-density Friedmann form:

$$H^2 \simeq \frac{\rho}{3}. \tag{39}$$

The formal matter-era claim is:

$$\boxed{\text{the locked expanding branch contains a standard low-density matter-dominated interval}} \tag{40}$$

E.10 The late residual claim

During expansion,

$$\Gamma_{\rm tr} = 0. \tag{41}$$

The field energy obeys

$$\dot{\rho}_C = -3HK^2. \tag{42}$$

For

$$H > 0, \tag{43}$$

one has

$$\dot{\rho}_C \leq 0. \tag{44}$$

The potential satisfies

$$V(C,\phi) \geq V_\infty. \tag{45}$$

The fields approach

$$C \to C_s, \qquad \phi \to 0, \qquad \Pi_C \to 0, \qquad \Pi_\phi \to 0. \tag{46}$$

Therefore

$$\rho_C \to V_\infty, \qquad p_C \to -V_\infty, \qquad w_C \to -1. \tag{47}$$

The present calculated value is

$$w_C(0) \simeq -0.99988. \tag{48}$$

The formal residual claim is:

$$\boxed{\text{the same canonical fields responsible for the saturation history settle into a stable late negative-pressure stat}} \tag{49}$$

E.11 The acoustic-scale result

The final CAMB result is

$$100\theta_* = 1.041033. \tag{50}$$

The comparison value is

$$1.0411. \tag{51}$$

The fractional difference is

$$-0.0064\%. \tag{52}$$

The drag horizon is

$$r_{\rm drag} = 149.960 \text{ Mpc}. \tag{53}$$

The comoving distance to last scattering is

$$D_M(z_*) = 14131.86 \text{ Mpc}. \tag{54}$$

The formal acoustic claim is:

$$\boxed{\text{the locked expansion history produces a near-reference acoustic angle through its own sound horizon and dis}} \tag{55}$$

The result does not use an acoustic ruler taken from a different background.

E.12 The BAO result

The BAO distances and drag horizon are computed from the same locked background.

The relevant ratios are

$$\frac{D_M}{r_{\rm drag}}, \qquad \frac{D_H}{r_{\rm drag}}, \qquad \frac{D_V}{r_{\rm drag}}. \tag{56}$$

The completed diagnostic statistic is approximately

$$\chi^2_{\rm BAO} \simeq 13.4. \tag{57}$$

The formal BAO claim is:

$$\boxed{\text{the retained locked sound horizon and distance ladder form an internally consistent BAO calculation}} \tag{58}$$

E.13 The growth result

The final matter fluctuation amplitude is

$$\sigma_8 = 0.779680. \tag{59}$$

The derived weak-lensing parameter is

$$S_8 = 0.79781. \tag{60}$$

The matched control gives

$$S_{8,\mathrm{control}} = 0.813037. \tag{61}$$

The reduction is approximately

$$1.87\%. \tag{62}$$

The ordinary matter source coefficient is retained.

No phenomenological reduction

$$G \to \mu G \tag{63}$$

is present in the final model.

The formal growth claim is:

$$\boxed{\text{the locked expansion history lowers the late linear growth amplitude without a retained algebraic weakening}} \tag{64}$$

E.14 The canonical-perturbation result

The final field amplitudes are

$$\max|\delta C| = 0.03514, \qquad \max|\delta\phi| = 0.04860. \tag{65}$$

The fields remain perturbative.

Their direct fractional influence on σ_8 is approximately

$$\mathcal{O}(5 \times 10^{-6}). \tag{66}$$

Their direct influence on the acoustic peak amplitudes is approximately

$$10^{-6}\text{–}10^{-7}. \tag{67}$$

The formal perturbative claim is:

$$\boxed{\text{the canonical saturation fields are linearly stable and act primarily as spectators after the background is lock}} \tag{68}$$

E.15 The primary-CMB result

The completed Boltzmann calculation produces the full

$$TT, \qquad TE, \qquad EE \tag{69}$$

spectra through the retained multipole range.

The primary-CMB workflow evaluates:

$$\begin{aligned}&\text{peak placement,}\\&\text{peak amplitudes,}\\&\text{polarization,}\\&\text{damping-tail behavior,}\\&\text{scale-dependent residuals,}\\&\text{the corresponding likelihood structure.}\end{aligned} \tag{70}$$

The differences from the matched control are largest at low multipoles and through the first acoustic region.

They decrease toward the high-ℓ tail.

The formal primary-CMB claim is:

$$\boxed{\text{the locked background produces complete, stable and numerically evaluated primary CMB spectra with a dis}} \tag{71}$$

E.16 The lensing-potential result

The weighted locked-to-control ratio over

$$40 \leq L \leq 2000 \tag{72}$$

is approximately

$$1.0209. \tag{73}$$

The physical CMB lensing-potential spectrum is therefore mildly enhanced.

The formal lensing-spectrum claim is:

$$\boxed{\text{the locked model produces slightly greater integrated CMB lensing power despite its lower present } S_8} \tag{74}$$

This follows because CMB lensing depends upon the full growth and distance history rather than upon σ_8 alone.

E.17 The Planck lensing result

Using the official Planck 2018 nine-bin lensing bandpowers, windows, and covariance gives

$$\chi^2_{\text{locked}} = 10.7379, \qquad \chi^2_{\text{control}} = 12.0022. \tag{75}$$

Therefore

$$\Delta\chi^2 = -1.2643. \tag{76}$$

The locked p-value is

$$p_{\text{locked}} = 0.294. \tag{77}$$

The formal Planck-lensing claim is:

$$\boxed{\text{the locked lensing-potential spectrum is compatible with the official nine-bin Planck lensing measurement}} \tag{78}$$

In the matched comparison, it produces the lower statistic.

E.18 The reconstruction-noise result

The PlanckLens TT-estimator calculation gives

$$\frac{N^{(0)}_{L,\text{locked}}}{N^{(0)}_{L,\text{control}}} \simeq 1.213. \tag{79}$$

The locked reconstruction noise is therefore approximately

$$21.3\% \tag{80}$$

higher under the common forecast assumptions.

The formal reconstruction claim is:

$$\boxed{\text{the altered locked TT spectrum reduces TT-only lensing-reconstruction efficiency even while the physical len}} \tag{81}$$

The physical signal and estimator noise are separate quantities.

E.19 The failed phenomenological-closure result

The tested effective modifications included:

$$f_{\text{sat}}(a,k), \qquad \mu(a,k), \tag{82}$$

and the slip relations

$$\eta = 1, \qquad \eta = \frac{1}{\mu}, \qquad \eta = 2-\mu. \tag{83}$$

These modifications increased

$$\sigma_8 \tag{84}$$

and distorted the alternating acoustic-peak structure.

They did not reproduce the canonical saturation-field calculation.

The formal closure result is:

$$\boxed{\text{the retained saturation model is not equivalent to a simple Poisson rescaling or algebraic gravitational slip}} \tag{85}$$

E.20 The robustness result

Seven principal directions were tested with approximately

$$\pm 0.5\% \qquad \text{and} \qquad \pm 1\% \tag{86}$$

one-at-a-time changes.

The test contained

$$28 \tag{87}$$

nearby parameter points.

Of these,

$$25 \tag{88}$$

passed the retained gates.

Therefore

$$\frac{25}{28} \simeq 0.893. \tag{89}$$

The formal robustness claim is:

$$\boxed{\text{the retained background occupies a locally stable parameter neighborhood rather than a single isolated nume}} \tag{90}$$

The most sensitive tested direction is

$$V_\infty. \tag{91}$$

E.21 Scope of the wave interpretation

The wave interpretation is grounded in the established propagation of gravitational disturbances.

The theory extends this fact into an organizational model containing a phase degree of freedom.

The phase field does not imply that the complete homogeneous Universe is a single sinusoidal gravitational wave.

The interpretation is instead that the gravitational system supports:

$$\begin{aligned}&\text{phase-dependent response,}\\&\text{restoration,}\\&\text{damping,}\\&\text{ringdown-like relaxation,}\\&\text{saturation.}\end{aligned} \tag{92}$$

The formal wave claim is:

$$\boxed{\text{the dynamics of gravitational organization are modeled through wave-compatible field behavior rather than t}} \tag{93}$$

E.22 Scope of the organizational field

The field C is a cosmological organizational degree of freedom.

It does not count stars, galaxies, or bound systems individually.

It measures the state of the effective gravitational organizational sector represented by the model.

The stable point is

$$C = C_s = 0.95. \tag{94}$$

The field is allowed to evolve continuously around that value.

The formal organizational claim is:

$$\boxed{C \text{ represents a dynamical approach to gravitational saturation rather than a hard algebraic limit}} \tag{95}$$

E.23 Scope of the phase field

The phase field has the potential

$$V_\phi = A_0(1 - \cos\phi). \tag{96}$$

It is periodic under

$$\phi \to \phi + 2\pi n. \tag{97}$$

Near the minimum,

$$V_\phi \simeq \frac{A_0}{2}\phi^2. \tag{98}$$

The formal phase claim is:

$$\boxed{\phi \text{ supplies a stable periodic degree of freedom through which phase restoration and cancellation can be repre}} \tag{99}$$

E.24 Scope of the transfer mechanism

The contraction-era transfer is active only when

$$H < 0. \tag{100}$$

Its purpose is to regulate the anti-friction generated by contraction and to transfer kinetic energy into radiation.

It is not retained during the expanding branch.

The formal transfer claim is:

$$\boxed{\text{the transfer is a conservative contraction-regulation mechanism, not a continuing late-time interaction}} \tag{101}$$

E.25 Scope of the CAMB splice

The transition ending at

$$a_{\text{lock}} = 2.05 \times 10^{-4} \tag{102}$$

is a numerical interface between:

$$\text{the independently integrated nonsingular trajectory} \tag{103}$$

and

$$\text{the expanding Boltzmann hierarchy.} \tag{104}$$

It does not create a new fluid.

It does not replace the physical field sector.

It allows the expanding branch to be represented inside a code designed around a monotonic expanding scale factor.

The formal splice claim is:

$$\boxed{\text{the smooth transition is the numerical representation of the resolved post-bounce background inside CAMB}} \tag{105}$$

E.26 Scope of the matched control

The matched control uses the same:

$$H_0, \quad \Omega_{m0}, \quad \Omega_{r0}, \quad \omega_b, \quad \omega_c, \quad A_s, \quad n_s, \quad \tau_{\text{reio}}, \tag{106}$$

and neutrino settings.

Its purpose is to isolate the effect of the locked expansion and saturation sector.

The control is not the globally optimized standard cosmology.

The formal comparison claim is:

$$\boxed{\text{locked-control differences measure the consequences of the retained saturation history under matched ordinar}} \tag{107}$$

E.27 Scope of the galactic relation

The representative galactic interpolation is

$$g\,\mu\left(\frac{g}{g_*}\right) = g_N, \tag{108}$$

with

$$\mu(x) = \frac{x}{\sqrt{1+x^2}}. \tag{109}$$

In the weak-acceleration limit,

$$g \simeq \sqrt{g_N g_*}. \tag{110}$$

For circular motion,

$$v^4 \simeq G M_b g_*. \tag{111}$$

This relation expresses the local weak-acceleration interpretation of gravitational organization.

The formal galactic claim is:

$$\boxed{\text{the organizational hypothesis naturally supports a screened high-acceleration limit and a baryonic weak-acce}} \tag{112}$$

The cosmological and galactic regimes are treated as scale-dependent manifestations of the same physical principle.

E.28 Claims not made by the theory

The completed theory does not claim that gravity propagates as pressure waves through ordinary matter.

It does not claim that gravitational waves require air, gas, or a conventional acoustic medium.

It does not claim that the cosmological background is literally an audible sound.

It does not claim that destructive interference eliminates gravitational energy.

It does not claim that local gravity becomes repulsive at the bounce.

It does not claim that the CMB lensing potential and the saturation phase field are the same quantity.

It does not claim that the successful canonical calculation is reproduced by the failed phenomenological μ-models.

It does not claim that one isolated statistic proves the entire physical interpretation.

The theory is supported by the internal agreement of its complete mathematical and numerical structure.

E.29 Distinction between a calculated result and an interpretation

A calculated result follows directly from the implemented equations and numerical pipeline.

Examples include:

$$100\theta_* = 1.041033, \qquad r_{\rm drag} = 149.960 \text{ Mpc}, \qquad S_8 = 0.79781, \qquad \chi^2_{\rm lens} = 10.7379. \tag{113}$$

An interpretation assigns physical meaning to the calculated result.

Examples include:

$$\text{the residual is stored saturation energy,} \tag{114}$$

and

$$\text{the bounce represents a limiting gravitational response.} \tag{115}$$

The calculations establish the behavior of the retained model.

The organizational interpretation explains why those equations were chosen and how their connected results are understood physically.

E.30 Distinction between mathematical stability and observational agreement

Mathematical stability requires:

$$\begin{aligned} &\text{positive kinetic coefficients,}\\ &\text{positive gradient coefficients,}\\ &\text{bounded perturbations,}\\ &\text{a potential bounded below,}\\ &\text{finite background variables.} \end{aligned} \tag{116}$$

Observational agreement requires the resulting predictions to match measured cosmological quantities.

The retained model satisfies both classes of test within the completed calculation.

These are separate requirements.

A stable model may fail observation.

A numerically fitted spectrum may conceal an unstable theory.

The locked model is evaluated through both.

E.31 Distinction between background viability and perturbative viability

Background viability requires:

$$\begin{aligned}
&a_{\min} > 0, \\
&E(1) = 1, \\
&\text{radiation and matter eras,} \\
&w_C \to -1, \\
&\text{acceptable cosmic distances.}
\end{aligned} \tag{117}$$

Perturbative viability requires:

$$\begin{aligned}
&\max|\delta C| < 1, \\
&\max|\delta\phi| < 1, \\
&c_s^2 > 0, \\
&\text{no ghost,} \\
&\text{acceptable CMB and lensing spectra.}
\end{aligned} \tag{118}$$

The completed model satisfies both levels.

The principal observable effects arise at the background level, but the perturbative sector was evolved rather than assumed away.

E.32 Distinction between physical lensing and reconstruction noise

The physical lensing spectrum is

$$C_L^{\varphi\varphi}. \tag{119}$$

The reconstruction noise is

$$N_L^{(0)}. \tag{120}$$

The locked model produces approximately

$$2.09\% \tag{121}$$

greater weighted physical lensing power and approximately

$$21.3\% \tag{122}$$

greater TT-estimator noise under the stated settings.

The two results describe different parts of the measurement process.

The first concerns the Universe.

The second concerns the estimator.

E.33 Distinction between S_8 and CMB lensing

The parameter

$$S_8 \tag{123}$$

compresses the present matter amplitude and matter fraction.

CMB lensing integrates the Weyl potential across redshift.

Therefore

$$S_8 \downarrow \tag{124}$$

does not require

$$C_L^{\varphi\varphi} \downarrow . \tag{125}$$

The locked calculation explicitly demonstrates this distinction:

$$S_8 = 0.79781, \tag{126}$$

while the weighted lensing ratio is

$$1.0209. \tag{127}$$

The result follows from altered geometry and potential evolution.

E.34 Distinction between saturation and weakened gravity

The final canonical calculation preserves the ordinary Einstein coupling to matter perturbations.

The lowered growth amplitude is produced by

$$H_{\text{locked}}(a), \tag{128}$$

not by a retained factor

$$\mu < 1. \tag{129}$$

The failed phenomenological tests showed that changing the scalar constraints directly produced the wrong growth and peak response.

The formal distinction is:

$$\boxed{\text{saturation changes the cosmic trajectory; it is not represented in the retained model as a simple reduction of}} \tag{130}$$

E.35 Distinction between a residual and a separate late fluid

The late field energy is

$$\rho_C = \frac{1}{2}K^2 + V. \tag{131}$$

The same fields existed during the earlier evolution.

As the fields settle,

$$\rho_C \to V_\infty. \tag{132}$$

No new density component is activated at late time.

The residual is the endpoint of the canonical field trajectory.

The formal distinction is:

$$\boxed{\text{the late accelerated state is dynamically inherited from the saturation sector rather than appended afterwar}} \tag{133}$$

E.36 Reproducibility claim

The background was checked through independent numerical solvers:

$$\text{DOP853} \qquad \text{and} \qquad \text{Radau.} \tag{134}$$

The observational pipeline used one frozen parameter set.

The final reproduction protocol specifies:

$$\begin{aligned}
&\text{the equations,}\\
&\text{the parameter values,}\\
&\text{the transition,}\\
&\text{the perturbation sources,}\\
&\text{the spectrum definitions,}\\
&\text{the statistical calculations.}
\end{aligned} \tag{135}$$

The formal reproducibility claim is:

$$\boxed{\text{the archived locked result is numerically defined well enough to be independently rerun without additional p}} \tag{136}$$

E.37 Falsification by background failure

The model fails if the locked equations cannot reproduce a finite bounce under independent integration.

Specifically, failure occurs if:

$$a_{\min} \leq 0, \qquad K_{\text{bounce}} \to \infty, \qquad \rho > \rho_s \tag{137}$$

on the accepted branch, or

$$\dot{H}_{\text{b}} \leq 0. \tag{138}$$

The model also fails if the post-bounce branch does not enter radiation and matter domination.

E.38 Falsification by instability

The canonical theory fails if an independent implementation finds:

$$\begin{aligned}&\text{a negative kinetic eigenvalue,}\\ &c_s^2 < 0,\\ &V_{,CC} < 0\end{aligned} \tag{139}$$

near the retained lock, or

$$\text{an uncontrolled linear runaway.} \tag{140}$$

The completed calculation finds none of these behaviors.

E.39 Falsification by acoustic geometry

The model fails if its own sound horizon and distance cannot jointly reproduce the observed acoustic angle.

The relevant requirement is not simply a chosen value of

$$r_s. \tag{141}$$

It is the internally calculated ratio

$$\theta_* = \frac{r_s(z_*)}{D_M(z_*)}. \tag{142}$$

The retained result is

$$100\theta_* = 1.041033. \tag{143}$$

E.40 Falsification by BAO

The model fails if the redshift-dependent ratios

$$D_M/r_{\text{drag}}, \qquad D_H/r_{\text{drag}}, \qquad D_V/r_{\text{drag}} \tag{144}$$

are rejected when computed from the same locked background.

The sound horizon cannot be replaced by an unrelated reference ruler to preserve the model.

E.41 Falsification by growth

The model fails if observed growth requires a redshift evolution incompatible with the locked background.

The test is not limited to the compressed value

$$S_8. \tag{145}$$

It includes:

$$f(z), \qquad f\sigma_8(z), \qquad P(k,z), \qquad \text{tomographic weak lensing.} \tag{146}$$

The locked model predicts a background-driven growth history.

E.42 Falsification by primary CMB

The model fails if the complete locked

$$TT, \qquad TE, \qquad EE \tag{147}$$

spectra cannot satisfy the retained primary-CMB likelihood structure.

The relevant test includes:

$$\begin{aligned}&\text{peak positions,}\\&\text{relative peak heights,}\\&\text{polarization,}\\&\text{low-multipole behavior,}\\&\text{the damping tail.}\end{aligned} \tag{148}$$

The completed workflow evaluates these spectra directly.

E.43 Falsification by lensing

The model fails if the predicted

$$C_L^{\varphi\varphi} \tag{149}$$

is rejected by lensing reconstruction or cross-correlation data.

The current retained prediction gives

$$\chi^2 = 10.7379 \tag{150}$$

for the official nine-bin Planck lensing comparison.

Future measurements can test the predicted mild enhancement with greater precision.

E.44 Falsification by local gravitational behavior

The galactic and local interpretation requires recovery of ordinary gravity at high acceleration.

For the representative interpolator,

$$\mu(x) = \frac{x}{\sqrt{1+x^2}}, \tag{151}$$

one has

$$\mu \to 1 \tag{152}$$

when

$$x \to \infty. \tag{153}$$

Therefore

$$g \to g_N. \tag{154}$$

The broader theory fails if a unified local implementation cannot preserve the high-acceleration limit.

E.45 Extensions of the retained model

The completed model provides a base for further calculations without altering the archived result.

Natural extensions include:

$$\begin{aligned}
&\text{nonlinear matter evolution,}\\
&\text{halo formation,}\\
&\text{cluster abundance,}\\
&\text{void statistics,}\\
&\text{tomographic cosmic shear,}\\
&\text{galaxy-galaxy lensing,}\\
&\text{redshift-space distortions,}\\
&\text{strong-lensing systems,}\\
&\text{gravitational-wave propagation across the locked background.}
\end{aligned} \tag{155}$$

These are extensions of the completed linear calculation, not corrections to its reported results.

E.46 Covariant completion of the saturation response

The canonical field sector is covariant.

The high-density homogeneous response is specified through the saturated Friedmann law.

A broader covariant completion may seek a tensor-level gravitational action whose homogeneous reduction gives

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{156}$$

Such a completion would connect the homogeneous saturation correction directly to a generalized geometric action.

The completed cosmological calculation uses the locked background response and canonical perturbation structure already stated.

The homogeneous saturation law is therefore treated as the retained effective gravitational limit of the theory.

E.47 Relation between effective and fundamental descriptions

The equation

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right) \tag{157}$$

is the effective high-density background response used in the numerical model.

The fields C and ϕ provide the canonical organizational dynamics.

A more fundamental derivation may connect ρ_s, C_s, and the field potential to a deeper geometric or quantum-gravitational structure.

The present calculation establishes the consequences of the locked effective realization.

It does not require that deeper derivation to evaluate the internal behavior of the model as implemented.

E.48 Claim hierarchy

The claims of the manuscript form the following hierarchy.

Level I: mathematical existence The equations possess a finite-density bouncing trajectory.

Level II: dynamical stability The canonical field sector is bounded, restorative, and perturbatively stable.

Level III: cosmological continuity The same trajectory passes through radiation, matter, and late residual eras.

Level IV: numerical implementation The background and canonical perturbations are implemented in CAMB.

Level V: observational calculation The model produces the completed acoustic, BAO, growth, primary-CMB, and lensing results.

Level VI: physical interpretation The connected behavior is interpreted as gravitational organization and saturation.

The higher-level interpretation is grounded in the lower-level mathematical and numerical results.

E.49 Formal statement of the completed result

The completed model demonstrates the existence of a retained parameter set for which:

$$
\begin{aligned}
a_{\min} &> 0, && (158)\\
\rho_{\mathrm{b}} &= \rho_s, && (159)\\
\dot{H}_{\mathrm{b}} &> 0, && (160)\\
K_{\mathrm{bounce}} &< \infty, && (161)\\
\Gamma_{\mathrm{tr}} &= 0 \quad \text{for} \quad H \geq 0, && (162)\\
w_C(0) &\simeq -0.99988, && (163)\\
100\theta_* &= 1.041033, && (164)\\
r_{\mathrm{drag}} &= 149.960 \text{ Mpc}, && (165)\\
\sigma_8 &= 0.779680, && (166)\\
S_8 &= 0.79781, && (167)\\
\max|\delta C| &= 0.03514, && (168)\\
\max|\delta\phi| &= 0.04860, && (169)\\
\chi^2_{\mathrm{BAO}} &\simeq 13.4, && (170)\\
\chi^2_{\mathrm{Planck\ lens}} &= 10.7379. && (171)
\end{aligned}
$$

These results are generated by one locked background and one canonical perturbation system.

E.50 Formal statement of theoretical identity

The theory is not identified by any one of the following alone:

$$\text{a bounce,} \qquad w \simeq -1, \qquad \text{a low } S_8, \qquad \text{a modified sound horizon.} \tag{172}$$

Its identity is the connected system:

$$\boxed{\text{wave-bearing gravitational organization + high-density saturation + canonical field locking + correlated cosm}} \tag{173}$$

Removing any major element changes the theory.

E.51 Final claim boundary

The strongest justified statement of the completed work is:

$$\boxed{\begin{array}{l}\text{A concrete gravitational-saturation model has been formulated,}\\ \text{integrated through a finite-density nonsingular bounce,}\\ \text{embedded in a full Boltzmann calculation,}\\ \text{evolved with canonical linear saturation fields,}\\ \text{and evaluated through acoustic, BAO, growth,}\\ \text{primary-CMB, reconstruction, and Planck-lensing calculations.}\end{array}} \tag{174}$$

The model remains stable in the completed linear calculation.
Its primary observable effects arise from the locked expansion history.
Its canonical perturbations remain small.
Its final numerical outputs form one coherent and reproducible cosmological result.

E.52 Closing scope statement

Gravitational saturation is presented as a physical and mathematical alternative to unrestricted gravitational evolution.

At high density, the gravitational response reaches a finite limit.
During contraction, kinetic growth is regulated through conservative transfer.
At the bounce, the Hubble rate crosses zero at finite scale factor.
During expansion, the canonical fields settle into their stable minimum.
Their residual energy drives the late accelerated background.
The Universe retains the memory of this history through:

$$r_s, \quad r_{\mathrm{drag}}, \quad D_M, \quad S_8, \quad C_\ell^{TT}, \quad C_\ell^{TE}, \quad C_\ell^{EE}, \quad C_L^{\varphi\varphi}. \tag{175}$$

The completed theory therefore stands as one continuous proposal:

$$\boxed{\text{gravity organizes, reaches saturation, rebounds, and locks,}} \tag{176}$$

$$\boxed{\text{while preserving a stable and observationally calculable cosmology.}} \tag{177}$$

End of Appendix E

Appendix F

Equation Index, Cross-Reference Map, and Model Dependency Structure

F.1 Purpose of the appendix

This appendix provides a consolidated map of the gravitational-saturation theory.

Its purpose is to show how the principal equations, numerical parameters, physical mechanisms, and observational results depend upon one another.

The theory is not a sequence of unrelated calculations.

It is a dependency chain:

$$\boxed{\text{physical premise} \rightarrow \text{field structure} \rightarrow \text{background evolution} \rightarrow \text{locked expansion} \rightarrow \text{linear perturbations} \rightarrow \text{ob}} \tag{1}$$

This appendix allows any equation or numerical result to be traced backward to the assumptions from which it follows.

Part I — Fundamental Model Definitions

F.2 Organizational and phase fields

The gravitational-saturation sector contains two canonical scalar variables:

$$\boxed{C(x^\mu)} \qquad \text{and} \qquad \boxed{\phi(x^\mu)}. \tag{2}$$

The field C represents gravitational organization.

The field ϕ represents the periodic phase response.

Their homogeneous components are

$$C = C(t), \qquad \phi = \phi(t). \tag{3}$$

Their cosmic-time momenta are

$$\boxed{\Pi_C = \dot{C}} \qquad \text{and} \qquad \boxed{\Pi_\phi = \dot{\phi}}. \tag{4}$$

The combined kinetic quantity is

$$\boxed{K^2 = \Pi_C^2 + \Pi_\phi^2}. \tag{5}$$

F.3 Saturation potential

The complete field potential is

$$\boxed{V(C,\phi) = V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4 + A_0(1 - \cos\phi)}. \tag{6}$$

The organizational derivative is

$$\boxed{V_{,C} = m_C^2(C - C_s) + \lambda(C - C_s)^3}. \tag{7}$$

The phase derivative is

$$\boxed{V_{,\phi} = A_0 \sin\phi}. \tag{8}$$

The organizational curvature is

$$\boxed{V_{,CC} = m_C^2 + 3\lambda(C - C_s)^2}. \tag{9}$$

The phase curvature is

$$\boxed{V_{,\phi\phi} = A_0 \cos\phi}. \tag{10}$$

The mixed curvature vanishes:

$$\boxed{V_{,C\phi} = 0}. \tag{11}$$

The stable minimum is

$$\boxed{C = C_s, \qquad \phi = 0}. \tag{12}$$

At that point,

$$\boxed{V(C_s, 0) = V_\infty}. \tag{13}$$

F.4 Locked potential parameters

The retained values are

$$\boxed{C_s = 0.95}, \tag{14}$$

$$\boxed{V_\infty = 2.187987}, \tag{15}$$

$$\boxed{m_C^2 = 1.827221}, \tag{16}$$

$$\boxed{\lambda = 0.802722}. \tag{17}$$

The phase amplitude satisfies

$$\boxed{A_0 > 0} \tag{18}$$

for stability at

$$\phi = 0. \tag{19}$$

These parameters determine:

$$\begin{aligned}&\text{the position of the field minimum,}\\&\text{the late residual energy,}\\&\text{the restoring force,}\\&\text{the linear field masses.}\end{aligned} \tag{20}$$

Part II — Stress-Energy and Conservation

F.5 Saturation density and pressure

The canonical field density is

$$\boxed{\rho_C = \frac{1}{2}K^2 + V}. \tag{21}$$

The field pressure is

$$\boxed{p_C = \frac{1}{2}K^2 - V}. \tag{22}$$

The equation of state is

$$\boxed{w_C = \frac{\frac{1}{2}K^2 - V}{\frac{1}{2}K^2 + V}}. \tag{23}$$

The enthalpy identity is

$$\boxed{\rho_C + p_C = K^2}. \tag{24}$$

The kinetic fraction is

$$\boxed{1 + w_C = \frac{K^2}{\rho_C}}. \tag{25}$$

These equations connect the canonical field dynamics to the late residual behavior.

F.6 Matter and radiation sectors

Pressureless matter satisfies

$$\boxed{p_m = 0} \qquad \text{and} \qquad \boxed{\dot{\rho}_m = -3H\rho_m}. \tag{26}$$

Radiation satisfies

$$\boxed{p_r = \frac{1}{3}\rho_r} \qquad \text{and} \qquad \boxed{\dot{\rho}_r = -4H\rho_r + Q}. \tag{27}$$

The total density is

$$\boxed{\rho = \rho_m + \rho_r + \rho_C}. \tag{28}$$

The total pressure is

$$\boxed{p = \frac{1}{3}\rho_r + p_C}. \tag{29}$$

Therefore

$$\boxed{\rho + p = \rho_m + \frac{4}{3}\rho_r + K^2}. \tag{30}$$

For the retained canonical components,

$$\rho + p \geq 0. \tag{31}$$

F.7 Contraction transfer

The transfer rate is

$$\boxed{\Gamma_{\rm tr} = W_-(H)\big[-3H + \Gamma_0 S_\rho\big]}. \tag{32}$$

The target density is

$$\boxed{\rho_{\rm tar} = \rho_m + \rho_r}. \tag{33}$$

The density activation function is

$$\boxed{S_\rho = \frac{(\rho_{\rm tar}/\rho_\Gamma)^m}{1 + (\rho_{\rm tar}/\rho_\Gamma)^m}}. \tag{34}$$

The transferred power is

$$\boxed{Q = \Gamma_{\rm tr} K^2}. \tag{35}$$

The field continuity equation is

$$\boxed{\dot\rho_C + 3H(\rho_C + p_C) = -Q}. \tag{36}$$

The radiation equation is

$$\boxed{\dot\rho_r + 4H\rho_r = +Q}. \tag{37}$$

The transfer therefore cancels internally:

$$\boxed{Q_{\rm fields} + Q_{\rm radiation} = 0}. \tag{38}$$

F.8 Total conservation

Adding matter, radiation, and field conservation gives

$$\boxed{\dot\rho + 3H(\rho + p) = 0}. \tag{39}$$

This equation is required for consistency between the saturated Friedmann constraint and the Raychaudhuri equation.

The dependency is

$$\boxed{\text{field equations} + \text{radiation transfer} + \text{matter conservation} \Rightarrow \text{total conservation}}. \tag{40}$$

Part III — Saturated Background Geometry

F.9 Saturated Friedmann constraint

The generalized response is

$$\boxed{H^2 = \frac{\rho}{3}\left[1 - \left(\frac{\rho}{\rho_s}\right)^q\right]}. \tag{41}$$

The retained model uses

$$\boxed{q = 1}. \tag{42}$$

Therefore

$$\boxed{H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right)}. \tag{43}$$

The locked saturation density is

$$\boxed{\rho_s = 1.58199 \times 10^{15}}. \tag{44}$$

This equation determines:

$$\begin{aligned} &\text{the bounded Hubble rate,} \\ &\text{the finite-density bounce,} \\ &\text{the low-density return to ordinary Friedmann evolution.} \end{aligned} \tag{45}$$

F.10 Low-density limit

When

$$\rho \ll \rho_s, \tag{46}$$

the correction becomes negligible:

$$1 - \frac{\rho}{\rho_s} \simeq 1. \tag{47}$$

Thus

$$\boxed{H^2 \simeq \frac{\rho}{3}}. \tag{48}$$

The fractional correction is

$$\boxed{\frac{H^2 - H^2_{\text{standard}}}{H^2_{\text{standard}}} = -\frac{\rho}{\rho_s}}. \tag{49}$$

The standard low-density expansion is recovered continuously.

F.11 Raychaudhuri equation

Differentiating the saturated constraint and using total conservation gives

$$\boxed{\dot{H} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right)}. \tag{50}$$

This equation determines the sign of the bounce crossing.

At

$$\rho = \rho_s, \tag{51}$$

one obtains

$$\boxed{\dot{H}_{\text{b}} = \frac{1}{2}(\rho_{\text{b}} + p_{\text{b}})}. \tag{52}$$

Since

$$\rho_\mathrm{b} + p_\mathrm{b} > 0, \tag{53}$$

the retained trajectory satisfies

$$\boxed{\dot{H}_\mathrm{b} > 0}. \tag{54}$$

F.12 Acceleration equation

Using

$$\frac{\ddot{a}}{a} = \dot{H} + H^2, \tag{55}$$

the acceleration equation becomes

$$\boxed{\frac{\ddot{a}}{a} = -\frac{1}{6}(\rho + 3p) + \frac{\rho}{\rho_s}\left(\frac{2}{3}\rho + p\right)}. \tag{56}$$

At the bounce,

$$H = 0, \tag{57}$$

so

$$\boxed{\frac{\ddot{a}_\mathrm{b}}{a_\mathrm{b}} = \frac{1}{2}(\rho_\mathrm{b} + p_\mathrm{b}) > 0}. \tag{58}$$

Therefore the bounce is a genuine minimum of the scale factor.

F.13 Maximum Hubble rate

Define

$$x = \frac{\rho}{\rho_s}. \tag{59}$$

Then

$$H^2 = \frac{\rho_s}{3}x(1 - x). \tag{60}$$

The maximum occurs at

$$x = \frac{1}{2}. \tag{61}$$

Thus

$$\boxed{H^2_\mathrm{max} = \frac{\rho_s}{12}} \qquad \text{and} \qquad \boxed{|H|_\mathrm{max} = \sqrt{\frac{\rho_s}{12}}}. \tag{62}$$

This is the central bounded-response result of the high-density theory.

Part IV — Complete Homogeneous Evolution System

F.14 State vector

The complete homogeneous state is

$$\boxed{\mathbf{y} = \big(a, H, C, \Pi_C, \phi, \Pi_\phi, \rho_m, \rho_r\big)}. \tag{63}$$

Every final background quantity follows from the evolution of this vector.

F.15 First-order equations

The scale factor evolves according to

$$\boxed{\dot{a} = aH}. \tag{64}$$

The Hubble rate evolves according to

$$\boxed{\dot{H} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right)}. \tag{65}$$

The organizational field evolves according to

$$\boxed{\dot{C} = \Pi_C} \qquad \text{and} \qquad \boxed{\dot{\Pi}_C = -(3H + \Gamma_{\rm tr})\Pi_C - V_{,C}}. \tag{66}$$

The phase field evolves according to

$$\boxed{\dot{\phi} = \Pi_\phi} \qquad \text{and} \qquad \boxed{\dot{\Pi}_\phi = -(3H + \Gamma_{\rm tr})\Pi_\phi - V_{,\phi}}. \tag{67}$$

Matter evolves according to

$$\boxed{\dot{\rho}_m = -3H\rho_m}. \tag{68}$$

Radiation evolves according to

$$\boxed{\dot{\rho}_r = -4H\rho_r + \Gamma_{\rm tr}K^2}. \tag{69}$$

F.16 Contraction branch

During contraction,

$$H < 0. \tag{70}$$

The ordinary term

$$3H \tag{71}$$

acts as anti-friction.

Deep in contraction,

$$W_- \simeq 1, \tag{72}$$

and

$$3H + \Gamma_{\rm tr} \simeq \Gamma_0 S_\rho. \tag{73}$$

Therefore

$$\boxed{\text{contraction anti-friction} \rightarrow \text{positive regulated damping}}. \tag{74}$$

Field kinetic energy is transferred into radiation.

F.17 Expanding branch

For

$$H \geq 0, \tag{75}$$

the transfer shuts off:

$$\boxed{\Gamma_{\text{tr}} = 0}. \tag{76}$$

The equations reduce to

$$\boxed{\ddot{C} + 3H\dot{C} + V_{,C} = 0} \quad \text{and} \quad \boxed{\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} = 0}. \tag{77}$$

Radiation obeys

$$\boxed{\rho_r \propto a^{-4}} \tag{78}$$

and matter obeys

$$\boxed{\rho_m \propto a^{-3}}. \tag{79}$$

Part V — Milestones of the Locked Background

F.18 Bounce scale

The minimum scale factor is

$$\boxed{a_{\text{min}} \simeq 2.068 \times 10^{-5}}. \tag{80}$$

The corresponding redshift is

$$\boxed{z_{\text{bounce}} \simeq 4.84 \times 10^4}. \tag{81}$$

At the bounce,

$$\boxed{H = 0}, \qquad \boxed{\rho = \rho_s}, \qquad \boxed{K_{\text{bounce}} < \infty}. \tag{82}$$

F.19 Lock transition

The retained transition endpoint is

$$\boxed{a_{\text{lock}} = 2.05 \times 10^{-4}}. \tag{83}$$

The retained width is

$$\boxed{W = 0.67 W_0}. \tag{84}$$

The transition lies after the bounce:

$$a_{\text{min}} < a_{\text{lock}}. \tag{85}$$

It also lies before matter-radiation equality:

$$a_{\text{lock}} < a_{\text{eq}}. \tag{86}$$

F.20 Matter-radiation equality

Equality is

$$\boxed{a_{\rm eq} = \frac{\Omega_{r0}}{\Omega_{m0}}}. \tag{87}$$

Using

$$\Omega_{r0} = 9 \times 10^{-5}, \qquad \Omega_{m0} = 0.314114, \tag{88}$$

one obtains

$$\boxed{a_{\rm eq} \simeq 2.865 \times 10^{-4}}. \tag{89}$$

The equality redshift is

$$\boxed{z_{\rm eq} \simeq 3489}. \tag{90}$$

The chronological ordering is

$$\boxed{a_{\rm min} < a_{\rm lock} < a_{\rm eq} < a_0}. \tag{91}$$

F.21 Late field lock

During expansion,

$$\dot{\rho}_C = -3HK^2. \tag{92}$$

Since

$$H > 0, \tag{93}$$

the field energy decreases toward its lower bound.

The asymptotic limits are

$$\boxed{C \to C_s}, \tag{94}$$

$$\boxed{\phi \to 0}, \tag{95}$$

$$\boxed{\Pi_C \to 0}, \tag{96}$$

$$\boxed{\Pi_\phi \to 0}, \tag{97}$$

$$\boxed{\rho_C \to V_\infty}, \tag{98}$$

$$\boxed{w_C \to -1}. \tag{99}$$

The present value is

$$\boxed{w_C(0) \simeq -0.99988}. \tag{100}$$

Part VI — Numerical Background Construction

F.22 Nonsingular time variable

The bounce is integrated in cosmic time or scaled affine time.

The variable

$$N = \ln a \tag{101}$$

is not used through the bounce because

$$\frac{d}{dN} = \frac{1}{H}\frac{d}{dt}. \tag{102}$$

At

$$H = 0, \tag{103}$$

this expression is singular.

The numerical rule is

$$\boxed{\text{no division by } H \text{ through the bounce}}. \tag{104}$$

F.23 Scaled variables

The scaled variables are

$$\boxed{h = \frac{H}{\sqrt{\rho_s}}}, \tag{105}$$

$$\boxed{u_C = \frac{\Pi_C}{\sqrt{\rho_s}}}, \tag{106}$$

$$\boxed{u_\phi = \frac{\Pi_\phi}{\sqrt{\rho_s}}}, \tag{107}$$

$$\boxed{\bar{\rho} = \frac{\rho}{\rho_s}}. \tag{108}$$

The scaled constraint is

$$\boxed{h^2 = \frac{\bar{\rho}}{3}(1 - \bar{\rho})}. \tag{109}$$

The scaled Raychaudhuri equation is

$$\boxed{\frac{dh}{d\tau_{\mathrm{aff}}} = -\frac{1}{2}(\bar{\rho} + \bar{p})(1 - 2\bar{\rho})}. \tag{110}$$

F.24 Solver verification

The background is evolved using

$$\boxed{\text{DOP853}} \qquad \text{and} \qquad \boxed{\text{Radau}}. \tag{111}$$

The final background is retained only after agreement in:

$$a_{\min}, \quad K_{\mathrm{bounce}}, \quad E(a), \quad \rho_C(0), \quad r_{\mathrm{drag}}, \quad 100\theta_*. \tag{112}$$

The dependency is

$$\boxed{\text{one physical equation set} + \text{two independent numerical methods} \Rightarrow \text{accepted locked trajectory}}. \tag{113}$$

F.25 Constraint residual

The numerical residual is

$$\boxed{\mathcal{C}_H = H^2 - \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right)}. \tag{114}$$

The constraint is monitored throughout the trajectory.

A sign-preserving checkpoint projection is

$$\boxed{H \to \operatorname{sgn}(H)\sqrt{\frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right)}}. \tag{115}$$

The projection controls numerical drift but does not replace the differential equation for H.

Part VII — CAMB Background Dependency

F.26 Expansion function

The expanding branch defines

$$\boxed{E(a) = \frac{H(a)}{H_0}}. \tag{116}$$

The normalization condition is

$$\boxed{E(1) = 1}. \tag{117}$$

The physical Hubble rate is

$$\boxed{H(a) = H_0 E(a)}. \tag{118}$$

The conformal Hubble parameter is

$$\boxed{\mathcal{H} = aH}. \tag{119}$$

These quantities enter every subsequent Boltzmann calculation.

F.27 Smooth transition

Let

$$S(a) \tag{120}$$

be the smooth transition function.

The background blend is

$$\boxed{E^2_{\text{blend}}(a) = [1 - S(a)]E^2_{\text{early}}(a) + S(a)E^2_{\text{locked}}(a)}. \tag{121}$$

The smoothstep form is

$$\boxed{S(x) = 3x^2 - 2x^3} \tag{122}$$

within the transition interval.

This equation determines the amount of the resolved post-bounce history passed into the Boltzmann integration.

F.28 Transition dependency

Changing

$$a_{\mathrm{lock}} \qquad \text{or} \qquad W \tag{123}$$

changes:

$$r_s, \quad r_{\mathrm{drag}}, \quad D_M(z_*), \tag{124}$$

and therefore

$$\theta_*. \tag{125}$$

The dependency is

$$\boxed{(a_{\mathrm{lock}}, W) \to E_{\mathrm{early}}(a) \to r_s,\ D_M \to \theta_*}. \tag{126}$$

The final retained values are

$$a_{\mathrm{lock}} = 2.05 \times 10^{-4}, \qquad W = 0.67 W_0. \tag{127}$$

Part VIII — Linear Saturation Perturbations

F.29 Perturbation variables

The field perturbation state is

$$\boxed{\left(\delta C, v_C, \delta\phi, v_\phi\right)} \tag{128}$$

with

$$v_C = \delta C', \qquad v_\phi = \delta\phi'. \tag{129}$$

F.30 Organizational perturbation equation

The equation is

$$\boxed{\delta C'' + 2\mathcal{H}\delta C' + \left(k^2 + a^2 V_{,CC}\right)\delta C + \frac{1}{2} C'_{\mathrm{bg}} h'_{\mathrm{syn}} = 0}. \tag{130}$$

In first-order form,

$$\boxed{\delta C' = v_C} \tag{131}$$

and

$$\boxed{v'_C = -2\mathcal{H} v_C - \left(k^2 + a^2 V_{,CC}\right)\delta C - \frac{1}{2} C'_{\mathrm{bg}} h'_{\mathrm{syn}}}. \tag{132}$$

F.31 Phase perturbation equation

The equation is

$$\boxed{\delta\phi'' + 2\mathcal{H}\delta\phi' + \left(k^2 + a^2 V_{,\phi\phi}\right)\delta\phi + \frac{1}{2}\phi'_{\rm bg}h'_{\rm syn} = 0}. \tag{133}$$

In first-order form,

$$\boxed{\delta\phi' = v_\phi} \tag{134}$$

and

$$\boxed{v'_\phi = -2\mathcal{H}v_\phi - \left(k^2 + a^2 V_{,\phi\phi}\right)\delta\phi - \frac{1}{2}\phi'_{\rm bg}h'_{\rm syn}}. \tag{135}$$

F.32 Perturbed stress-energy

The density perturbation is

$$\boxed{\delta\rho_C = \frac{1}{a^2}\left(C'_{\rm bg}v_C + \phi'_{\rm bg}v_\phi\right) + V_{,C}\delta C + V_{,\phi}\delta\phi}. \tag{136}$$

The pressure perturbation is

$$\boxed{\delta p_C = \frac{1}{a^2}\left(C'_{\rm bg}v_C + \phi'_{\rm bg}v_\phi\right) - V_{,C}\delta C - V_{,\phi}\delta\phi}. \tag{137}$$

The momentum perturbation is

$$\boxed{\delta q_C = \frac{k}{a^2}\left(C'_{\rm bg}\delta C + \phi'_{\rm bg}\delta\phi\right)}. \tag{138}$$

The intrinsic anisotropic stress is

$$\boxed{\pi_C = 0}. \tag{139}$$

F.33 Suppression dependency

Near the lock,

$$C'_{\rm bg} \to 0, \qquad \phi'_{\rm bg} \to 0, \qquad V_{,C} \to 0, \qquad V_{,\phi} \to 0. \tag{140}$$

Therefore

$$\boxed{\delta\rho_C, \delta p_C, \delta q_C \to 0} \tag{141}$$

for finite field perturbations.

The dependency is

$$\boxed{\text{background locking} \to \text{small stress-energy coefficients} \to \text{spectator perturbations}}. \tag{142}$$

The maximum amplitudes are

$$\boxed{\max|\delta C| = 0.03514} \qquad \text{and} \qquad \boxed{\max|\delta\phi| = 0.04860}. \tag{143}$$

F.34 Stability dependency

The kinetic matrix is

$$\boxed{K_{IJ} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}. \tag{144}$$

The sound speeds are

$$\boxed{c_{s,C}^2 = c_{s,\phi}^2 = 1}. \tag{145}$$

The mass matrix is

$$\boxed{M^2 = \begin{pmatrix} V_{,CC} & 0 \\ 0 & V_{,\phi\phi} \end{pmatrix}}. \tag{146}$$

Near the lock,

$$V_{,CC} = m_C^2 > 0, \qquad V_{,\phi\phi} = A_0 > 0. \tag{147}$$

Thus

$$\boxed{\text{positive kinetic matrix} + \text{positive gradients} + \text{positive mass curvature} \Rightarrow \text{stable linear fields}}. \tag{148}$$

Part IX — Acoustic and Distance Dependency

F.35 Sound speed

The baryon-loaded sound speed is

$$\boxed{c_s(z) = \frac{c}{\sqrt{3[1 + R(z)]}}} \tag{149}$$

with

$$\boxed{R(z) = \frac{3\rho_b}{4\rho_\gamma}}. \tag{150}$$

F.36 Sound horizons

The recombination sound horizon is

$$\boxed{r_s(z_*) = \int_{z_*}^{\infty} \frac{c_s(z)}{H(z)}\, dz}. \tag{151}$$

The drag horizon is

$$\boxed{r_{\rm drag} = \int_{z_{\rm drag}}^{\infty} \frac{c_s(z)}{H(z)}\, dz}. \tag{152}$$

The final drag horizon is

$$\boxed{r_{\rm drag} = 149.960\ \text{Mpc}}. \tag{153}$$

F.37 Cosmological distances

The Hubble distance is

$$\boxed{D_H(z) = \frac{c}{H(z)}}. \tag{154}$$

The comoving transverse distance is

$$\boxed{D_M(z) = \int_0^z \frac{c\,dz'}{H(z')}}. \tag{155}$$

The angular-diameter distance is

$$\boxed{D_A(z) = \frac{D_M(z)}{1+z}}. \tag{156}$$

The luminosity distance is

$$\boxed{D_L(z) = (1+z)D_M(z)}. \tag{157}$$

The volume distance is

$$\boxed{D_V(z) = \left[zD_M^2(z)D_H(z)\right]^{1/3}}. \tag{158}$$

F.38 Acoustic angle

The acoustic angle is

$$\boxed{\theta_* = \frac{r_s(z_*)}{D_M(z_*)}}. \tag{159}$$

The final result is

$$\boxed{100\theta_* = 1.041033}. \tag{160}$$

The dependency chain is

$$\boxed{H(z) \to r_s(z_*), D_M(z_*) \to \theta_*}. \tag{161}$$

The final comoving distance is

$$\boxed{D_M(z_*) = 14131.86 \text{ Mpc}}. \tag{162}$$

F.39 BAO dependency

The BAO observables are

$$\boxed{D_M/r_{\text{drag}}}, \qquad \boxed{D_H/r_{\text{drag}}}, \qquad \boxed{D_V/r_{\text{drag}}}. \tag{163}$$

Both the numerator and denominator depend upon the locked expansion history.

The correct dependency is

$$\boxed{H(z) \to \begin{cases} D_M,\ D_H,\ D_V, \\ r_{\text{drag}} \end{cases} \to \text{BAO ratios}}. \tag{164}$$

The final diagnostic statistic is

$$\boxed{\chi^2_{\rm BAO} \simeq 13.4}. \tag{165}$$

Part X — Growth Dependency

F.40 Growth equation

On the expanding branch,

$$N = \ln a \tag{166}$$

may be used.

The growth equation is

$$\boxed{\frac{d^2D}{dN^2} + \left(2 + \frac{d\ln H}{dN}\right)\frac{dD}{dN} - \frac{3}{2}\Omega_m(a)D = 0}. \tag{167}$$

The time-dependent matter fraction is

$$\boxed{\Omega_m(a) = \frac{\Omega_{m0}a^{-3}}{E^2(a)}}. \tag{168}$$

The locked background modifies:

$$\frac{d\ln H}{dN} \qquad \text{and} \qquad \Omega_m(a). \tag{169}$$

It does not replace the gravitational coefficient

$$\frac{3}{2} \tag{170}$$

with a retained phenomenological μ.

F.41 Growth observables

The final fluctuation amplitude is

$$\boxed{\sigma_8 = 0.779680}. \tag{171}$$

The weak-lensing combination is

$$\boxed{S_8 = \sigma_8\sqrt{\frac{\Omega_{m0}}{0.3}}}. \tag{172}$$

The final value is

$$\boxed{S_8 = 0.79781}. \tag{173}$$

The dependency is

$$\boxed{H(a) \to D(a) \to P(k,a) \to \sigma_8 \to S_8}. \tag{174}$$

Part XI — Primary CMB Dependency

F.42 Primordial spectrum

The scalar primordial spectrum is

$$\boxed{\mathcal{P}_{\mathcal{R}}(k) = A_s \left(\frac{k}{k_*}\right)^{n_s-1}}. \tag{175}$$

The retained values are

$$\boxed{A_s = 2.0 \times 10^{-9}} \qquad \text{and} \qquad \boxed{n_s = 0.965}. \tag{176}$$

The optical depth is

$$\boxed{\tau_{\text{reio}} = 0.054}. \tag{177}$$

These values are matched between the locked model and control.

F.43 Angular spectra

The primary spectra are

$$\boxed{C_\ell^{TT}, \quad C_\ell^{TE}, \quad C_\ell^{EE}}. \tag{178}$$

The plotted form is

$$\boxed{D_\ell^{XY} = \frac{\ell(\ell+1)}{2\pi} C_\ell^{XY}}. \tag{179}$$

The spectra depend upon:

$$\begin{gathered}\text{the primordial spectrum,}\\ \text{the locked } H(a),\\ \text{recombination,}\\ \text{metric evolution,}\\ \text{canonical field sources,}\\ \text{lensing.}\end{gathered} \tag{180}$$

The dependency is

$$\boxed{\left[\mathcal{P}_{\mathcal{R}}, H(a), \delta T_{\mu\nu}\right] \to \text{transfer functions} \to C_\ell^{TT,TE,EE}}. \tag{181}$$

F.44 Spectral residuals

For TT and EE,

$$\boxed{R_\ell^{XX} = \frac{D_{\ell,\text{locked}}^{XX} - D_{\ell,\text{control}}^{XX}}{D_{\ell,\text{control}}^{XX}}}. \tag{182}$$

For TE,

$$\boxed{R_\ell^{TE} = \frac{D_{\ell,\mathrm{locked}}^{TE} - D_{\ell,\mathrm{control}}^{TE}}{\sqrt{D_{\ell,\mathrm{control}}^{TT} D_{\ell,\mathrm{control}}^{EE}}}}. \tag{183}$$

The largest residuals occur at:

$$\boxed{\text{low multipoles and the first acoustic region}}. \tag{184}$$

The residuals decrease toward the high-ℓ tail.

F.45 Primary-CMB dependency map

The principal dependencies are:

$$\boxed{a_{\mathrm{lock}}, W \to r_s, \theta_* \to \text{peak positions}}, \tag{185}$$

$$\boxed{H(a) \to \text{potential decay} \to \text{acoustic driving}}, \tag{186}$$

$$\boxed{H(a) \to \text{distance projection} \to \ell\text{-space peak mapping}}, \tag{187}$$

$$\boxed{C'_{\mathrm{bg}}, \phi'_{\mathrm{bg}}, V_{,C}, V_{,\phi} \to \delta T_{\mu\nu}^{C\phi} \to \text{small direct field correction}}. \tag{188}$$

Part XII — CMB Lensing Dependency

F.46 Lensing potential

The projected lensing potential is

$$\boxed{\varphi_{\mathrm{lens}}(\hat{\mathbf{n}}) = -2 \int_0^{\chi_*} d\chi \frac{\chi_* - \chi}{\chi_* \chi} \Phi_W}. \tag{189}$$

The Weyl potential is

$$\boxed{\Phi_W = \frac{\Phi + \Psi}{2}}. \tag{190}$$

The angular spectrum is

$$\boxed{C_L^{\varphi\varphi}}. \tag{191}$$

F.47 Lensing dependency map

CMB lensing depends upon:

$$\boxed{\text{matter growth}}, \quad \boxed{\text{Weyl-potential evolution}}, \quad \boxed{D_M(z_*)}, \quad \boxed{\text{the full line-of-sight kernel}}. \tag{192}$$

The dependency is

$$\boxed{H(a) \to \begin{cases} D(a), \\ \chi(z), \\ \chi_* \end{cases} \to C_L^{\varphi\varphi}}. \tag{193}$$

This is why a lower present-day S_8 does not require lower CMB lensing power.

F.48 Final lensing result

The weighted locked-control ratio is

$$\boxed{\left\langle \frac{C_{L,\text{locked}}^{\varphi\varphi}}{C_{L,\text{control}}^{\varphi\varphi}} \right\rangle = 1.0209}. \tag{194}$$

The physical lensing spectrum is therefore enhanced by approximately

$$\boxed{2.09\%}. \tag{195}$$

F.49 Planck bandpower statistic

The binned prediction is

$$\boxed{P_b = \sum_L W_{bL} C_L^{\varphi\varphi}}. \tag{196}$$

The statistic is

$$\boxed{\chi^2 = (\mathbf{P} - \mathbf{D})^{\mathrm{T}} \Sigma^{-1} (\mathbf{P} - \mathbf{D})}. \tag{197}$$

The final results are

$$\boxed{\chi^2_{\text{locked}} = 10.7379} \qquad \text{and} \qquad \boxed{\chi^2_{\text{control}} = 12.0022}. \tag{198}$$

Therefore

$$\boxed{\Delta\chi^2 = -1.2643}. \tag{199}$$

F.50 Reconstruction-noise dependency

The quadratic-estimator reconstruction noise is

$$\boxed{N_L^{(0)}}. \tag{200}$$

It depends on the primary CMB spectra and filtering assumptions.

The dependency is

$$\boxed{C_\ell^{TT} + \text{beam} + \text{instrument noise} \to N_L^{(0)}}. \tag{201}$$

The final mean ratio is

$$\boxed{\frac{N^{(0)}_{L,\mathrm{locked}}}{N^{(0)}_{L,\mathrm{control}}} \simeq 1.213}. \tag{202}$$

Thus the locked primary spectrum produces approximately

$$\boxed{21.3\%} \tag{203}$$

higher TT reconstruction noise under the stated forecast settings.

Part XIII — Failed Closure Dependency

F.51 Phenomenological Poisson modification

The tested form was

$$\boxed{k^2\Psi = -4\pi G a^2 \mu(a,k)\rho\Delta}. \tag{204}$$

The tested slip was

$$\boxed{\eta(a,k) = \frac{\Phi}{\Psi}}. \tag{205}$$

The tested relations included

$$\boxed{\eta = 1}, \qquad \boxed{\eta = \frac{1}{\mu}}, \qquad \boxed{\eta = 2 - \mu}. \tag{206}$$

A shared factor

$$f_{\mathrm{sat}}(a,k) \tag{207}$$

was also inserted into the scalar constraints.

F.52 Failure chain

The direct constraint modifications altered:

$$\begin{aligned}&\text{metric driving,}\\ &\text{potential decay,}\\ &\text{matter growth,}\\ &\text{odd-even peak amplitudes.}\end{aligned} \tag{208}$$

The numerical outcome was

$$\boxed{\mu, \eta, f_{\mathrm{sat}} \to \sigma_8 \uparrow + \text{peak distortion}}. \tag{209}$$

The canonical model instead gives

$$\boxed{\text{locked } H(a) + \text{small additive canonical stress-energy} \to S_8 \downarrow + \text{stable fields}}. \tag{210}$$

Therefore

$$\boxed{\text{phenomenological force rescaling} \neq \text{canonical gravitational saturation}}. \tag{211}$$

Part XIV — Parameter-to-Observable Dependency Matrix

F.53 V_∞

The parameter

$$V_\infty \tag{212}$$

primarily controls:

$$\begin{aligned} &\text{the residual field density,}\\ &w_C(a),\\ &\text{late } H(a),\\ &D_M(z),\\ &\theta_*. \end{aligned} \tag{213}$$

Its dependency chain is

$$\boxed{V_\infty \to \rho_C(a) \to H(a) \to \text{distance and growth observables}}. \tag{214}$$

It is the most sensitive direction in the retained local robustness scan.

F.54 ρ_s

The saturation density controls:

$$\begin{aligned} &\text{the density of the bounce,}\\ &|H|_{\max},\\ &a_{\min},\\ &\text{the early nonsingular trajectory.} \end{aligned} \tag{215}$$

The dependency is

$$\boxed{\rho_s \to \text{high-density response} \to \text{bounce geometry} \to \text{post-bounce history}}. \tag{216}$$

F.55 m_C^2

The organizational curvature controls:

$$\begin{aligned} &\text{the restoration rate,}\\ &\text{the approach to } C_s,\\ &V_{,CC},\\ &\text{the mass of } \delta C. \end{aligned} \tag{217}$$

The dependency is

$$\boxed{m_C^2 \to V_{,C}, V_{,CC} \to C_{\rm bg}(a), \delta C}. \tag{218}$$

F.56 λ

The quartic coupling controls the nonlinear restoration away from the minimum.

Its dependency is

$$\boxed{\lambda \to \text{large-displacement field dynamics} \to \text{contraction and settling trajectory}}. \tag{219}$$

Near the final lock, the quadratic curvature dominates.

F.57 $C_{\rm initial}$ and $\Pi_{C,\rm initial}$

These parameters determine the starting position and motion of the organizational field.

The dependency is

$$\boxed{(C_{\rm initial}, \Pi_{C,\rm initial}) \to \text{field trajectory} \to \rho_C(a) \to H(a)}. \tag{220}$$

The retained values are

$$\boxed{C_{\rm initial} = 0.918318} \qquad \text{and} \qquad \boxed{\Pi_{C,\rm initial} = 0.055327}. \tag{221}$$

F.58 $a_{\rm lock}$ and W

These determine the numerical matching of the resolved post-bounce history into CAMB.

The dependency is

$$\boxed{(a_{\rm lock}, W) \to E_{\rm CAMB}(a) \to r_s, \theta_*, \text{peak structure}}. \tag{222}$$

The retained values are

$$\boxed{a_{\rm lock} = 2.05 \times 10^{-4}} \qquad \text{and} \qquad \boxed{W = 0.67 W_0}. \tag{223}$$

F.59 Primordial parameters

The primordial quantities

$$A_s, \quad n_s, \quad \tau_{\rm reio} \tag{224}$$

control the initial perturbation amplitude, tilt, and reionization response.

They are matched between the locked and control runs.

Therefore

$$\boxed{\text{locked-control differences} \not\leftarrow \text{independent primordial retuning}}. \tag{225}$$

Part XV — Complete Dependency Tree

F.60 Physical-to-mathematical chain

The physical proposition is

$$\boxed{\text{wave-bearing gravity can organize and saturate}}. \tag{226}$$

This motivates

$$\boxed{C, \phi, V(C, \phi), \rho_s}. \tag{227}$$

These define

$$\boxed{\rho_C, p_C, \Gamma_{\rm tr}}. \tag{228}$$

Those determine

$$\boxed{\rho, p, H, \dot{H}}. \tag{229}$$

These produce

$$\boxed{a(t), H(a), C(a), \phi(a)}. \tag{230}$$

The expanding branch produces

$$\boxed{E_{\rm locked}(a)}. \tag{231}$$

CAMB then produces

$$\boxed{r_s, r_{\rm drag}, D_M, \sigma_8, C_\ell^{XY}, C_L^{\varphi\varphi}}. \tag{232}$$

The data comparison produces

$$\boxed{\chi^2_{\rm BAO}, \chi^2_{\rm CMB}, \chi^2_{\rm lens}}. \tag{233}$$

F.61 Early-universe dependency tree

$$\boxed{\rho_s + \Gamma_{\rm tr} + V(C, \phi)} \tag{234}$$

determine

$$\boxed{\text{contraction regulation}} \tag{235}$$

which determines

$$\boxed{K_{\rm bounce} + a_{\rm min}} \tag{236}$$

which determines

$$\boxed{\text{post-bounce radiation branch}} \tag{237}$$

which contributes to

$$\boxed{H(a) \text{ before recombination}} \tag{238}$$

which determines

$$\boxed{r_s,\ r_{\rm drag},\ \theta_*}. \tag{239}$$

F.62 Late-universe dependency tree

$$\boxed{V_\infty + C \to C_s + K^2 \to 0} \tag{240}$$

determine

$$\boxed{w_C \to -1} \tag{241}$$

which determines

$$\boxed{H(a) \text{ at late time}} \tag{242}$$

which determines

$$\boxed{D_M,\ D_H,\ D_V} \tag{243}$$

and

$$\boxed{D(a), \sigma_8, S_8} \tag{244}$$

which jointly determine

$$\boxed{\text{BAO, weak lensing, ISW, and CMB lensing}}. \tag{245}$$

F.63 Perturbation dependency tree

$$\boxed{C_{\rm bg}, \phi_{\rm bg}, V_{,CC}, V_{,\phi\phi}} \tag{246}$$

determine

$$\boxed{\delta C, \delta\phi} \tag{247}$$

which determine

$$\boxed{\delta\rho_C, \delta p_C, \delta q_C} \tag{248}$$

which enter

$$\boxed{\delta G_{\mu\nu} = 8\pi G \delta T_{\mu\nu}} \tag{249}$$

which contribute to

$$\boxed{TT, TE, EE, \text{growth, and lensing}}. \tag{250}$$

Near the lock, the coefficients are suppressed, making the direct effect small.

Part XVI — Result Cross-Reference Card

F.64 Background results

$$\boxed{a_{\rm min} \simeq 2.068 \times 10^{-5}}, \tag{251}$$

$$\boxed{z_{\rm bounce} \simeq 4.84 \times 10^{4}}, \tag{252}$$

$$\boxed{a_{\rm lock} = 2.05 \times 10^{-4}}, \tag{253}$$

$$\boxed{a_{\rm eq} \simeq 2.865 \times 10^{-4}}, \tag{254}$$

$$\boxed{w_C(0) \simeq -0.99988}. \tag{255}$$

F.65 Acoustic results

$$\boxed{100\theta_* = 1.041033}, \tag{256}$$

$$\boxed{r_{\rm drag} = 149.960 \text{ Mpc}}, \tag{257}$$

$$\boxed{z_* = 1089.691}, \tag{258}$$

$$\boxed{z_{\rm drag} = 1058.511}, \tag{259}$$

$$\boxed{D_M(z_*) = 14131.86 \text{ Mpc}}. \tag{260}$$

F.66 Growth results

$$\boxed{\sigma_8 = 0.779680}, \qquad \boxed{S_8 = 0.79781}, \tag{261}$$

and

$$\boxed{S_{8,\rm control} = 0.813037}. \tag{262}$$

The locked reduction is approximately

$$\boxed{1.87\%}. \tag{263}$$

F.67 Field results

$$\boxed{\max|\delta C| = 0.03514}, \tag{264}$$

$$\boxed{\max|\delta\phi| = 0.04860}, \tag{265}$$

$$\boxed{\pi_C = 0}, \tag{266}$$

$$\boxed{c_{s,C}^2 = c_{s,\phi}^2 = 1}. \tag{267}$$

F.68 Statistical results

$$\boxed{\chi^2_{\rm BAO} \simeq 13.4}, \tag{268}$$

$$\boxed{\chi^2_{\rm Planck\ lens,locked} = 10.7379}, \tag{269}$$

$$\boxed{\chi^2_{\rm Planck\ lens,control} = 12.0022}, \tag{270}$$

$$\boxed{\Delta\chi^2_{\rm lens} = -1.2643}. \tag{271}$$

F.69 Lensing results

$$\boxed{\left\langle C^{\varphi\varphi}_{L,\rm locked}/C^{\varphi\varphi}_{L,\rm control}\right\rangle = 1.0209} \tag{272}$$

and

$$\boxed{\left\langle N^{(0)}_{L,\rm locked}/N^{(0)}_{L,\rm control}\right\rangle = 1.213}. \tag{273}$$

Thus the model gives:

$$\boxed{2.09\% \text{ more weighted physical lensing power}} \tag{274}$$

and

$$\boxed{21.3\% \text{ more mean TT reconstruction noise}}. \tag{275}$$

Part XVII — Logical Consistency Map

F.70 Bounce consistency

$$\boxed{\rho = \rho_s \Rightarrow H = 0} \tag{276}$$

and

$$\boxed{\rho + p > 0 \Rightarrow \dot{H}_{\rm b} > 0}. \tag{277}$$

Therefore

$$\boxed{\dot{a}_{\rm b} = 0, \qquad \ddot{a}_{\rm b} > 0}. \tag{278}$$

F.71 Conservation consistency

$$\boxed{\text{field loss} = -Q} \qquad \text{and} \qquad \boxed{\text{radiation gain} = +Q}. \tag{279}$$

Therefore

$$\boxed{\nabla_\mu T^{\mu\nu}_{\rm total} = 0}. \tag{280}$$

F.72 Stability consistency

$$\boxed{K_{IJ} > 0}, \qquad \boxed{c_s^2 > 0}, \qquad \boxed{M^2_{\rm lock} > 0}, \qquad \boxed{V \geq V_\infty}. \tag{281}$$

Therefore the retained canonical field sector is stable around the locked state.

F.73 Spectator consistency

$$\boxed{C'_{\rm bg}, \phi'_{\rm bg} \to 0} \qquad \text{and} \qquad \boxed{V_{,C}, V_{,\phi} \to 0}. \tag{282}$$

Therefore

$$\boxed{\delta T^{C\phi}_{\mu\nu} \to 0} \tag{283}$$

for finite

$$\delta C, \delta\phi. \tag{284}$$

This matches the numerical result that the direct field contribution is negligible.

F.74 Observable consistency

The same function

$$H(a) \tag{285}$$

determines:

$$r_s, \quad r_{\rm drag}, \quad D_M, \quad D_H, \quad D_V, \quad D(a), \quad C_\ell^{XY}, \quad C_L^{\varphi\varphi}. \tag{286}$$

Therefore no final observable is calculated from an unrelated background.

The central observational rule is

$$\boxed{\text{one locked history} \to \text{all final cosmological observables}}. \tag{287}$$

Part XVIII — Final Equation Index

F.75 Core equation list

Potential

$$V = V_\infty + \frac{m_C^2}{2}(C - C_s)^2 + \frac{\lambda}{4}(C - C_s)^4 + A_0(1 - \cos\phi). \tag{288}$$

Field density

$$\rho_C = \frac{1}{2}(\Pi_C^2 + \Pi_\phi^2) + V. \tag{289}$$

Field pressure

$$p_C = \frac{1}{2}(\Pi_C^2 + \Pi_\phi^2) - V. \tag{290}$$

Transfer

$$\Gamma_{\rm tr} = W_-(H)[-3H + \Gamma_0 S_\rho]. \tag{291}$$

Transfer power

$$Q = \Gamma_{\rm tr}(\Pi_C^2 + \Pi_\phi^2). \tag{292}$$

Saturated Friedmann constraint

$$H^2 = \frac{\rho}{3}\left(1 - \frac{\rho}{\rho_s}\right). \tag{293}$$

Raychaudhuri equation

$$\dot{H} = -\frac{1}{2}(\rho + p)\left(1 - 2\frac{\rho}{\rho_s}\right). \tag{294}$$

Organizational field

$$\dot{C} = \Pi_C, \qquad \dot{\Pi}_C = -(3H + \Gamma_{\rm tr})\Pi_C - V_{,C}. \tag{295}$$

Phase field

$$\dot{\phi} = \Pi_\phi, \qquad \dot{\Pi}_\phi = -(3H + \Gamma_{\rm tr})\Pi_\phi - V_{,\phi}. \tag{296}$$

Matter

$$\dot{\rho}_m = -3H\rho_m. \tag{297}$$

Radiation

$$\dot{\rho}_r = -4H\rho_r + Q. \tag{298}$$

Organizational perturbation

$$\delta C'' + 2\mathcal{H}\delta C' + (k^2 + a^2 V_{,CC})\delta C + \frac{1}{2}C'_{\rm bg}h'_{\rm syn} = 0. \tag{299}$$

Phase perturbation

$$\delta\phi'' + 2\mathcal{H}\delta\phi' + (k^2 + a^2 V_{,\phi\phi})\delta\phi + \frac{1}{2}\phi'_{\rm bg}h'_{\rm syn} = 0. \tag{300}$$

Acoustic angle

$$\theta_* = \frac{r_s(z_*)}{D_M(z_*)}. \tag{301}$$

Growth

$$D'' + \left(2 + \frac{d\ln H}{dN}\right)D' - \frac{3}{2}\Omega_m D = 0. \tag{302}$$

BAO statistic

$$\chi^2_{\rm BAO} = \Delta^{\rm T}_{\rm BAO}C^{-1}_{\rm BAO}\Delta_{\rm BAO}. \tag{303}$$

Lensing statistic

$$\chi^2_{\rm lens} = \Delta^{\rm T}_{\rm lens}\Sigma^{-1}_{\rm lens}\Delta_{\rm lens}. \tag{304}$$

F.76 Final dependency statement

The complete theory can be read in one direction:

$$\boxed{C, \phi, V, \rho_s, \Gamma_{\rm tr}} \tag{305}$$

determine

$$\boxed{\rho, p, H, \dot{H}} \tag{306}$$

which determine

$$\boxed{a(t), H(a), C(a), \phi(a)} \tag{307}$$

which determine

$$\boxed{\delta C, \delta\phi, \delta T_{\mu\nu}} \tag{308}$$

and

$$\boxed{r_s, r_{\mathrm{drag}}, D_M, D_H, D_V, D(a)} \tag{309}$$

which determine

$$\boxed{TT, TE, EE, \sigma_8, S_8, C_L^{\varphi\varphi}} \tag{310}$$

which determine the final statistical comparisons.

The theory can also be tested in reverse.

A failed observable can be traced backward to the background, perturbation source, transition, or parameter responsible for it.

F.77 Closing cross-reference statement

The gravitational-saturation model is one coupled mathematical structure.

The bounce is not independent of the high-density constraint.

The radiation branch is not independent of the contraction transfer.

The late residual is not independent of the field potential.

The acoustic scale is not independent of the locked expansion history.

The reduced growth amplitude is not independent of the same history.

The CMB lensing result is not independent of the growth and distance kernels.

The small direct field contribution is not an assumption; it follows from the locked background coefficients and is confirmed numerically.

The complete dependency structure is therefore:

$$\boxed{\text{gravitational organization} \to \text{saturation} \to \text{bounce} \to \text{lock} \to \text{observable cosmic history}}. \tag{311}$$

This index completes the internal cross-reference framework of the manuscript.

End of Appendix F

References and Bibliography

The following references document the principal published observational, computational, and theoretical foundations explicitly identified in the manuscript, including Planck 2018, CAMB, PlanckLens, DESI BAO, galactic scaling relations, gravitational-wave observations, Type Ia supernova work, inhomogeneous cosmology, and nonsingular cosmology.

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Bibliography correction: this section was added to the deposited manuscript to supply the omitted scholarly reference list. The model equations, locked parameter values, numerical results, appendices, and scientific conclusions in the preceding manuscript are unchanged.